Liberty

ARRANGER

BUILD | CONTROL | MANAGE

UI CREATOR

User Manual

Liberty's Arranger Control system (UI Creator) is a stand alone web based control system to be used in applications where DigiIP AV over IP systems are NOT present. If a DigiIP system IS present, please refer to the control system options of the DigiIP version of Arranger.Arranger control can control third party devices via IP or via RS232, IR, contact closure, sensor, and more when using Global Cache. Presets and events can be created and executed, and custom designed user interfaces (UI's) can be created to used on any device with a browser such as tablets, phones, or computers. QR codes can also be created giving an easy path to a custom UI or to execute a preset.

Rev 251029

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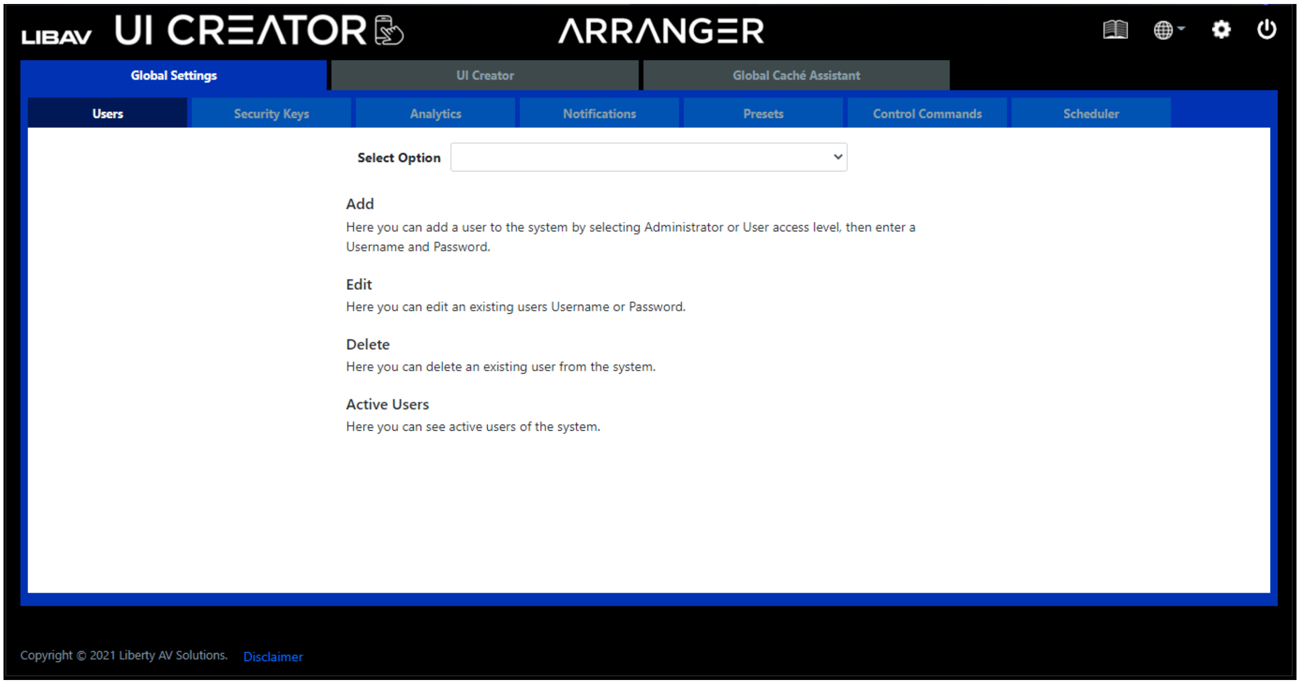
Global Settings

In the *Global Settings* tab you will find all the global settings of the software.

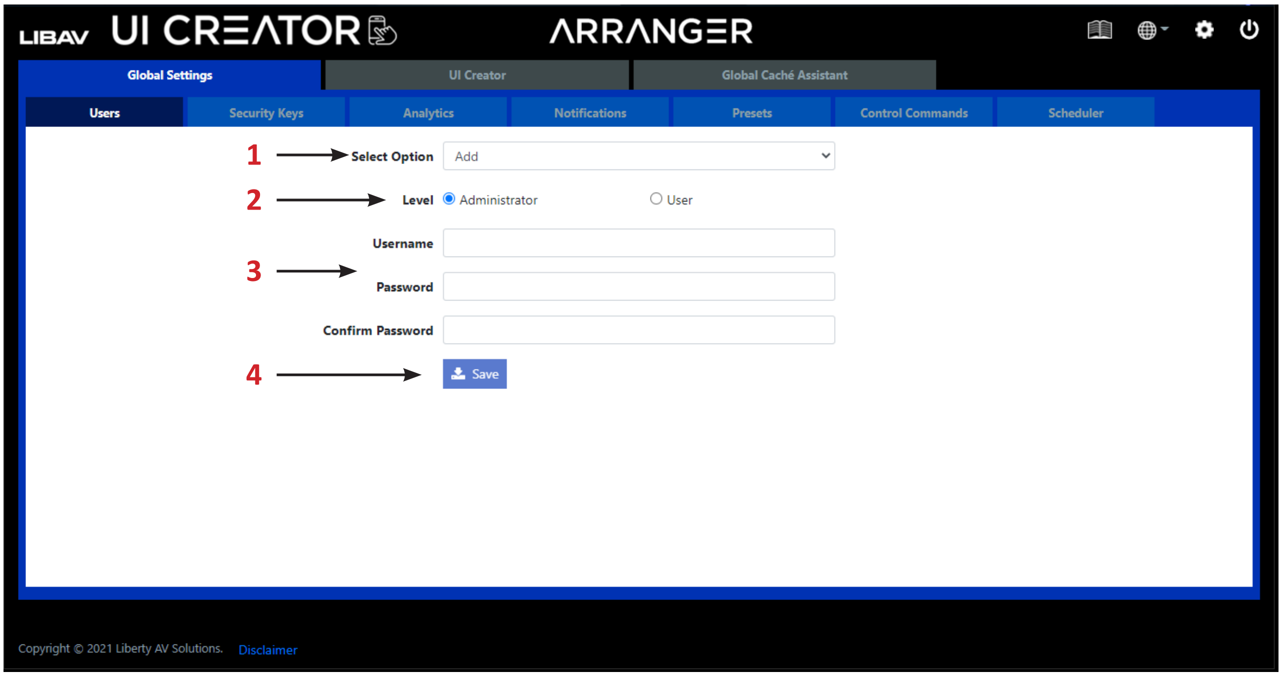
Users

The system can be configured for user access control. Two levels of system access are available, *administrator* and *user*. *Administrator* logins receive all menu options while the *user* login only receives the UI Creator and Global Cache Assistant menus.



Add User

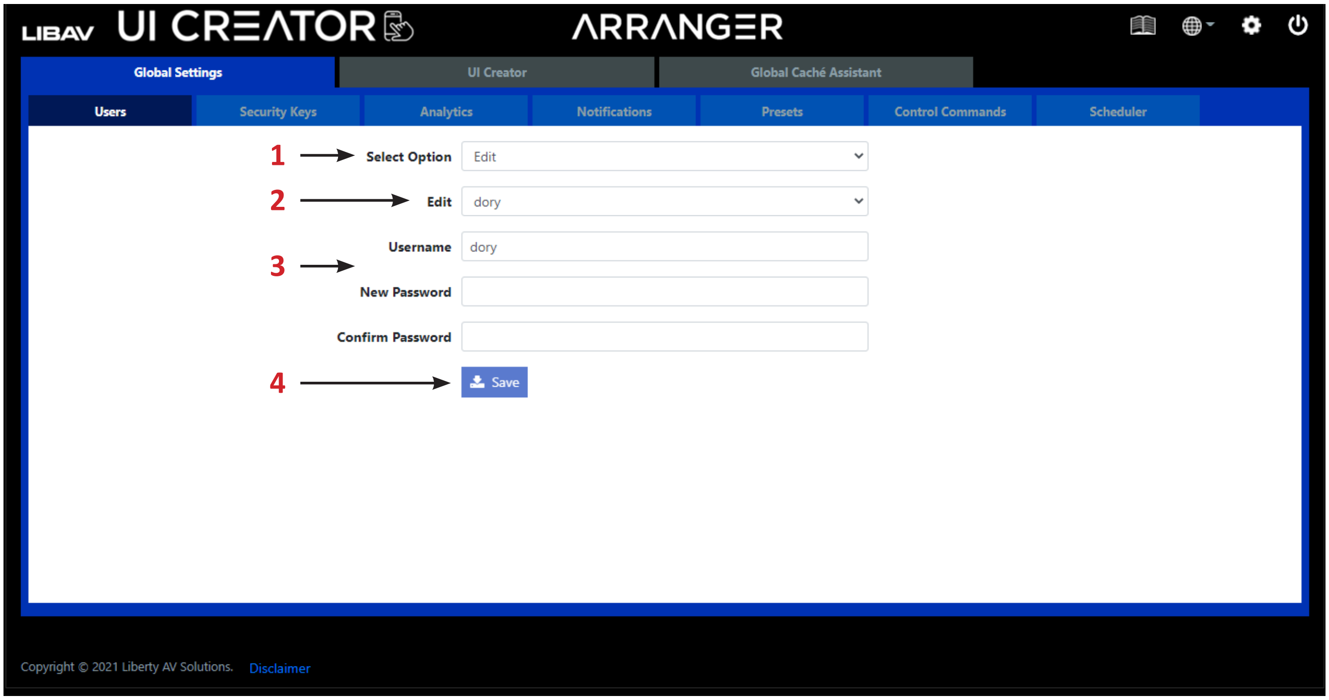
Here you can add a user to the system by selecting ***Administrator*** or ***User*** access level, then enter a name and password for the new user.



1. Select *Add.*
2. Select *Level.*
3. Enter in *username, password* and then *confirm password.*
4. Click *Save.*

Edit User

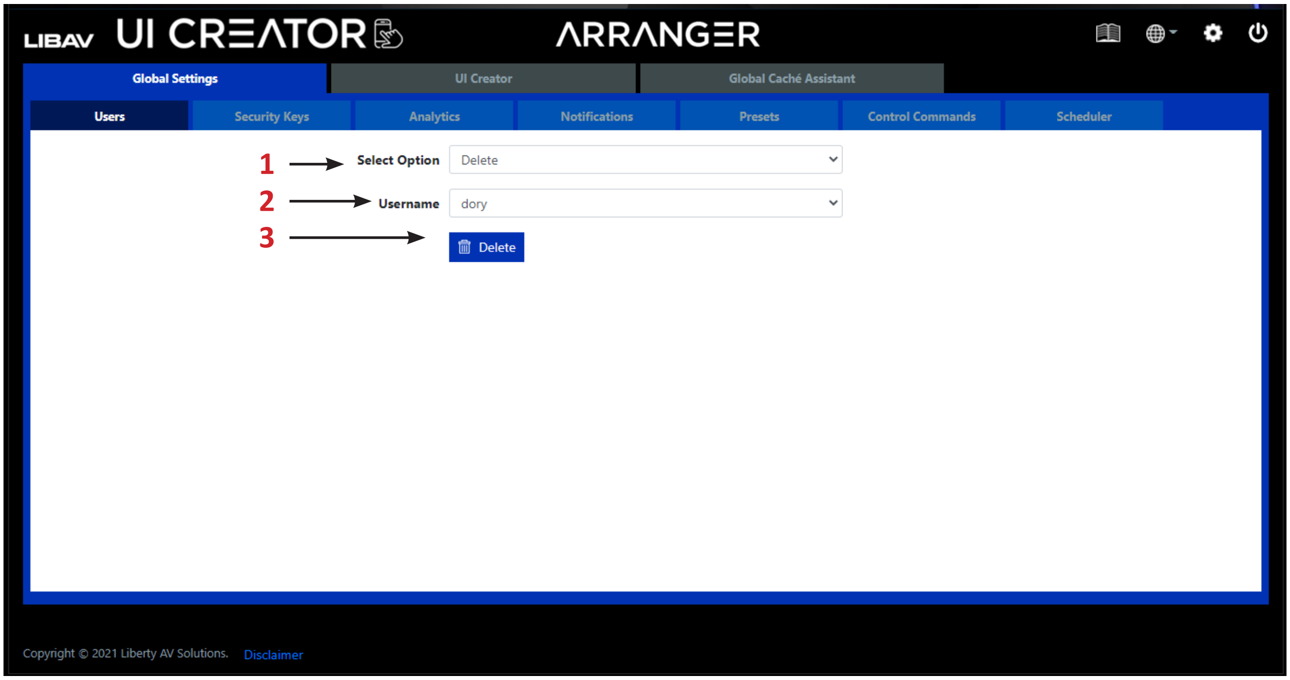
Here you can edit an existing user's username and password.



1. Select *Edit.*
2. Select *User.*
3. Enter in *new password* and then *confirm password.*
4. Click *Save.*

Delete User

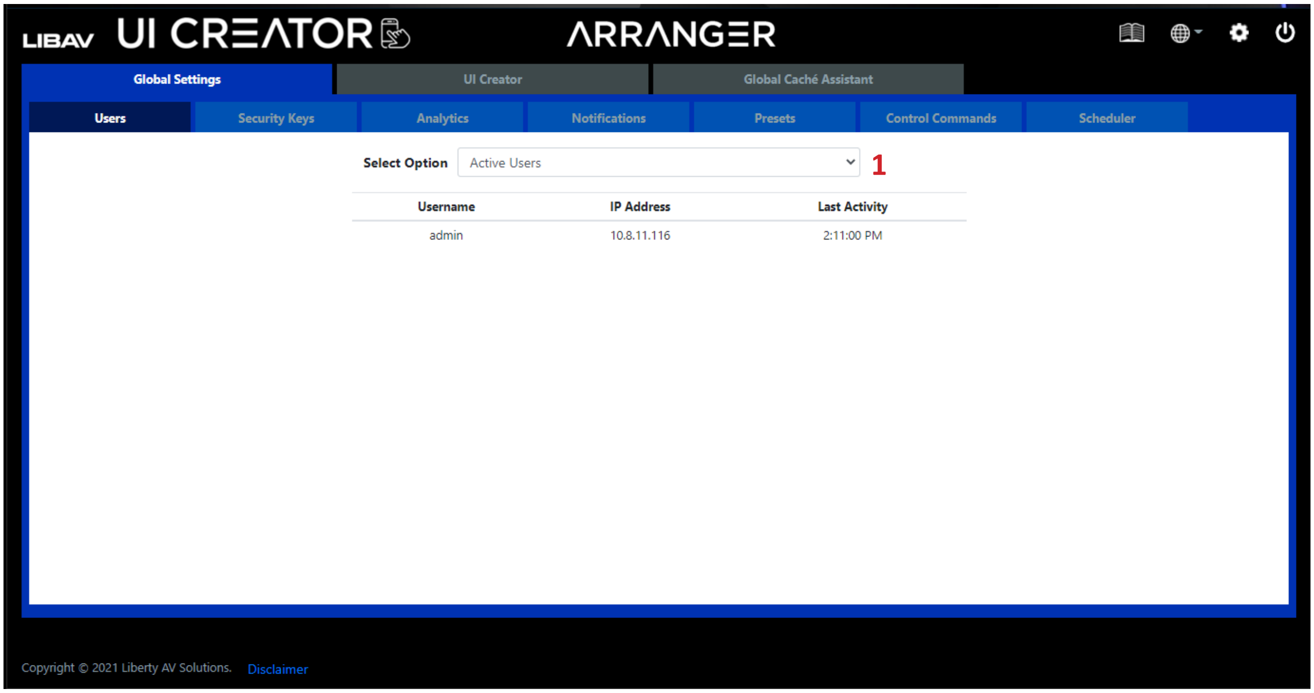
Here you can delete an existing user from the system.



1. Select *Delete.*
2. Select *Username.*
3. Click *Delete.*

Active Users

Here you can see all the active users logged into the system and the time of their last activity.



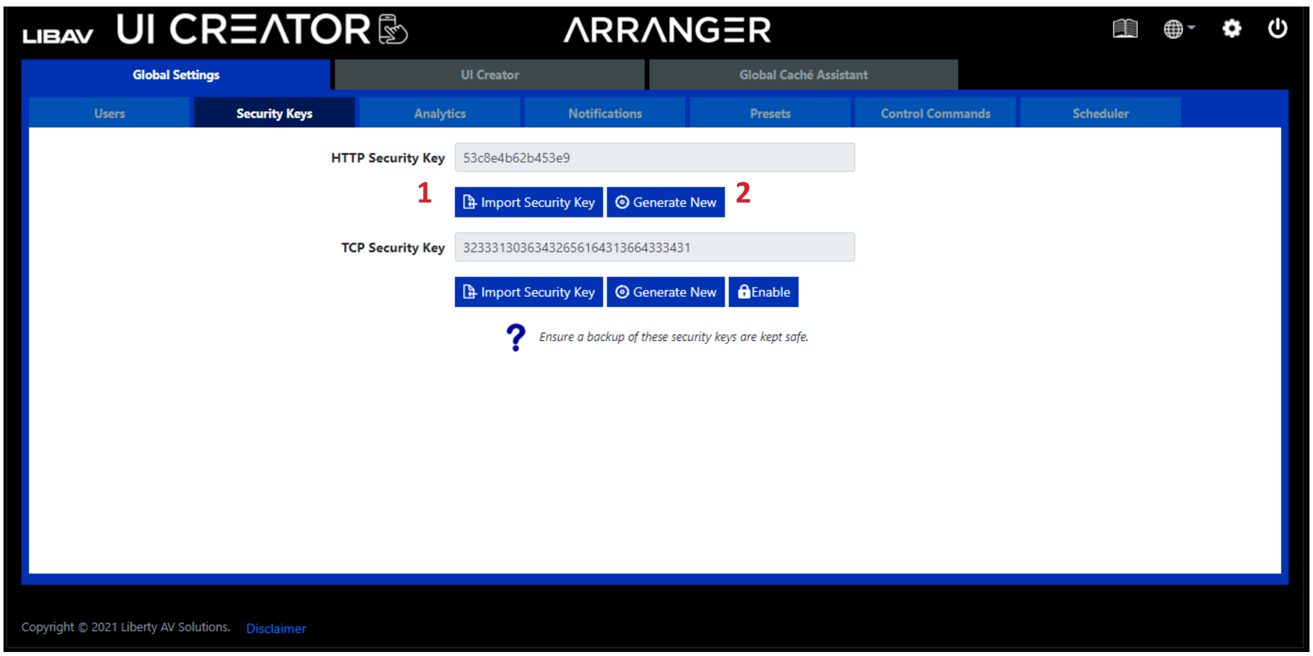
1. Select *Active Users.*

Security Keys

Security keys are required with all HTTP level requests and optional for TCP commands on port 6980. Only keys generated from the software can be used.

HTTP API Security Key

The Arranger Controller API can be accessed via HTTP GET and POST requests. To ensure security over the network a HTTP security key is required to be passed with all such requests. Here you can generate a new key or import a saved key that had been previously generated.



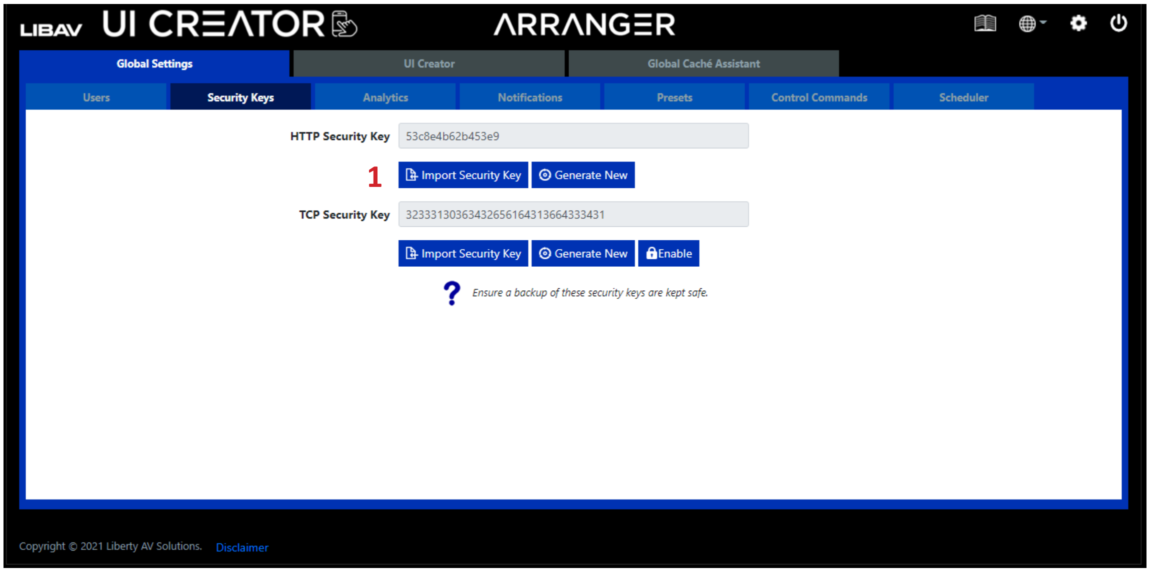
1. Select *Import Security Key* button to import a new key.

OR

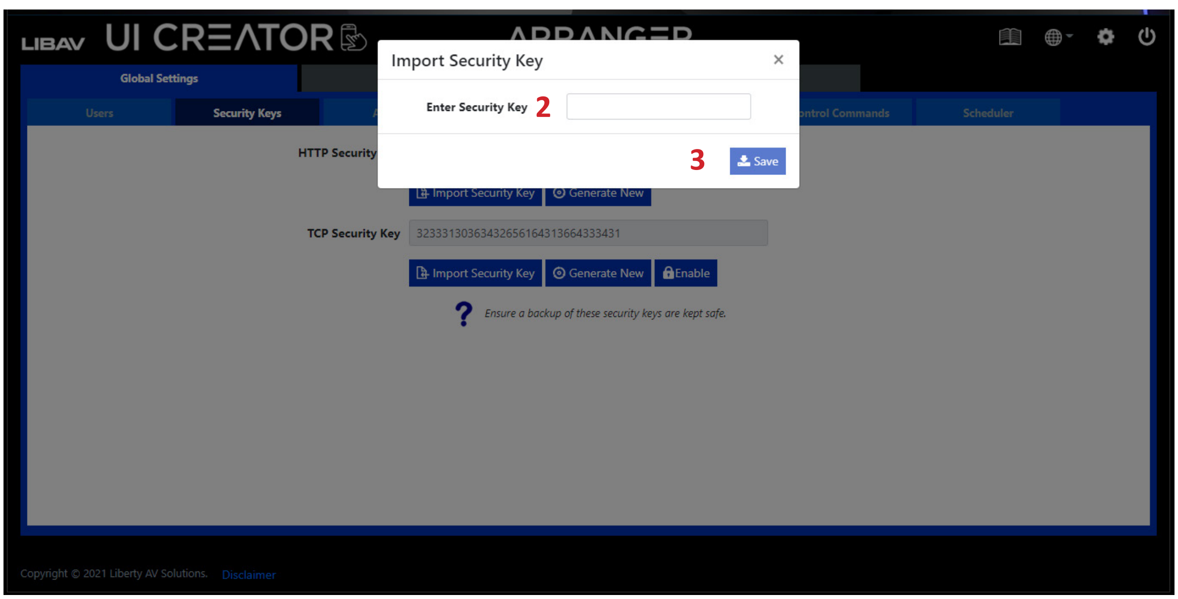
2. Select *Generate New* button to allow Arranger to generate a new key.

Security Keys continued.....

Importing a HTTP API Security Key



1. Select *Import Security Key* button.

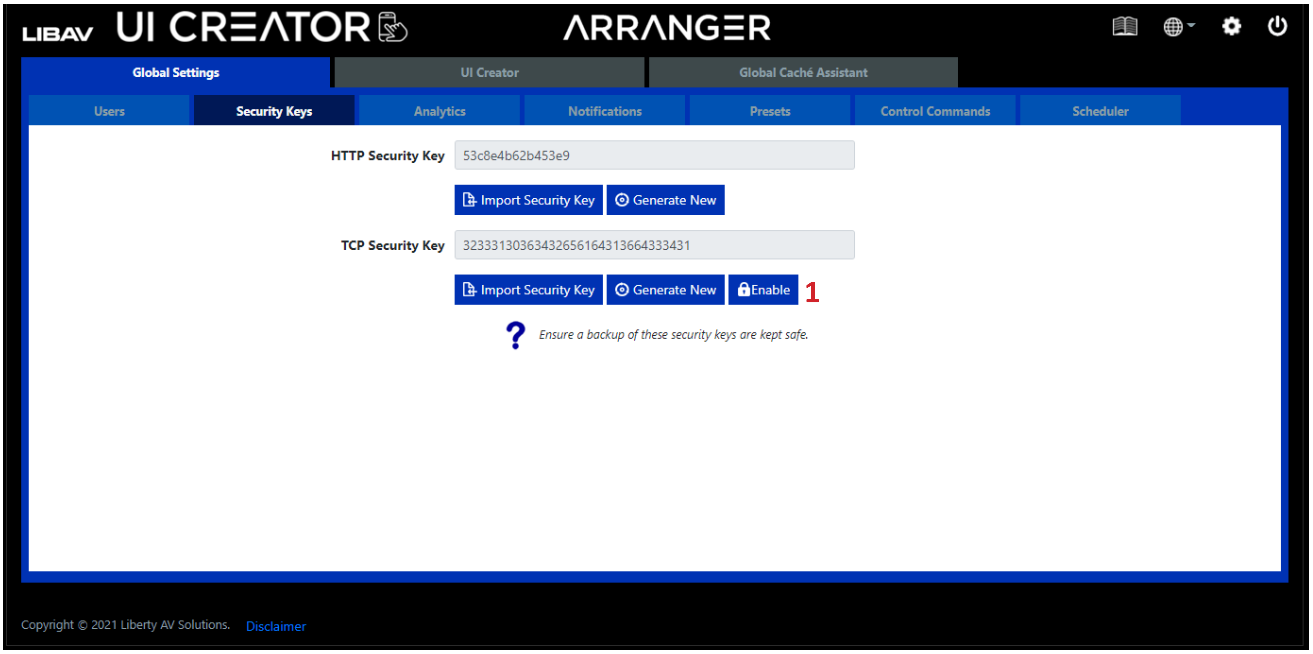


2. Enter Security Key.
3. Click *Save.*

Security Keys continued.....

TCP Security Key

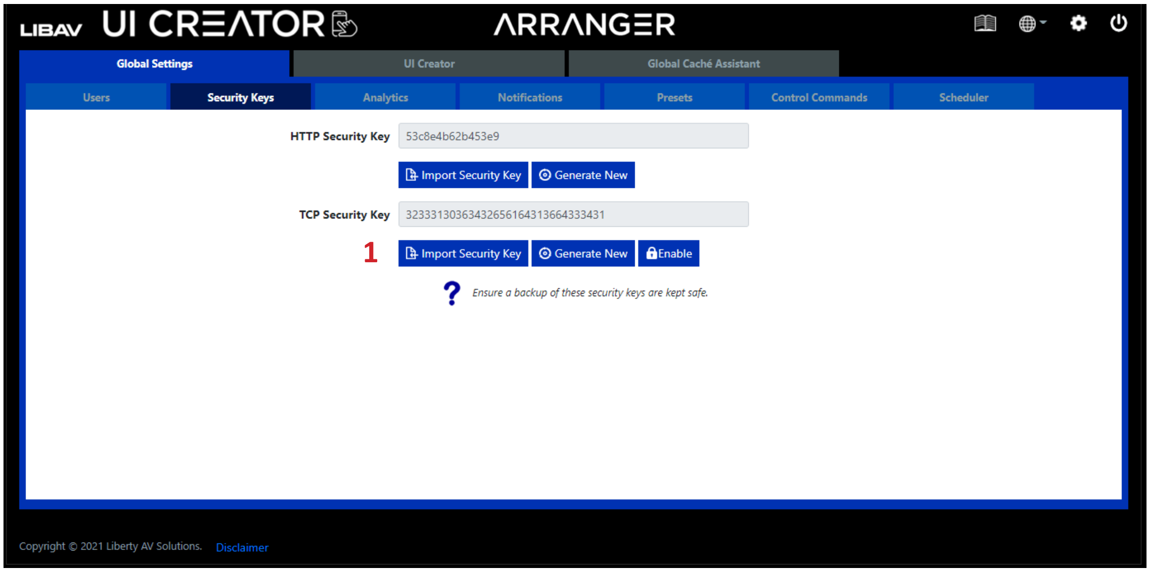
The Arranger Controller API can be accessed via Telnet requests on TCP port 6980. To ensure security over the network a TCP security key can be passed with all such commands. Here you can generate a new key or import a saved key that had been previously generated. As the TCP security key is optional its use can be enabled or disabled from here.



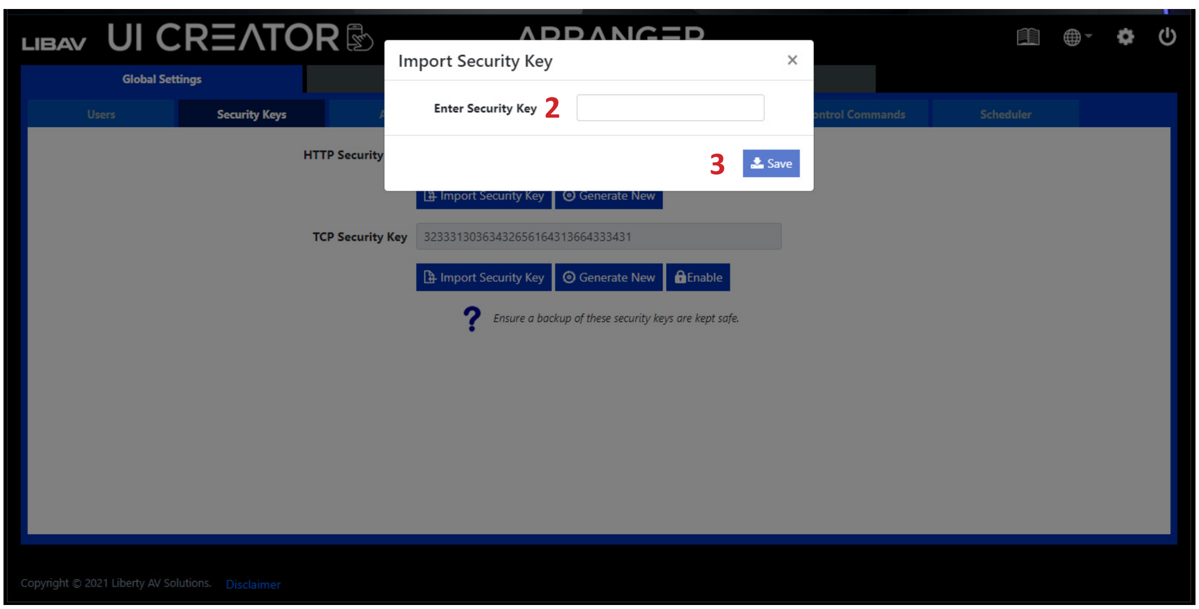
1. Click *Enable* to enable the TCP security key option.

Security Keys continued.....

Importing a TCP API Security Key



1. Select *Import Security Key* button.

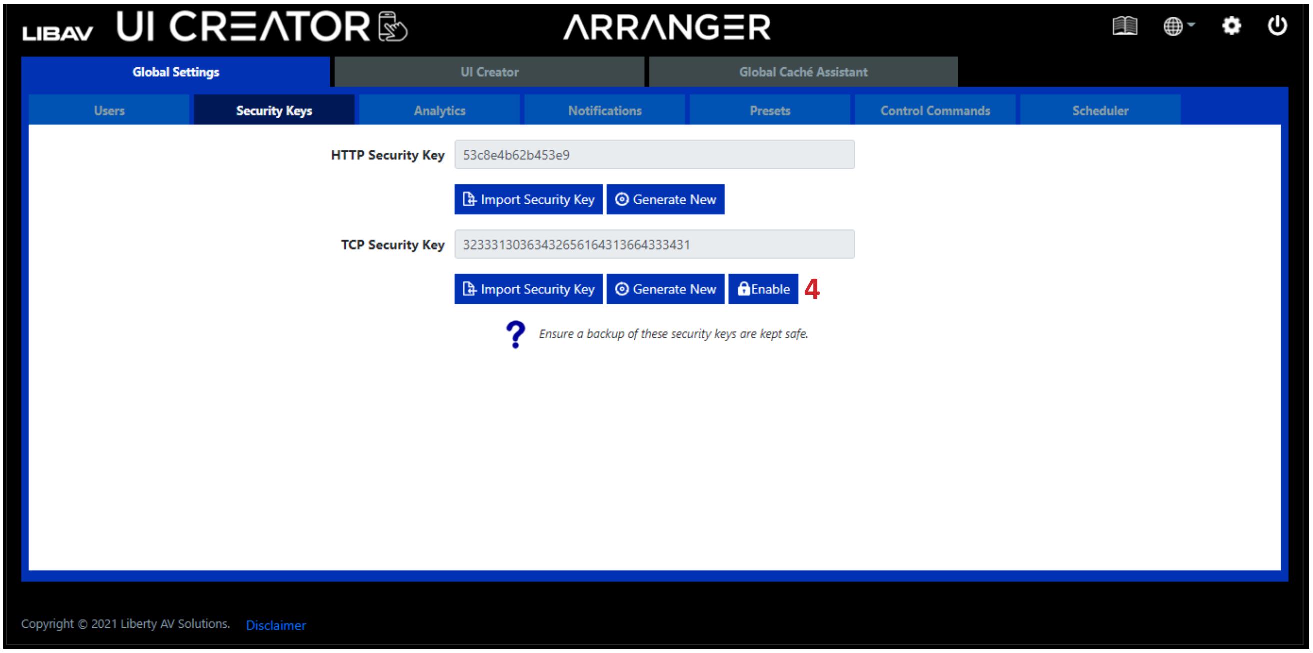


2. Enter Security Key.

3. Click *Save.*

Security Keys continued.....

Importing a TCP API Security Key Continued....



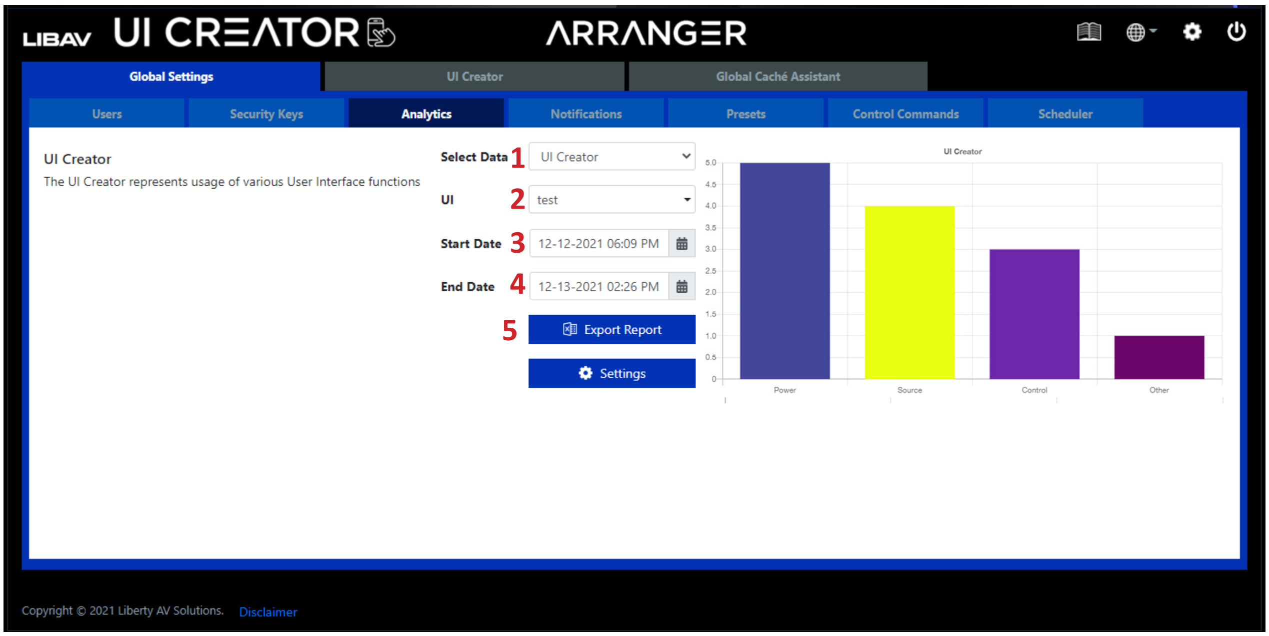
4. Click *Enable.*

Analytics

Analytical data is constantly being stored on the system. By default data will be maintained for 1 month, but this can be changed up to 12 months.

UI Creator

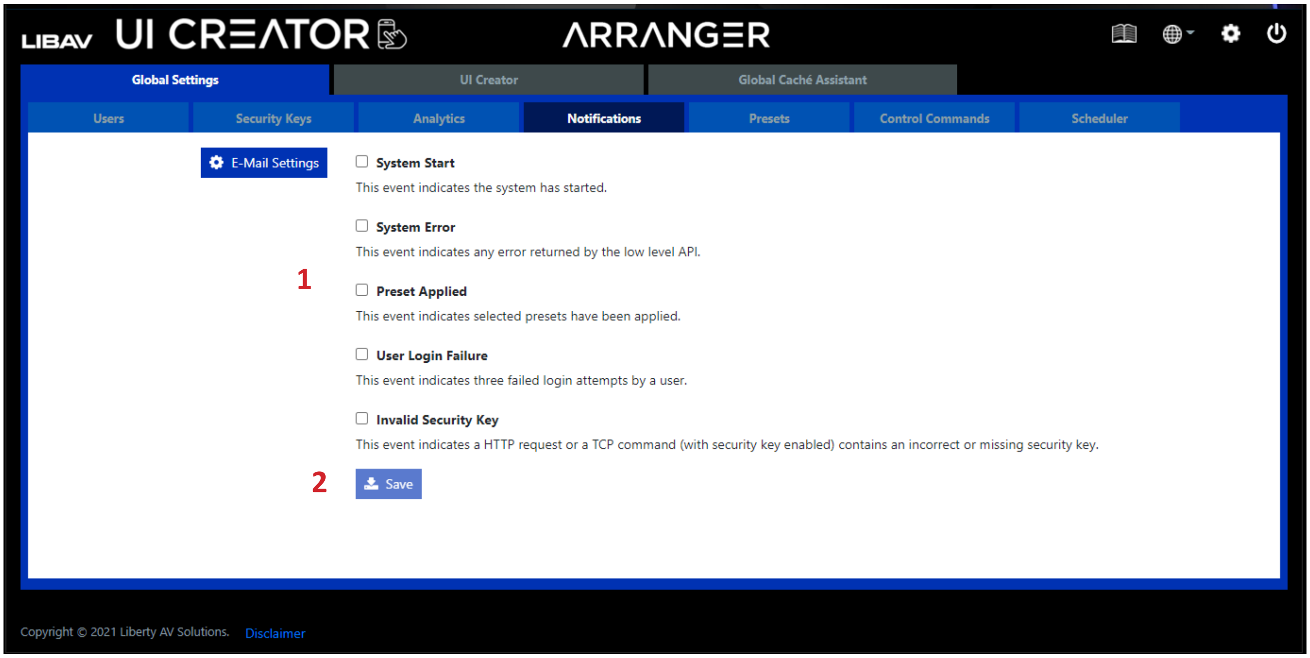
The *UI Creator* metric represents usage of various user defined interface functions.



1. Select *UI Creator* from the *Select Data* drop-down.
2. Select *User Interface.*
3. Select *Start Date.*
4. Select *End Date.*
5. Click *Export Report.*

Notifications

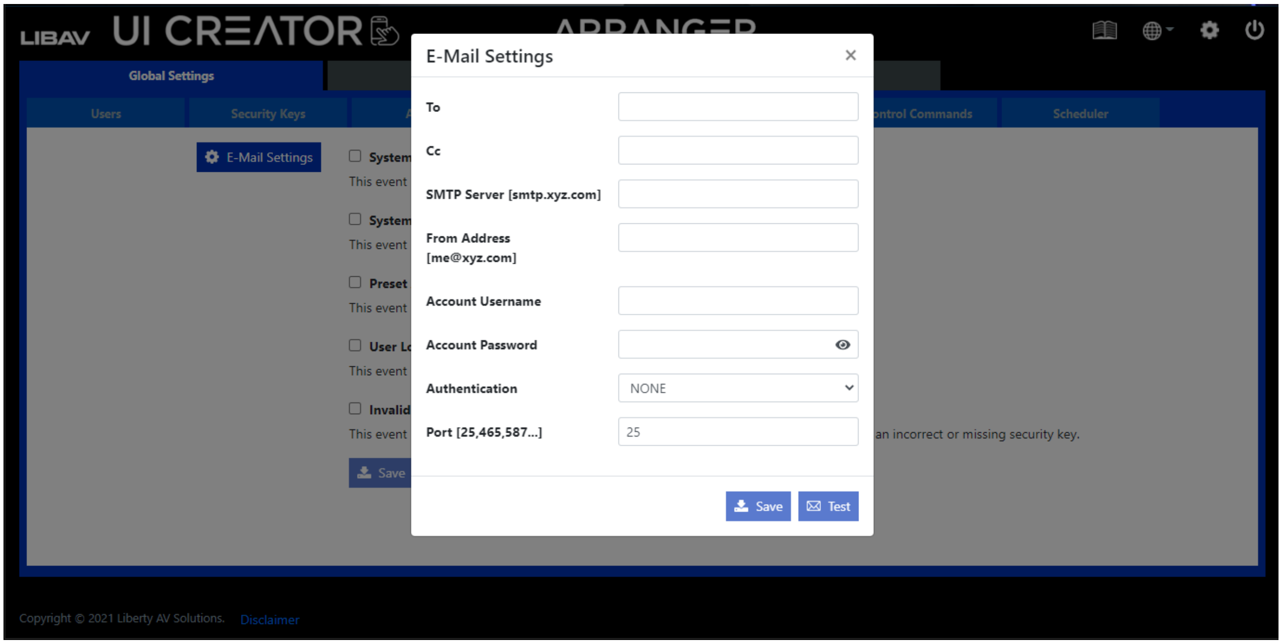
Notifications will send e-mail alerts whenever a selected event occurs on the system.



1. Select Event.
2. Select *Save.*

Email Settings

Here you configure the e-mail client to allow notification alerts to be sent from a specified e-mail account. The *Test* button sends a confirmation e-mail to confirm the settings are correct.

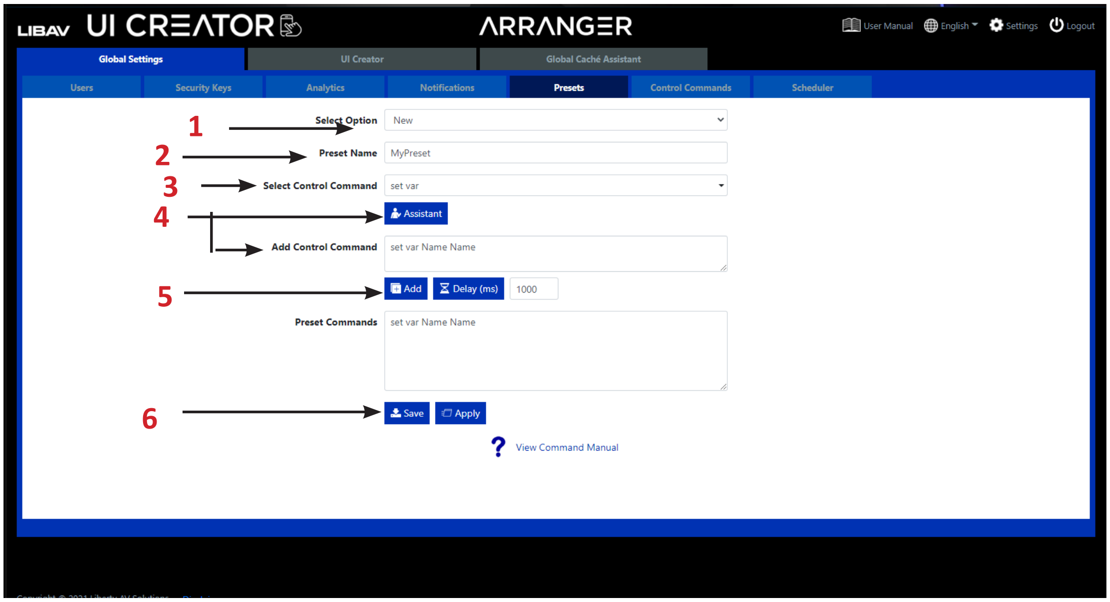


Presets

The system can store a virtually unlimited number of presets. A preset can be applied with a single ***preset load*** command. The preset can contain a virtually unlimited number of commands.

Make a New Preset

Here you can create a new preset to be stored on the system. Give the preset a name and then start adding control commands as required by either entering commands directly or using the *Assistant* wizard.



1. Select New
2. Enter a name for the preset.
3. Select a command from the drop-down menu.

   Note: Click on the View Command Manual link next to the ? for command definitions.

4. Assistant is a command wizard that will allow you to easily input the information required for the command in order to render the finished code correctly.

   Click on Assistant to build the command or enter the command in manually in the Add Control Command field.

5. Click Add.

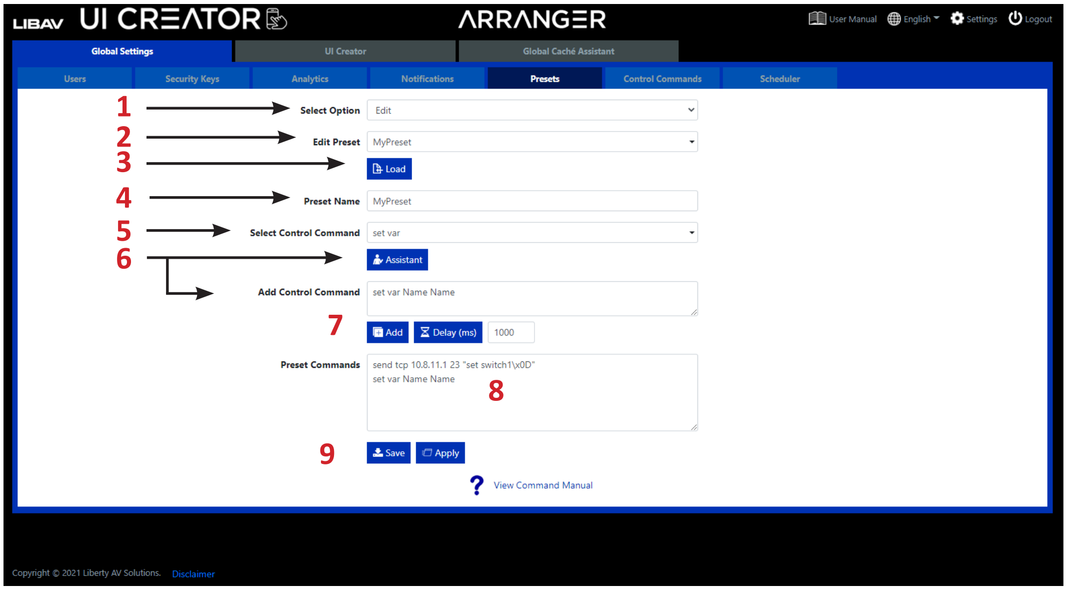
   Notice that the command is now entered into the field labeled Preset Commands. If you need to enter in more commands to the preset repeat steps 3-5.

   Best practice! When using multiple commands in a preset, add the preset delay command to the preset. Preset delay will allow you to separate each command execution in a preset in milliseconds.

6. Click Save or Apply to test the preset.

Edit a Preset

Here you can edit any existing preset by adding, deleting or changing control commands as required.



1. Select *Edit.*

2. Select *Preset.*

3. Click *Load.*

   Notice that the commands in the current preset have now populated in the *Preset Commands* field which can be edited manually.

4. Change *preset name.*

5. Select a system command from the *Select Control Command* drop-down menu.

   Note: Click on the *View Command Manual* link next to the ? for command definitions.

6. *Assistant* is a command wizard that will allow you to easily input the information required for the command in order to render the finished code correctly.

   Click on *Assistant* to build the command or enter the command in manually in the *Add Control Command* field.

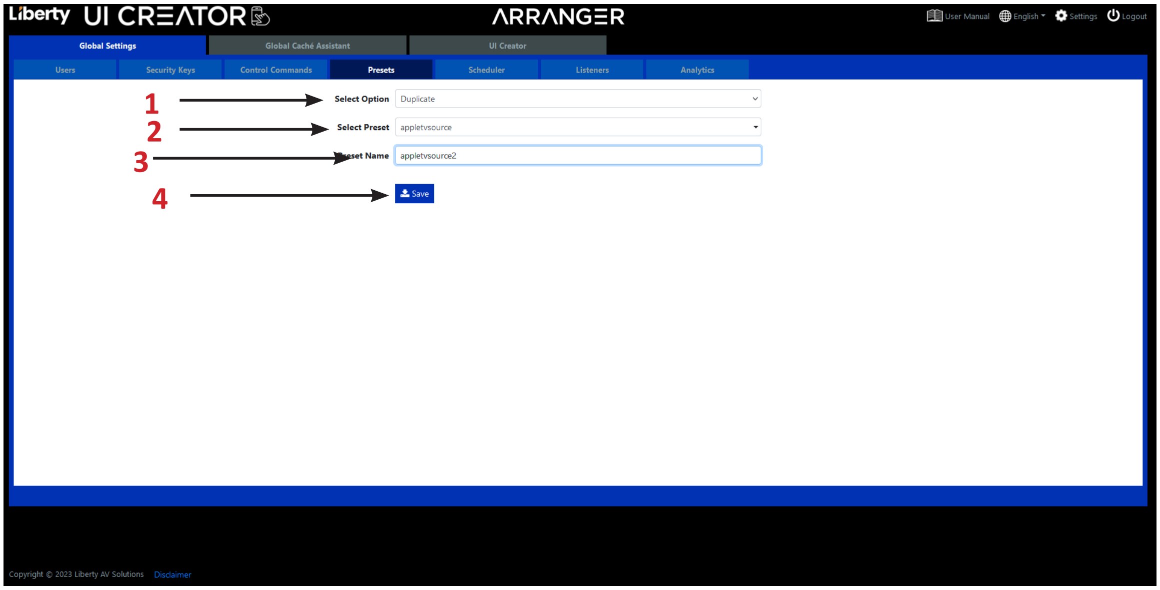
7. Click *Add.*

8. Commands can be edited manually here.

9. Click *Save* or *Apply* to test the edited preset.

Duplicate a Preset

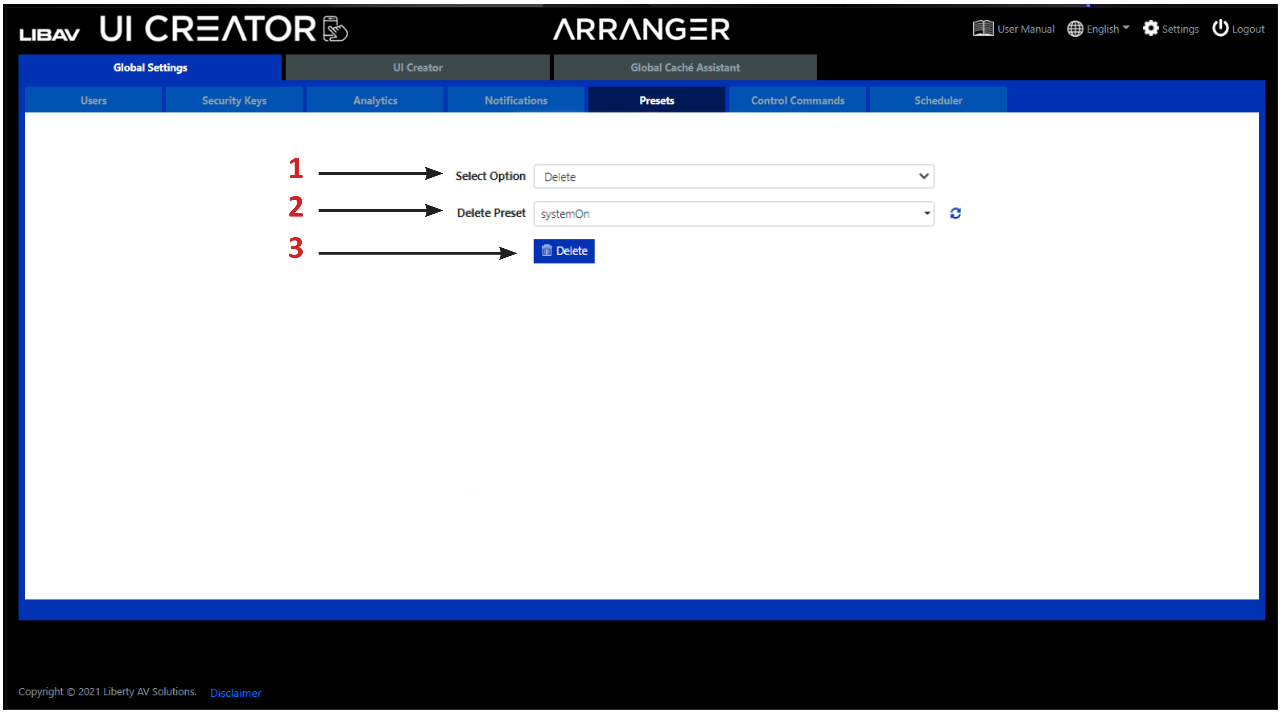
Here you can duplicate any existing preset.



1. Select *Duplicate*
2. Select *Preset.*
3. Type in the new name of the preset
4. Press Save once done.

Delete a Preset

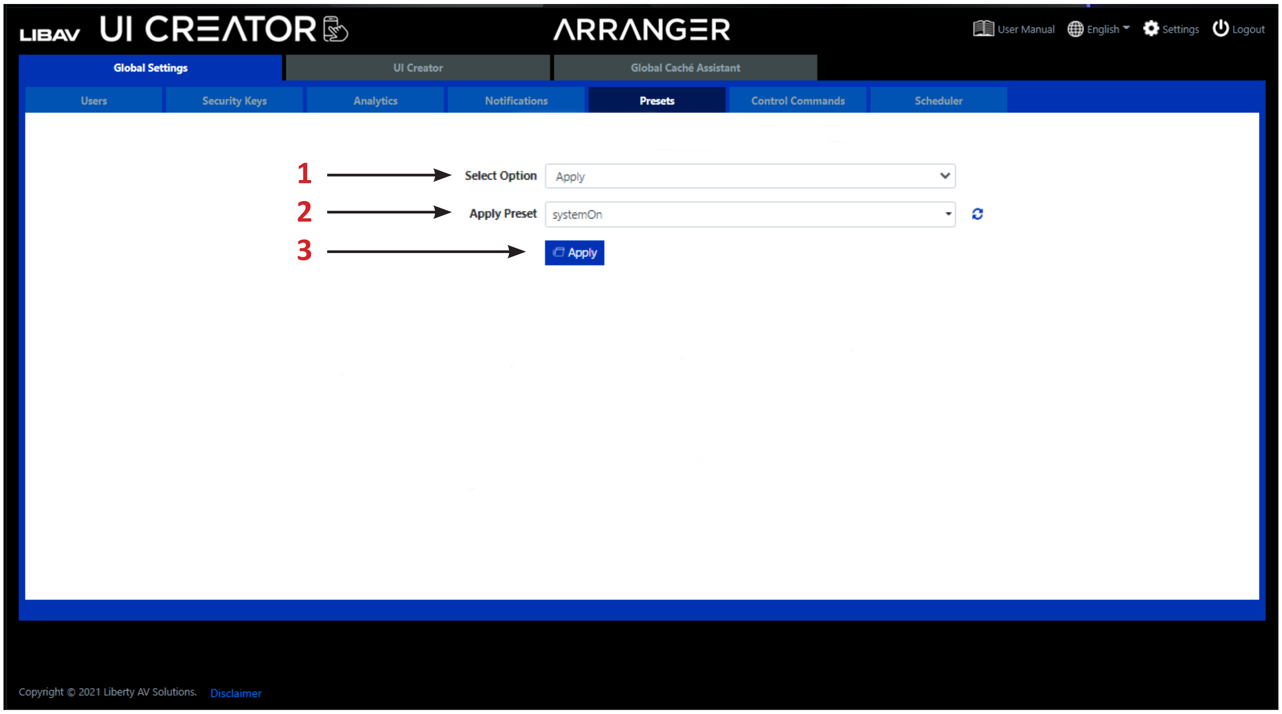
Here you can delete any existing preset from the system.



1. Select *Delete.*
2. Select *Preset.*
3. Click *Delete.*

Apply a Preset

Here you can apply any existing preset on the system.



1. Select *Apply.*

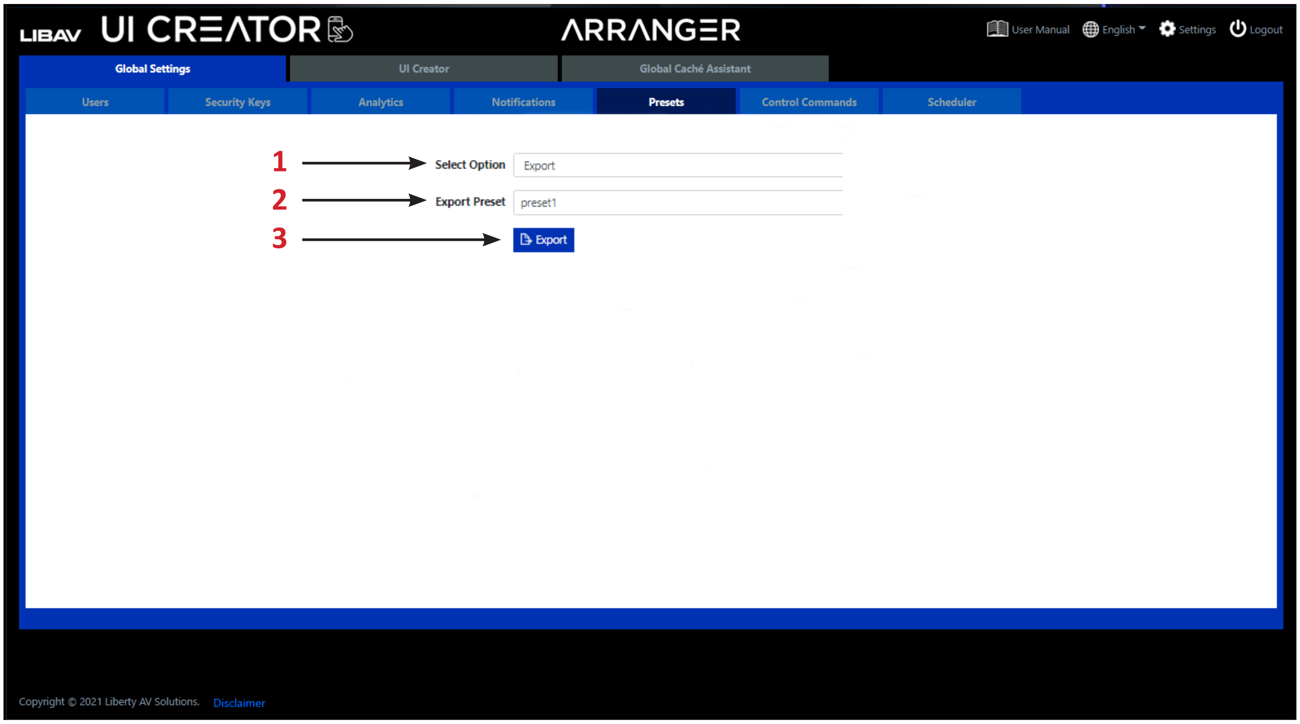
2. Select *Preset.*

3. Click *Apply.*

NOTE: To initiate a *Preset* using API commands use the command ***preset load {presetname}.***

Export a Preset

Here you can export an existing preset on the system which can then be used as a backup or edited. The preset will be saved to your Downloads folder on your PC as an ini file, example; *preset1.ini.* The export preset can be edited with an application like Notepad++; right click the file and select "Open with..."



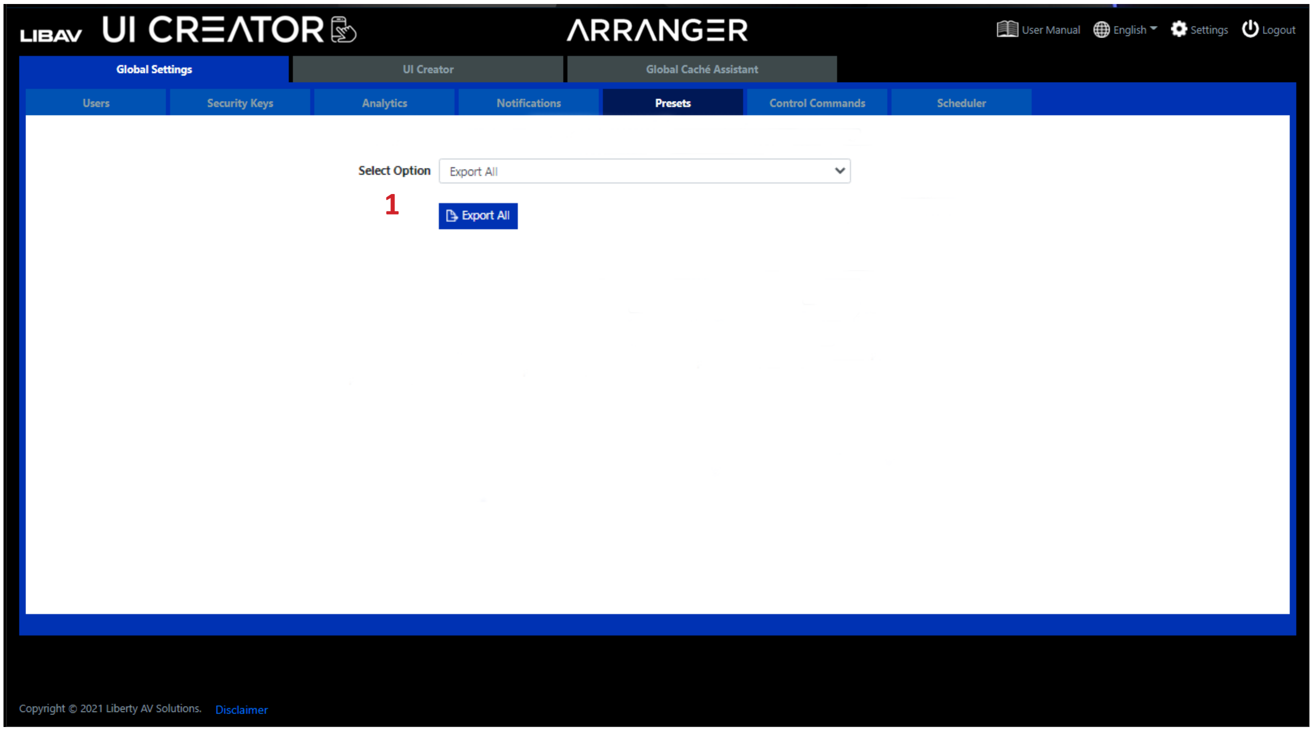
1. Select *Apply.*

2. Select *Preset.*

3. Click *Export.*

Export ALL Presets

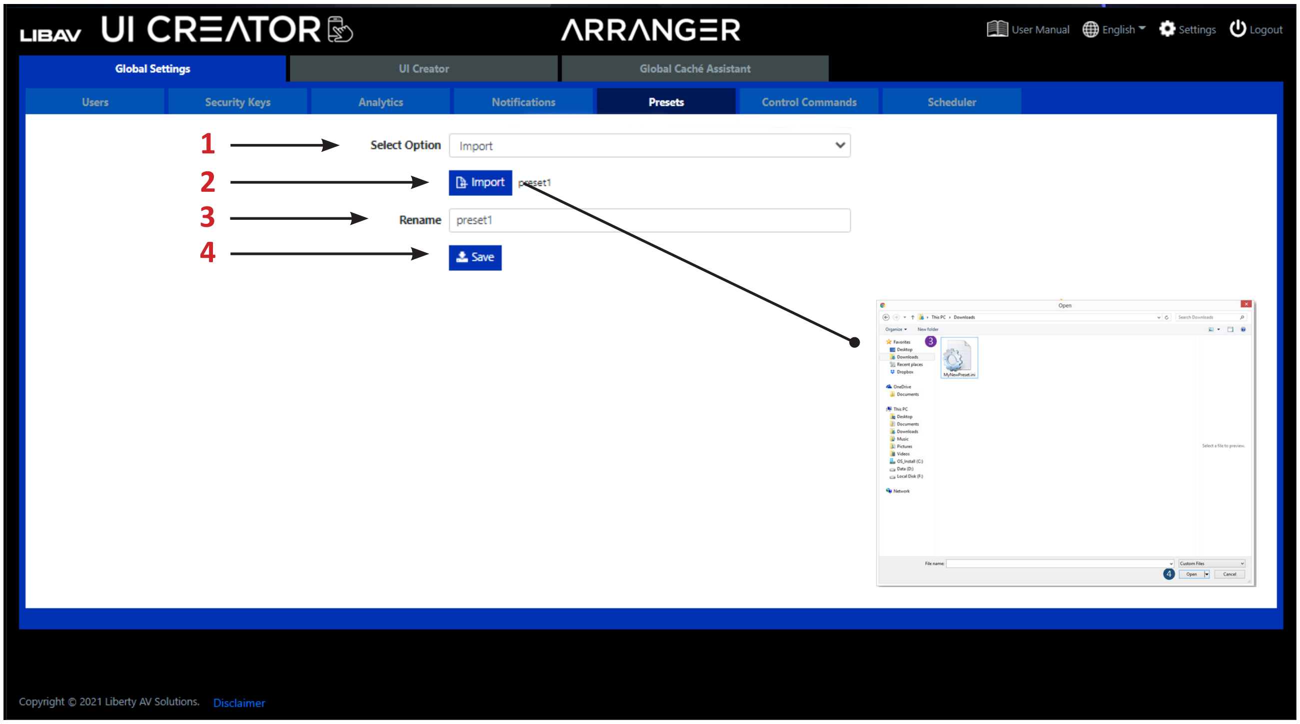
Here you can export all existing presets on the system which can then be used as a backup or edited. The presets will be saved to your Downloads folder as an exp file:presets.exp.



1. Click *Export All.*

Import a Preset

Here you can import a preset into the system.



1. Select *Import.*

2. Click *Import.*

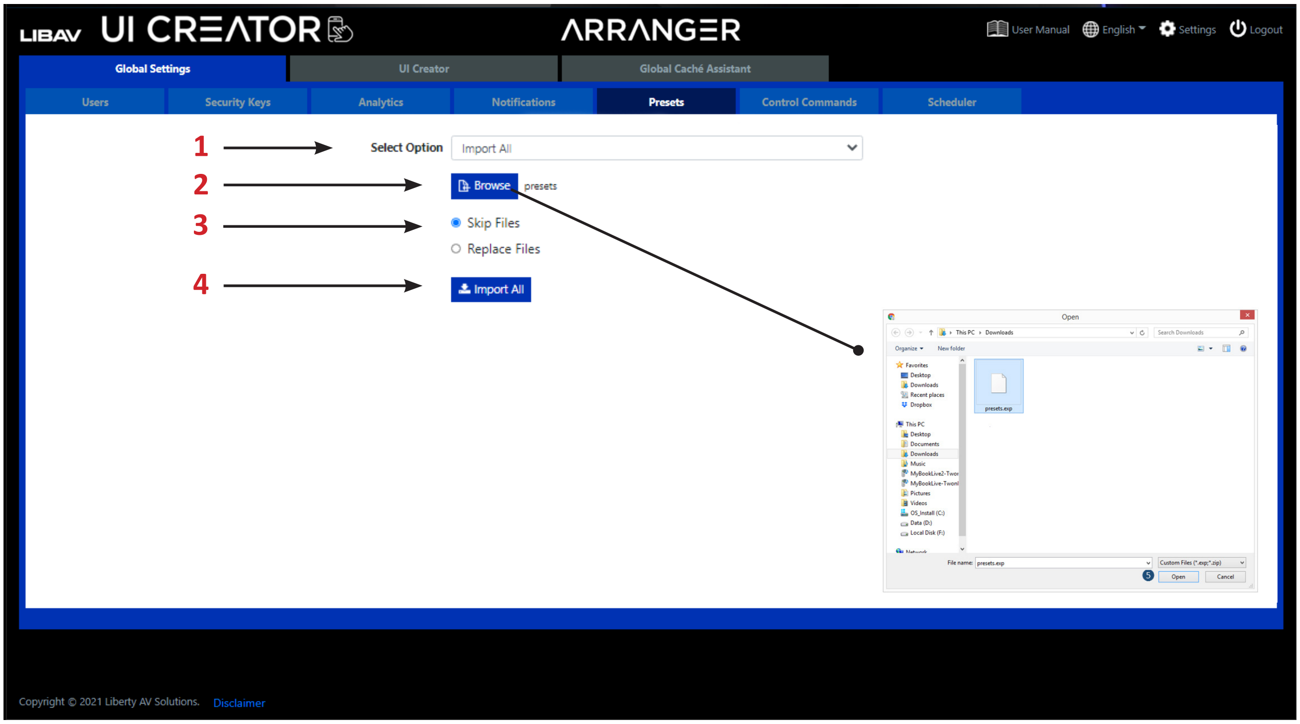
   A pop-up window on your PC will appear; find the appropriate .ini file to import from your PC.

3. Rename the preset if required.

4. Click *Save.*

Import ALL Presets

Here you can import all presets into the system from an all preset export.



1. Select *Import ALL.*

2. Click *Browse.*

   A pop-up window on your PC will appear; find the appropriate .exp file to import from your PC.

3. Select *Skip files* or *Replace Files.*

4. Click *Import All.*

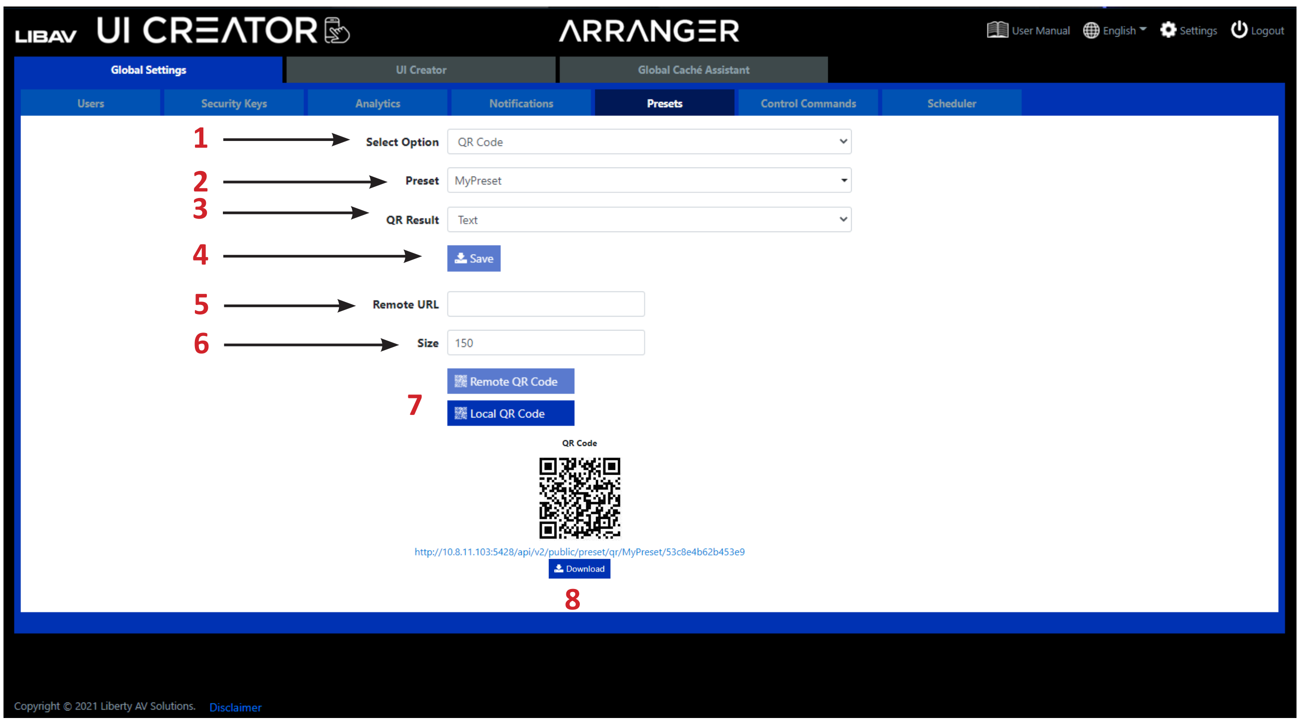
QR Code Preset

You can create a QR code to directly execute a preset. After the QR code has been scanned and opened in a browser the preset will be executed and the result displayed in the user's default web browser.

Here you can set how the result of a preset when scanned from a QR code will be displayed on the user's browser. This includes QR code buttons used in UI Creator.

The QR result can be *Text* for a standard API text response. Select *Static Image* to display a user uploaded image on success or failure of the preset. Or, select *User Interface* to be redirected to a *QR Results User Interface*. See *Global Settings > UI Creator > QR Code Result mode* for more information.

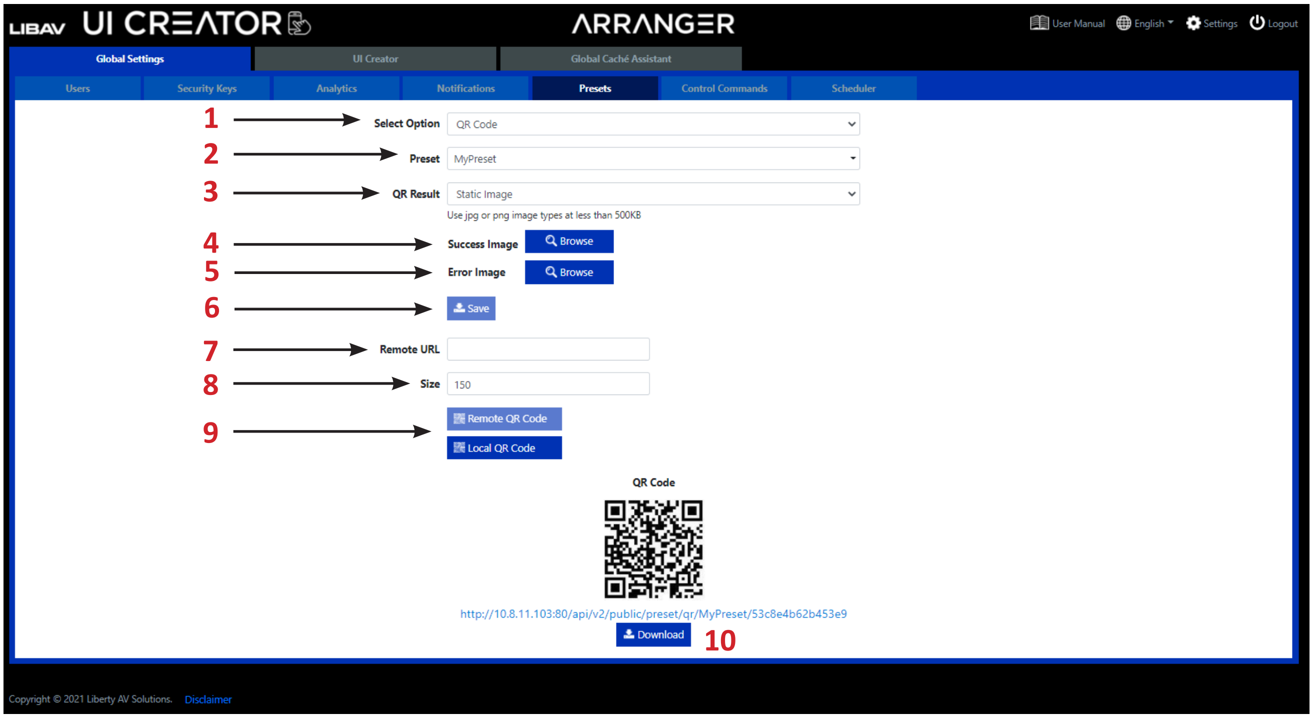
The below will provide a text response: *preset load preset1 success.*



1. Select *QR Code.*
2. Select *Preset.*
3. Select *Text* from the *QR Result* drop-down menu.
4. *Click Save button.*
5. Enter *Remote URL* if QR must be accessed over the Internet.
6. Enter *QR Graphics* Size.
7. Click either *Remote or Local QR Code* button.
8. Click *Download* button.

QR Code Preset continued....

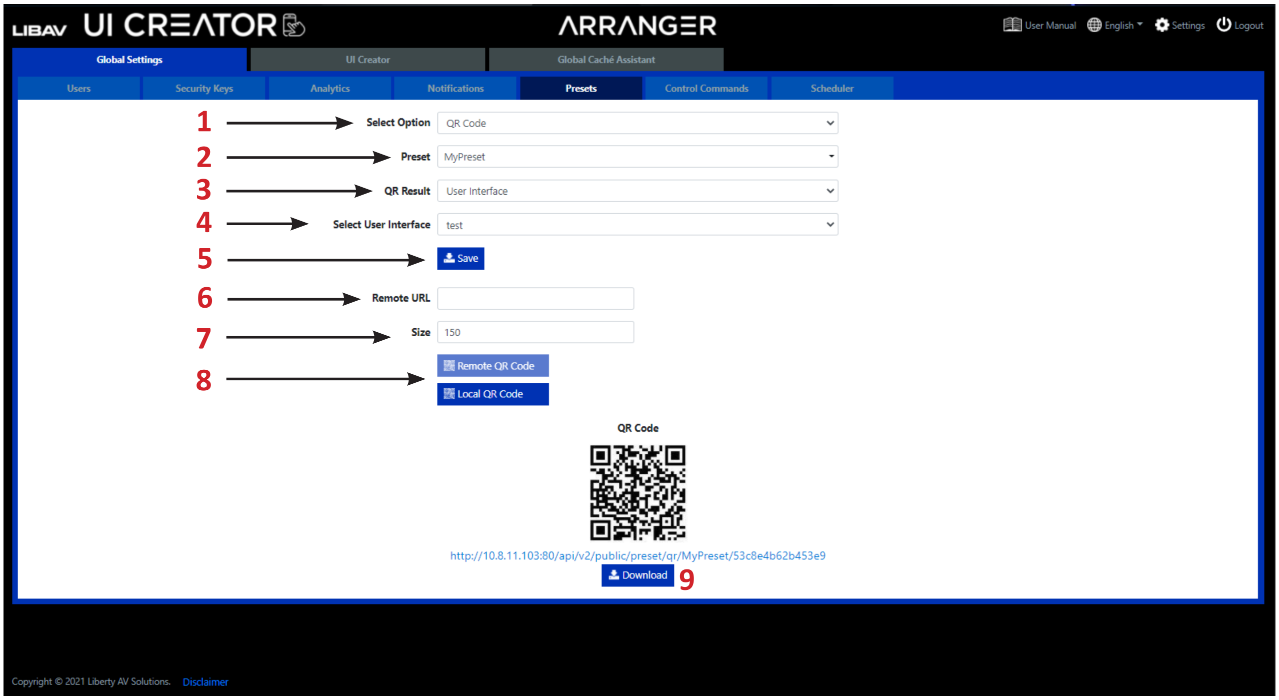
The below will provide a static image response:



1. Select *QR Code.*
2. Select *Preset.*
3. Select *Static Image* from the *QR Result* drop-down menu.
4. Select *Success Image.*
5. Select *Error Image.*
6. *Click Save button.*
7. Enter *Remote URL* if QR must be accessed over the Internet.
8. Enter *QR Graphics* Size.
9. Click either *Remote* or *Local QR Code* button
10. Click *Download* button.

QR Code Preset continued....

The below will provide a QR Results user interface response:



1. Select *QR Code.*

2. Select *Preset.*

3. Select *User Interface* from the *QR Result* drop-down menu.

4. Select *User Interface.*

5. *Click Save button.*

6. Enter *Remote URL* if QR must be accessed over the Internet.

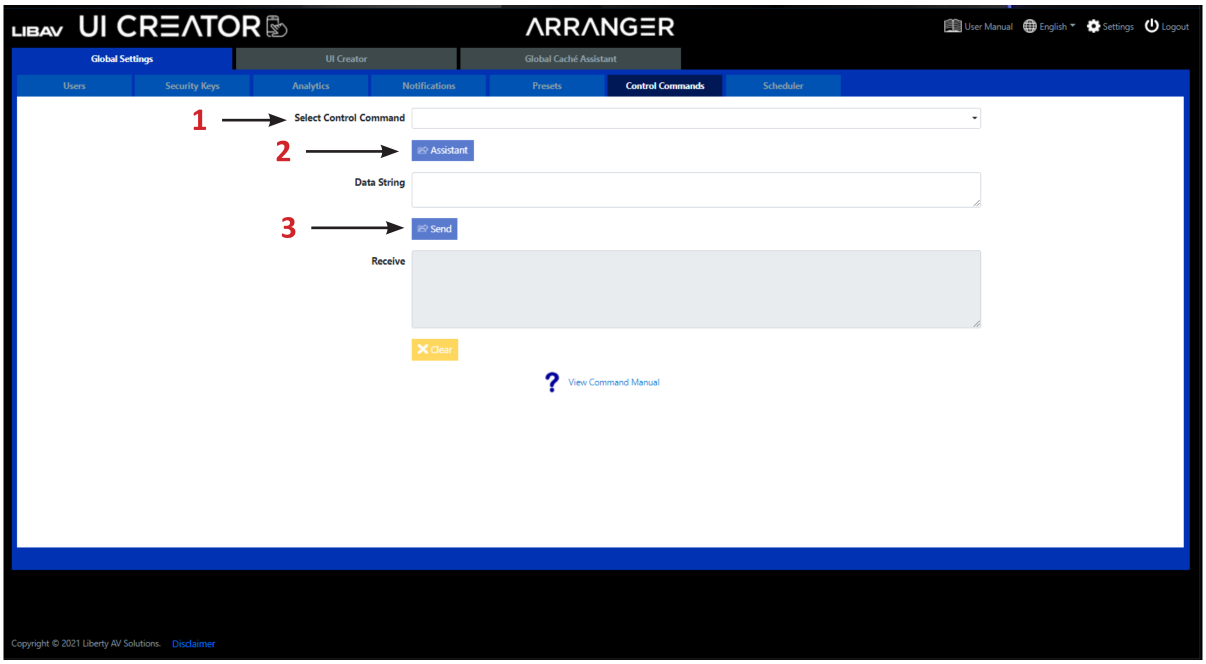
7. Enter *QR Graphics* Size.

8. Click either *Remote* or *Local QR Code* button.

9. Click *Download* button.

Control Commands

The *Control Command* is used to send any of the API commands available to the system for testing purposes.



1. Select a command from the *Select Control Command* drop-down menu.

   Note: Click on the *View Command Manual* link next to the ? for system command definitions.

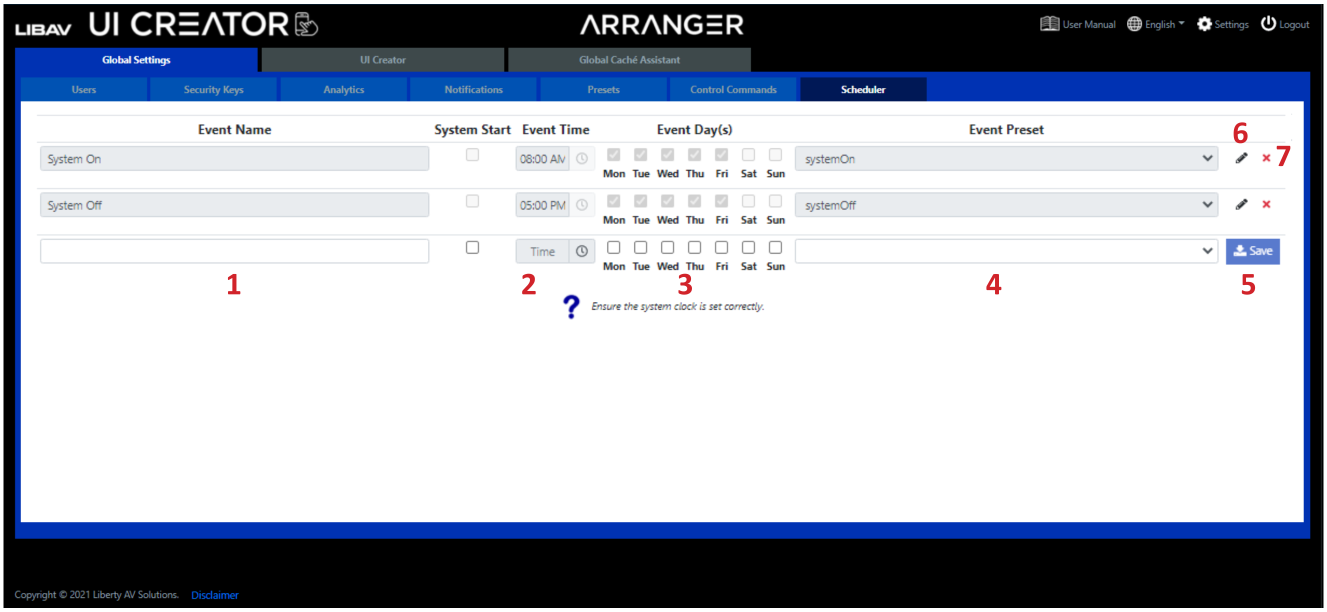
2. *Assistant* is a command wizard that will allow you to easily input the information required for the command in order to render the finished code correctly.

   Use *Assistant* to configure the command or change the parameters manually in the *Data String* field surrounded with "<" & ">"

3. Click *Send.*

Scheduler

The scheduler is used to apply presets at a particular time of a given day or on system start.



1. Enter *Event Name.*
2. Select a time.
3. Select day(s).
4. Select *Preset.*
5. Click *Save.*
6. Once saved, the pen icon is used to edit the event.
7. Click the cross icon to delete the event

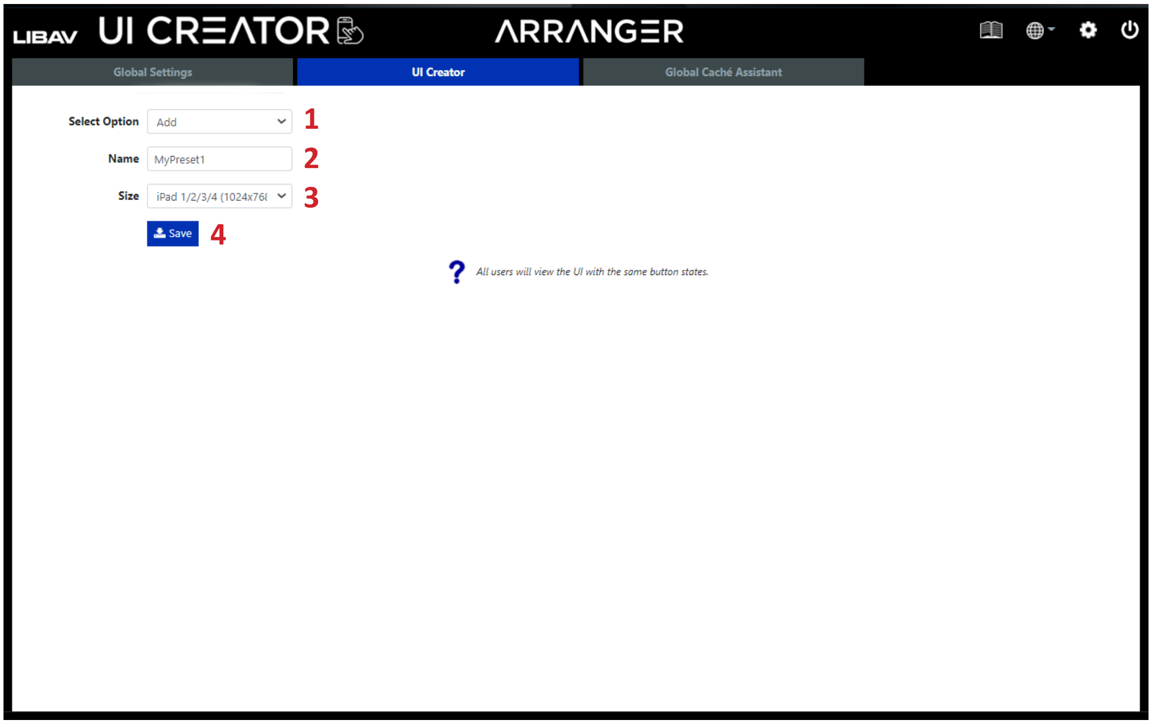
UI Creator

UI Creator can be used to fully control the functions of the system and much more. Here you can design your own user interface (UI) to recall functions that have been saved as presets.

UI Creator lets you create a virtually unlimited number of UI's which can be viewed on any device's supported web browser such as Google Chrome or Safari. Demo UI's of custom pin code and polycom phone system are included.

Create a New User Interface

Here you can add a new UI to the system ready to be edited as required. The UI name must be specified along with the UI resolution. UI and Element name can not exceed 20 characters.A selection of standard-sized displays are available or user can enter their own size from 100x100 to 3820x2160.

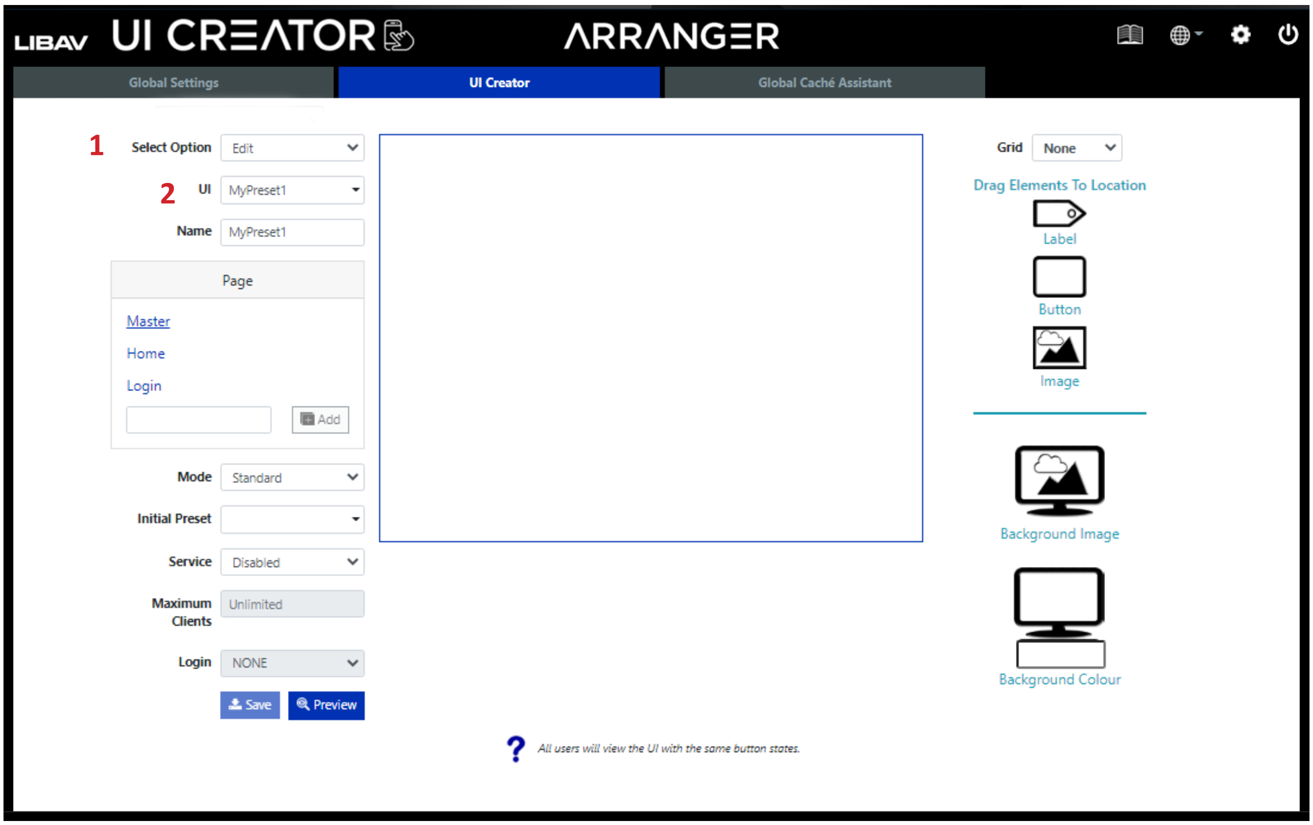


1. Select *Add.*
2. Enter UI *Name.*
3. Select UI *Size.*
4. Click *Save.*

You can now *edit* this preset to build the UI.

Edit a User Interface

Here you can change the UI name or edit and preview an existing UI on the system. The UI service and login requirements can also be set from here.

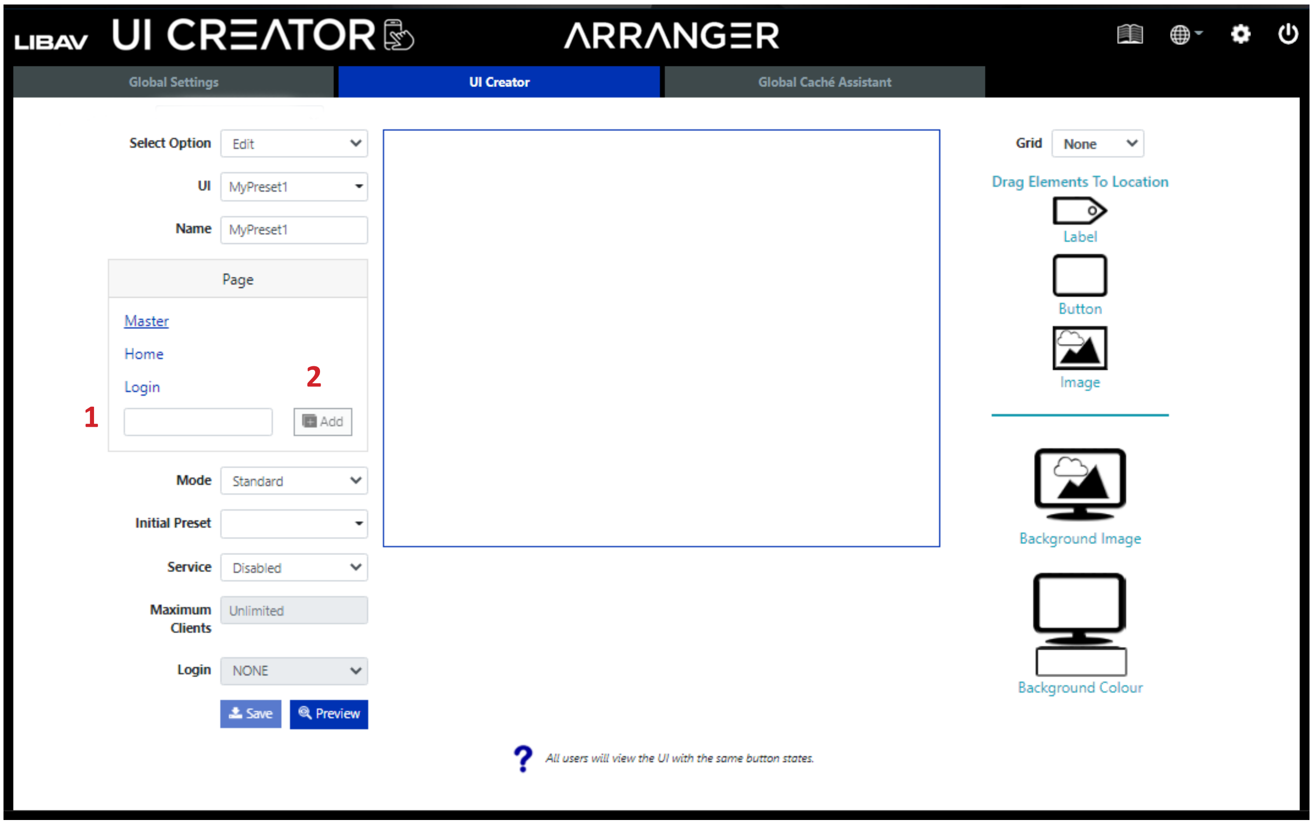


1. Select *Edit.*

2. Select UI *Name.*

Edit a User continued.....

Initially only 3 pages are available, *Master, Home and Login*. The *Master* page is used for elements to be displayed on all other pages that do not have a background set. The *Home* page is the displayed page when the UI is loaded. The *Login* page is displayed when a pin code is required to access the UI. From here you can add and remove pages whenever required.



1. Enter new page name

2. Click *Add*

UI Modes

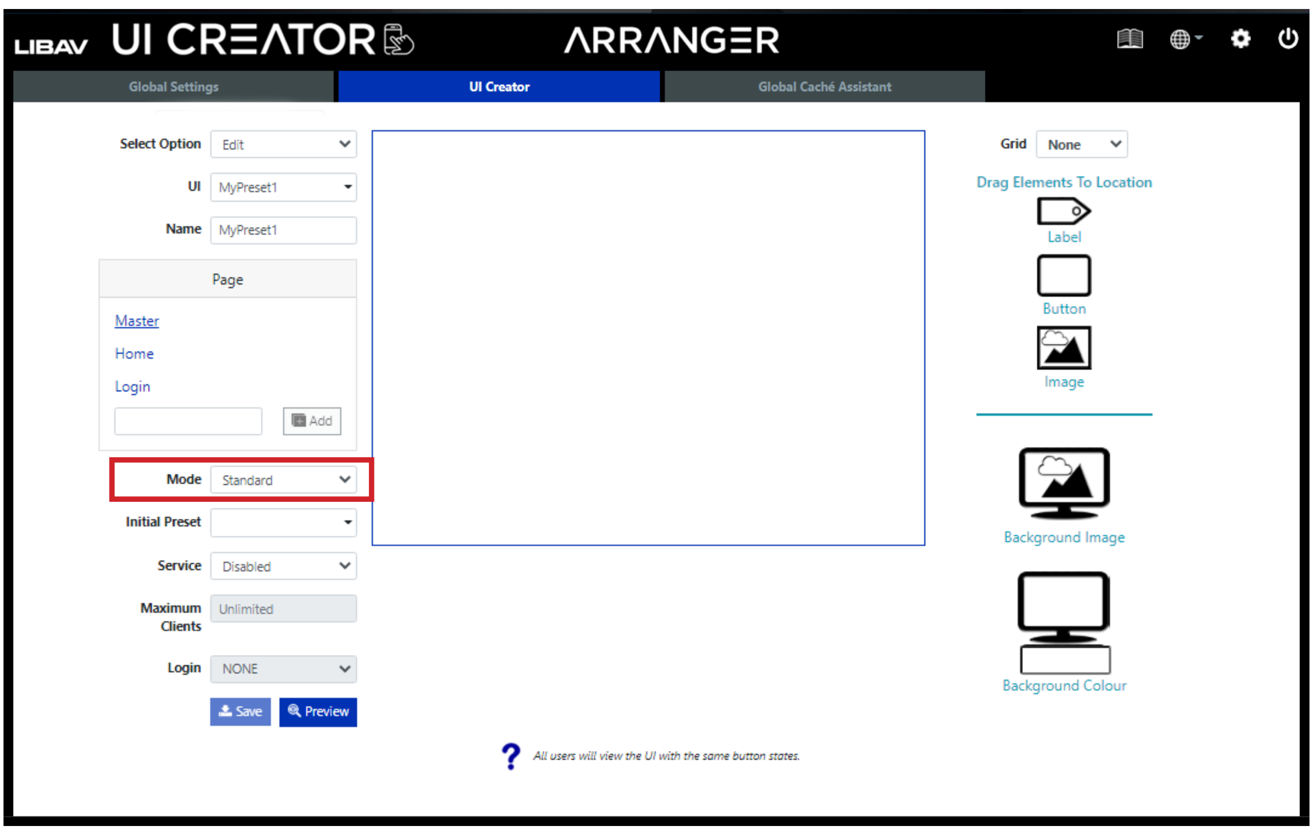
In the highlighted section below is how you will define your UI mode.

There are two modes:

1) Standard - *Standard* mode provides the default pages *Master Page, Home Page and Login Page*. The *Master Page* is used to display the elements on all other pages without a background applied. The *Home Page* is the initial page to be displayed. The *Login page* is shown when a login code is required. *Standard mode* provides options for limiting the maximum allowed clients and login with fixed or random number with a session timeout.

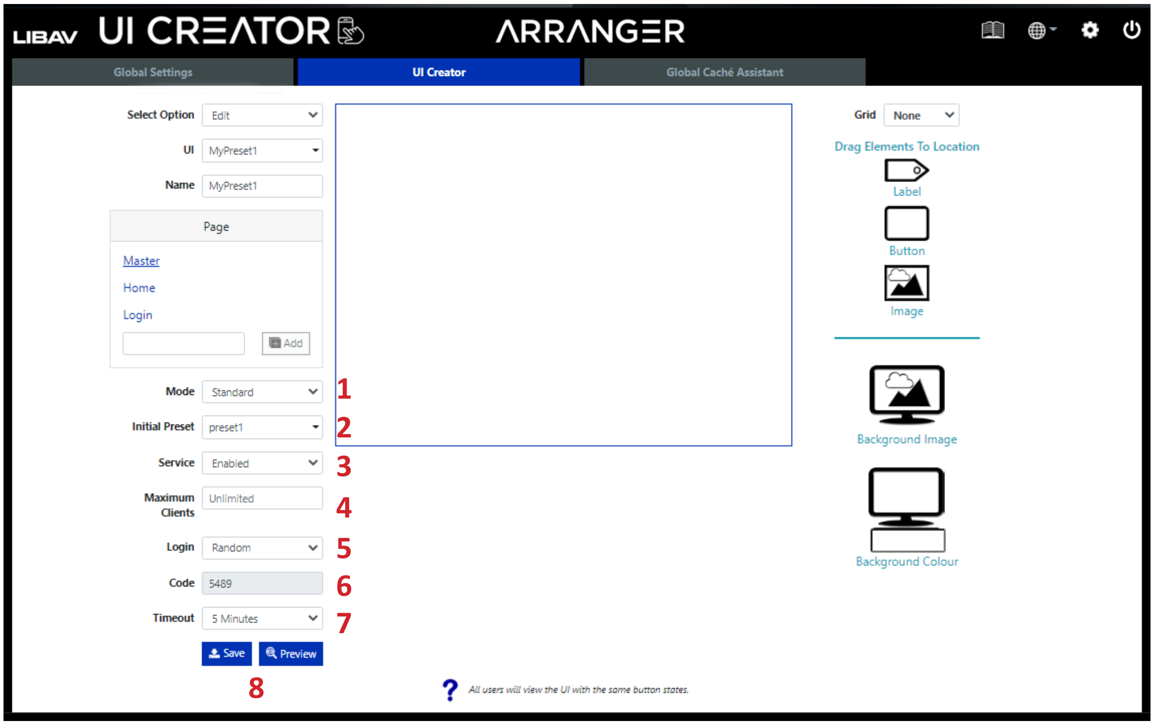
2) QR Code Result - *QR Code Result* mode provides the default pages *Master, Success and Error*. The Master page is used to display the elements on all other pages without a background applied. The *Success page* is shown after a scanned QR code preset is executed successfully. The *Error page* is shown after a scanned QR Code preset is executed with an error. The resulting user interface can be used to display a single page message or a multipage user interface with the same abilities as standard mode.

See *Global Settings > Presets > QR Code Preset* or more details.



Standard Mode

In the section below a standard configuration has been made.



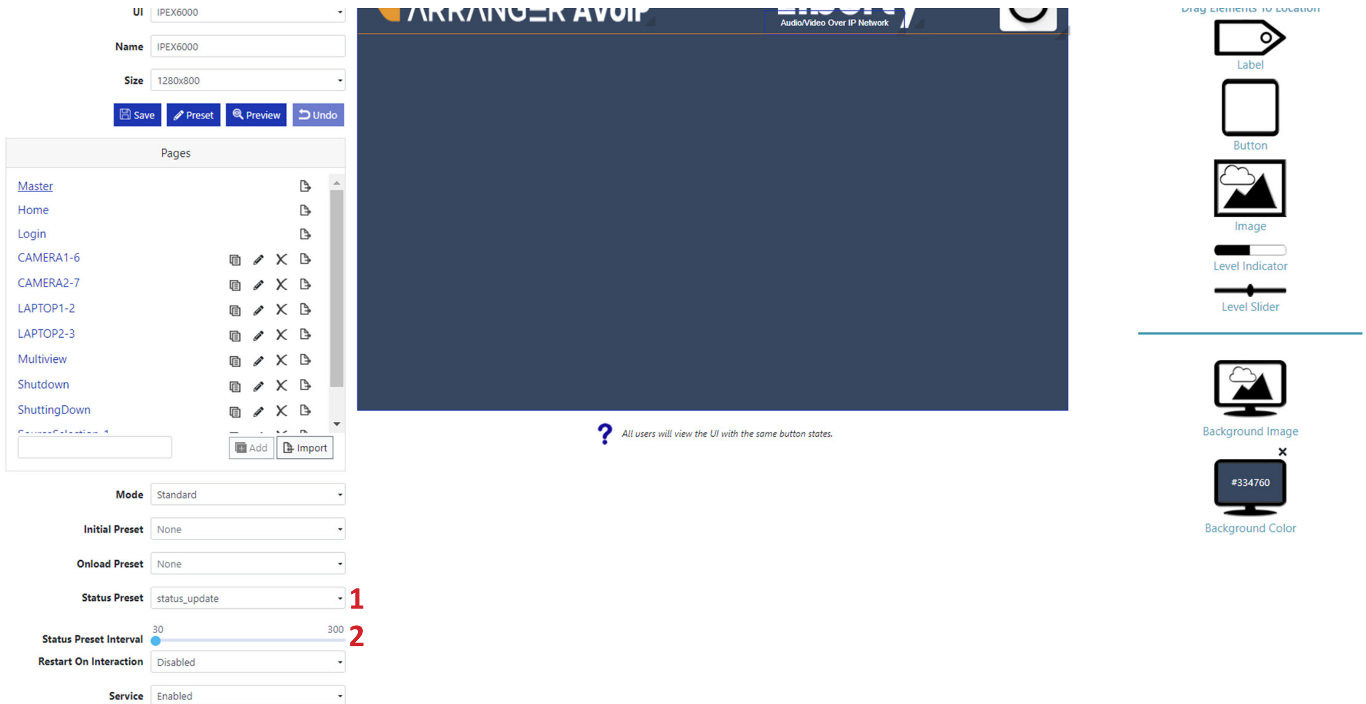
1. Select *Standard* mode.
2. Select a *Preset* from the Initial Preset drop-down.

   The *Initial Preset* is used to select a preset to be executed when the UI service is enabled. This preset can be used to set a default configuration to match the user interface initial button states. The control command ***set ui*** can be used to toggle the service state.
3. Select *Enable* from *Service* drop-down.

   When *Service* is *disabled* the UI will not be able to be accessed by a client.
4. Select a maximum number of control clients that can access the UI simultaneously.
5. Select *Login* option; *NONE, Random, User defined Login.*

   If a pin code to access the user interface is not required then leave the login as *NONE*. The login page will not be used or shown in the case.
6. If the *Random* login code option is selected in Step 5, a random number will be displayed here, this is the login code. If *User Defined option* was selected in Step 5 then you will be able to define the code in this field.
7. A *timeout* can also be applied here when using a login pin code that will prevent the client access after the selected time has elapsed.
8. Click *Save* to save UI or *Preview* to preview your UI in a web browser.

Standard Mode : Status Preset

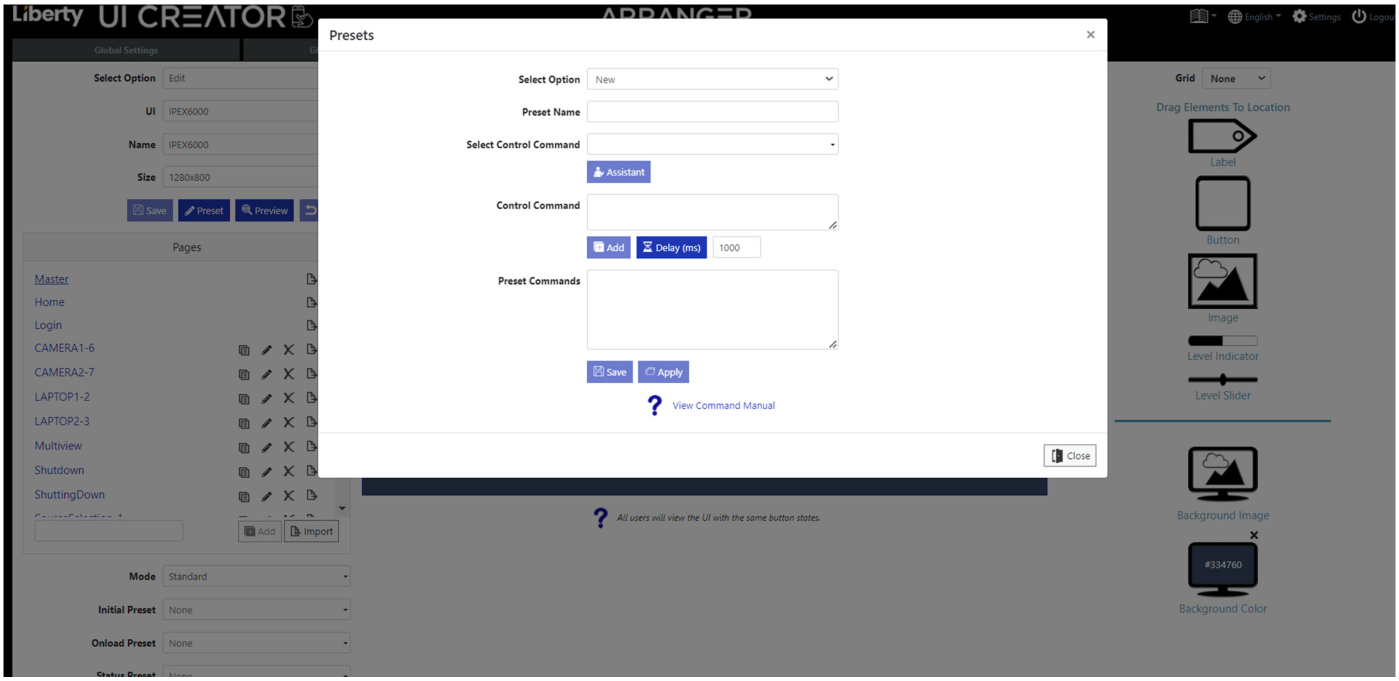


1. Select a preset that is used to to update the UI on a regular interval.
2. Select the interval from 30 to 300 seconds in 30 seconds increments with a default of 60 seconds to run the status preset.

Preset Add / Edit

1. Preset button will open a pop up that allows you to add / edit a preset without leaving the UI tab..

Preset Add / Edit

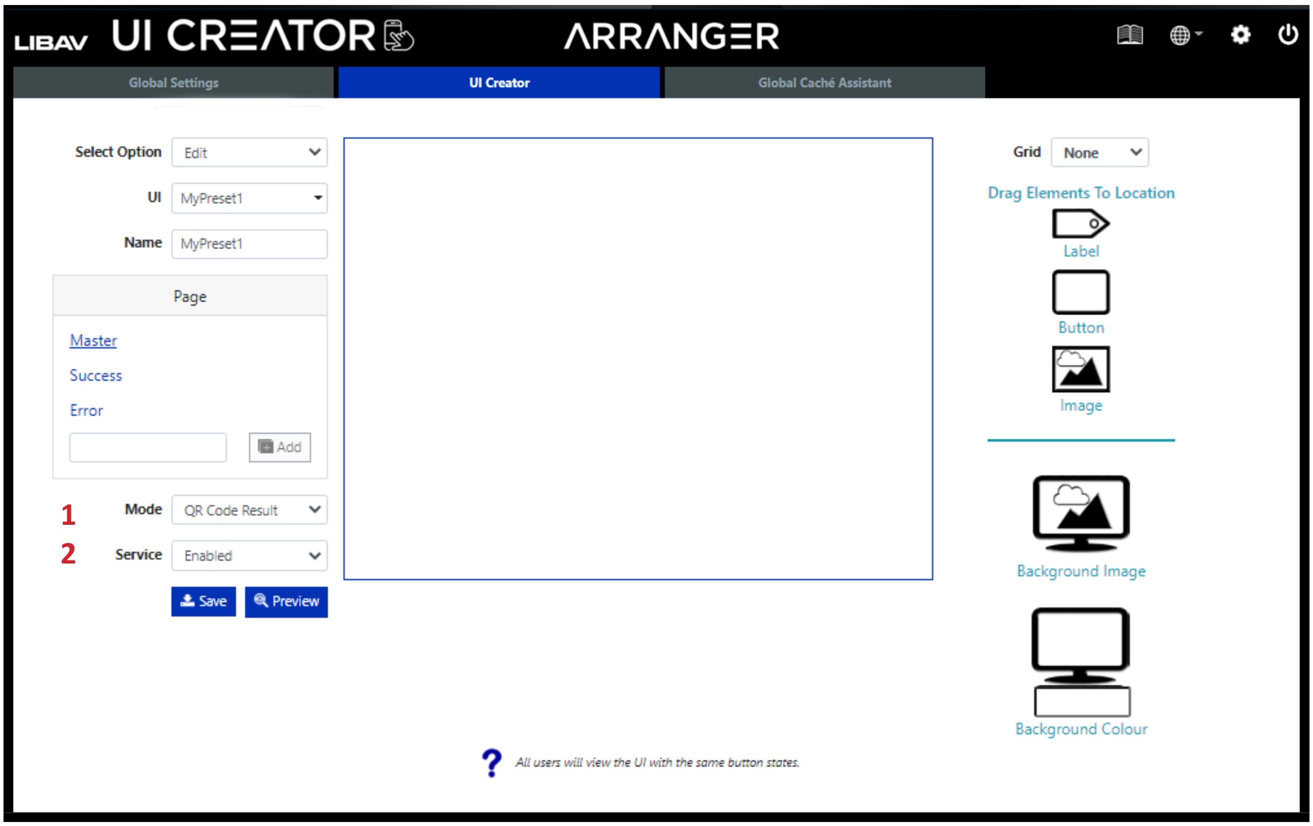


1. This popup has the full options in preset to allow editing or adding a preset without leaving the UI tab. The preset will show up in the dropdown selections on buttons..

QR Code Result Mode

In the section below a *QR Code Result* configuration has been made.

See *Global Settings > Presets > QR Code Preset* for more information on linking QR Code presets to a QR Code Result UI.

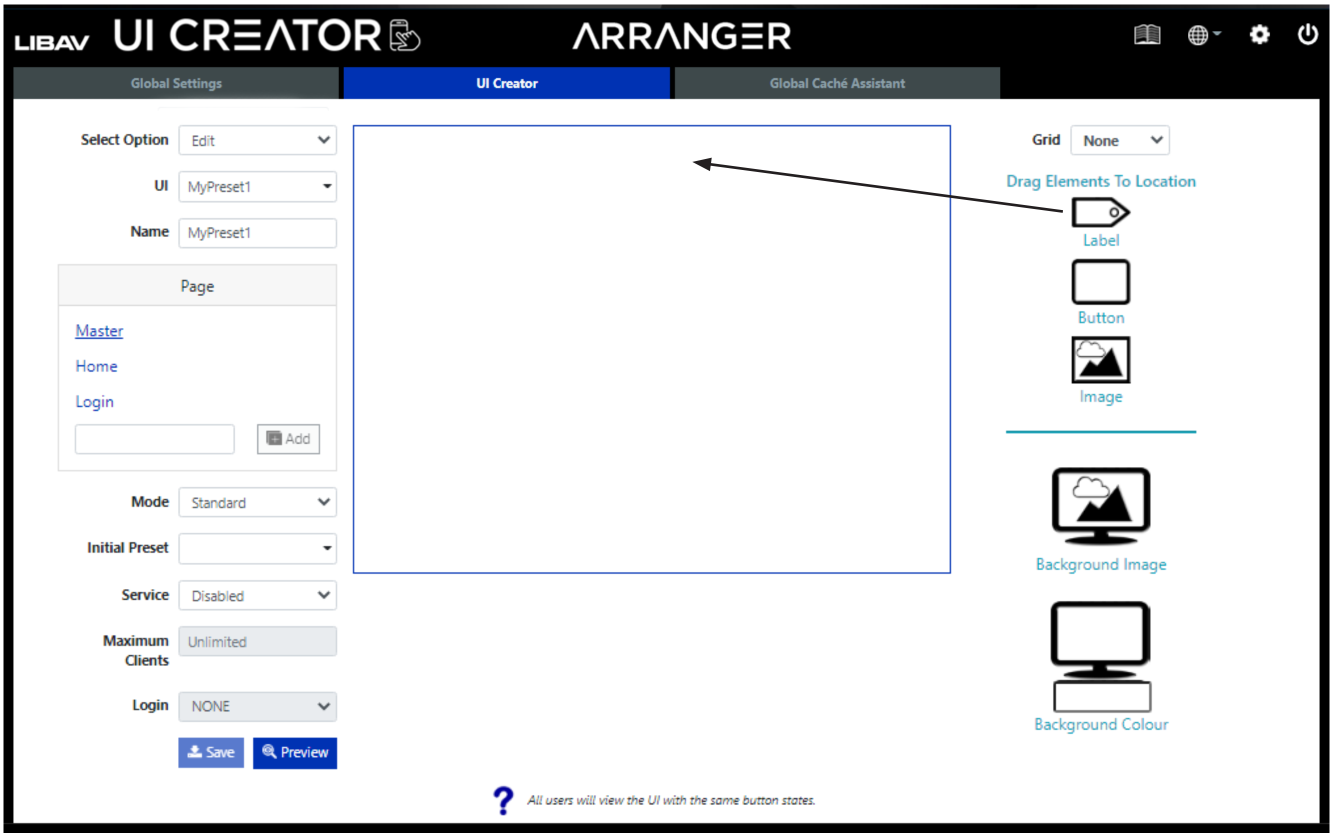


1. Select *QR Code Result* mode.
2. Select *Enable* from *Service* drop-down.
   When *Service* is *disabled* the UI will not be able to be accessed by a client.

Adding Labels

A *label* can be dragged to any location and used as a heading, label or wherever text is required on the UI. The label must be given a name to change the color, text and visibility via the control command ***set ui_label.***

Here we are adding a title for the UI on the *master page* by dragging and dropping the label icon into the UI layout.

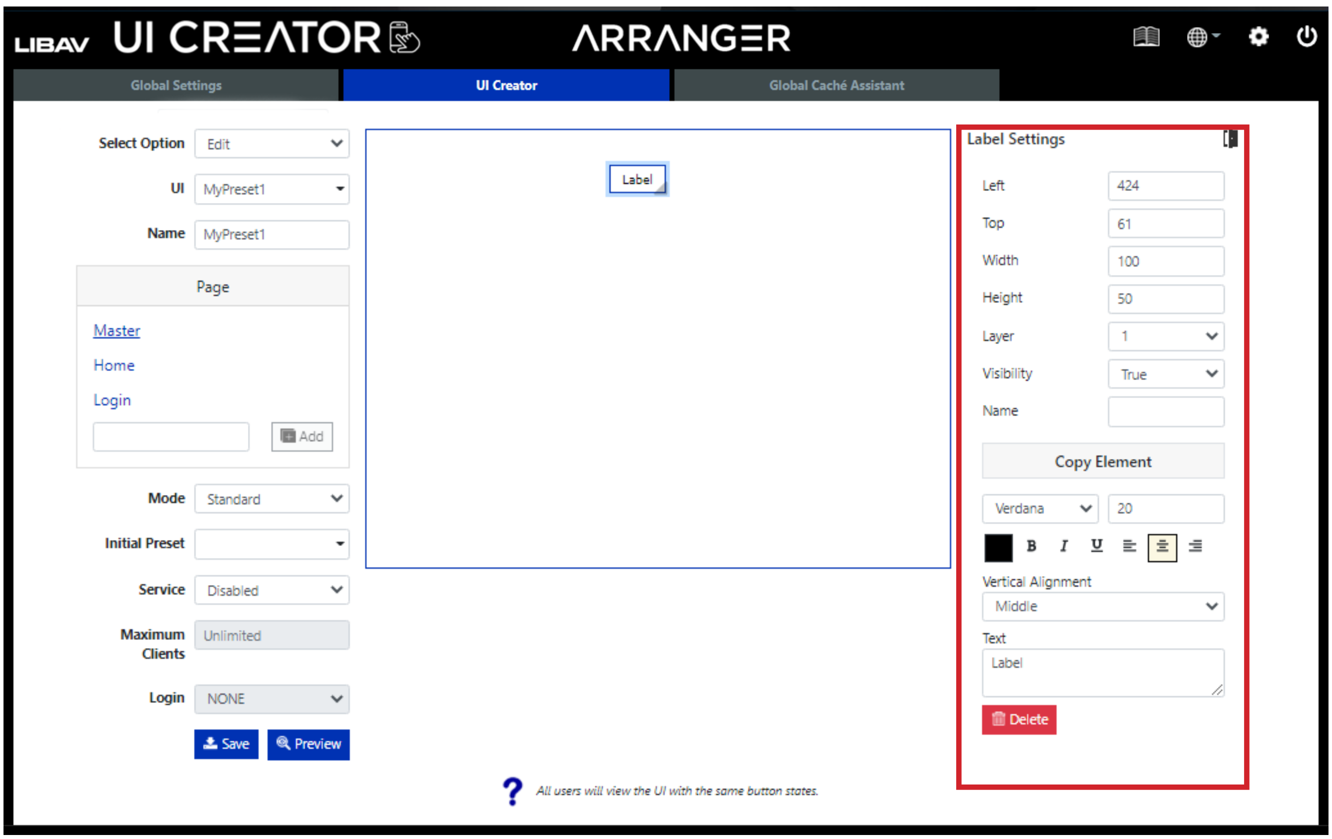


Adding Labels continued....

Once a label has been added the *Label Settings* will load to the right of the screen, see highlighted section below.

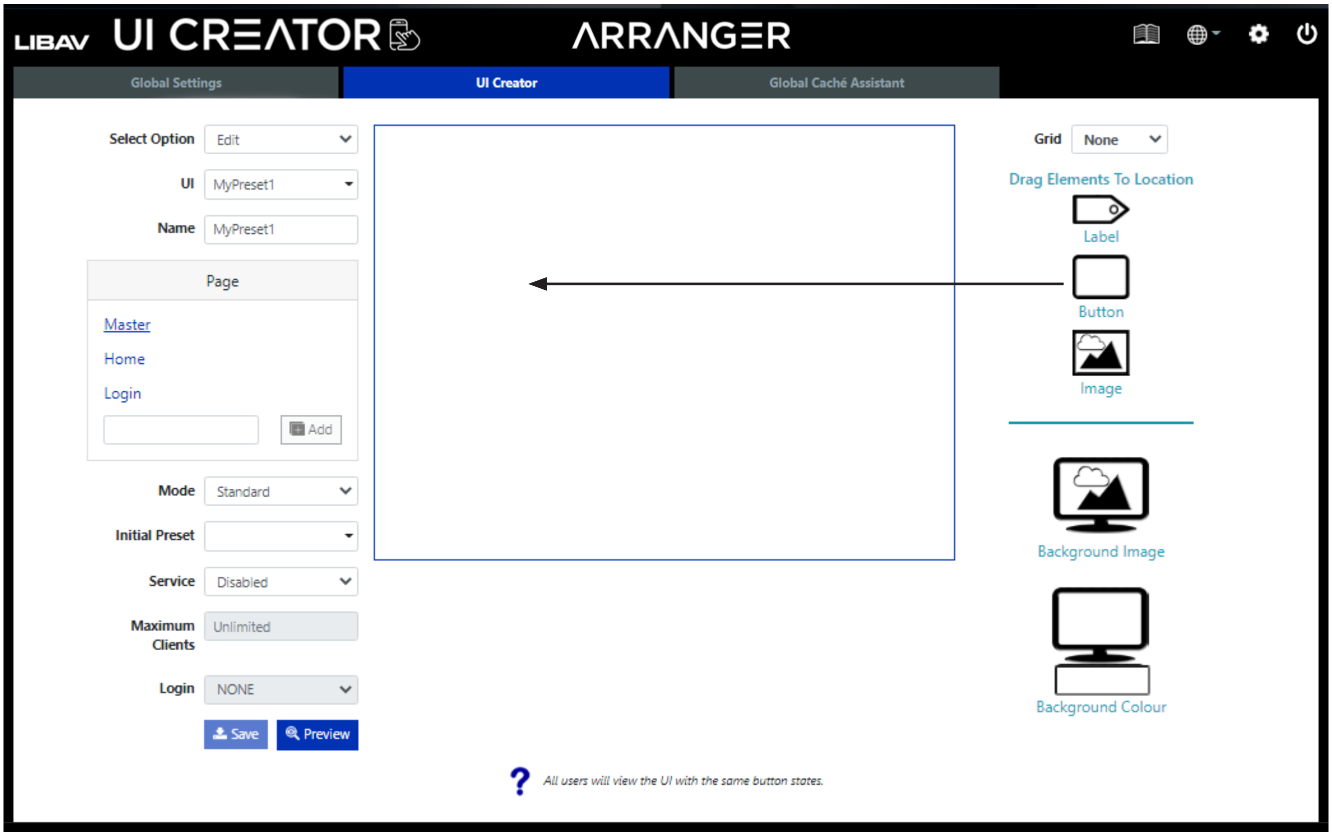
From here you can edit the text font, size, style, alignment and position, given the label a name or remove it from the UI.

Note: You can always populate the *Label Settings* by clicking on any label in the UI.



Adding Buttons

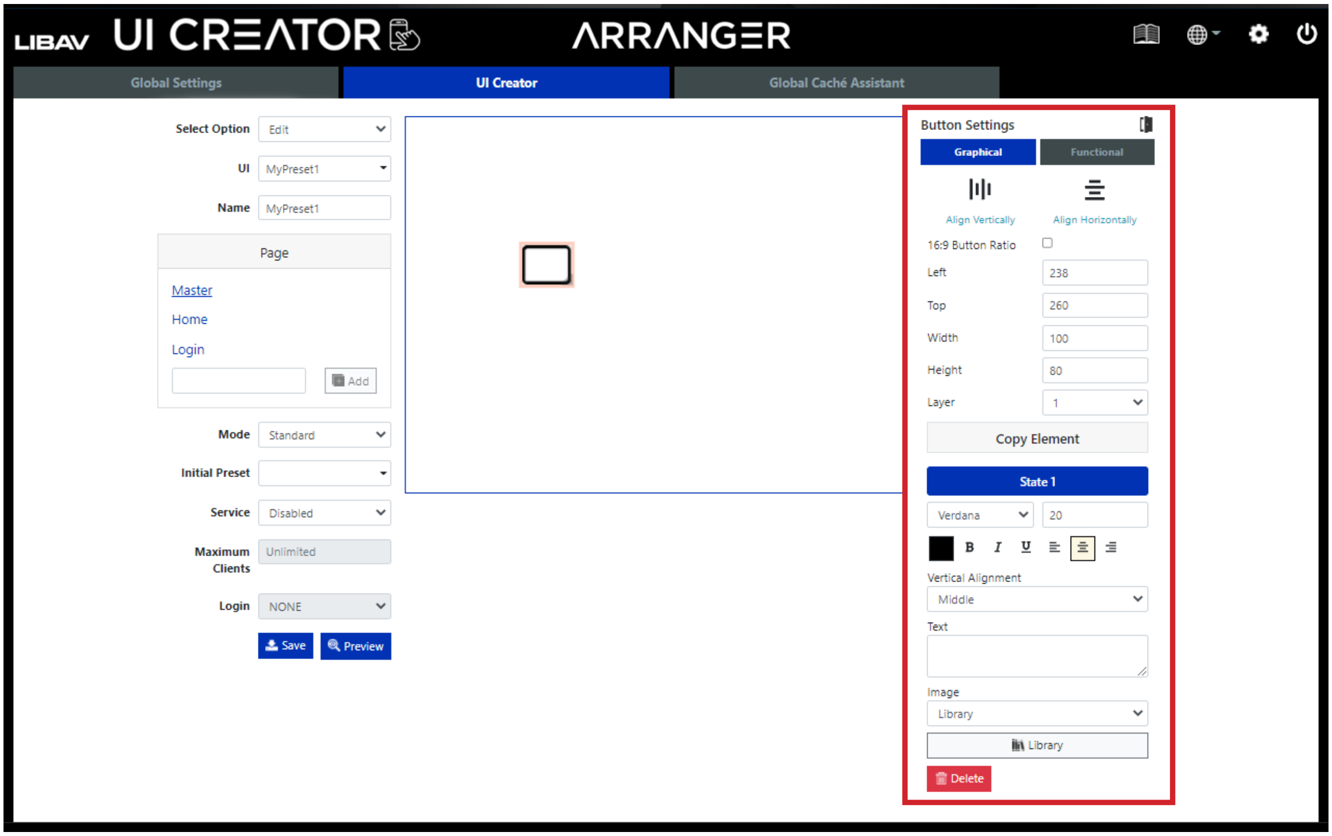
A *button* can be dragged to any location and used as a press button, QR Code or an indicator.



Adding buttons continued....

Once a button has been placed in the UI the *Button Settings* will populate to right of the UI Creator. See highlighted section below.

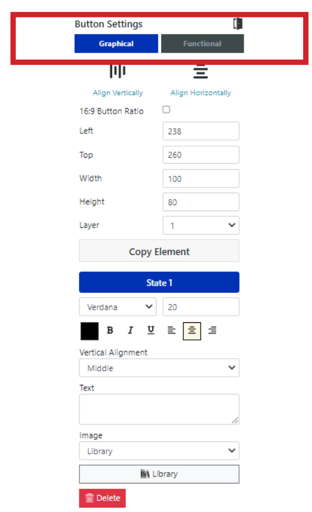
Note: You can always populate the *Button Settings* by clicking on any button in the UI.



Note: There are two selectable menus in *Button Settings.*

1. Graphical - Will allow you to edit the button size, position, text font, size, style and alignment, or remove the button from the UI altogether.

2. Functional - Allows you to configure the *Preset* triggers once a button has been pressed.

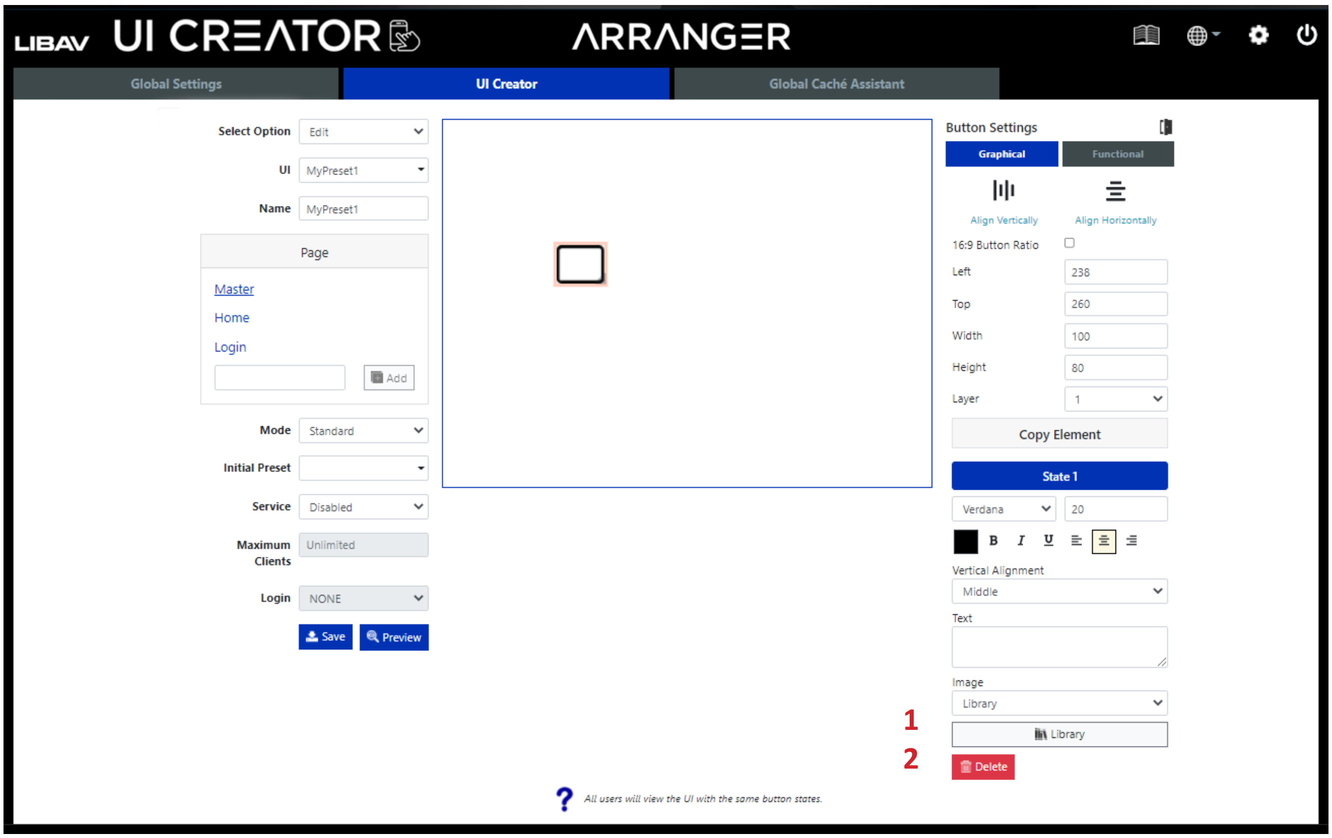


Adding buttons continued....

There are three choices for a button image:

1) Use an external pre made button image (External File)
2) Use internal Arranger button image (Library)
3) Use a static or streamed preview image from the specified endpoint (Preview)

Select an image for the button by selecting either *External File* and browsing your own images or selecting *Library* to choose one from the button library. When selecting an image from the button library, both state 1 and state 2 images will be assigned when required. When using external file, an image must be assigned for each button state.

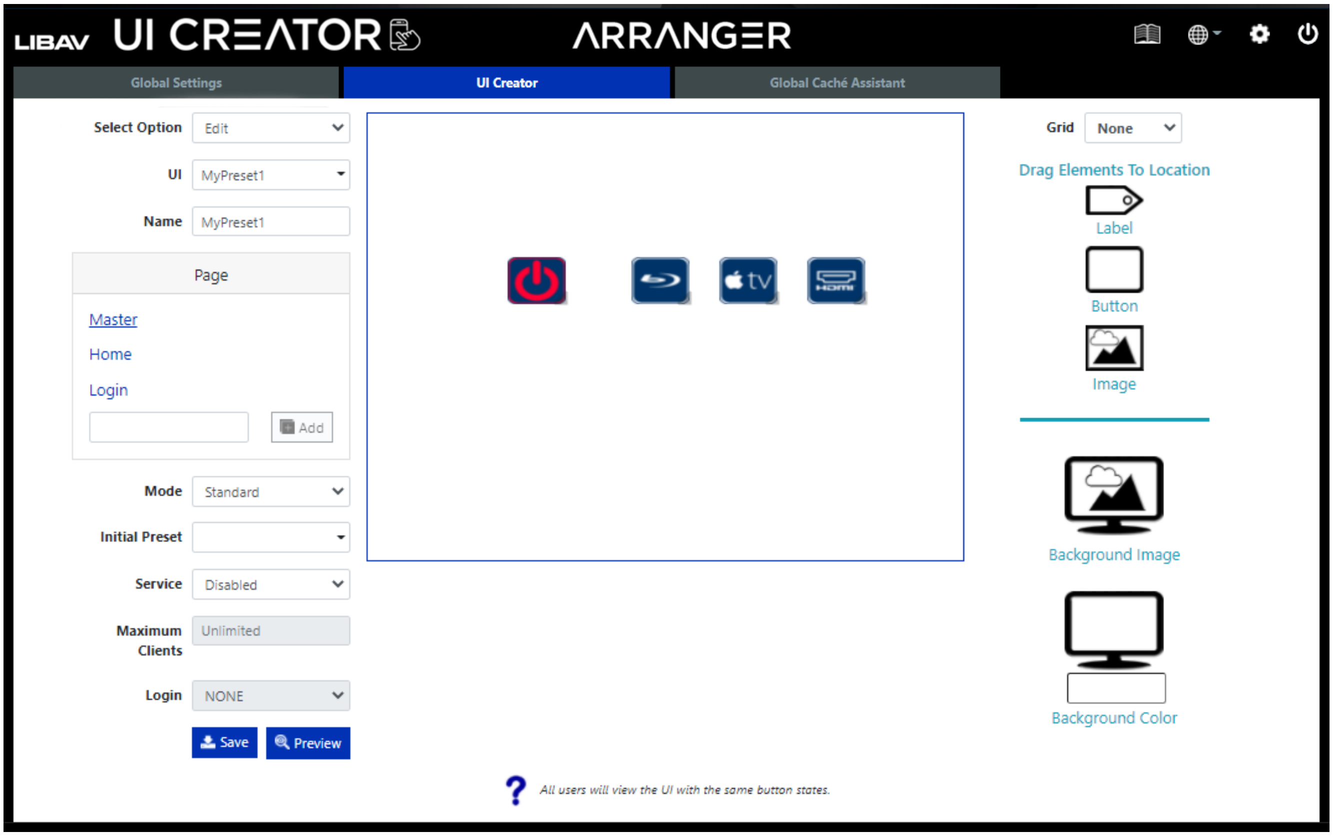


1. Select *Library, External File or Preview* from image drop-down.

2. If you select *Library* from Step 1, click *Library* to access the internal library of button graphics. If you select *External File* from Step 1. A pop will show up with all images you have imported before select one or click *Browse* to access your files on your PC to use as the button graphic. If you choose *Preview* then you will need to select the *Device* and *Function* associated with the preview in the *Graphical* and *Functional Properties*.

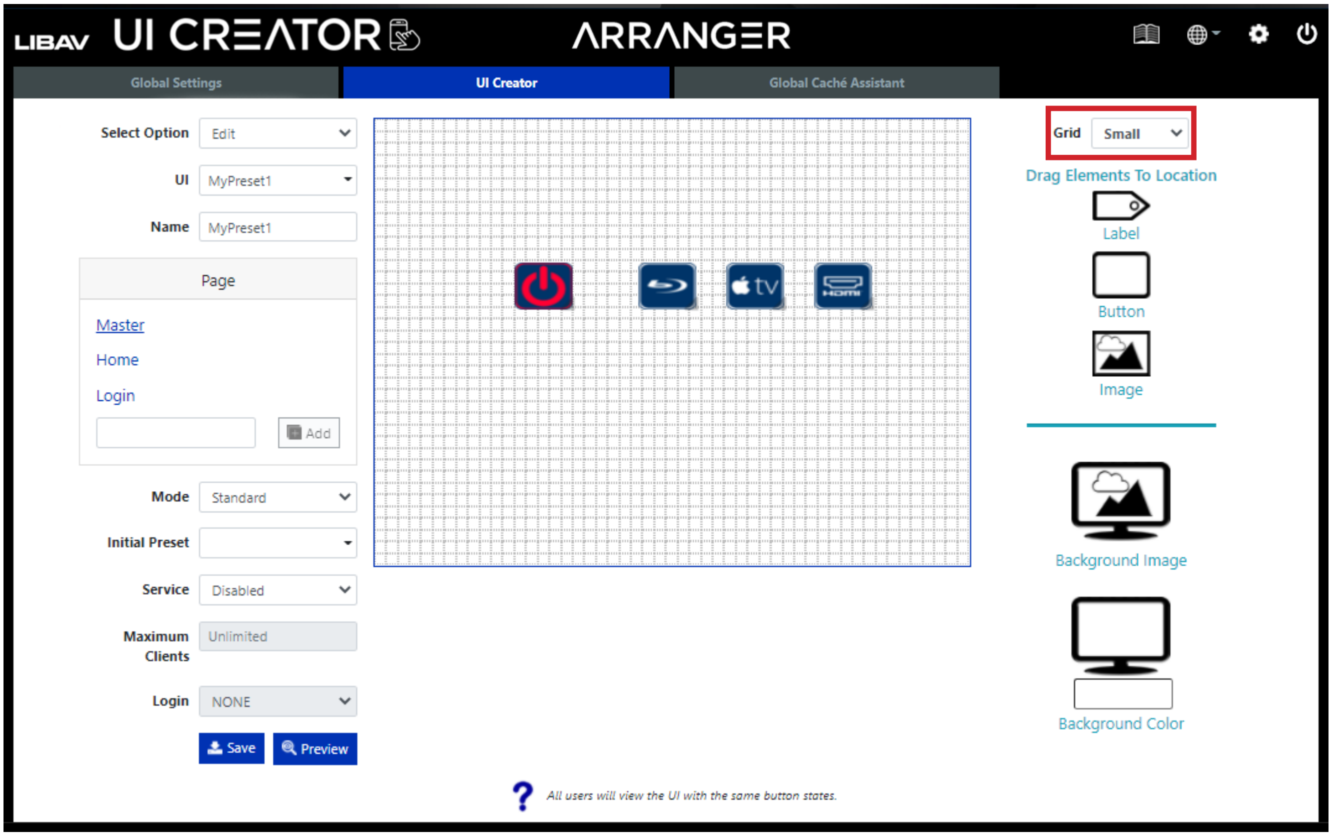
Adding buttons continued....

In the examples below, common buttons from the internal graphical library have been used.



Adding buttons continued....

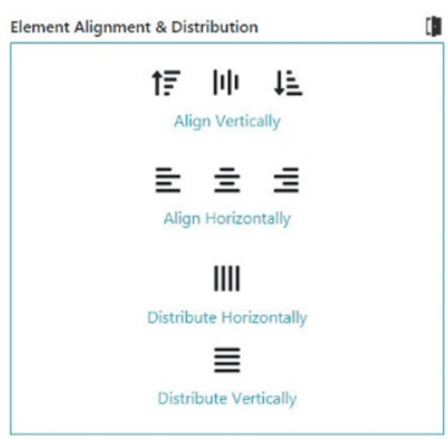
Note: To help align buttons on the UI layout, click on any open and unused space in the UI and select *Grid* type near the top right of UI Creator. The buttons will align with the grid selections you have made. In the example to the right, small grid size has been used. Also the selected element can be moved using the arrow keys on the PC.



Multiple elements can be aligned with respect to the first selected element.

Click on the first referenced element to select it, then hold Ctrl while selecting further elements to be aligned. An *Element Alignment & Distribution* panel will then be shown to Align Vertically, Align Horizontally, Distribute Horizontally or Distribute Vertically.

Clicking in white space will deselect selected elements.

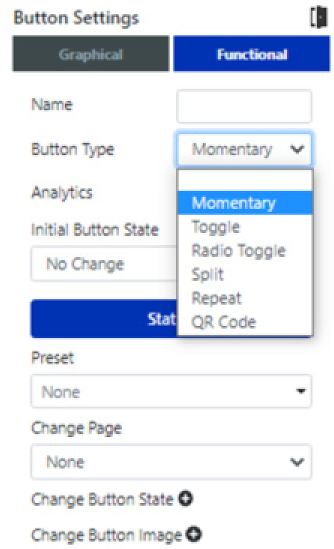


Button Logic

A *button* can be configured to operate with 6 different functions as explained here:

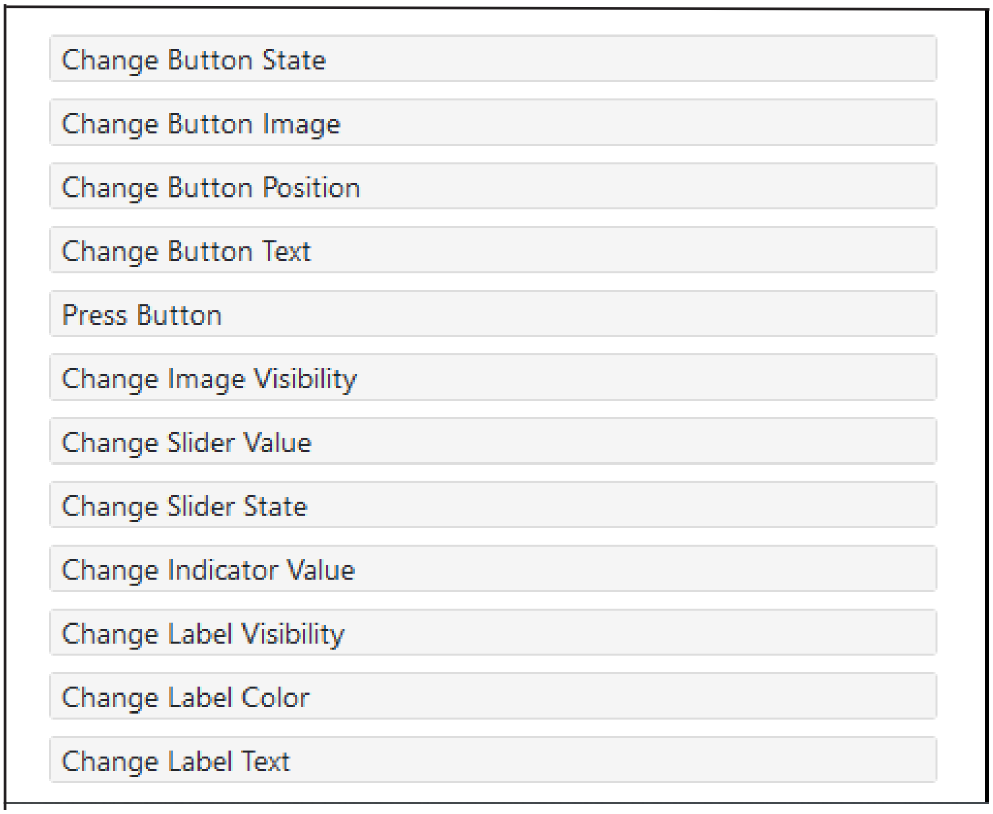
1. Momentary button - A *Momentary* button will operate in a push-button single-state fashion where a preset is executed once for every press.
2. Repeat button - A *Repeat* button will operate in a momentary fashion where only a preset is assigned to state 1 but the preset will be repeated while the button is held down. The preset will be executed as soon as the button is pressed, then there is a configurable repeat delay before repeating begins and a configurable repeating interval which sets the delay time between preset execution.
3. Toggle button - A *Toggle* button will operate in a *push on, push off* fashion so a preset can be assigned to both state 1 and state 2. First press of the button executes state 1 preset and puts the button into state 2 showing a state 2 button image. Second press of the button then executes state 2 preset and returns the button back to state 1.
4. Radio Toggle button (Exclusive Toggle button) - A *Radio Toggle* group of buttons will operate in an exclusive toggle fashion and must be assigned to a *Button Group*. When you want a group of buttons to work together as radio toggle buttons where only one button of the group can be in state 2 (down), such as a radio station selector or source selection, then define the same button group name for each of those buttons.
5. QR Code Button - Adds touchless functionality to a touchscreen control panel. A QR code button will operate in a momentarily fashion where only a preset is assigned to state 1 and a QR code replaces the button image. When the QR code is scanned a virtual press of the button is performed.

Button logic options can be accessed and set in the *Functional* menu of *Button Settings* under *Button Type* for the selected button in the UI layout.



Button Logic- UI elements control

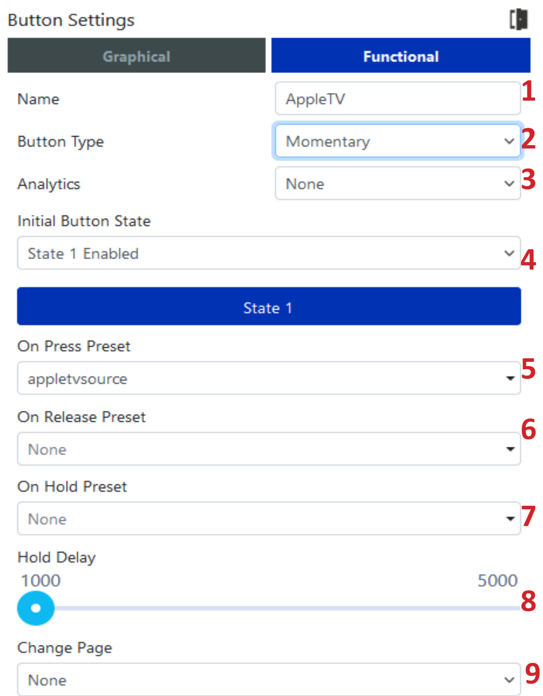
With in each button logic is the ability to change all UI elements. This allows a button basic manipulation of the UI without any coding required.



Button Logic - Momentary

A *Momentary* button will operate in a push-button single-state fashion where a preset is executed once for every press.

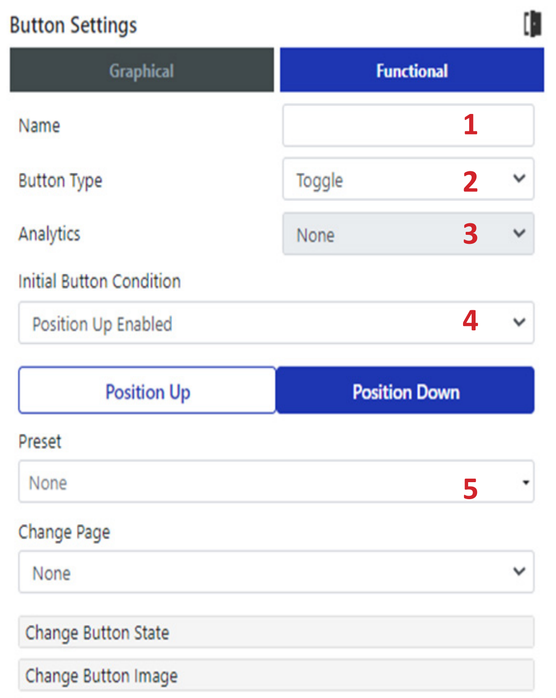
Select a button in the UI layout to populate the *Button Settings*, then navigate to the *Functional* menu.



1. Enter a *Name* for the button. A button name is required as a reference for *Analytics* or when manipulating the button from another buttons functionality or via the API. See *Global Settings > Analytics > UI Creator* for reference.
2. Select *Momentary.*
3. Select a button function from the *Analytics* list that best matches the operation of the button.
4. Select *Initial Button State:* This is the initial state of the button when the UI is loaded.
5. Select the *Preset* to be executed on button press.
6. Select the *Preset* to be executed on button release
7. Select the *Preset* to be executed on button hold.
8. Select the hold trigger delay time.
9. Change Page: This allows you to change to another page.

Button Logic - Toggle

A *Toggle* button will operate in a push on, push off fashion so a preset can be assigned to both state 1 and state 2. First press of the button executes state 1 preset and puts the button into state 2 showing a state 2 button image. Second press of the button then executes state 2 preset and returns the button back to state 1.



1. Enter a *Name* for the button. A button name is required as a reference for analytics or when manipulating the button from another buttons functionality or via the API. *See Global Settings > Analytics > UI Creator for reference.*

2. Select *Toggle.*

3. Select a button function from the *Analytics* list that best matches the operation of the button.

4. Select *Initial Button State:* This is the initial state of the button when the UI is loaded.

5. State 1 / State 2: These buttons allow you to select the following for each button state:

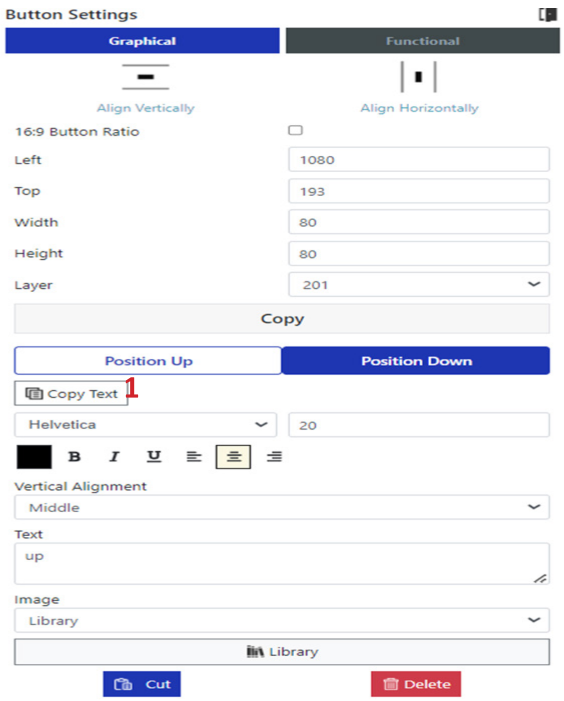
    Preset: Select the preset to be executed on button press.

    Change Page: This allows you to change to another page.

    Change Button State: This allows you to change the state of a button.

    Change Button Image: This allows you to change the image of a button.

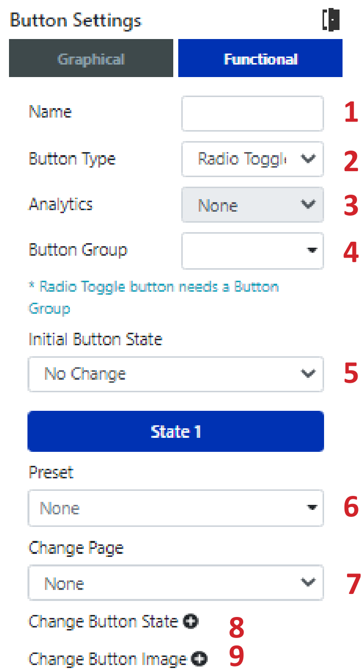
Button text copy



The Copy text option allows you to copy the text from Position Up text field and paste into the Position down text field automatically.

Button Logic - Radio Toggle

A *Radio Toggle* group of buttons will operate in an exclusive toggle fashion and must be assigned to a *Button Group*. When you want a group of buttons to work together as radio toggle buttons where only one button of the group can be in state 2 (down), such as a radio station selector or source selection, then define the same button group name for each of those buttons.



1. Enter a *Name* for the button. A button name is required as a reference for analytics or when manipulating the button from another buttons functionality or via the API. *See Global Settings > Analytics > UI Creator for reference.*

2. Select *Radio Toggle.*

3. Select a button function from the *Analytics* list that best matches the operation of the button.

4. *Button Group:* A group name must be provided to combine individual buttons to function together.

5. Select *Initial Button State:* This is the initial state of the button when the UI is loaded.

6. Select the *Preset* to be executed on button press.

7. Change Page: This allows you to change to another page.

8. Change Button State: This allows you to change the state of a button.

9. Change Button Image: This allows you to change the image of a button.

Button Logic - Split

A Split button group will work in a matrix type way whereby buttons are configured as either an Encoder button or Decoder button. Encoder buttons will operate as Radio toggle (Exclusive) buttons so only one can be selected, while the Decoder buttons will operate as a Toggle button so multiple can be selected.
Once a button is configured as an Encoder button the actual Encoder to be used is selected. Buttons configured as a Decoder button will also have the required Decoder selected along with two presets that are executed when the Decoder button is pressed from up position and also down position.

A single All Decoders button can also be assigned to the Split button group. This type of button is used to join the selected Encoder to all the Decoders. No device selection is required for this button and a preset can be optionally assigned to each button position. The reason the presets are optional is depending on how the system is to be used. When all the Decoders of the system are allocated to a Split button group then the use of join all_rx command can simplify things and in some cases ensure all displays change seamlessly (depending on the system capabilities). So then the All Decoders button can be set with presets to control all devices. If no presets are set on the All Decoders button then the individual enabled Decoder button presets will be sequentially executed, as if pressing the Decoder buttons one after each other.

A preset has to be created for the Decoder button position up as a join command and position down as a leave command. In the majority of cases All Decoders buttons will use the same preset. In the preset assistant you will notice in the device list <<encoder_name>> and <<decoder_name>>. These are the device selections required to create presets for Split button functionality.
<<encoder_name>> will be replaced by the selected Encoder button device and <<decoder_name>> will be replaced by the selected Decoder button device. A Decoder button preset can also use <<button_name>> to identify the actual Decoder button pressed by name.

Preset example Decoder button position up join:
 join av <<encoder_name>> <<decoder_name>>
Preset example Decoder button position down leave:
 leave av <<decoder_name>>
Preset example Decoder All button position up join:
 join av <<encoder_name>> all_rx
Preset example Decoder button position down leave:
 leave av all_rx

Notes:
- <<decoder_name>> is not supported for a All Decoders button preset.
- If you need to interact with buttons then a unique name for the button must be specified. The position of the buttons can then be changed with the button functionality settings or via the control command set ui_button. The set ui_button command can also be used to change the buttons state, text or be virtually pressed.
- All Decoders will bypass disabled Decoder button individual presets.

Button Logic - Split

Split buttons and split button groups can be controlled in a unique way to set and clear their routing connections.

- To clear all routing connections and restore a split button group to default with all button positions up use the function clear:

    set ui_button <<ui_name>> SplitButtonGroup clear

- To set all split buttons in a group to position up use the function position up:

    set ui_button <<ui_name>> SplitButtonGroup position up

- To create a routing connection between an Decoder and Encoder use the enc_btn_name option along with function position down:

    set ui_button <<ui_name>> SplitButtonDecoder01 position down SplitButtonEncoder01
    set ui_button <<ui_name>> SplitButtonDecoder02 position down SplitButtonEncoder01
    set ui_button <<ui_name>> SplitButtonDecoder03 position down SplitButtonEncoder01
    set ui_button <<ui_name>> SplitButtonDecoder01 position down SplitButtonEncoder01
    set ui_button <<ui_name>> SplitButtonDecoder02 position down SplitButtonEncoder02
    set ui_button <<ui_name>> SplitButtonDecoder03 position down SplitButtonEncoder03

- To remove a routing connection between a Decoder and Encoder use the enc_btn_name option with null along with function position down:

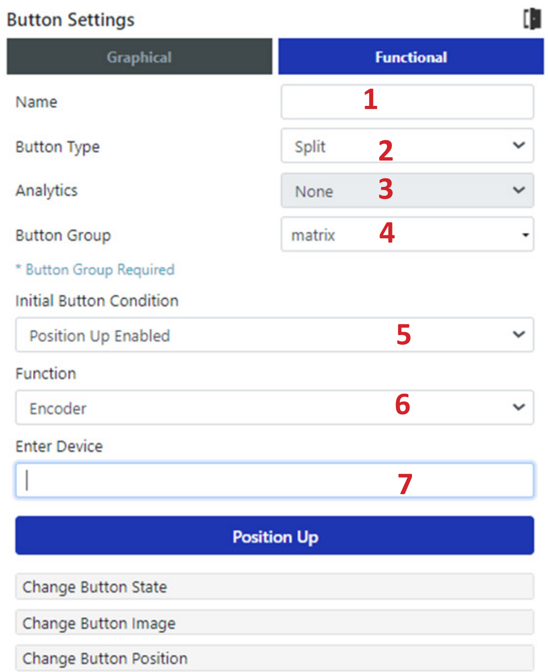
    set ui_button <<ui_name>> SplitButtonDecoder01 position down null

- To create a routing connection between an Encoder and all Decoders use the enc_btn_name option along with function position down:

    set ui_button <<ui_name>> SplitButtonAllDecoders position down SplitButtonEncoder01

Button Logic - Split

Here you can see the functionality of an encoder split button.



1. Enter a *Name* for the button. A button name is required as a reference for analytics or when manipulating the button from another button's functionality or via the API. *See Global Settings > Analytics > UI Creator for reference.*

2. Select *Split.*

3. Select a button function from the *Analytics* list that best matches the operation of the button.

4. Enter *Button Group* name: A group name must be provided to combine individual buttons to function together.

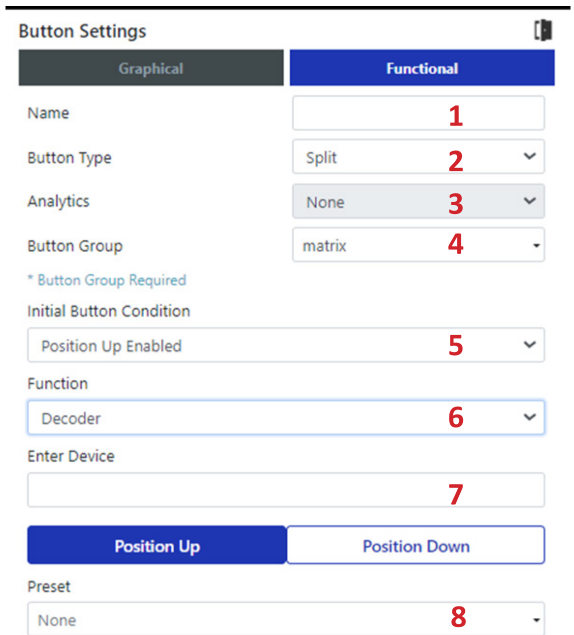
5. Select *Initial Button State:* This is the initial state of the button when the UI is loaded.

6. Function: Select the button to operate the encoder.

7. Type in the Encoder name.

Button Logic - Split

Here you can see the functionality of an decoder split button.



1. Enter a *Name* for the button. A button name is required as a reference for analytics or when manipulating the button from another button's functionality or via the API. *See Global Settings > Analytics > UI Creator for reference.*
2. Select *Split.*
3. Select a button function from the analytics list that best matches the operation of the button.
4. Enter *Button Group* name: A group name must be provided to combine individual buttons to function together.
5. Select *Initial Button State:* This is the initial state of the button when the UI is loaded.
6. Function: Select the button to operate the decoder.
7. Type in the decoder name.
8. Position Up / Position Down: These buttons allow you to select the following for each button state:

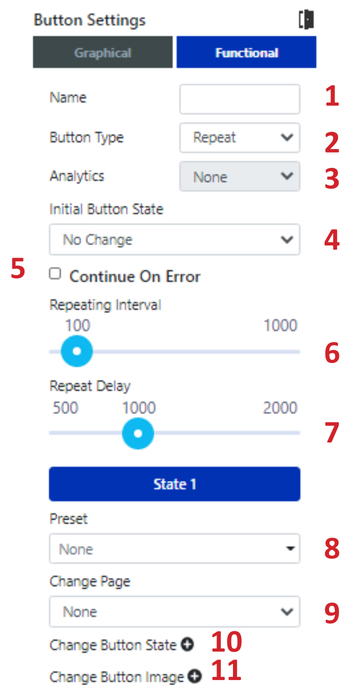
    Preset: Select the preset to be executed on button press.

    Change Button State: This allows you to change the state of a button.

    Change Button Image: This allows you to change the image of a button.

Button Logic - Repeat

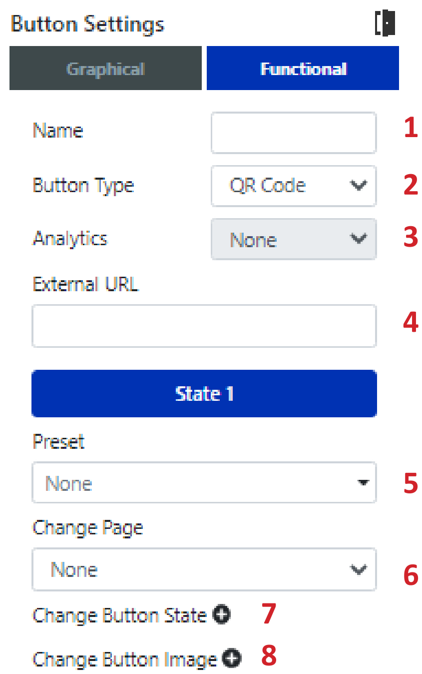
A *Repeat* button will operate in a momentary fashion where only a preset is assigned to state 1 but the preset will be repeated while the button is held down. The preset will be executed as soon as the button is pressed, then there is a configurable repeat delay before repeating begins and a configurable repeating interval which sets the delay time between preset execution.



1. Enter a *Name* for the button. A button name is required as a reference for analytics or when manipulating the button from another button's functionality or via the API. *See Global Settings > Analytics > UI Creator for reference.*
2. Select *Repeat.*
3. Select a button function from the *Analytics* list that best matches the operation of the button.
4. Select *Initial Button State:* This is the initial state of the button when the UI is loaded.
5. Continue On Error: This is an option to continue executing the preset if it returns failed.
6. Repeating Interval: This is the time delay in milliseconds the button preset repeats while being held down.
7. Repeat Delay: This is the time in milliseconds the button must remain held down before the preset starts repeating.
8. Preset: Select the preset to be executed on button press.
9. Change Page: This allows you to change to another page.
10. Change Button State: This allows you to change the state of a button.
11. Change Button Image: This allows you to change the image of a button.

Button Logic - QR Code

Adds touchless functionality to a touchscreen control panel. A *QR Code* button will operate in a momentary fashion where only a preset is assigned to state 1 and a QR code replaces the button image. When the QR code is scanned a virtual press of the button is performed.

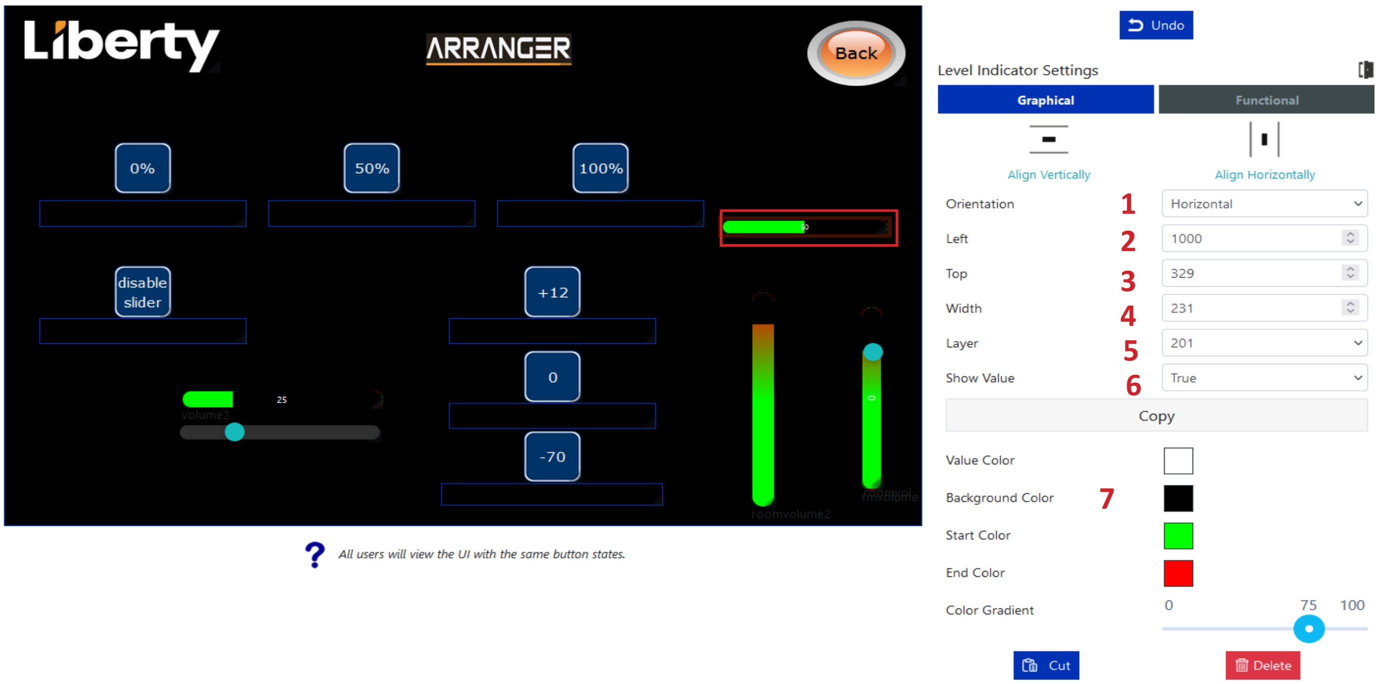


1. Enter a *Name* for the button. A button name is required as a reference for analytics or when manipulating the button from another button's functionality or via the API. *See Global Settings > Analytics > UI Creator for reference.*
2. Select *Repeat.*
3. Select a button function from the *Analytics* list that best matches the operation of the button.
4. External URL: Enter the controller's external URL if working outside of the local network.
5. Preset: Select the preset to be executed on button press.
6. Change Page: This allows you to change to another page.
7. Change Button State: This allows you to change the state of a button.
8. Change Button Image: This allows you to change the image of a button.

Level Indicator

A Level Indicator can be dragged to any location then resized by dragging the placeholder or changing the size and location values directly. The Level Indicator must be given a name to change the value with button functionality and via API control command set ui_indicator.

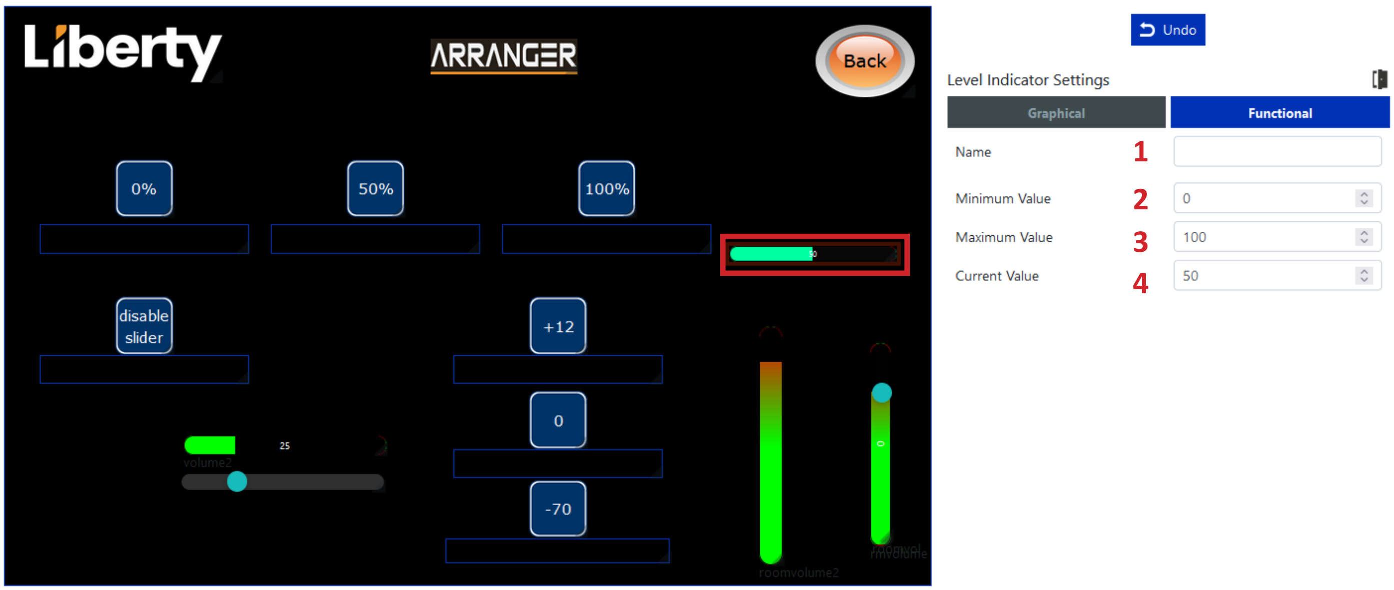
Some examples of uses of level indicators are room volume, camera zoom, and camera focus.



1. Select horizontal or vertical orientation.
2. Enter in value for starting left location.
3. Enter in value for starting top location.
4. Enter the width of the indicator.
5. Select the layer of the indicator on the page.
6. Select if you want the text value of the level to show or not.
7. Select Background, Start, End, and Gradient color. .

Level Indicator continued

Functional tab.



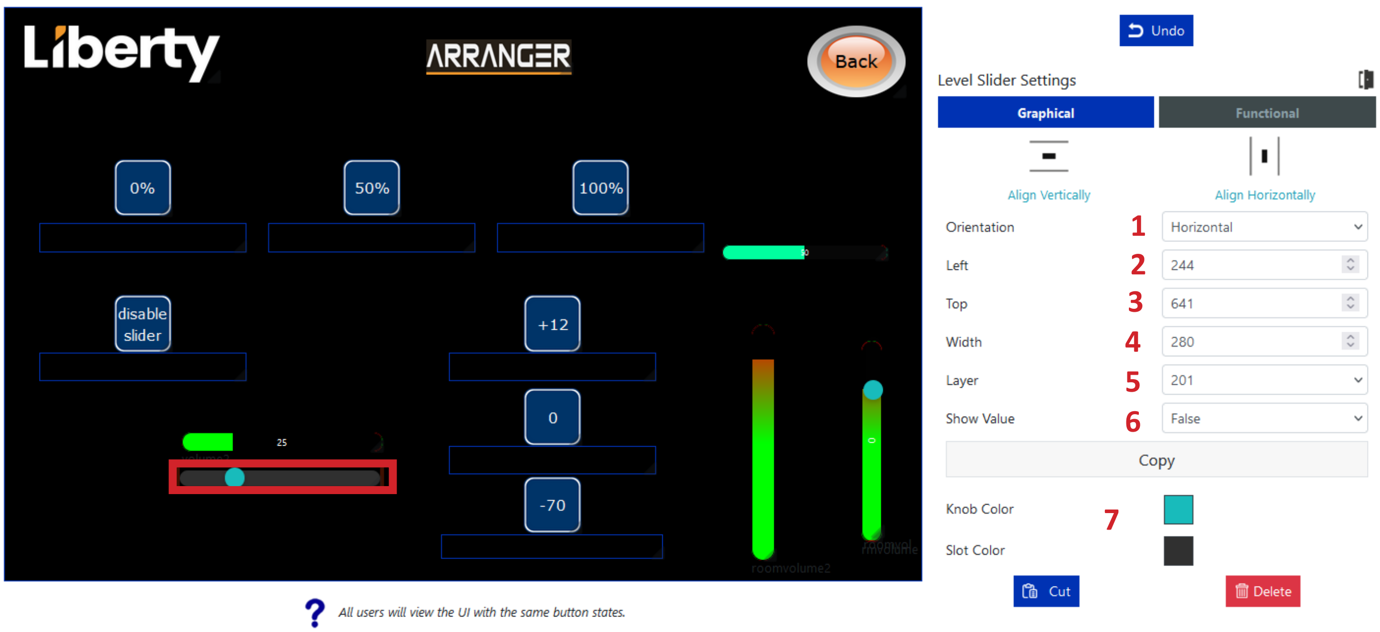
1. Enter in name for the level Indicator.
2. Enter in minimum value.
3. Enter in maximum value.
4. Enter in current value.

The range of the level can be from -20000 to 20000. A name must be entered to control the indicator from a button, API, sliders. Once a level indicator has been combined with a slider the slider values will be applied to the level indicator.

Level Slider

A Level Slider can be dragged to any location then resized by dragging the placeholder or changing the size and location values directly. The Level Slider must be given a name to change the value with button functionality and via API control command set ui_slider.

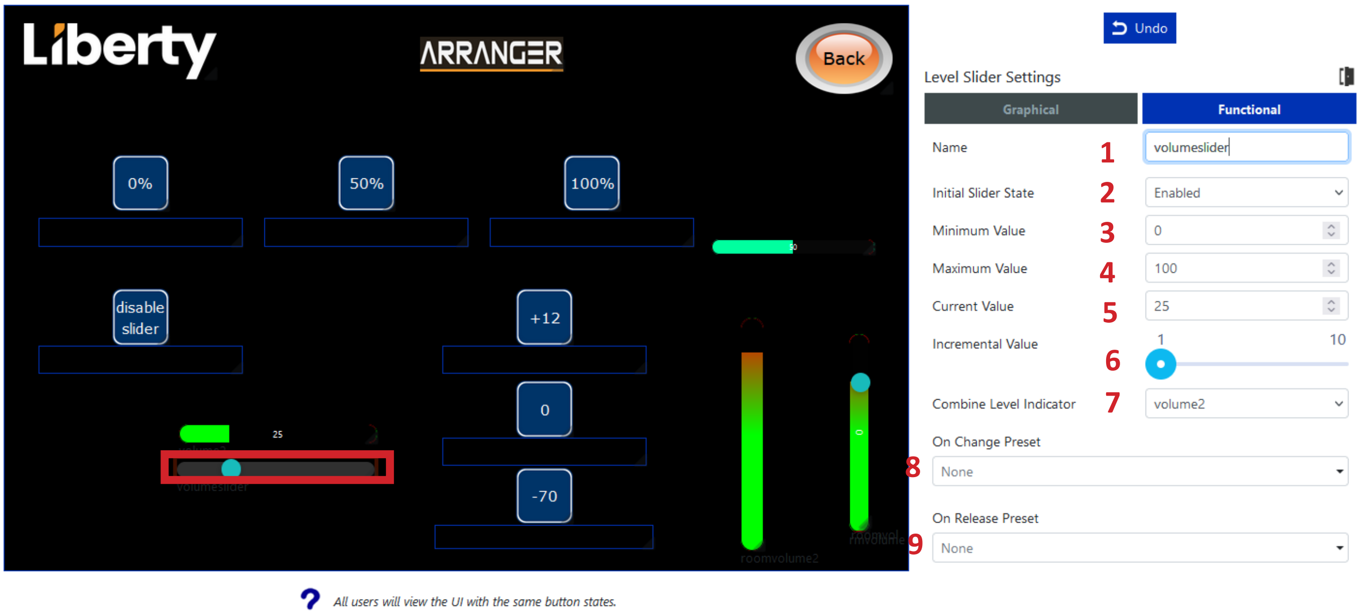
Some examples of uses of level slider are room volume, camera zoom, and camera focus.



1. Select horizontal or vertical orientation.
2. Enter in value for starting left location.
3. Enter in value for starting top location.
4. Enter the width of the slider
5. Select the layer of slider on the page.
6. Select if you want the text value of the level to show or not.
7. Select Color of the knob and slot. .

Level Slider continued

Functional tab.



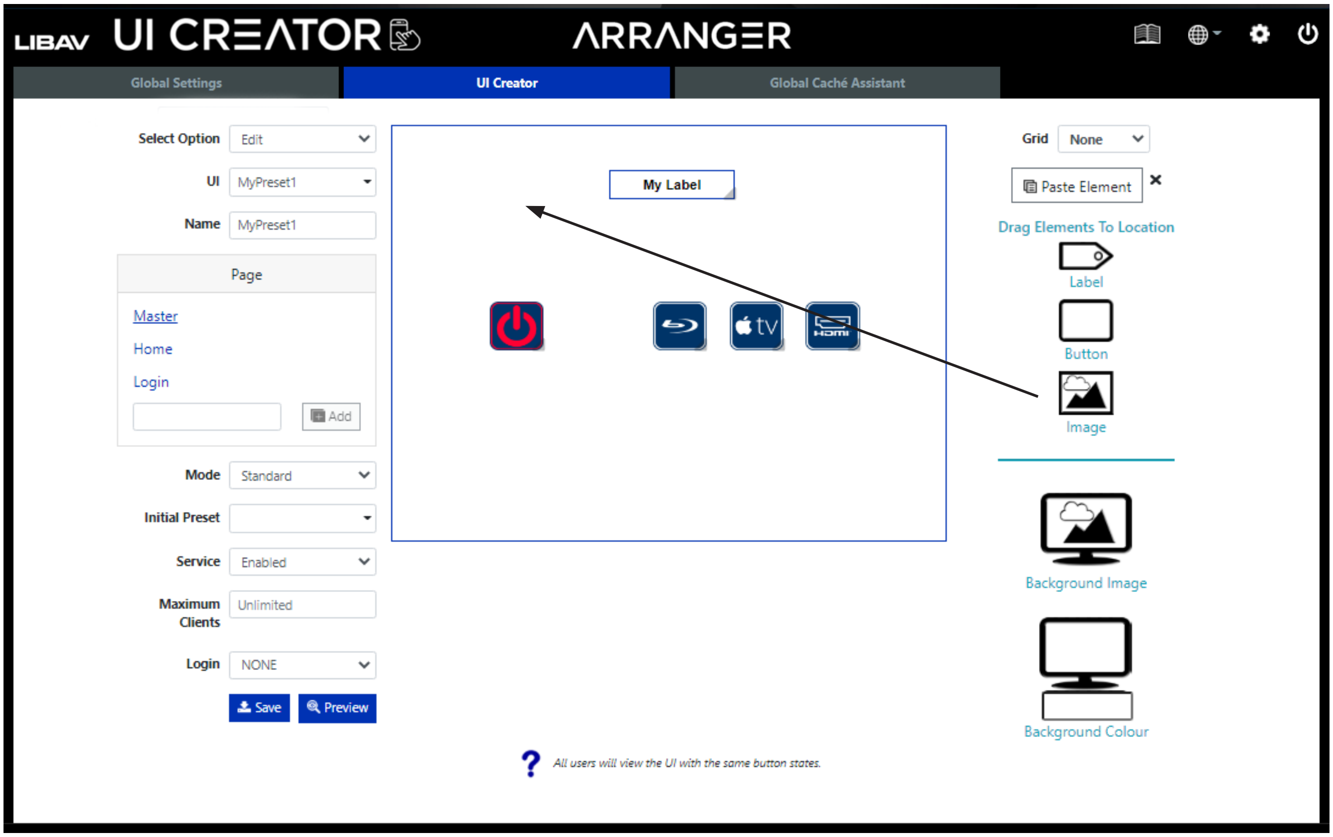
1. Enter in name for the level slider.
2. Select to enable or disable for the initial state of the slider when the UI is loaded.
3. Enter in minimum value.
4. Enter in maximum value..
5. Enter in current value.
6. Select value the slider will increment.
7. Enter in the level Indicator the slider will control if needed.
8. Select a preset to be applied on a change of the slider value.
9. Select a preset to be applied on the release of the slider.

The range of the level can be from -20000 to 20000. A name must be entered to control the slider from a button, API. Once a level indicator has been combined with a slider the slider values will be applied to the level indicator.

Adding an Image

An *Image* can be dragged to any location then resized by dragging the image placeholder or changing the image settings directly. The selected image will be resized to fit the size of the image placeholder. It is recommended to use only the same sized images as the size being displayed.

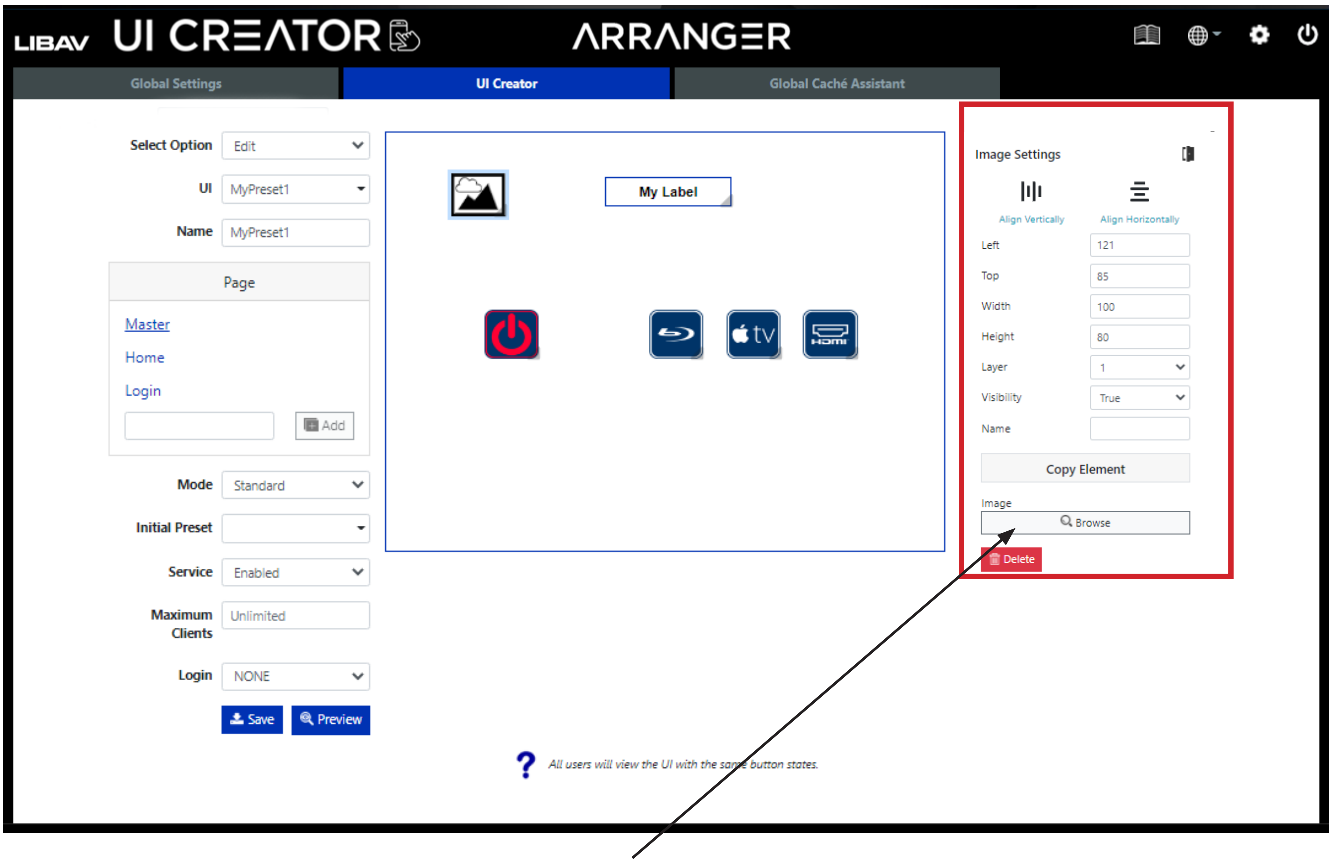
The image must be given a name to change the visibility via the control command ***set ui_image.***



Adding an Image continued....

Once an image has been placed in the UI the *Image Settings* will populate to right of the UI Creator. Image settings gives you the ability to size and name the image as required. See highlighted section below. Settng the visibility to false will remove it from working layers allowing them to be placed over buttons

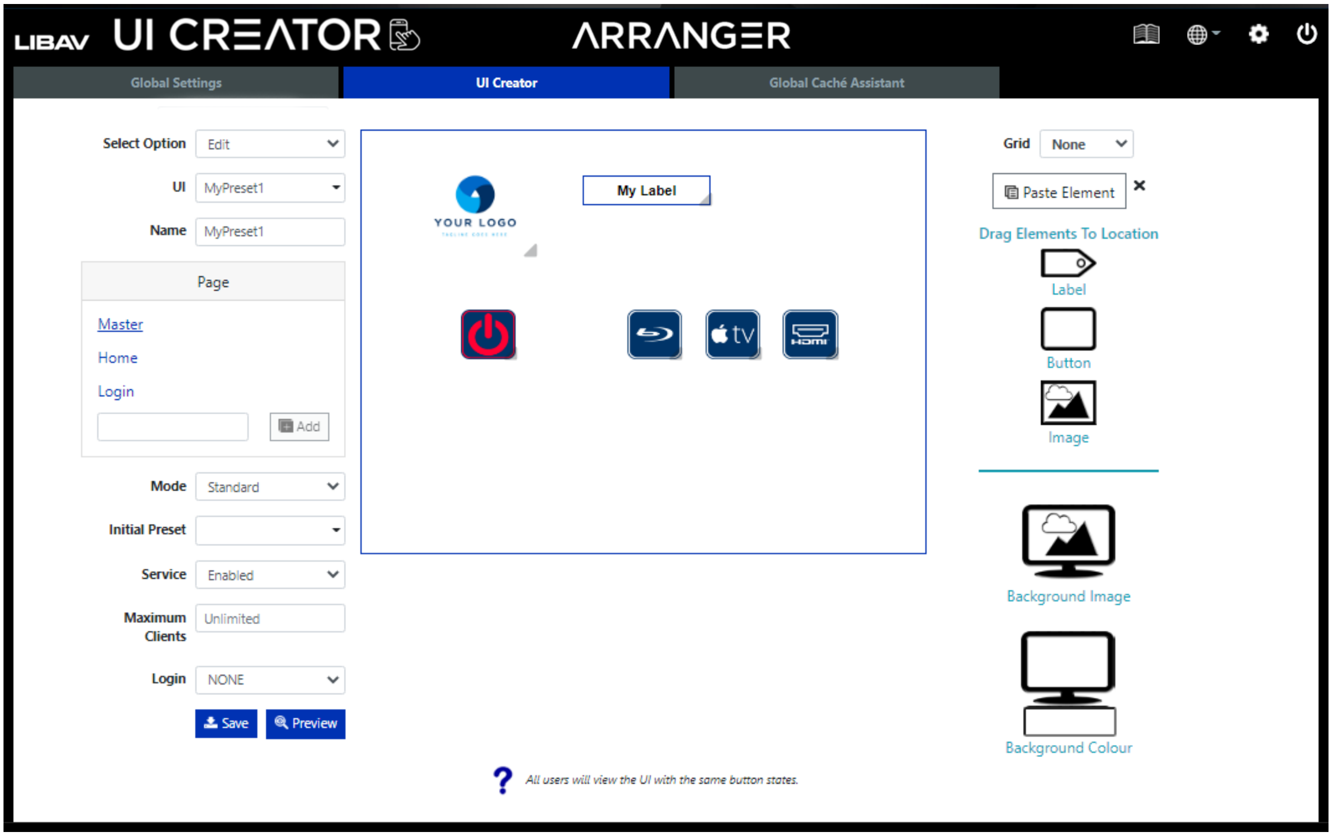
Note: You can always populate the *Image Settings* by clicking on any image in the UI.



To select an image from your PC, select the *Image > Browse* button to browse local folders on your PC.

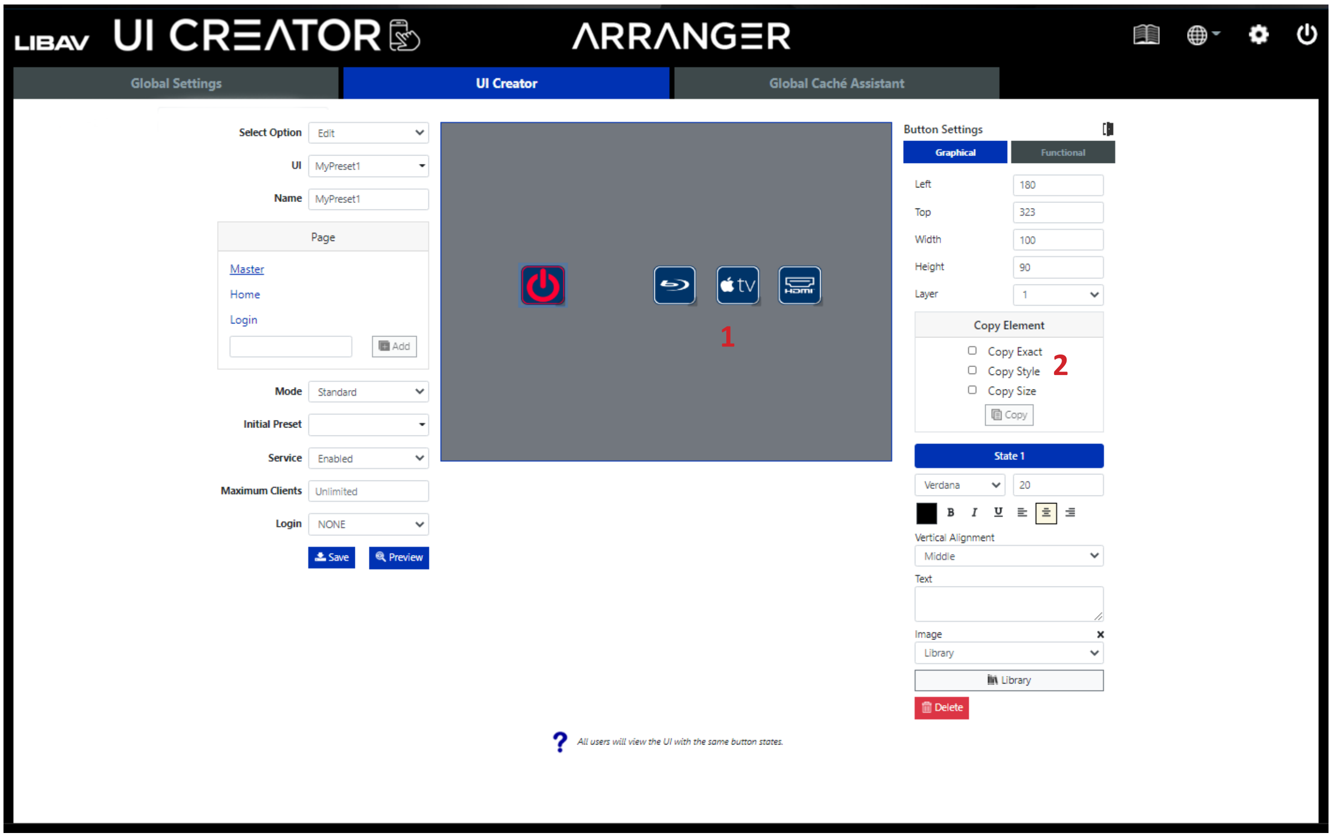
Adding an Image continued....

Here a logo image has now been assigned.



Copying and Pasting Elements

Any button label or image can be copied and pasted onto the UI. You copy the style, size or the element exactly.

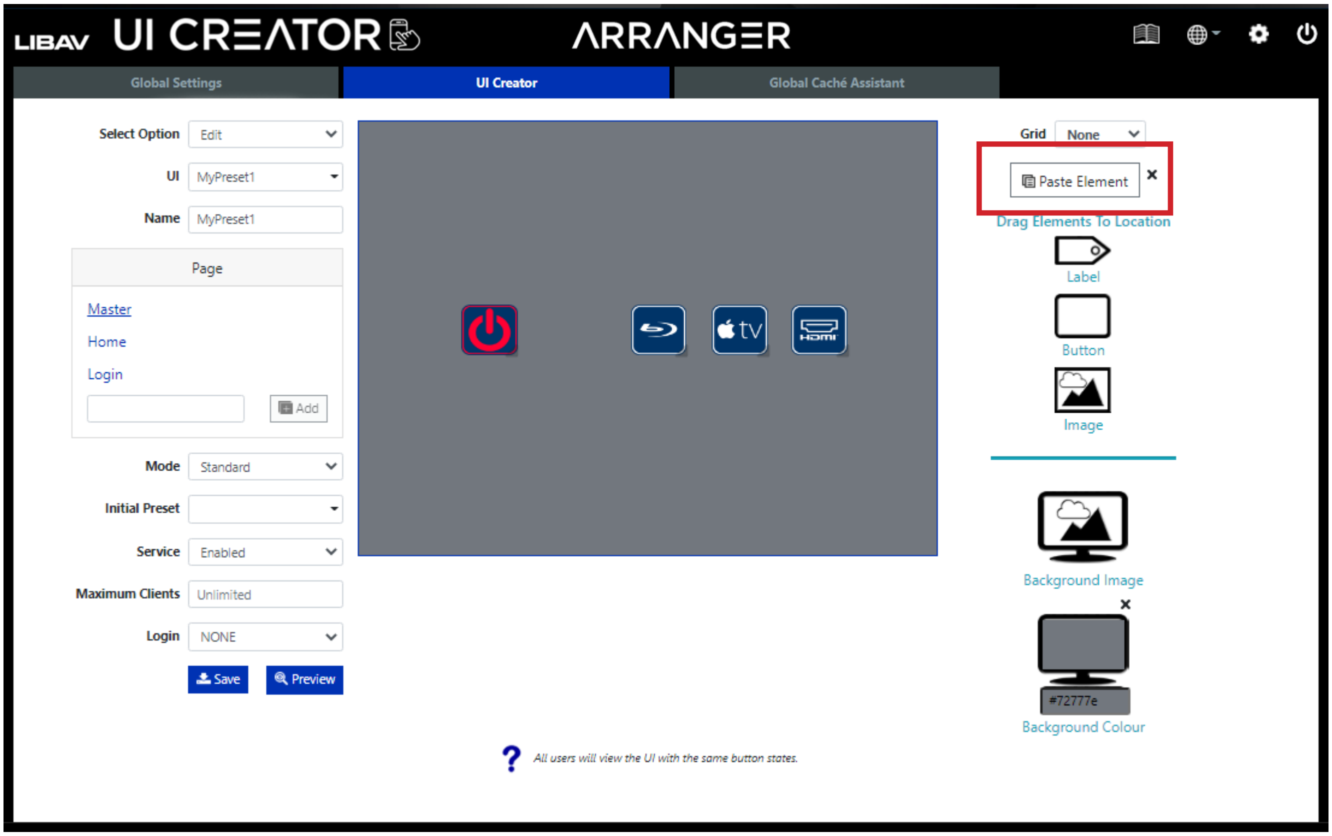


1. Click on any button, label or image element in the UI.

2. Click *Copy Element*, then choose what copy type and click *Copy.*

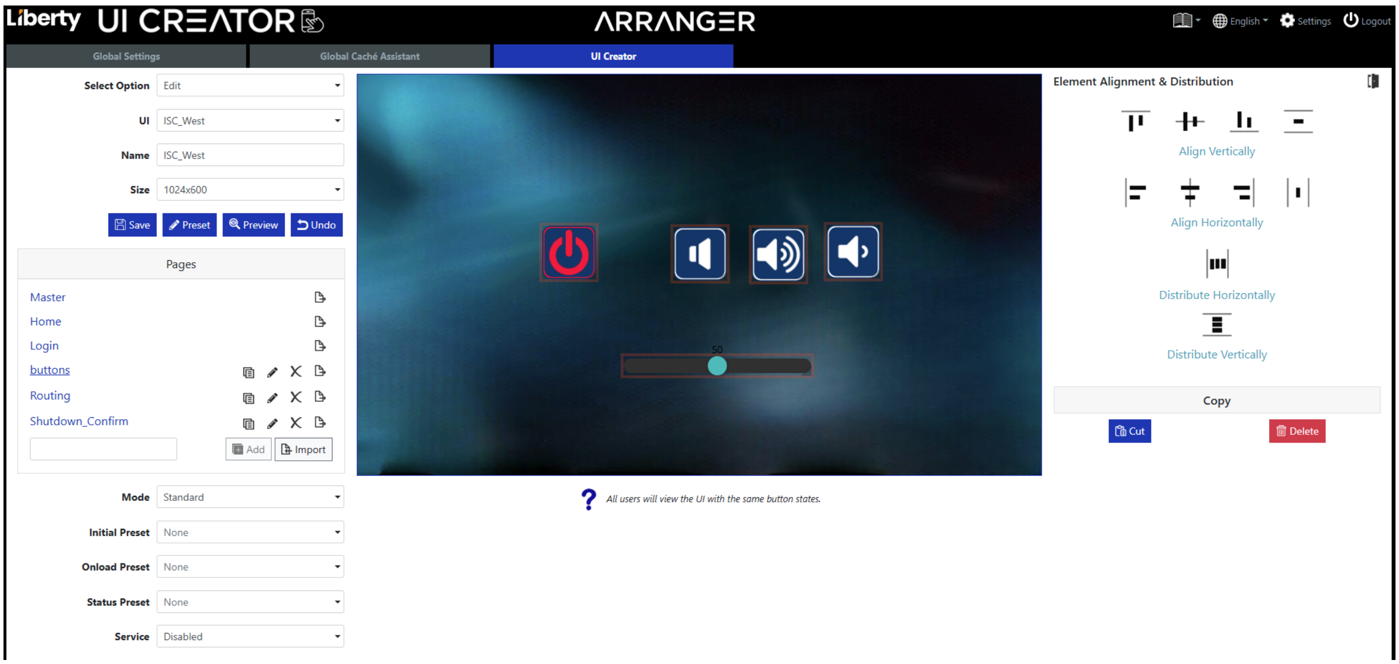
Copying and Pasting continued.....

Once an element has been copied, you will now see in the upper right-hand corner of the UI Creator an icon that says *Paste Element*. Click this icon to paste the copy type selection into the UI.



Element Alignment

An elements graphical tab provides vertically and horizontally page alignment. Multiple selected elements can also be centred on the page.

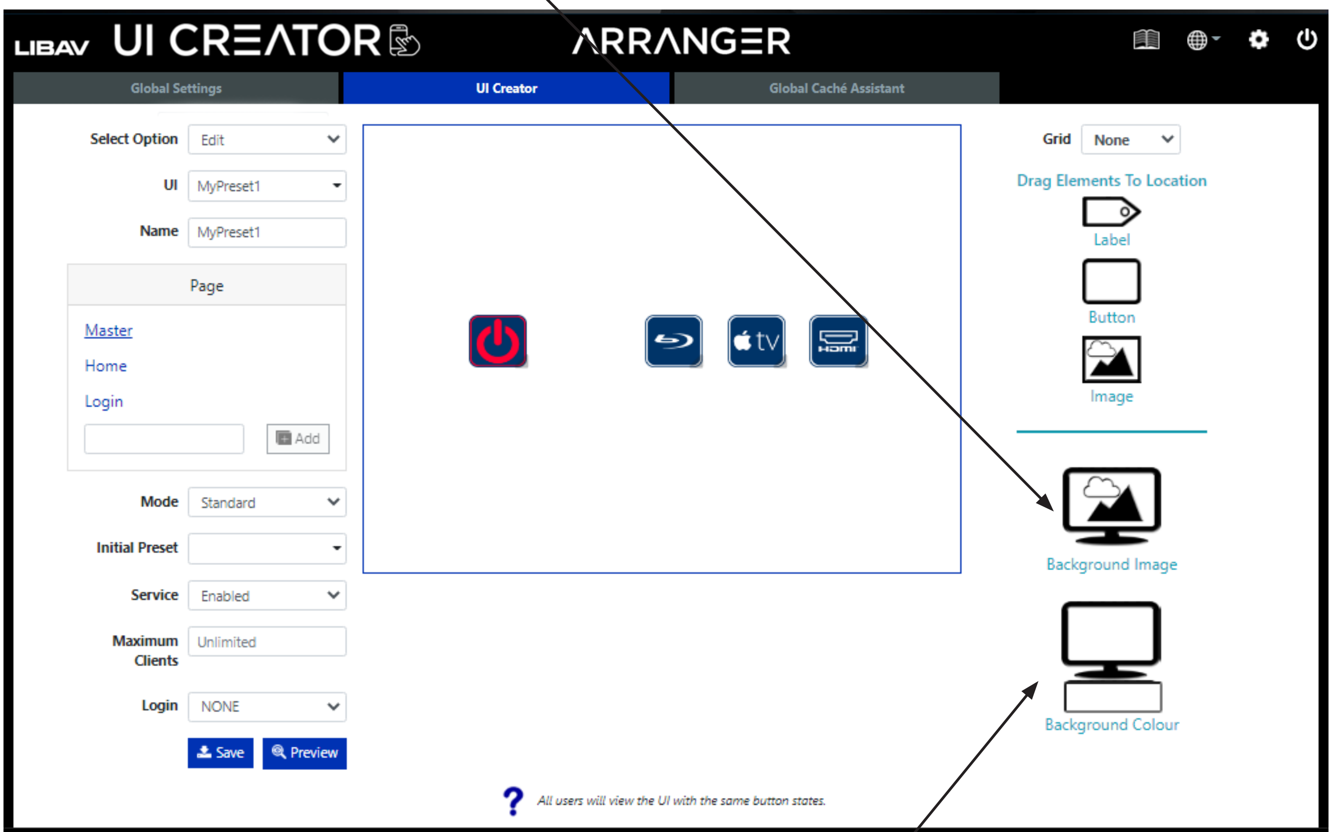


Multiple elements can be aligned with respect to the first selected element, or the page.
Click on the first referenced element to select it, then hold Ctrl while selecting further elements to be aligned.
An Element Alignment and Distribution panel will then be shown to Align Vertically, Align Horizontally, Distribute Horizontally or Distribute Vertically.
Clicking in white space will deselect selected elements.

Adding a Background

Either an image or solid color can be selected for the page background. Applying a background on the *Master Page* will be seen on all other pages without a background. Applying a background to any other page than the *Master Page* will hide the *Master Page* altogether.

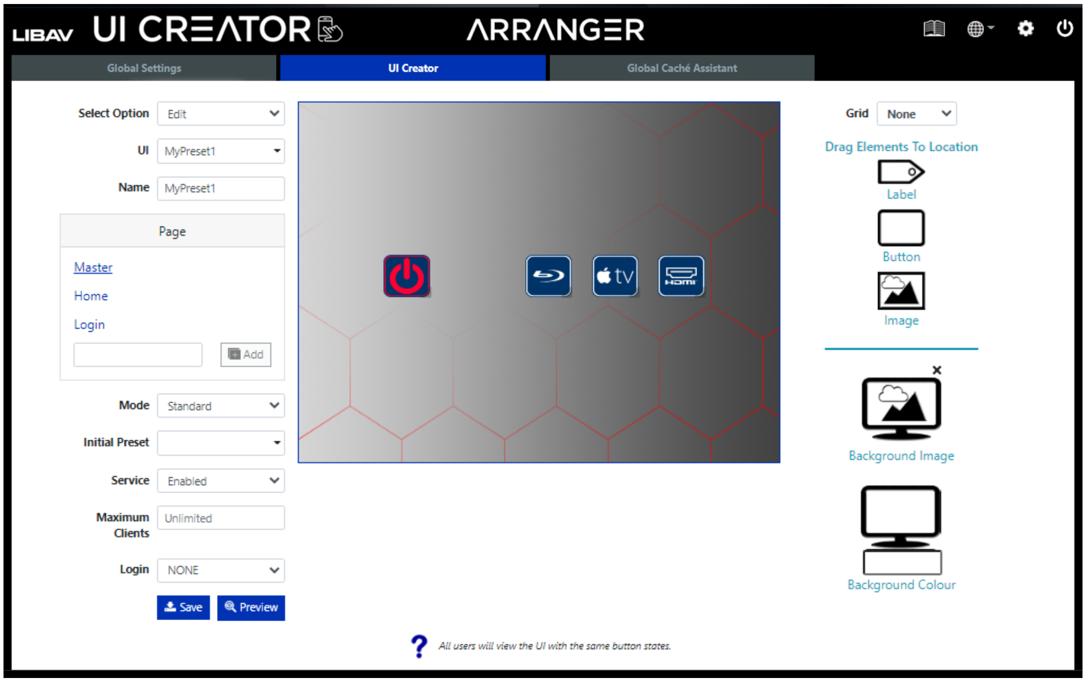
Click the *Background Image* icon to the right of UI creator to browse your local folders on your PC for a background image to use as a background.



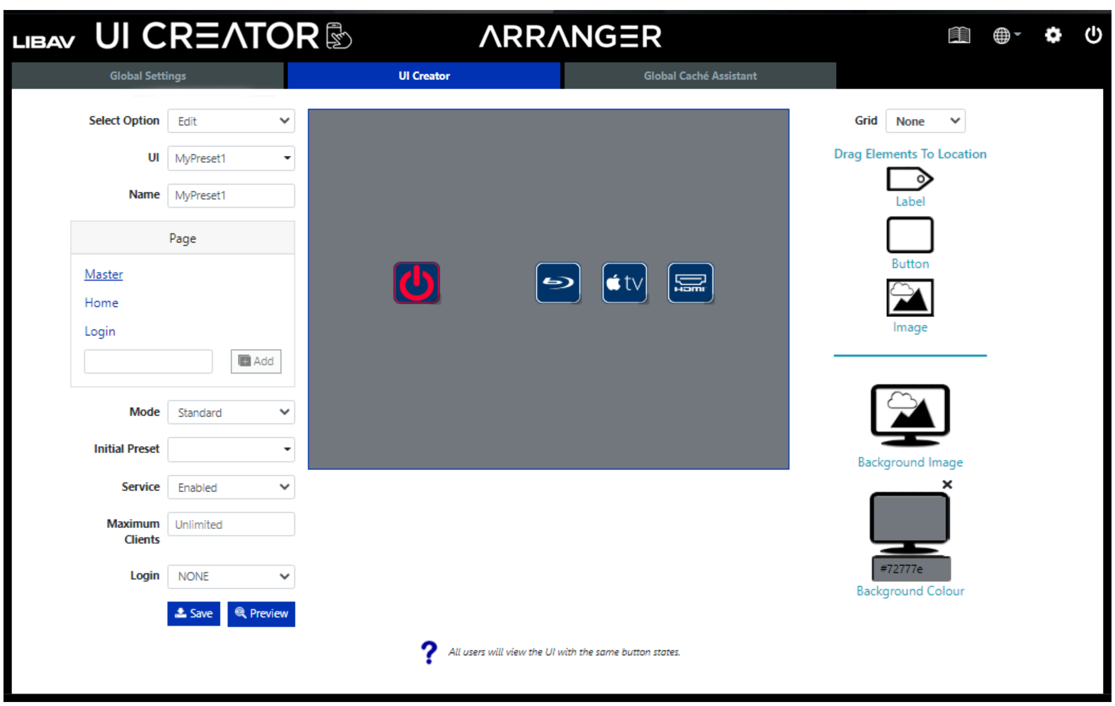
If a solid color for the background is required then select the *Background Color* icon and select a color from the pop-up color picker.

Adding a Background continued.....

Here a background image has been applied to the *Master Page.*

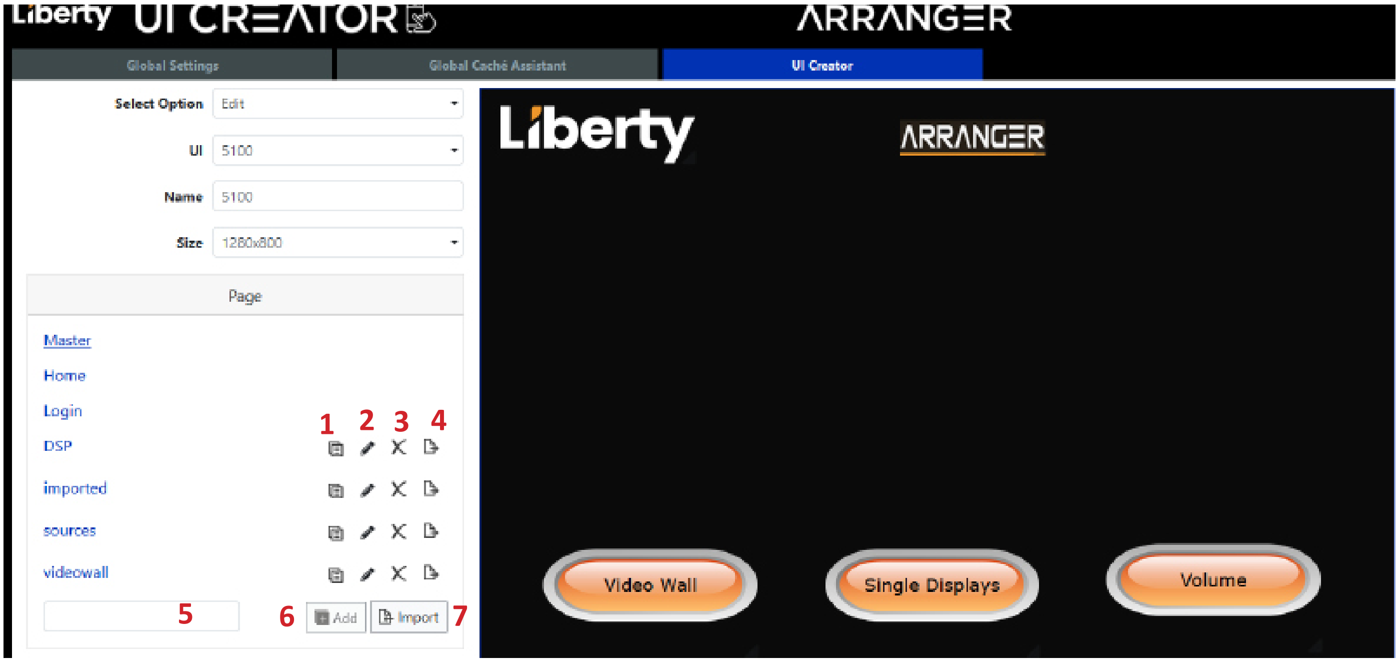


Here a background color has been applied to the *Master Page.*



Editing pages

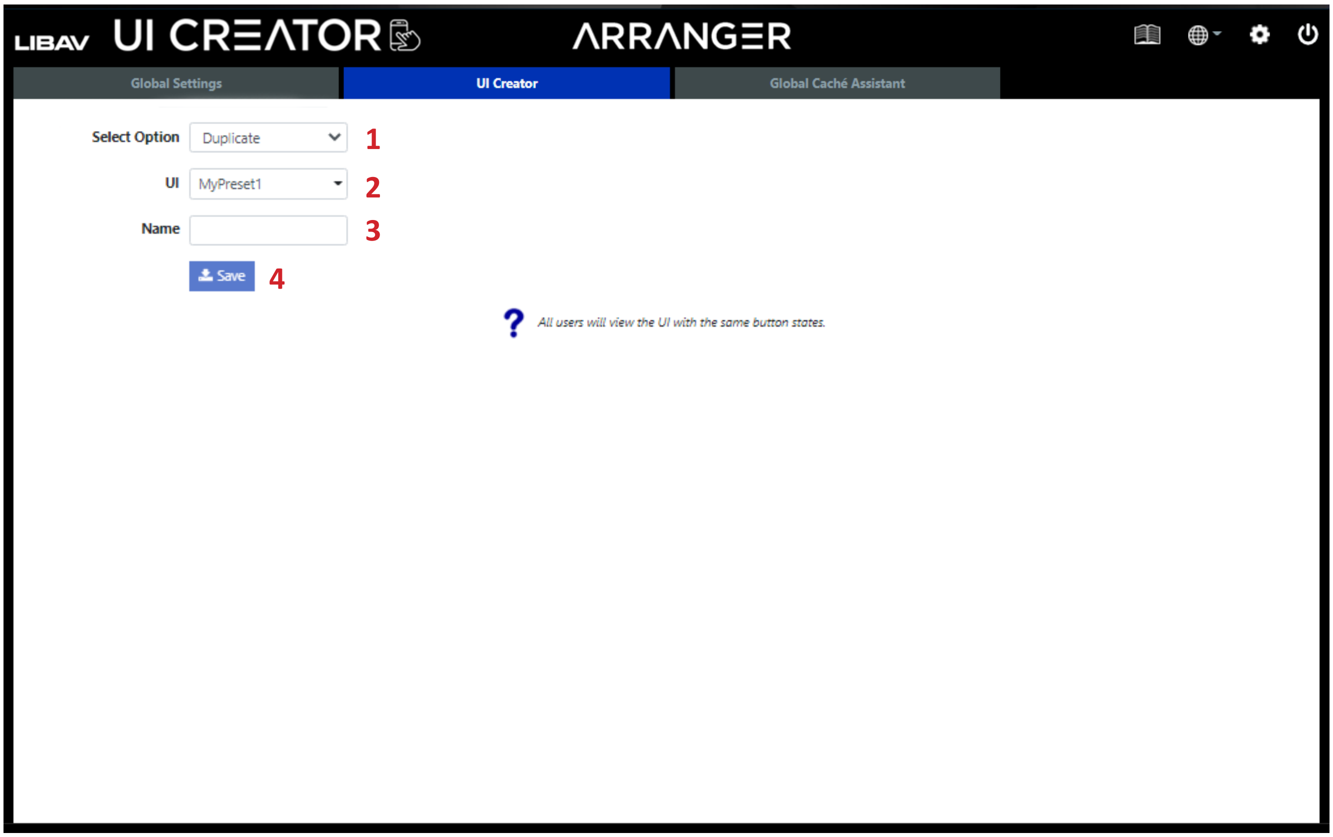
Pages can be duplicated, deleted,exported or imported including Master, Home and login.



1. Press to Duplicate a page
2. Press to Rename a page.
3. Press to Delete a page.
4. Press to Export a page.
5. Type in a name before clicking add.
6. Press to add a page but first type in a name in the box to the left.
7. Press to import in a page. After pressing a file explorer window will open, browse to the file and click open. A popup window will allow renaming, overwriting presets.

Duplicate a User Interface

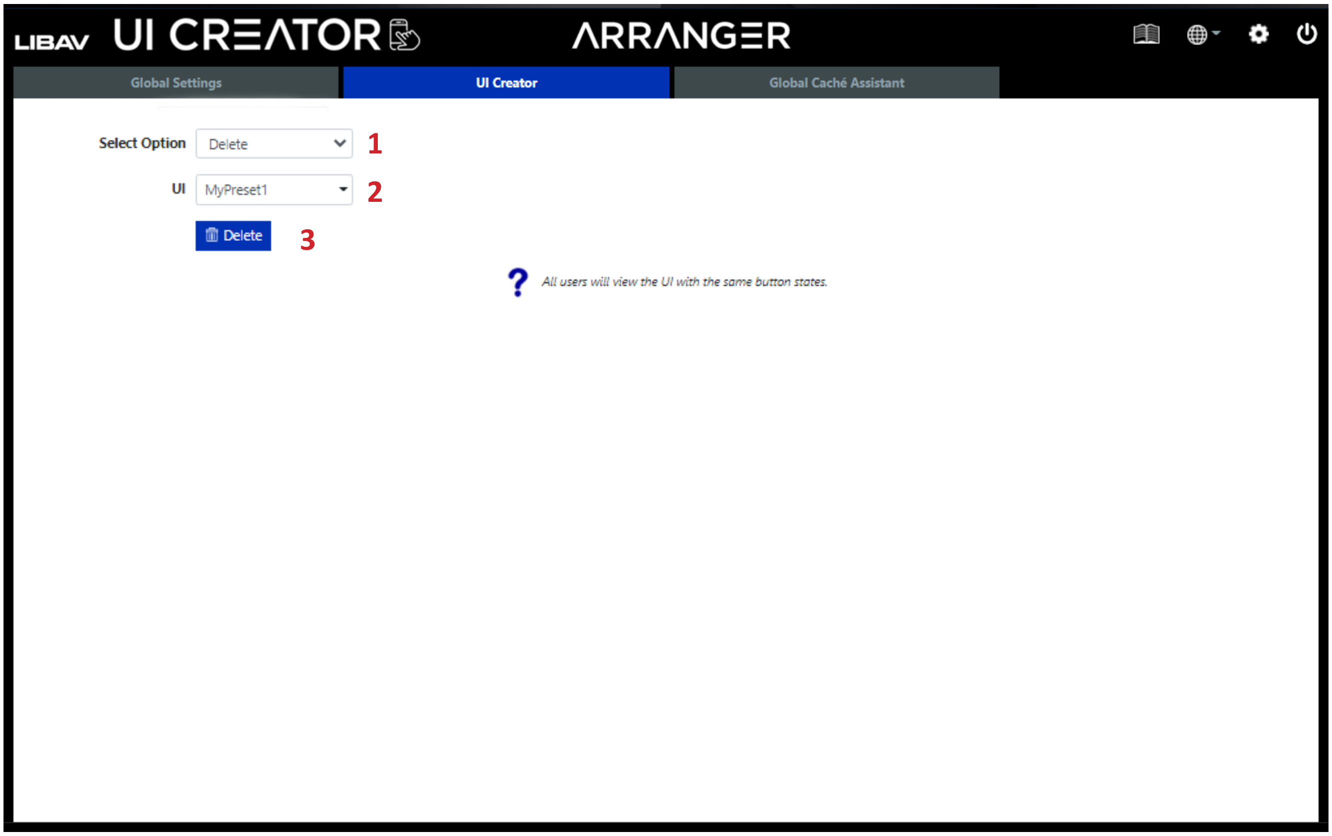
Here you can duplicate an existing user interface to be used as a backup or duplicated from a template file that can then be edited.



1. Select *Duplicate.*
2. Select *UI.*
3. Enter name of duplicate UI.
4. Click *Save.*

Delete a User Interface

To delete an existing UI select option *Delete*, select the user interface and then click the delete button.



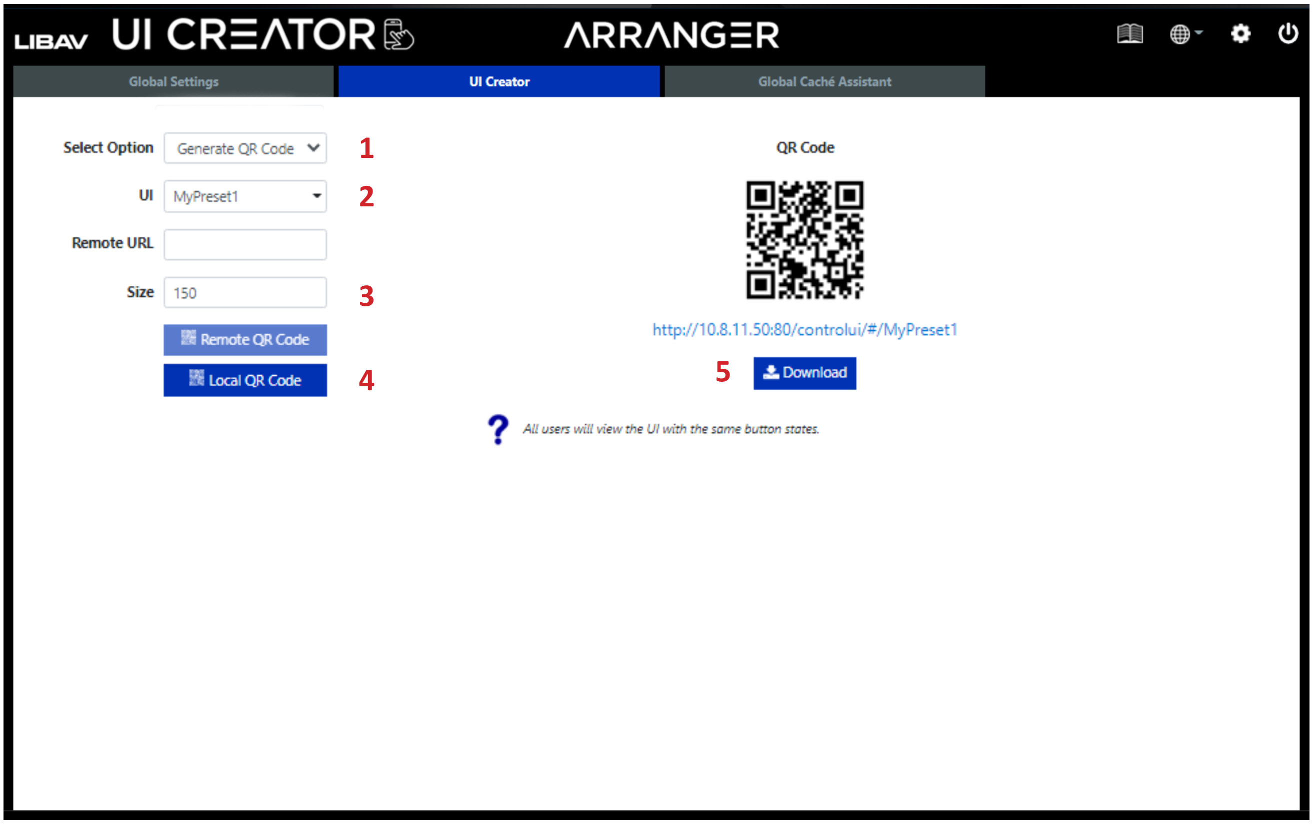
1. Select *Delete.*

2. Select *UI.*

3. Click *Save.*

Generating a Local UI QR Code and URL

QR codes and URL links can be generated and downloaded to easily create the URL required to browse to the *User Interface* webpage. The size of the QR code can be set between 100 – 2000px.



1. Select *Generate QR Code.*

2. Select *UI.*

3. Enter QR code *Size.*

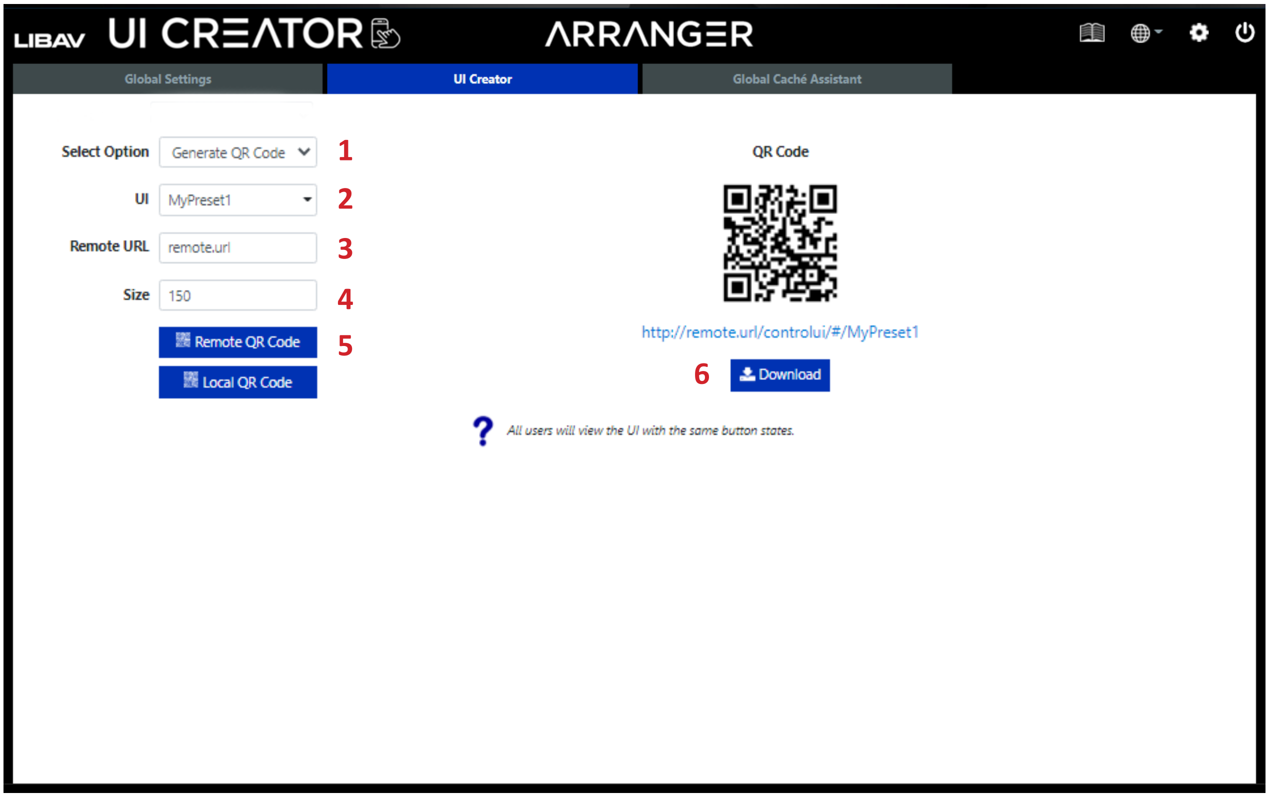
4. Click *Local QR Code.*

5. **Click** *Download.*

   The URL for the UI is displayed under the QR code.

Generating a Remote UI QR Code and URL

To browse to the UI via an external URL enter the details in the external URL box and select remote QR code. The size of the QR code image can be changed then downloaded to be used in manuals or printed as required.



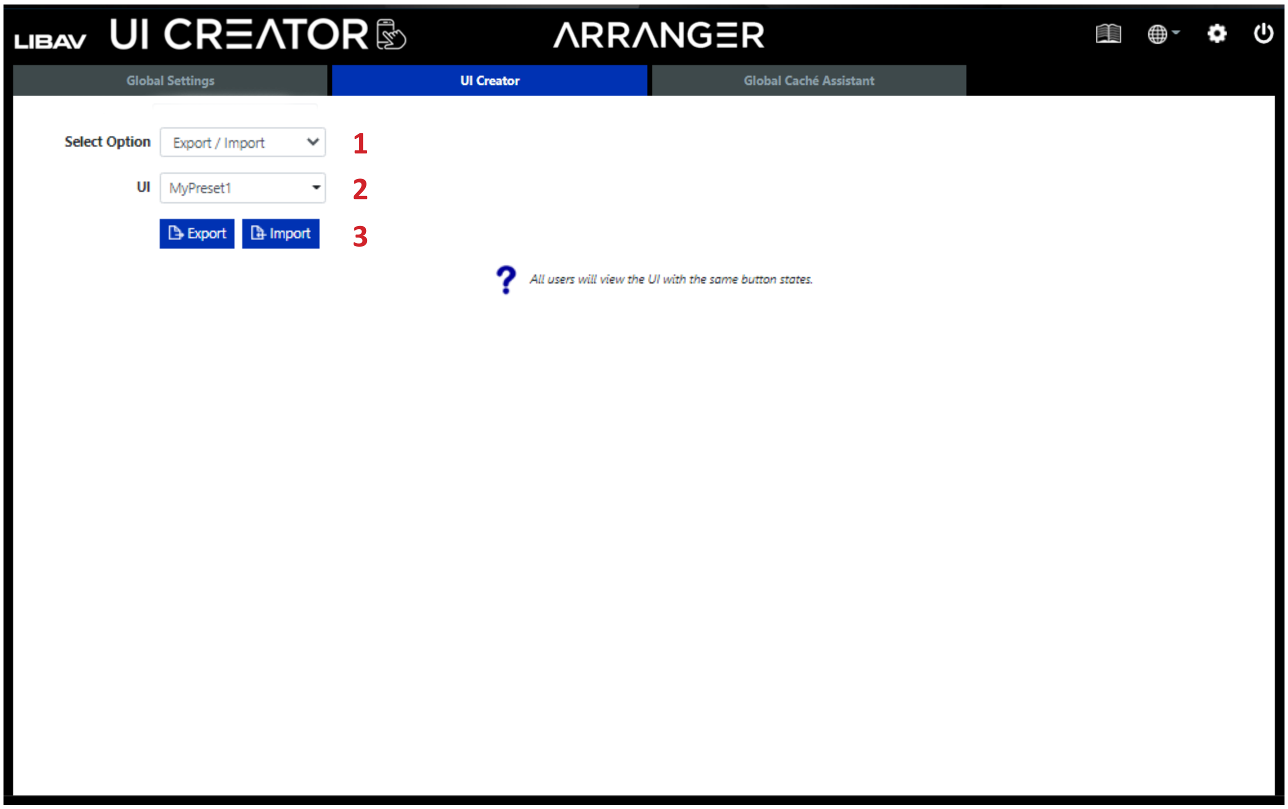
1. Select *Generate QR Code.*
2. Select *UI.*
3. Enter *Remote URL.*
4. Enter QR code *Size.*
5. Click *Remote QR Code.*
6. **Click** *Download.*

   The URL for the UI is displayed under the QR code.

Exporting / Importing User Interface

To keep a backup of your UI work select *Export / Import* then click the *Export* button.

A *.exp file will be saved to your downloads folder.



1. Select *Export / Import.*

2. Select *UI.*

3. Click *Export* to export .exp file to your Downloads folder.

   OR

   Click *Import* to import an .exp file from your local PC.

Global Cache Assistant

The *Global Cache Assistant* tab is used to configure and communicate with Global Cache endpoints to communicate with third-party devices over a network via serial, infrared (IR), contact closures, and relays using the API control command ***send gc.***

Sending Serial Signals

The following wired and wireless Global Cache devices can be used to send serial signals to 3rd party devices from Arranger.

iTach Flex PoE
PN: FLEXIP-IP

iTach Flex WiFi
PN: FLEXIP-IP

The following Global Cache accessories can be used on all devices listed above for serial endpoint communication.

FLC-SL-232

FLC-SL-485

FLC-SL-MJ

Sending Serial Signals continued.....

The following wired Global Cache devices can be used to send serial signals to 3rd party devices from Arranger and does not require proprietary cabling for serial / RS232 connections.

iTach Ethernet to Serial
PN: IP2SL

iTach Ethernet to Serial
PN: IP2SL-P (PoE)

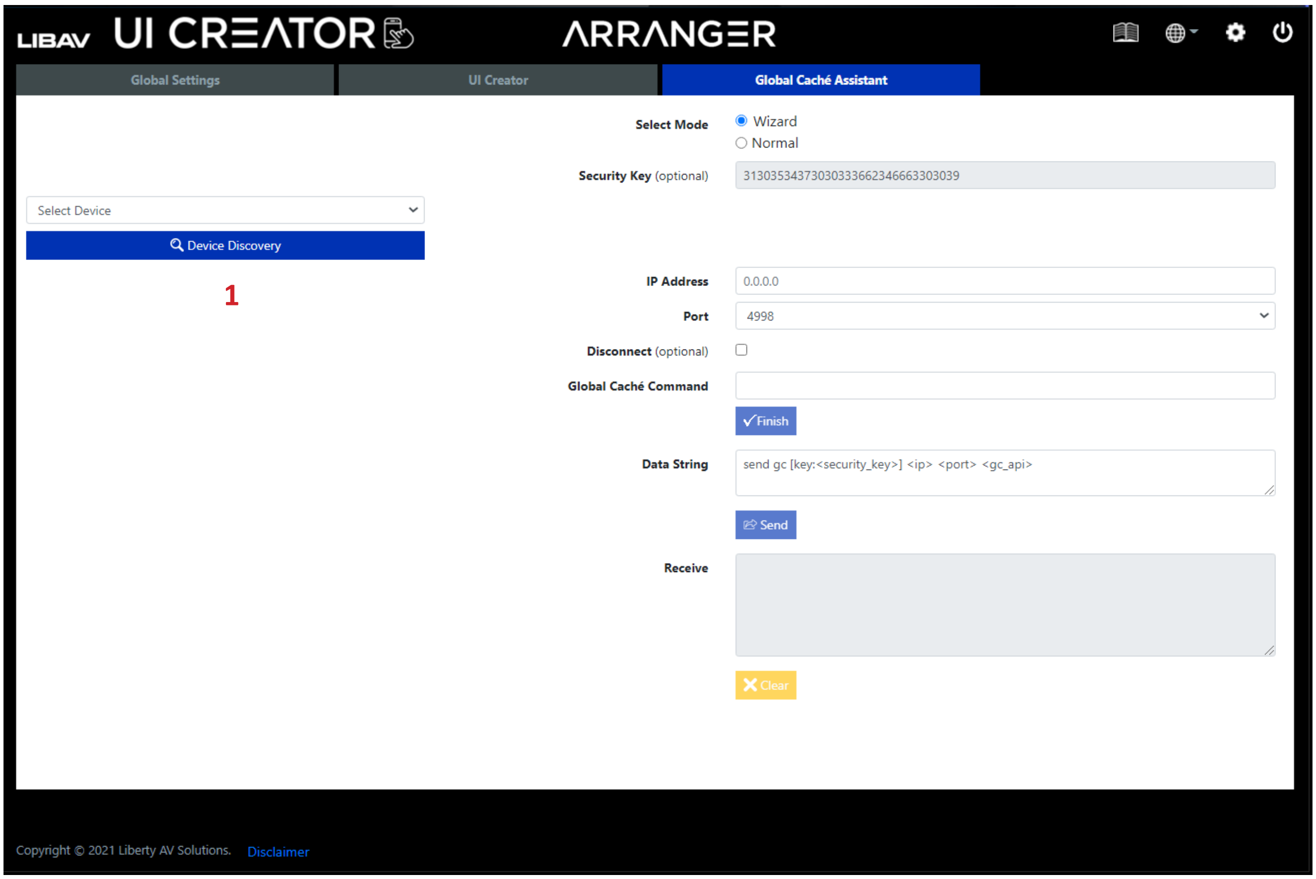
iTach Ethernet to Serial
PN: WF2SL (WiFi)

Sending Serial Signals continued.....

In this example we will configure the iTach IP2SL TCP/IP to serial endpoint which will allow us to send a serial command string to a third-party device connected to the IP2SL via DB9 / RS232 connection.

In order to communicate with a 3rd party device, gather device details for the serial connection of the device, i.e. baud rate, parity, stop bits and ASCII control commands that you wan to use. This unit will not generate those commands or settings for you automatically.

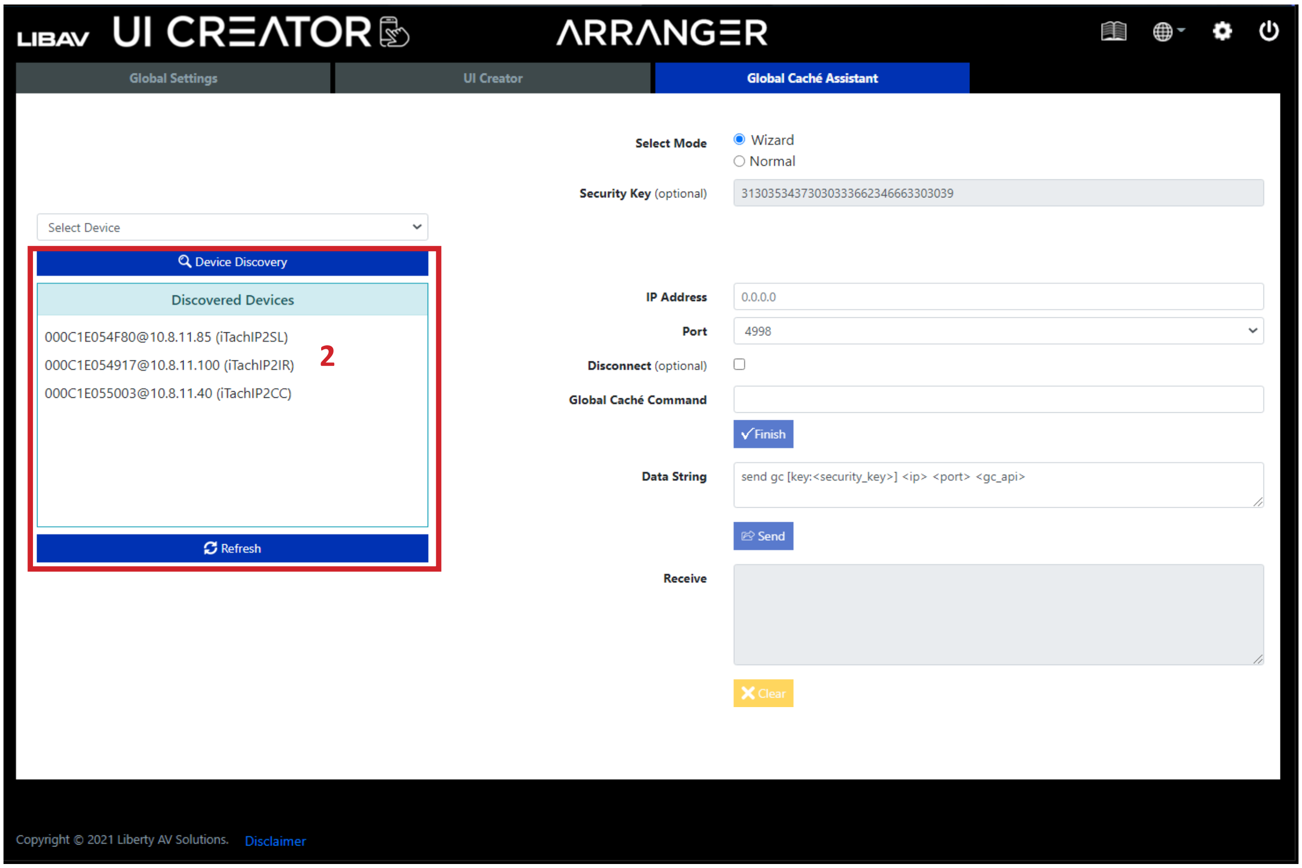
NOTE: The Global Cache *Assistant* is also used in the *PRESETS* menu so you can save the generated strings for future use



1. To identify Global Cache devices on your network, click *Device Discovery*; this will search for the Global Cache devices and will return a list like below.

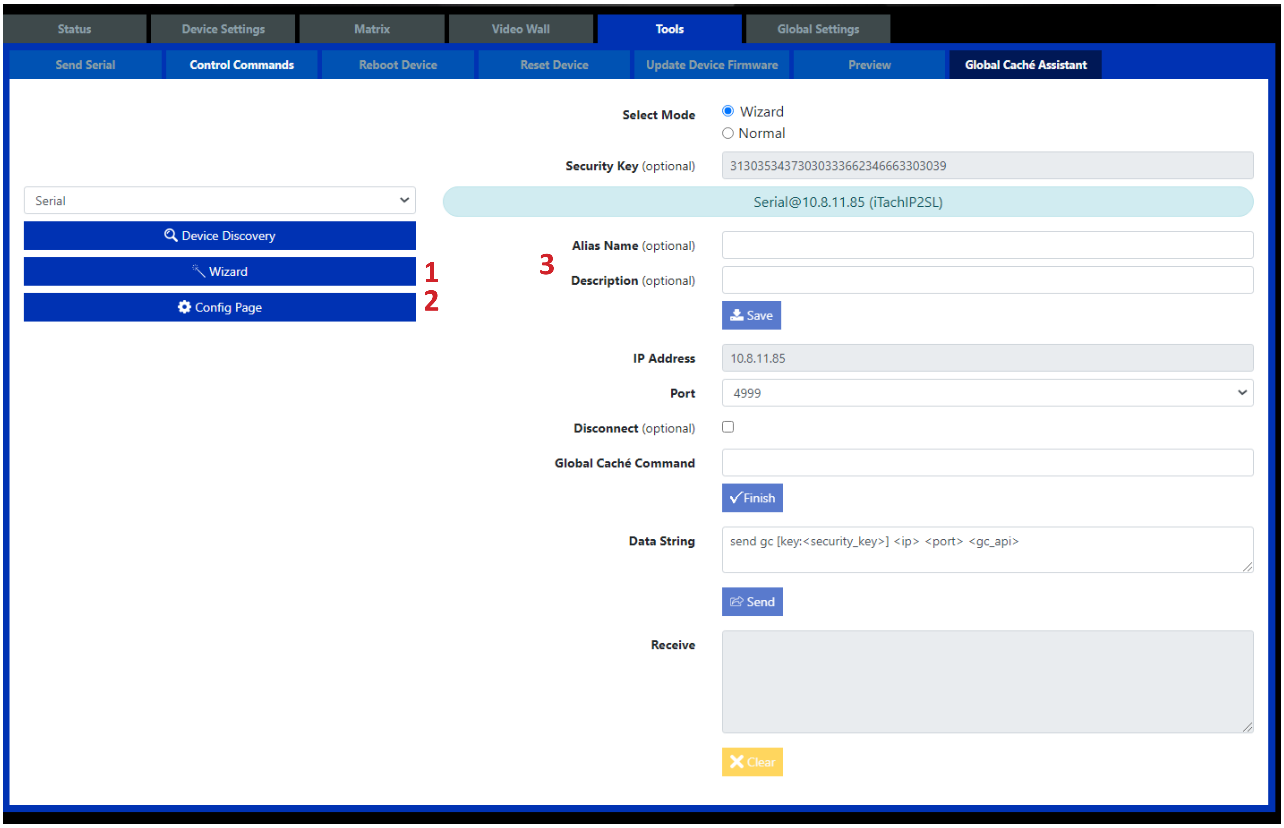
Sending Serial Signals continued.....

2. After initial discovery of Global Cache devices click on the iTachIP2SL in the menu below. Click on the desired Global Cache device to configure.



Sending Serial Signals continued.....

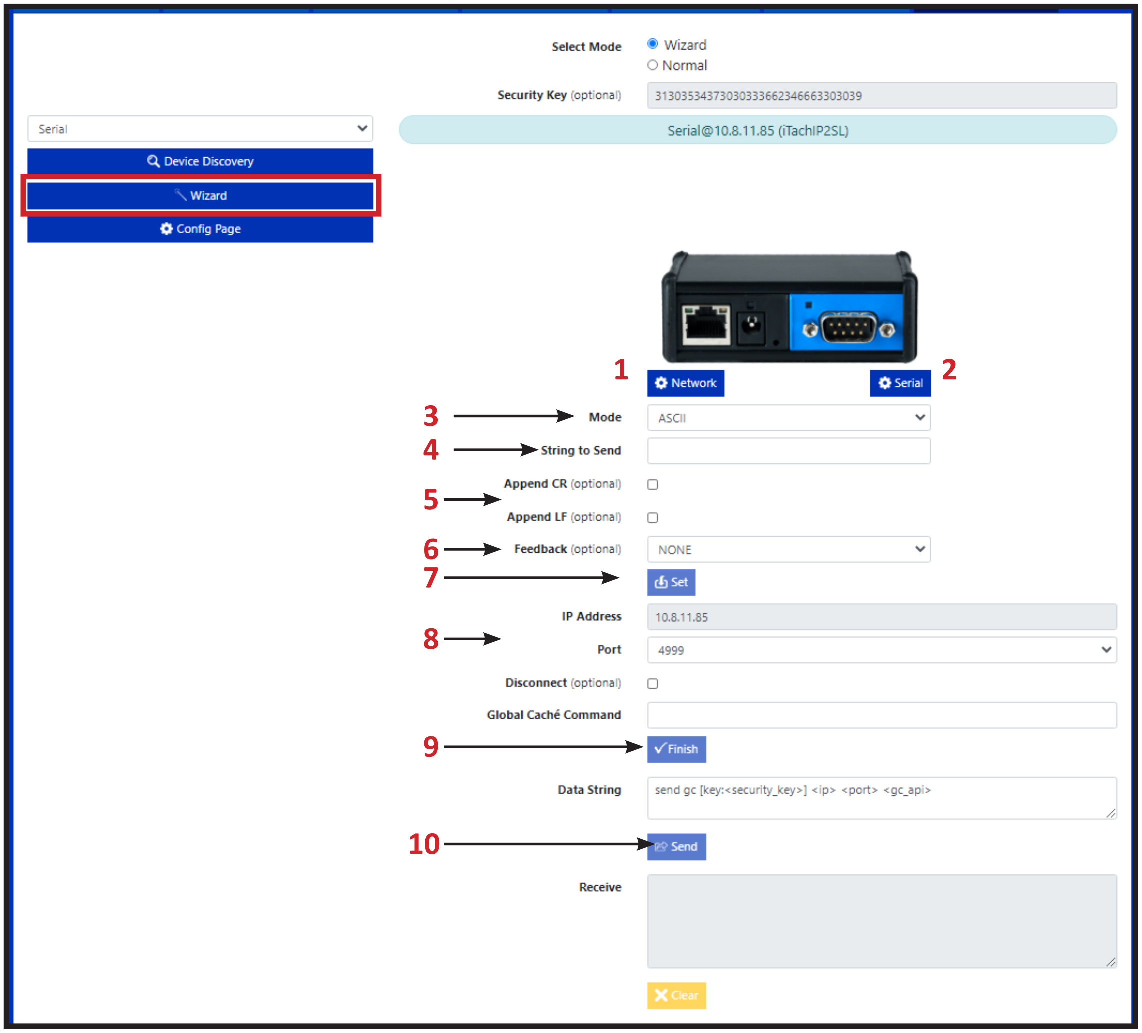
The following menus appear after the iTach device is selected. This menu is referred to as *Wizard mode*; to enter information in manually select *Normal* as the *Select Mode* option.



1. The *Wizard* menu provides access to configure the device to send serial strings.
2. The *Config Page* will direct you to the Global Caches internal web server; this will pop-up in another browser menu.
3. Apply an alias name and description of the device and click *Save* so the device can be accessed more quickly in the future, otherwise you will need to discover the device again. These named devices will appear via drop-down above the *Device Discovery* menu.

Sending Serial Signals continued.....

The following appears after the *Wizard* menu is selected in the previous step. Here we will configure the serial command to send.



1. *Network* option allows you to update the Global Caches devices network settings; by default the IP2SL is set to DHCP and falls back on APIPA 169.254.0.0/16 Network ID in absence of a DHCP server / router.

2. *Serial* option allows you to update the serial communication parameters of the device i.e. baud rate.

3. Apply an *alias name* and *description* of the device and click *Save* so the device can be accessed more quickly in the future otherwise you will need to discover the device again. These named devices will appear via drop-down above the *Device Discovery* menu.

Sending Serial Signals continued.....

4. Enter in the serial string you would like to transmit (see 3rd party device documentation for commands)

5. Add terminator <CR> and/or <LF> if required.

6. Enter desired *Feedback* (optional).

7. Once all the above parameters are entered click *Set*; this will format the entered command in the *Global Cache Command* field automatically for use.

8. Select port and disconnect if required (optional).

9. Once the information above has been entered in click *Finish.*

10. Click *Send* to test the code.

You can now use this generated command in a preset to be used with UI control, scheduling and automation OR send this command from a 3rd party control system to the Arranger controller to activate the current configuration. Arranger controller listens on TCP port 6980.

Sending IR Signals

The following wired and wireless Global Cache devices can be used to send IR signals to 3rd party devices from Arranger.

iTach Flex PoE
PN: FLEXIP-IP

iTach Flex WiFi
PN: FLEXIP-IP

The following Global Cache accessories can be used on all devices listed above for IR endpoint communication.

FLC-1E

FLC-3E

FLC-SL-MJ

FLC-BL

FLC-T3

Sending IR Signals continued.....

The following wired Global Cache devices can be used to send IR signals to 3rd party devices from Arranger.

iTach WiFi to IR
PN: WF2IR

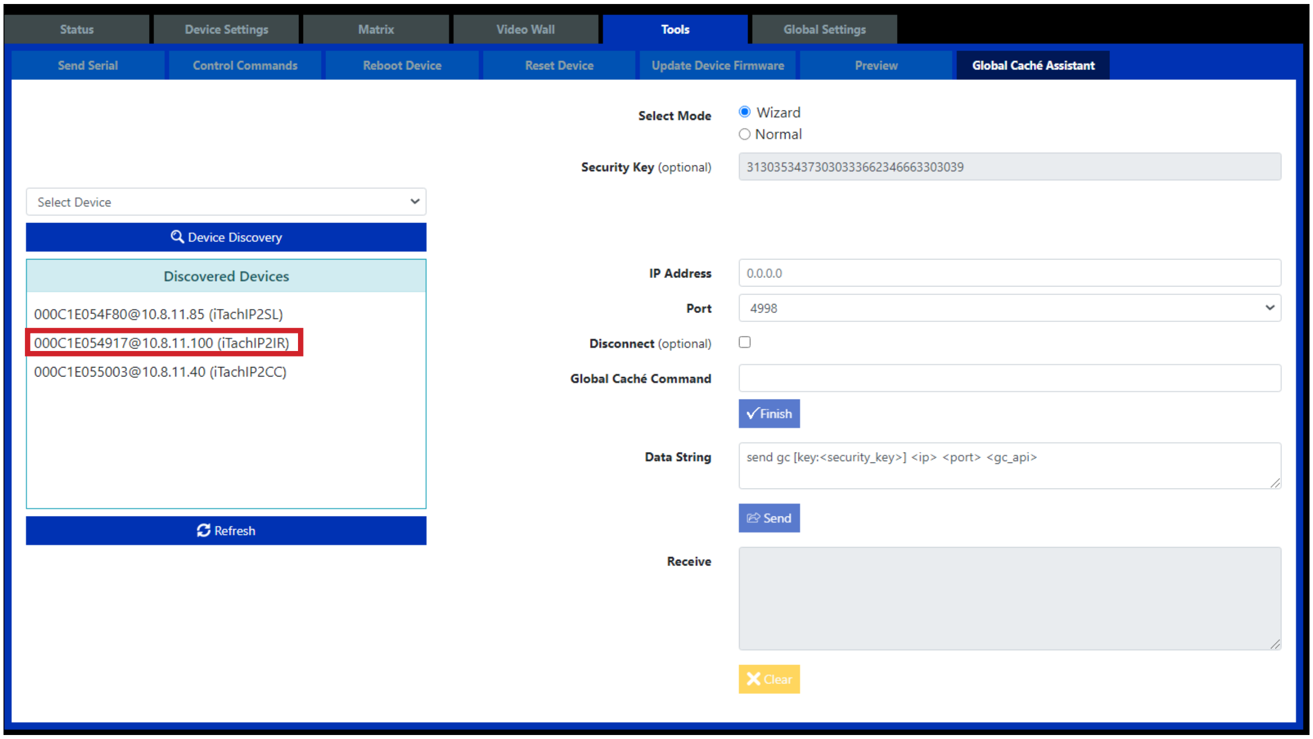
iTach Ethernet to IR
PN: IP2SL-P (PoE)

iTach Ethernet to IR
PN: IP2SL

Sending IR Signals continued.....

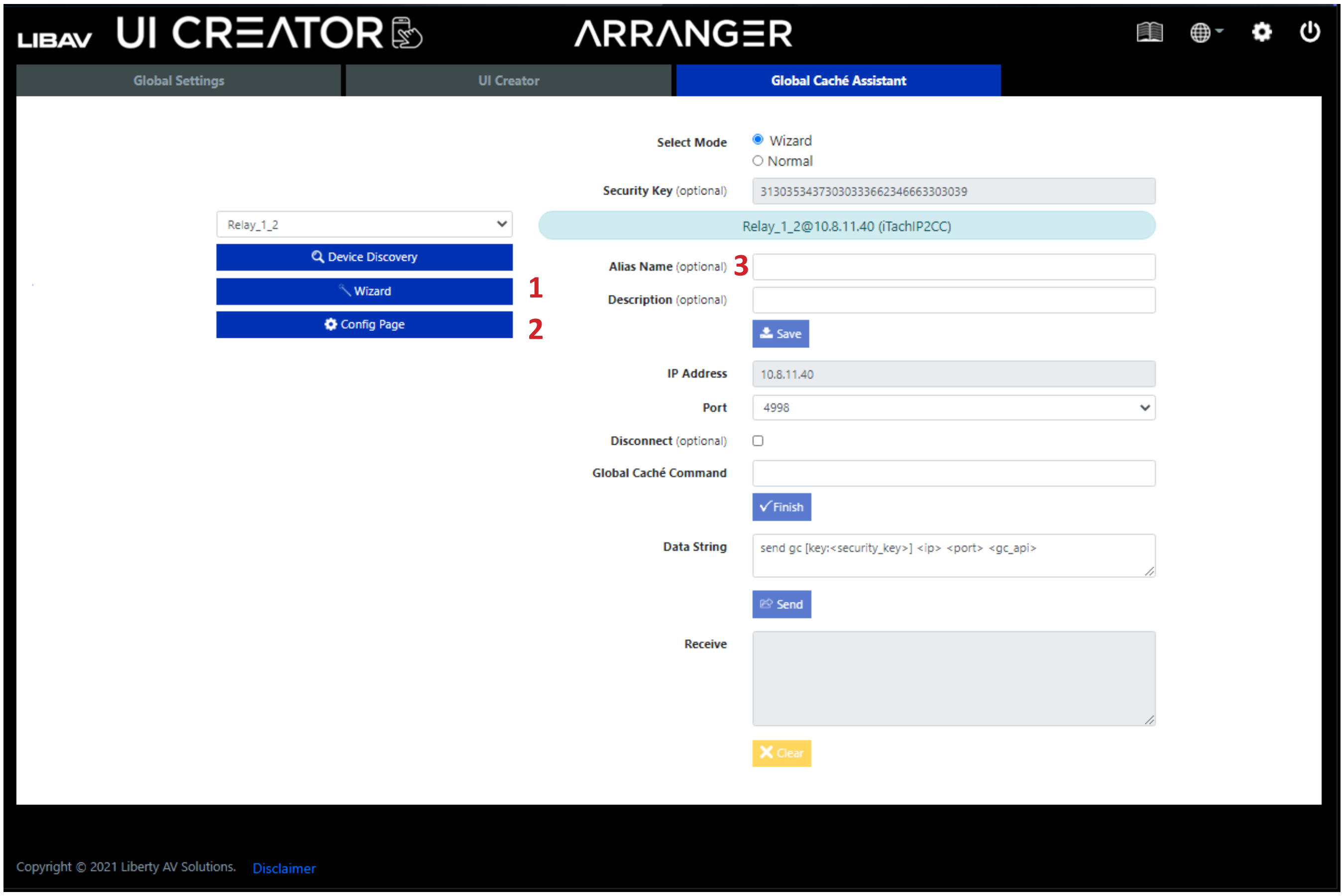
In this example we will configure the iTach IP2IR TCP/IP to IR endpoint. The IP2IR can connect, monitor, and control infrared devices over a network.

After initial discovery of Global Cache devices click on the iTach IP2IR in the menu below.



Sending IR Signals continued.....

The follow menus appear after the iTach device is selected. This menu is referred to as *Wizard mode*; to enter information in manually select *Normal* as the *Select Mode* option.



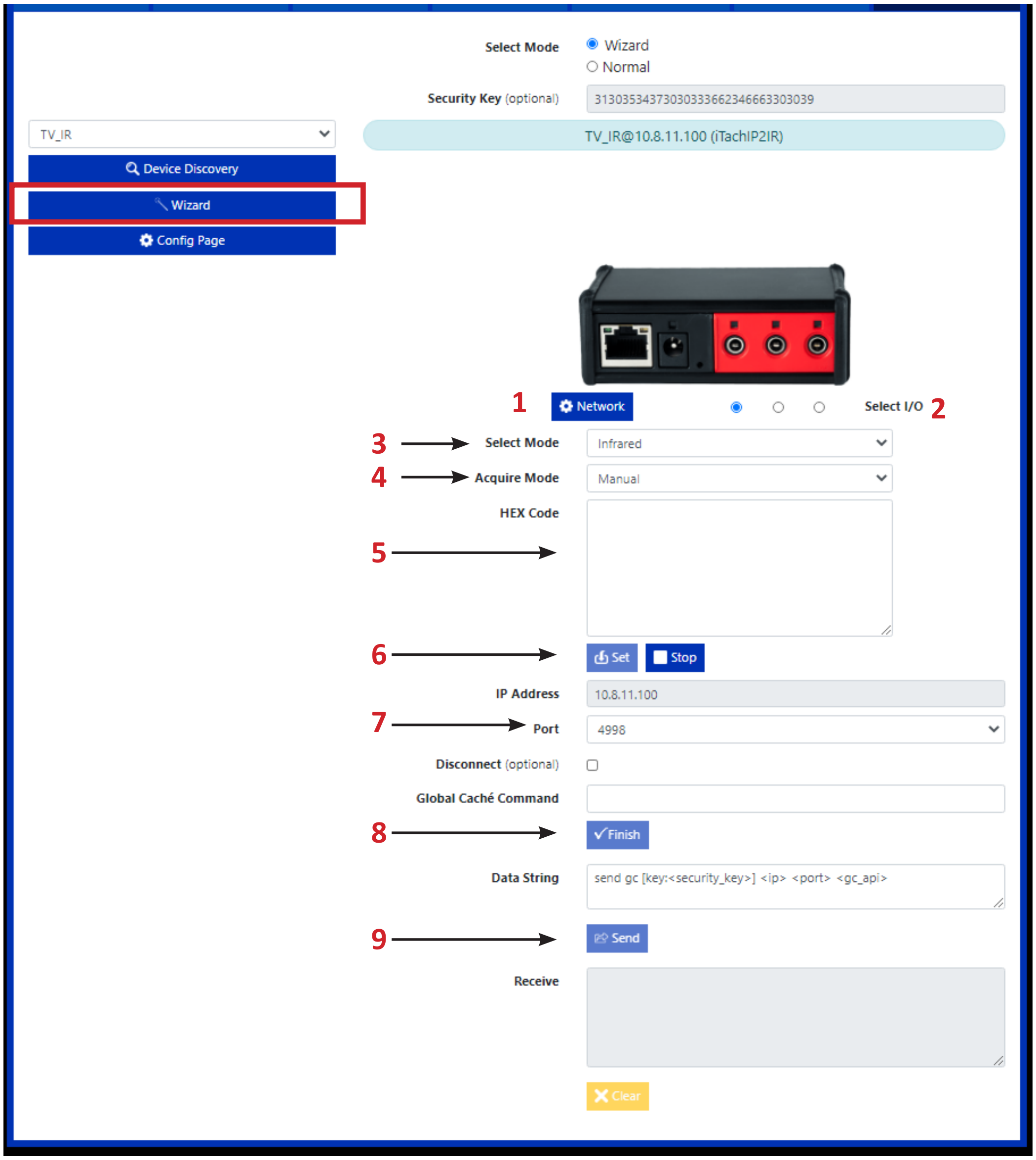
1. The *Wizard* menu provides access to configure the device to send serial strings.

2. The *Config Page* will direct you to the Global Caches internal web server; this will pop-up in another browser menu.

3. Apply an alias name and description of the device and click *Save* so the device can be accessed more quickly in the future otherwise you will need to discover the device again. These named devices will appear via drop-down above the *Device Discovery* menu.

Sending IR Signals continued.....

The following appears after the *Wizard* menu is selected in the previous step. Here we will configure the serial command to send.



1. *Network* option allows you to update the Global Caches devices network settings; by default the IP2IR is set to DHCP and falls back on APIPA 169.254.0.0/16 Network ID in absence of a DHCP server / router.

2. *Select I/O* option allows you to select the IR port you wish to use.

Sending IR Signals continued.....

3. *Select Mode* drop-down menu gives you the following options:

Infrared
- *The selected connection will be used as an IR output.*

Sensor
- *The selected connection will be used as a sensor input such as an occupancy sensor.*

Sensor Notify
- *The selected connection will be used as sensor input for notification purposes.*

LED Lighting
- *The selected connection will be used as output to control LED lighting.*

4. *Acquire Mode* drop-down menu gives you the following options:

Manual
- *Enter IR HEX codes manually.*

Cloud Based
- *Use the Global Cache Device IR Cloud Database.*
  NOTE: *This option requires the Arranger controller to have an Internet connection.*

Learn
- *Use the Global Cache IP2IR IR leaner to learn codes from remotes*

5. Enter desired IR HEX code in this field if using the *Infrared Mode* from step 3.

6. Once all the above parameters are entered in click *Set/Stop.*

7. Select port and disconnect if required (optional).

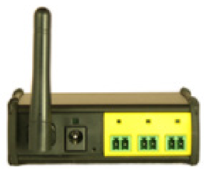
8. Once the information above has been entered in click *Finish.*

9. Click *Send* to set or test.

You can now use this generated command in a preset to be used with UI control, scheduling and automation OR send this command from a 3rd party control system to the Arranger controller to activate the current configuration. Arranger controller listens on TCP port 6980.

Controlling Contact Closures

The following wired Global Cache devices can be used to open/close contact closures connected to 3rd party devices from Arranger.



iTach WiFi to IR
PN: WF2IR

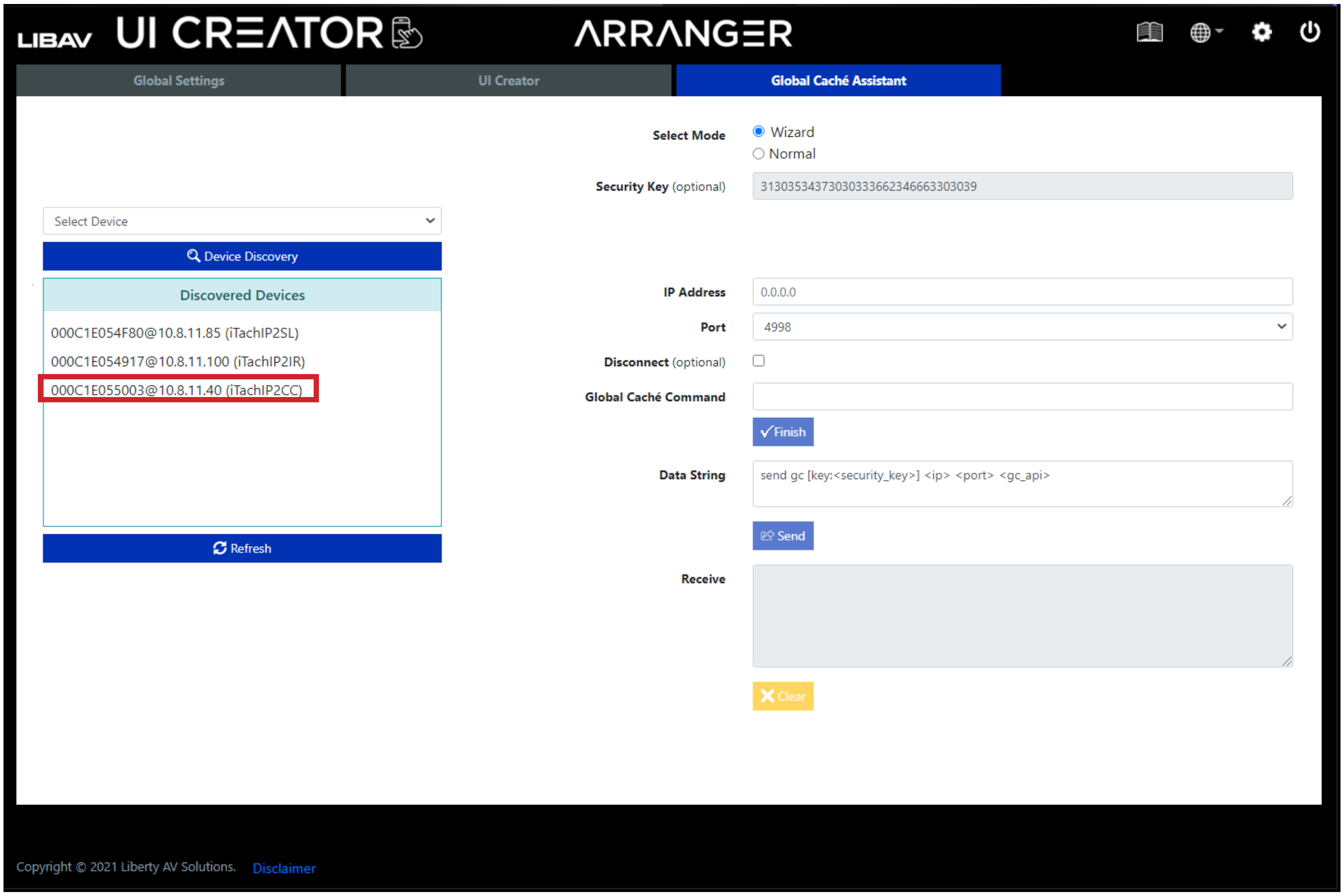
iTach Ethernet to IR
PN: IP2SL-P (PoE)

iTach Ethernet to IR
PN: IP2SL

Contact Closures continued.....

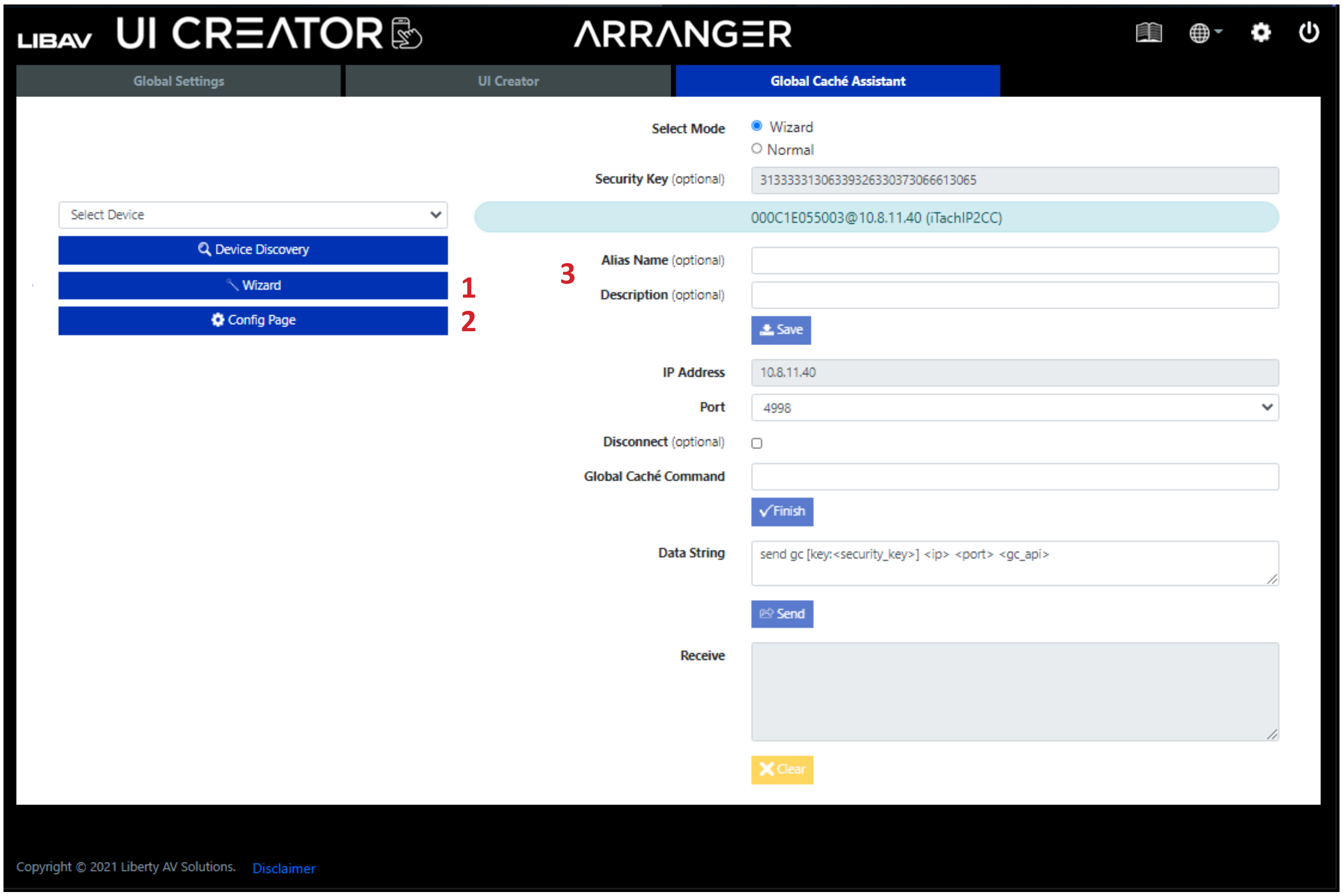
In this example we will configure the iTach IP2CC TCP/IP to contact closure endpoint.

After initial discovery of Global Cache devices click on the iTach IP2CC in the menu below.



Contact Closures continued.....

The following menus appear after the iTach device is selected. This menu is referred to as *Wizard mode*; to enter information in manually select *Normal* as the *Select Mode* option.



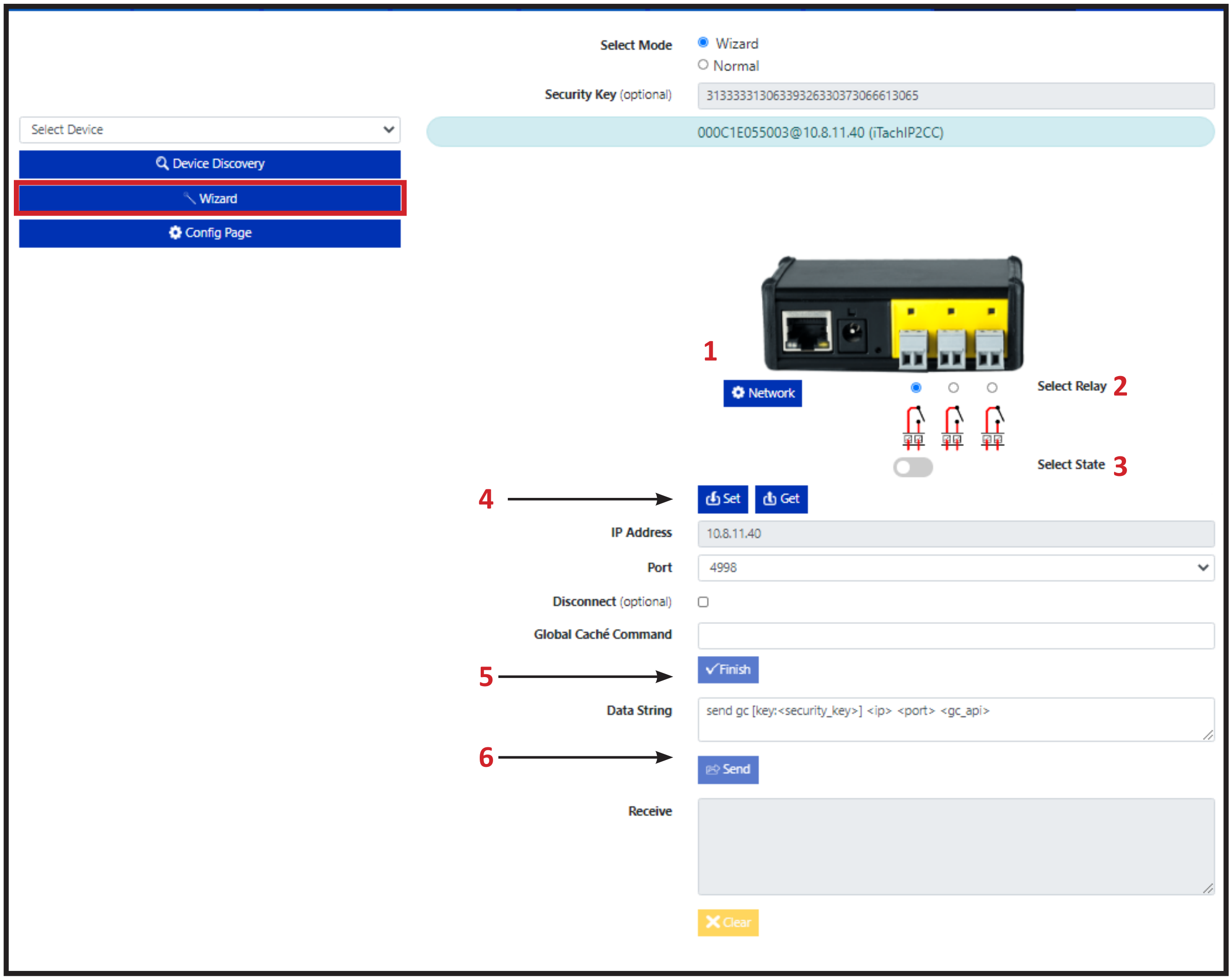
1. The *Wizard* menu provides access to configure the device to send serial strings.

2. The *Config Page* will direct you to the Global Caches internal web server, this will pop-up in another browser menu.

3. Apply an alias name and description of the device and click *Save* so the device can be accessed more quickly in the future, otherwise you will need to discover the device again. These named devices will appear via drop-down above the *Device Discovery* menu.

Contact Closures continued.....

The following appears after the *Wizard* menu is selected in the previous step. Here we will configure which contact closure to open or close.



1. *Network* option allows you to update the Global Caches devices network settings; by default the IP2CC is set to DHCP and falls back on APIPA 169.254.0.0/16 Network ID in absence of a DHCP server / router.

2. *Select Relay* option allows you to select the port you wish to use.

3. *Select State* allows you to select open or closed status of selected port.

Contact Closures continued.....

4. Once all the above parameters are entered in click *Set/Get* (***Set*** will configure the port as desired, ***Get*** will retrieve current status of selected port.

5. Once the information above has been entered in click *Finish.*

6. Click *Send.*

You can now use this generated command in a preset to be used with UI control, scheduling and automation OR send this command from a 3rd party control system to the Arranger controller to activate the current configuration. Arranger controller listens on TCP port 6980.

Controlling Relays

The following wired and wireless Global Cache devices, paired with the FLC-RS adapter, can control relay outputs conntect to device from Arranger.

iTach Flex PoE
PN: FLEXIP-IP

iTach Flex WiFi
PN: FLEXIP-IP

FLC-RS

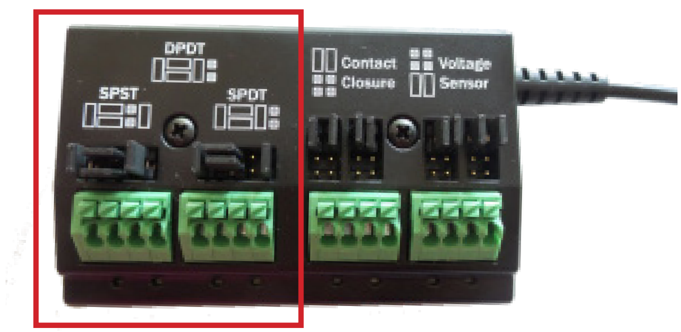
Controlling Relays continued.....

In this example we will configure the iTach Flex PoE TCP/IP endpoint coupled with the Global Cache FLC-RS cable that supports four configurable relay outputs.

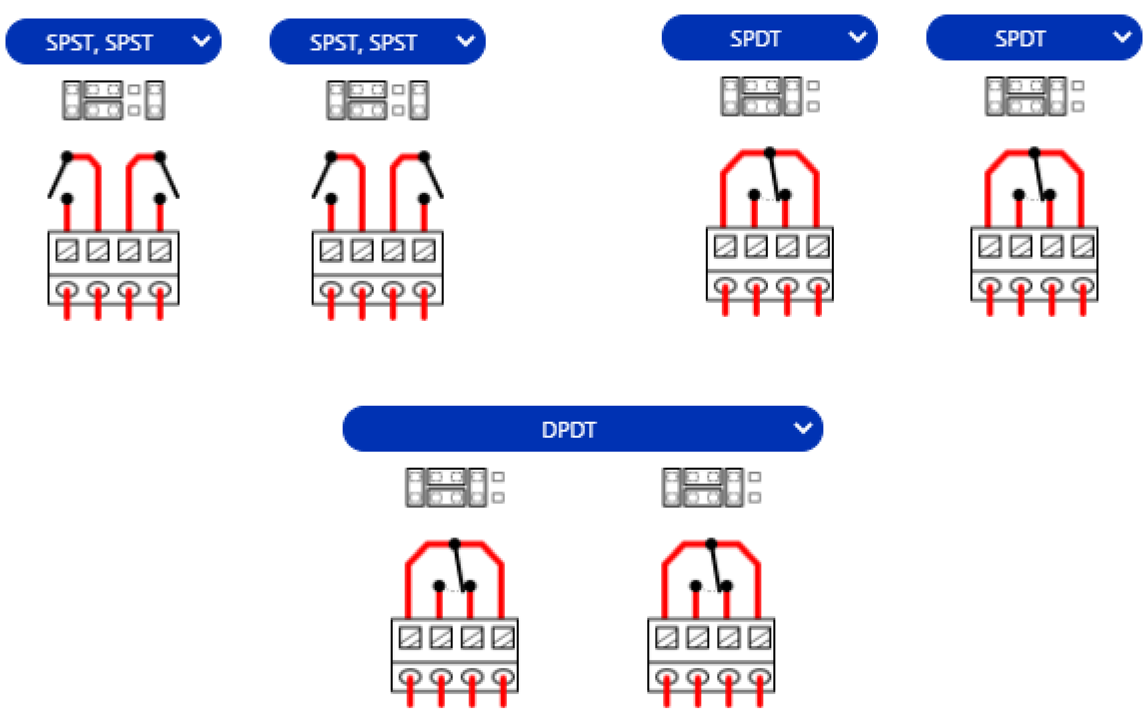
Relay outputs are configurable into common relay types

-Single Pole Single Throw (SPST)
-Single Pole Double Throw (SPDT)
-Double Pole Double Throw (DPDT)

When using the FLC-RS relay cable, be sure to set the pins on the relays for the desired configuration
.

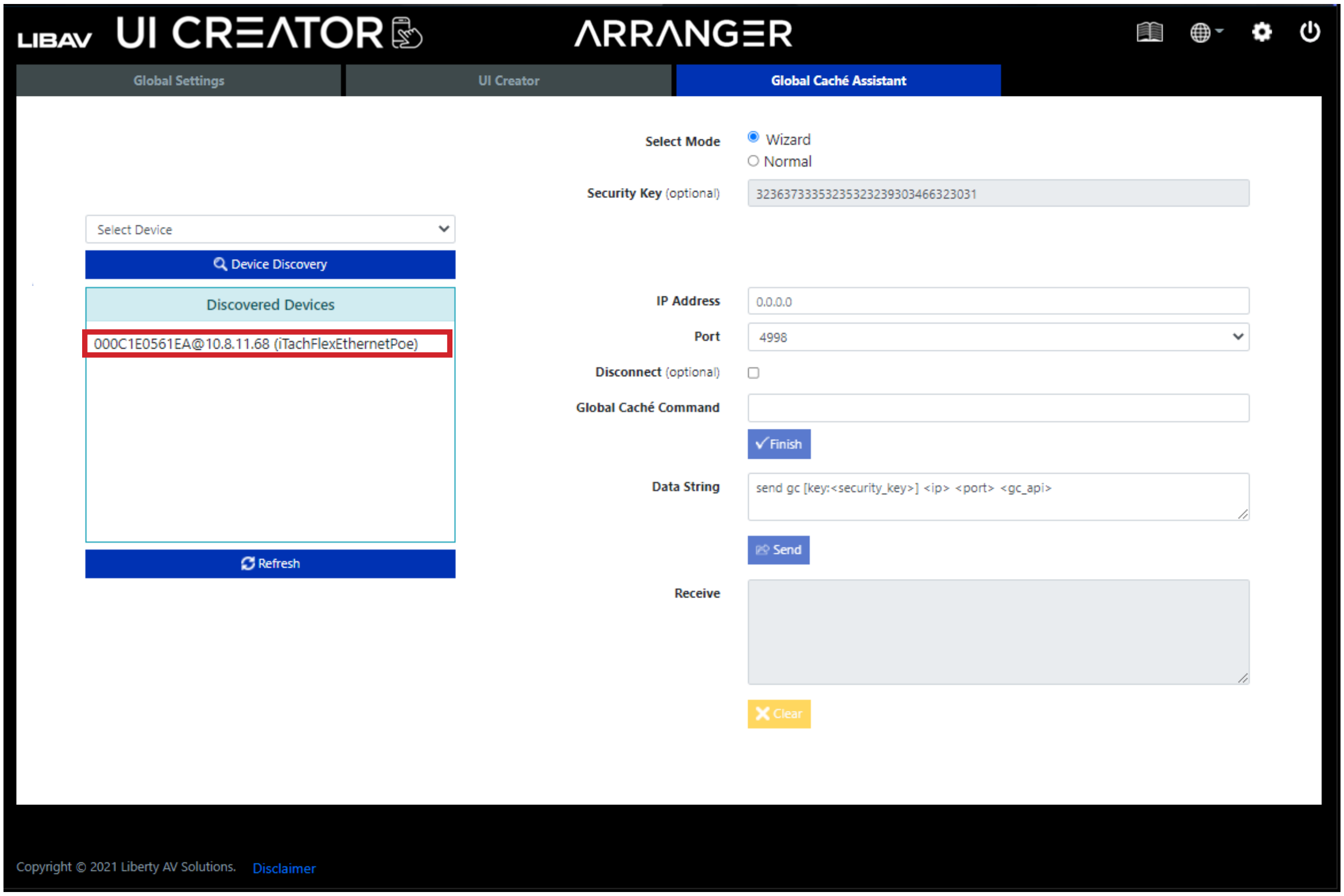


Wiring and jumper configuration examples...



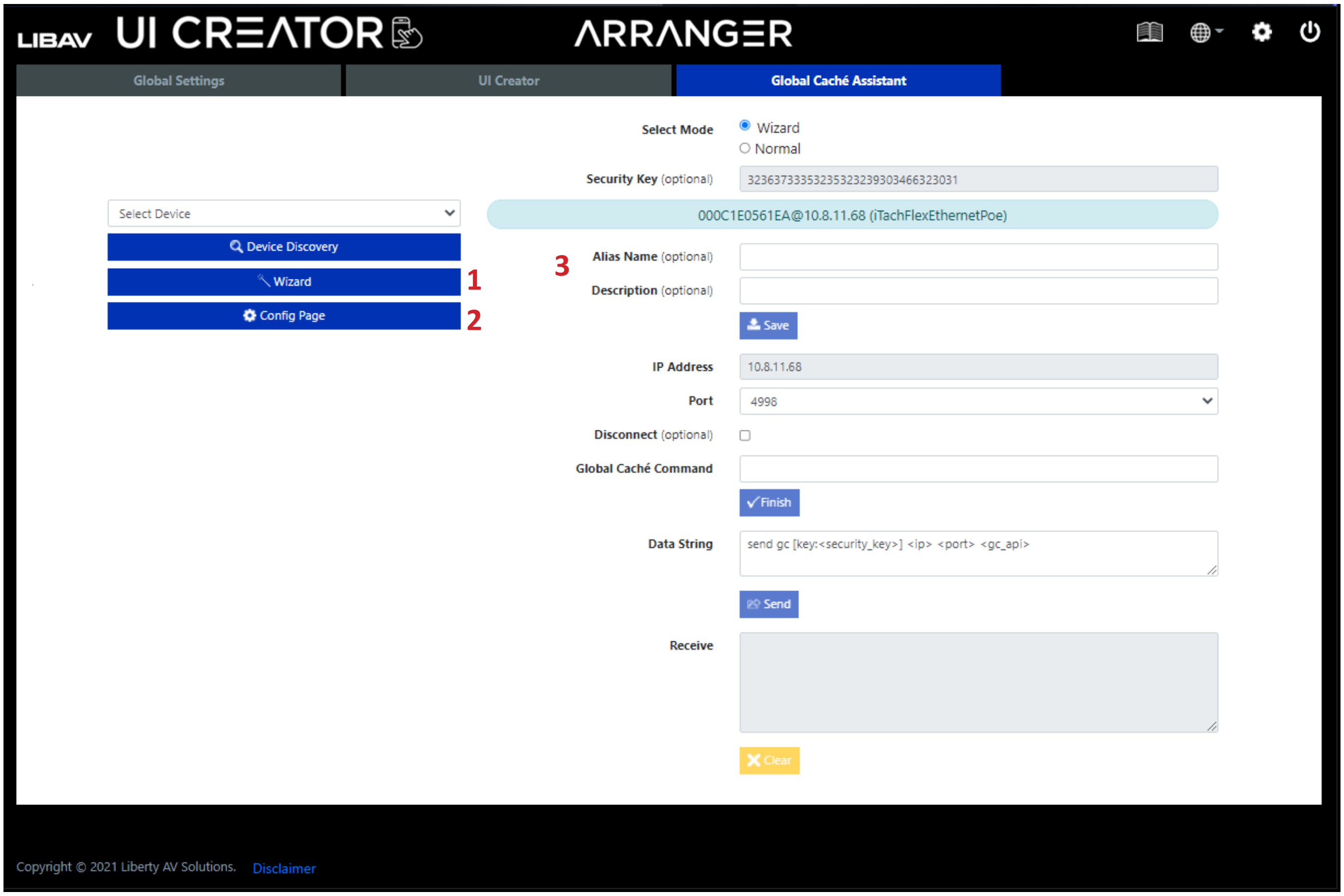
Controlling Relays continued.....

After initial discovery of Global Cache devices click on the iTach Flex PoE in the menu below.



Controlling Relays continued.....

The follow menus appear after the iTach device is selected. This menu is referred to as *Wizard mode*; to enter information in manually select *Normal* as the *Select Mode* option.



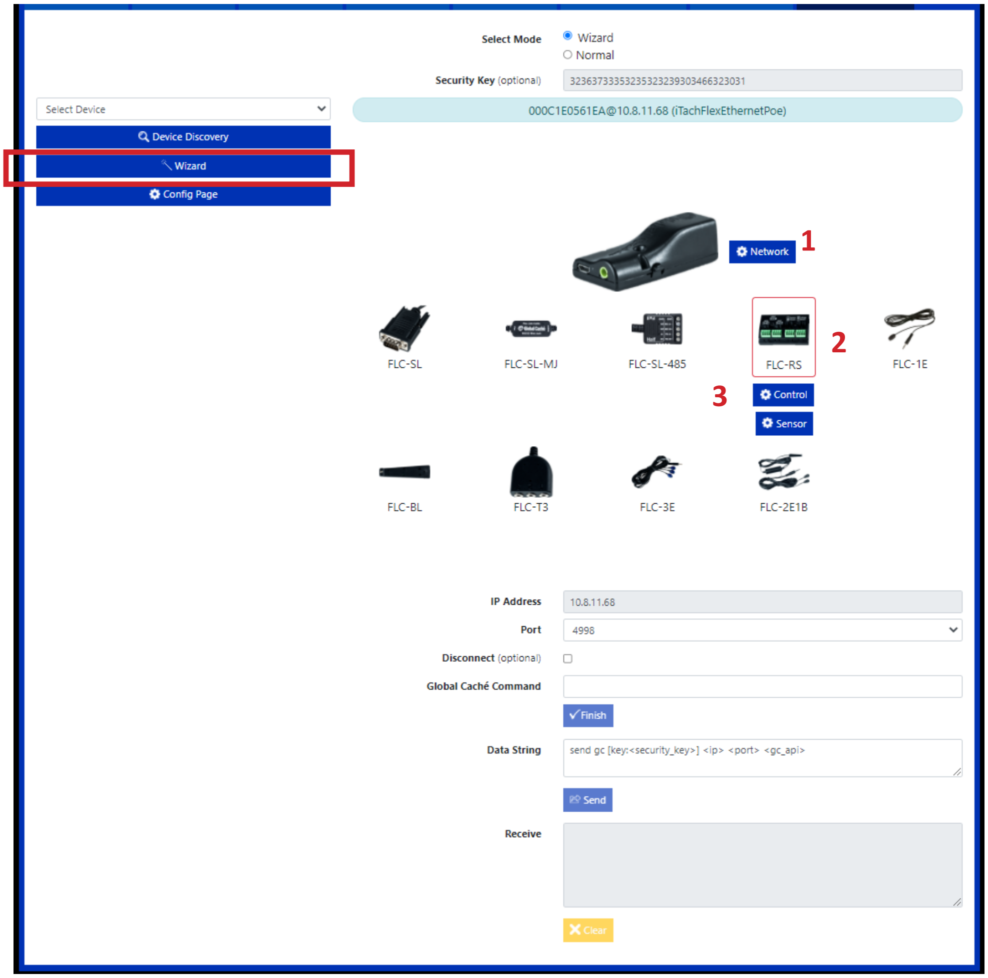
1. The *Wizard* menu provides access to configure the device to send serial strings.

2. The *Config Page* will direct you to the Global Caches internal web server; this will pop-up in another browser menu.

3. Apply an alias name and description of the device and click *Save* so the device can be accessed more quickly in the future otherwise you will need to discover the device again. These named devices will appear via drop-down above the *Device Discovery* menu.

Controlling Relays continued.....

The following appears after the *Wizard* menu is selected in the previous step. Here we will configure which relay to open or close.

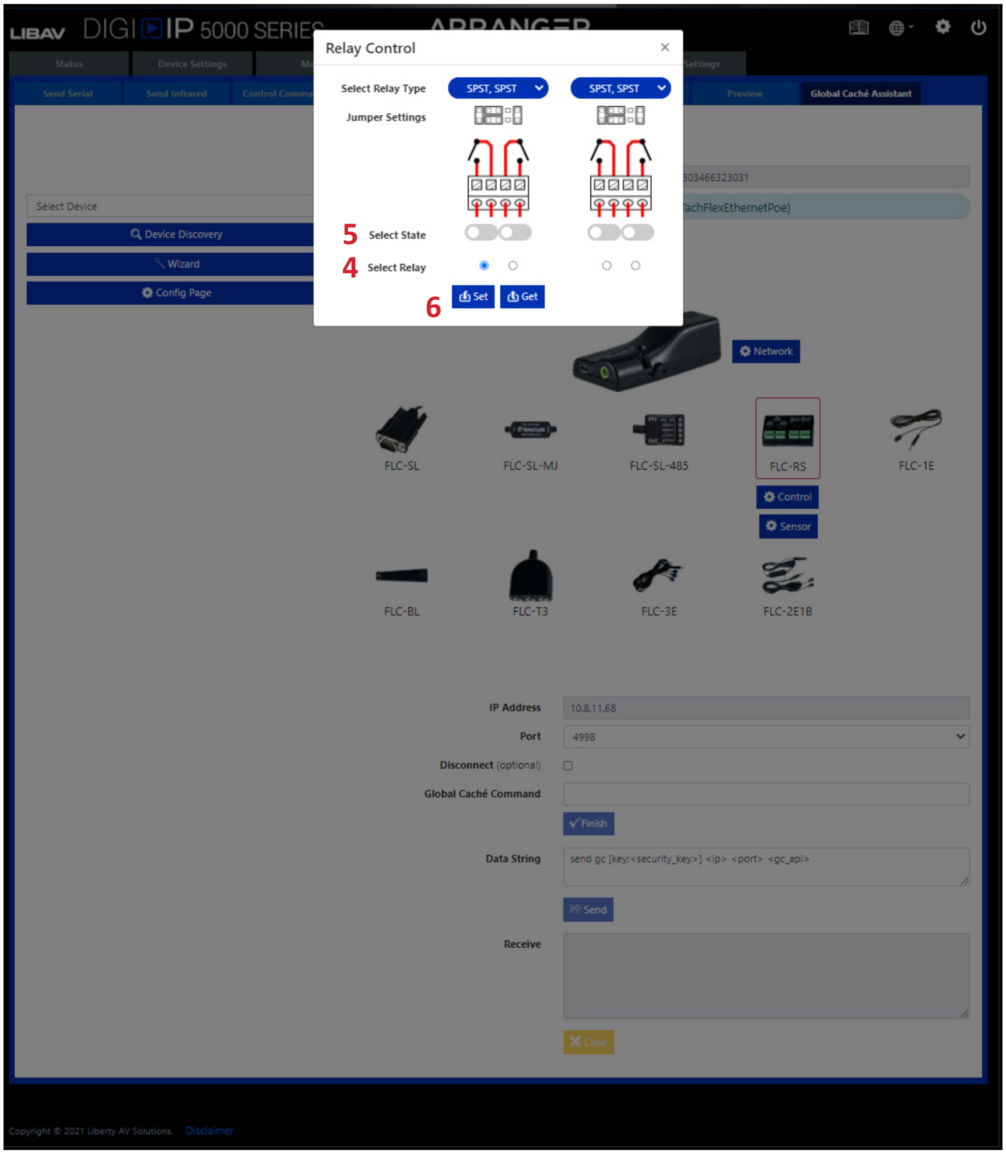


1. *Network* option allows you to update the Global Caches devices network settings; by default the IP2CC is set to DHCP and falls back on APIPA 169.254.0.0/16 Network ID in absence of a DHCP server / router.

2. Select the *FLC-RS relay cable* in the accessories option.

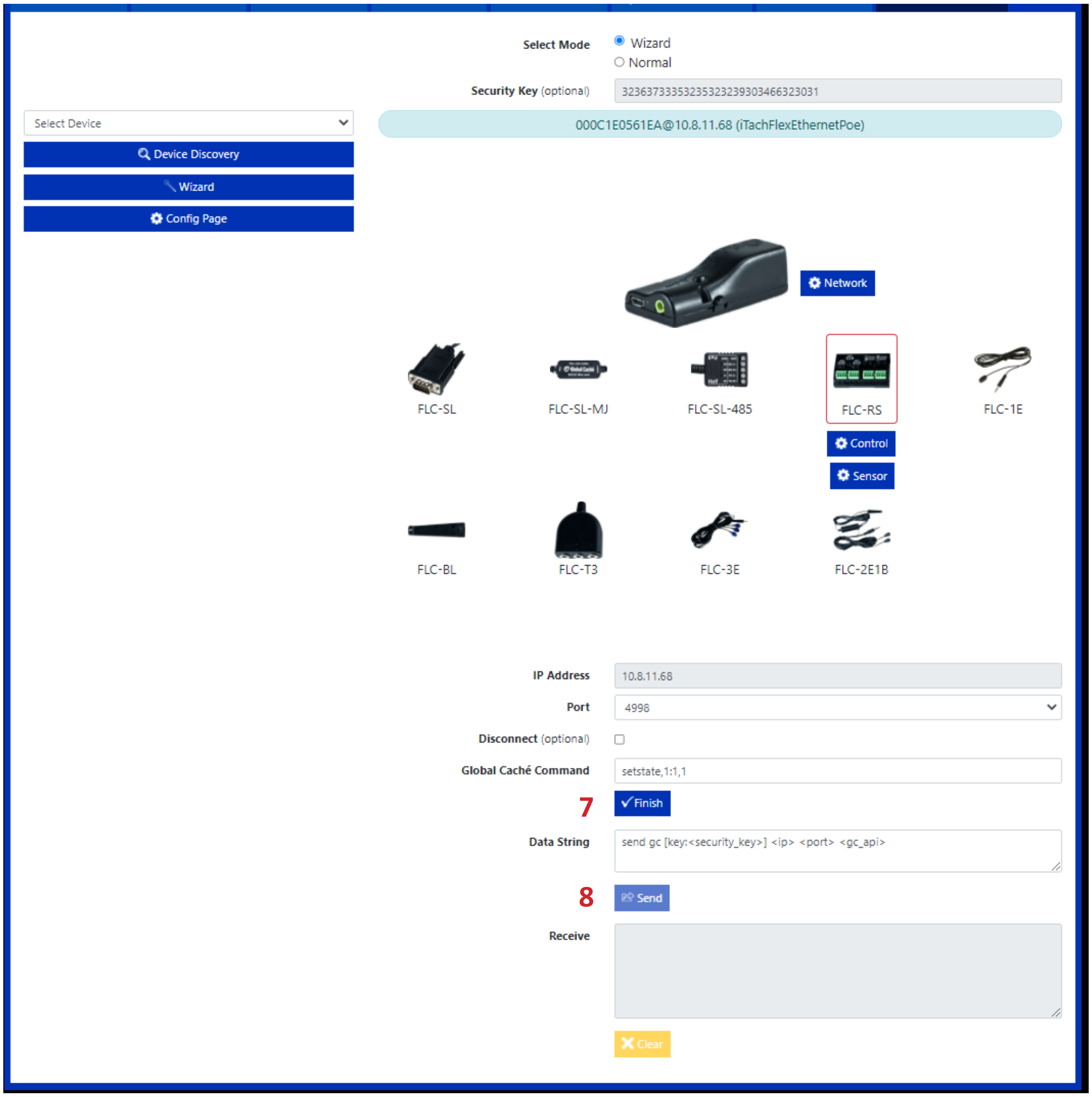
3. Click on the *Control* settings

Controlling Relays continued.....



4. Select *Relay* that you wish to control.
5. Select desired *State*.
6. Once all the above parameters are entered in click *Set/Get* (***Set*** will configure the port as desired, ***Get*** will retrieve current status of selected port.
7. Once the information above has been entered in click *Finish.*

Controlling Relays continued.....



8. Click *Send.*

You can now use this generated command in a preset to be used with UI control, scheduling and automation OR send this command from a 3rd party control system to the Arranger controller to activate the current configuration. Arranger controller listens on TCP port 6980.

Configuring Sensor and Contact Closure Triggers

The following wired and wireless Global Cache devices, paired with the FLC-RS adapter allows configuration of sensor and contact closure input triggers in Arranger

iTach Flex PoE
PN: FLEXIP-IP

iTach Flex WiFi
PN: FLEXIP-IP

FLC-RS

The following wired Global Cache devices, paired with sensor accessories, can be used to configure sensor and contact closure input triggers.

iTach WiFi to IR
PN: WF2IR

iTach Ethernet to IR
PN: IP2SL-P (PoE)

iTach Ethernet to IR
PN: IP2SL

IT-SP1 AC/DC Voltage Sensor

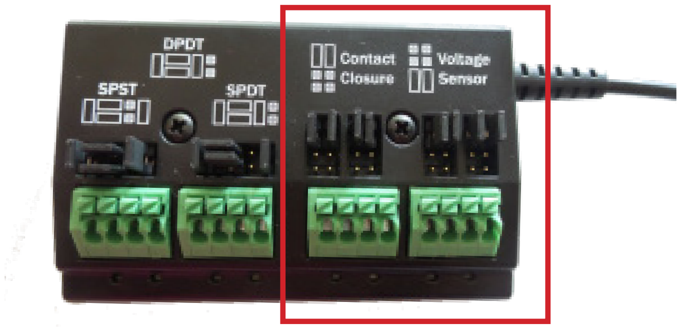
IT-SC1 Contact Closure Sensor

Contact Closure and Sensor Triggers continued.....

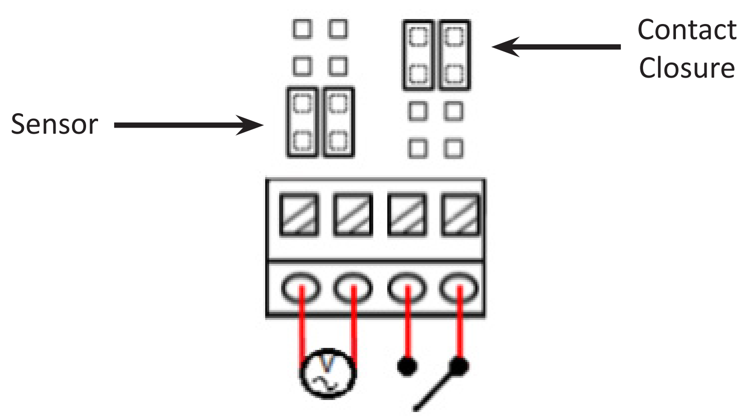
In this example we will configure the iTach Flex PoE TCP/IP endpoint coupled with the Global Cache FLC-RS cable to set up sensor and contact closure input triggers.

Once the wiring configuration is complete we will set up the Arranger API command [set listener} that will 'listen' for the sensor or contact closures to activate, this event can then trigger a *PRESET* in Arranger. See *Global Settings > Presets* for more information on creating Presets in Arranger.

When using the FLC-RS relay cable, be sure to set the pins on the closures are set for the desired configuration .

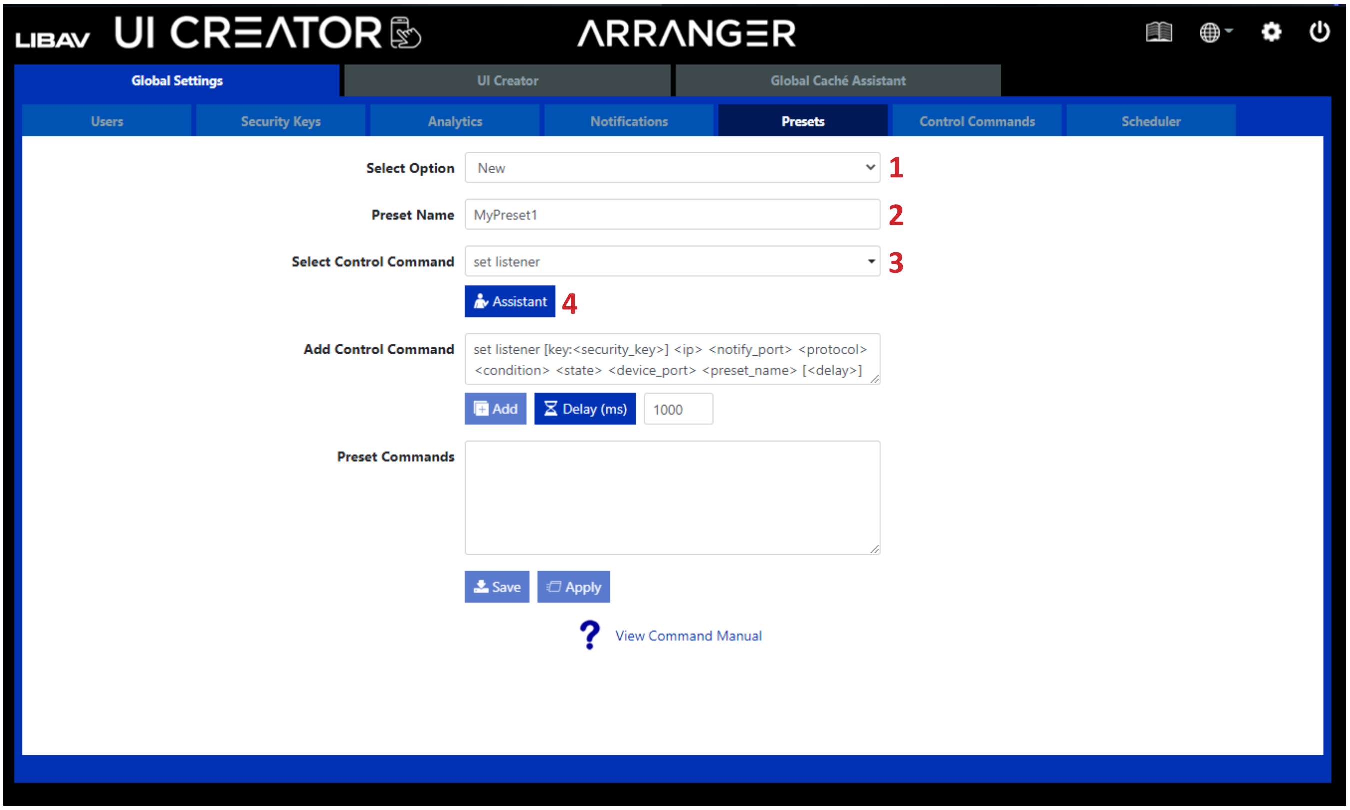


Wiring and jumper configuration examples for both a sensor and a contact closure.....



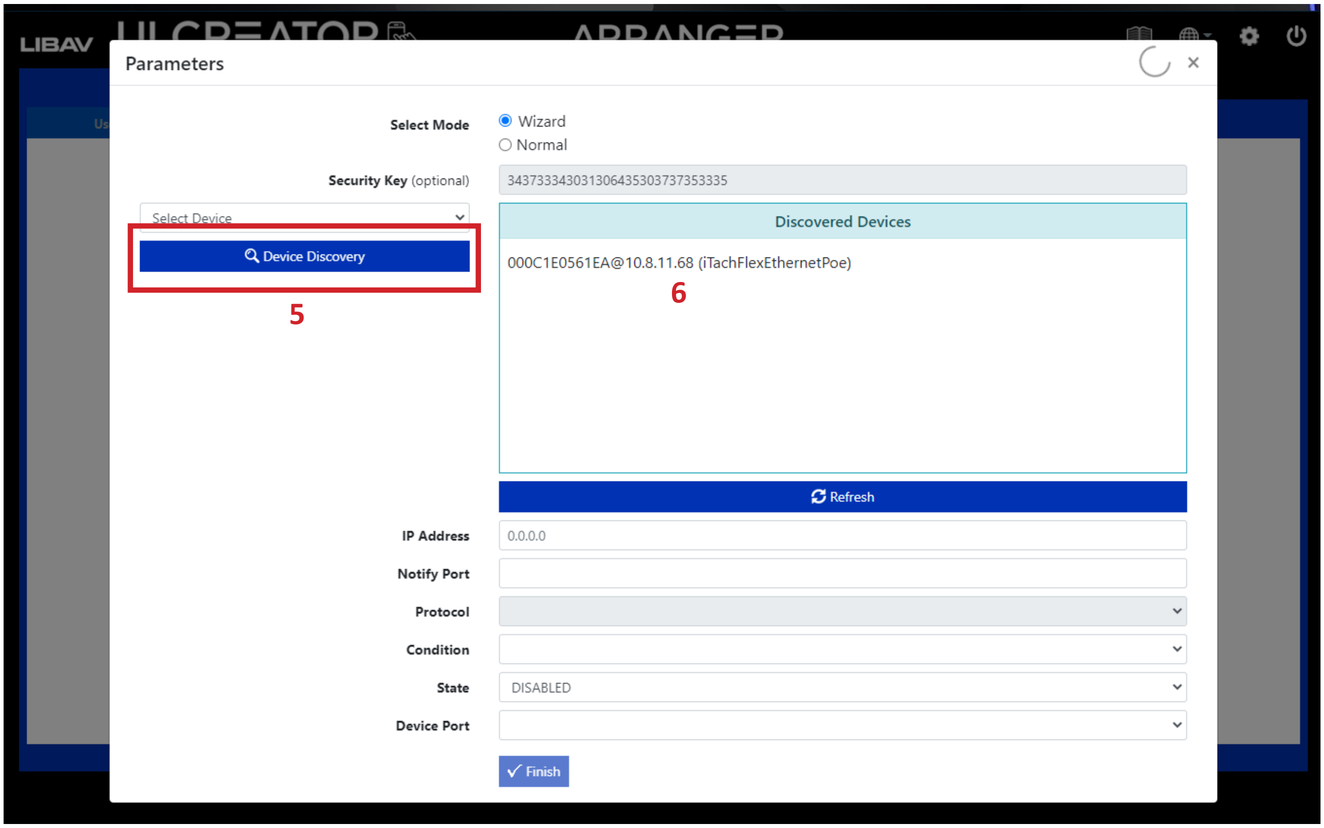
Contact Closure and Sensor Triggers continued.....

Navigate to *Global Settings*, then click on the *Presets* sub menu



1. Choose *New* preset.
2. Name the preset.
3. Choose the command [set listener] from the drop down list.
4. Click on *Assistant.*

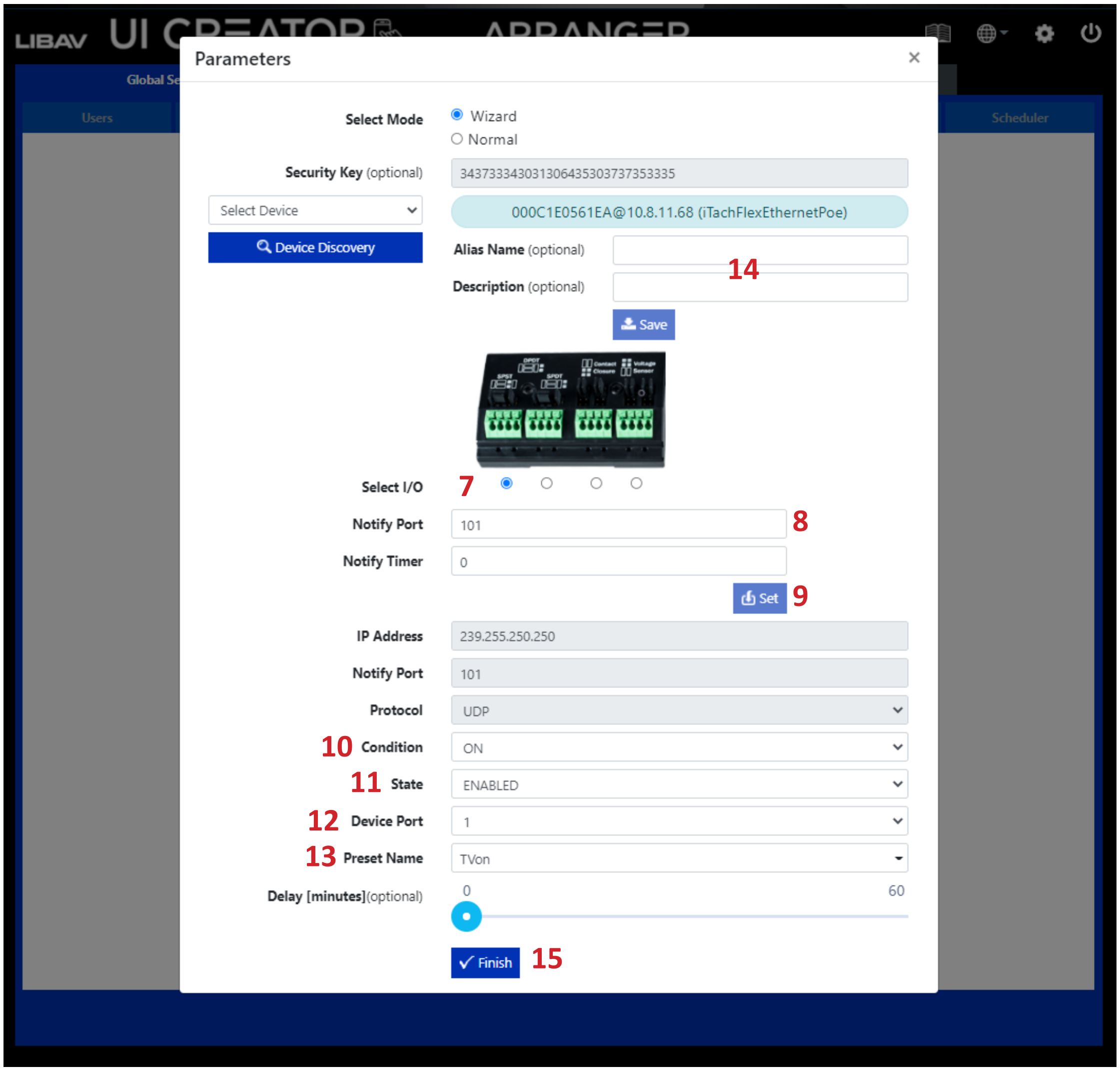
Contact Closure and Sensor Triggers continued.....



5. Click *Device Discovery* to locate Global Cache devices on the network.

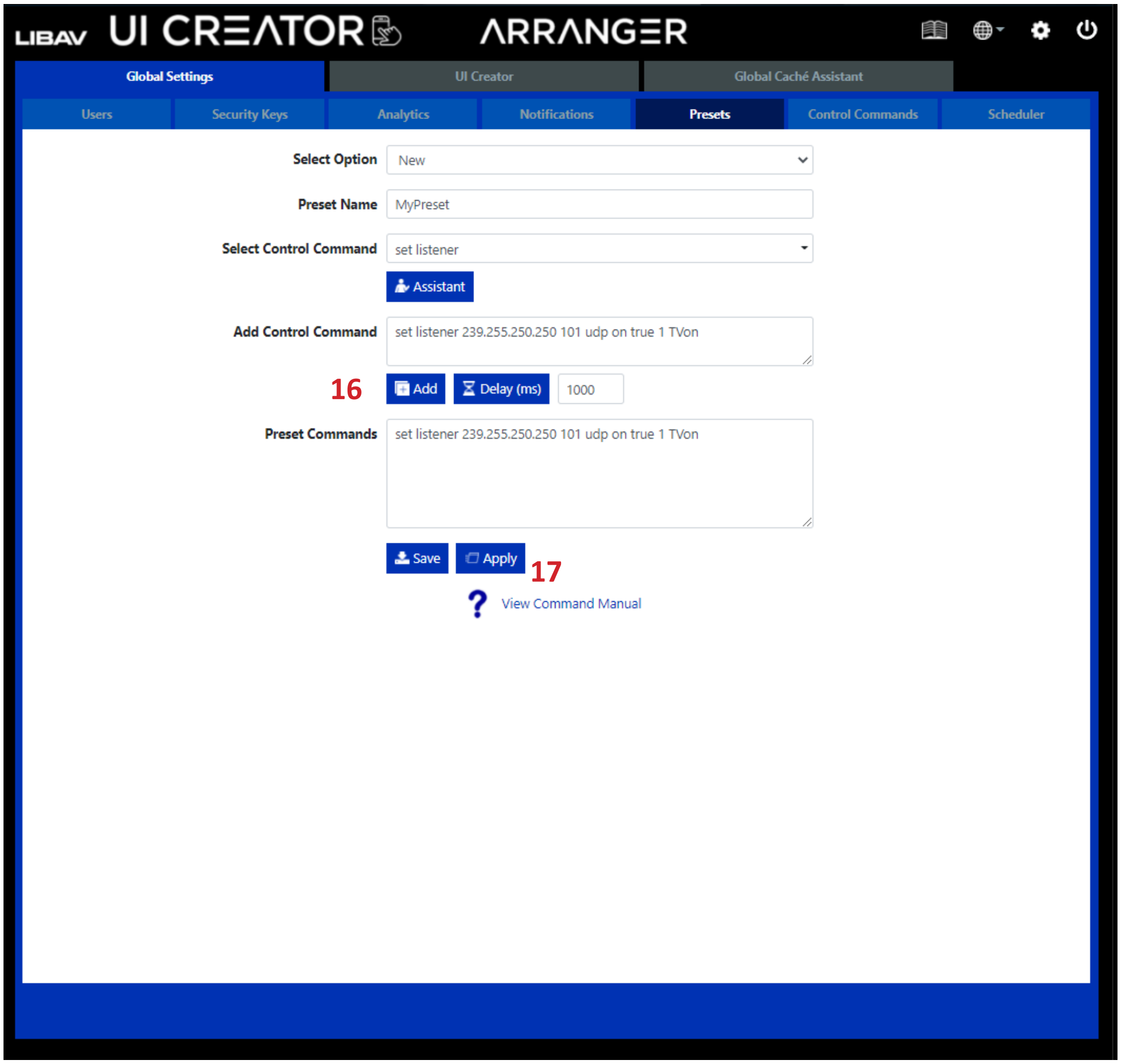
6. Click on the Flex device in the list once discovered.

Contact Closure and Sensor Triggers continued.....



7. Select the input port you wish to use, in this example we chose the first input option which is set for a contact closure according to the physical pin-out configuration on the FLC-RS.
8. Configure the *Notify Port*, numbers start at 101.
9. Click *Set*.
10. Choose *ON* for the *Condition*.
11. Choose *ENABLED* for the *State*.
12. Change the *Device Port* (I/O port if you want to change the port number in the configuration)
13. Choose the appropriate *Preset Name* you want to execute when the contact closure closes.
14. Apply an alias name and description of the device and click *Save* so the device can be accessed more quickly in the future, otherwise you will need to discover the device again. These named devices will appear via drop-down above the *Device Discovery* menu.
15. Click *Finish.*

Contact Closure and Sensor Triggers continued.....



16. Click *ADD* to add the generated code to the preset.

17. Click *Apply* to test, then click *Save*.

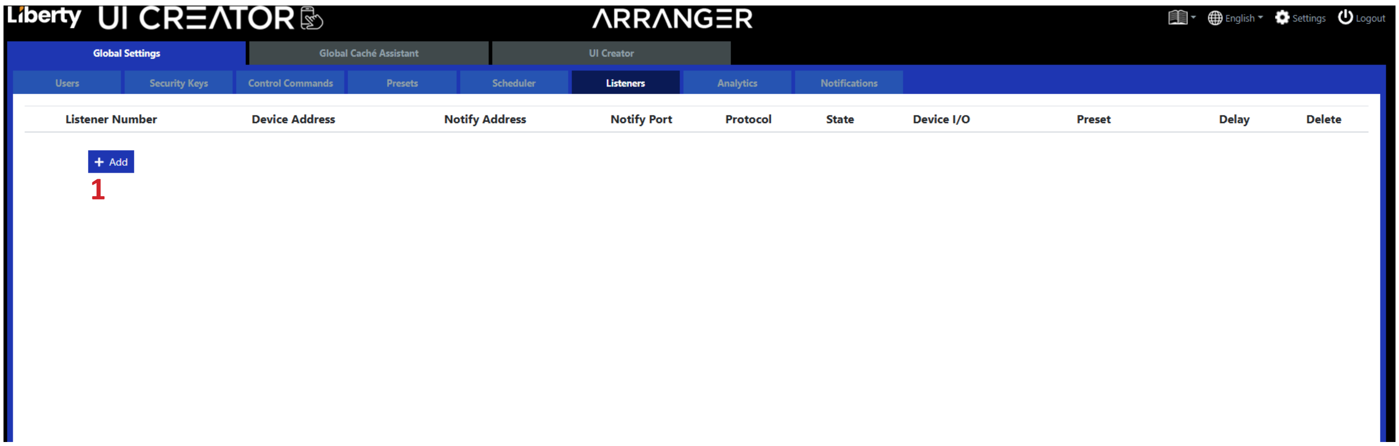
The configuration for using sensors, like a PIR sensor, to execute a preset in Arranger is the SAME as above accept the input ports physical configuration will need to be set to *VOLTAGE SENSOR*.

You can now schedule this preset to always be ON and listening when the Arranger controller is on or has been rebooted. See *Global Settings > Scheduler* for more information on how to use the *System Start* scheduler option.

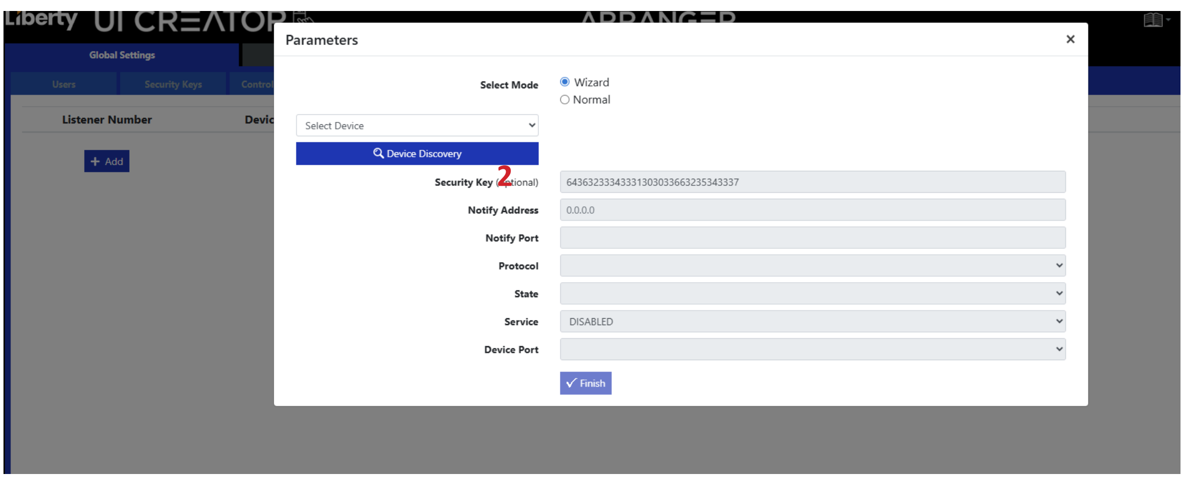
Listeners

The Listeners are Global Caché functions to apply presets when 'sensor' notifications are received from a Global Caché device as the sensor input state changes.

The listeners can be established either via the API command ***'set listener'*** or directly from here.

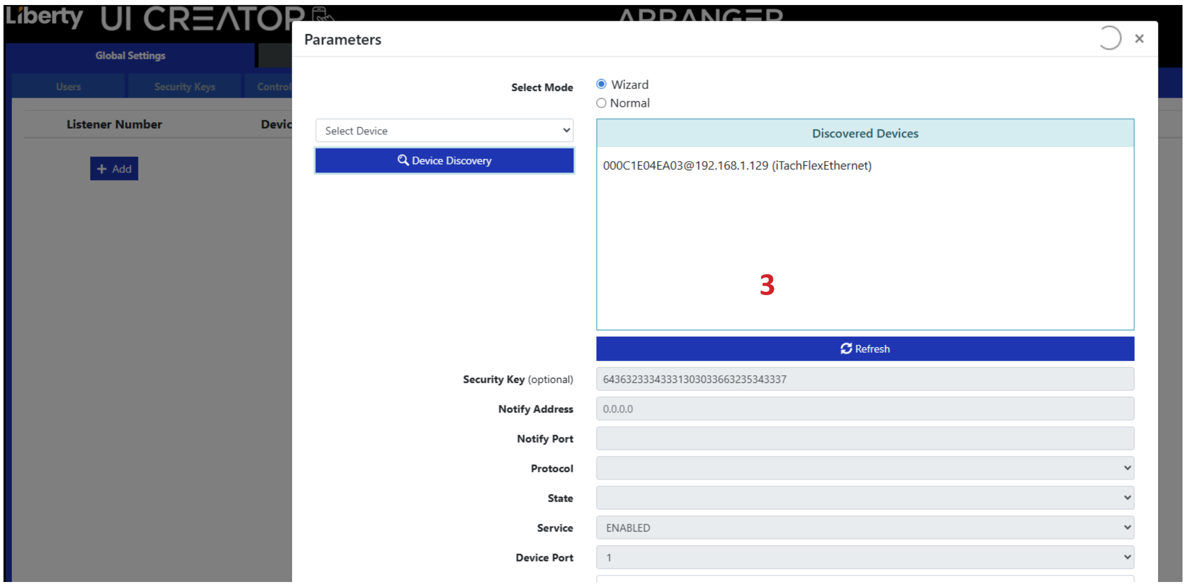


1. Choose New preset.



2. Either select a previously saved device or click the *Device Discovery* button

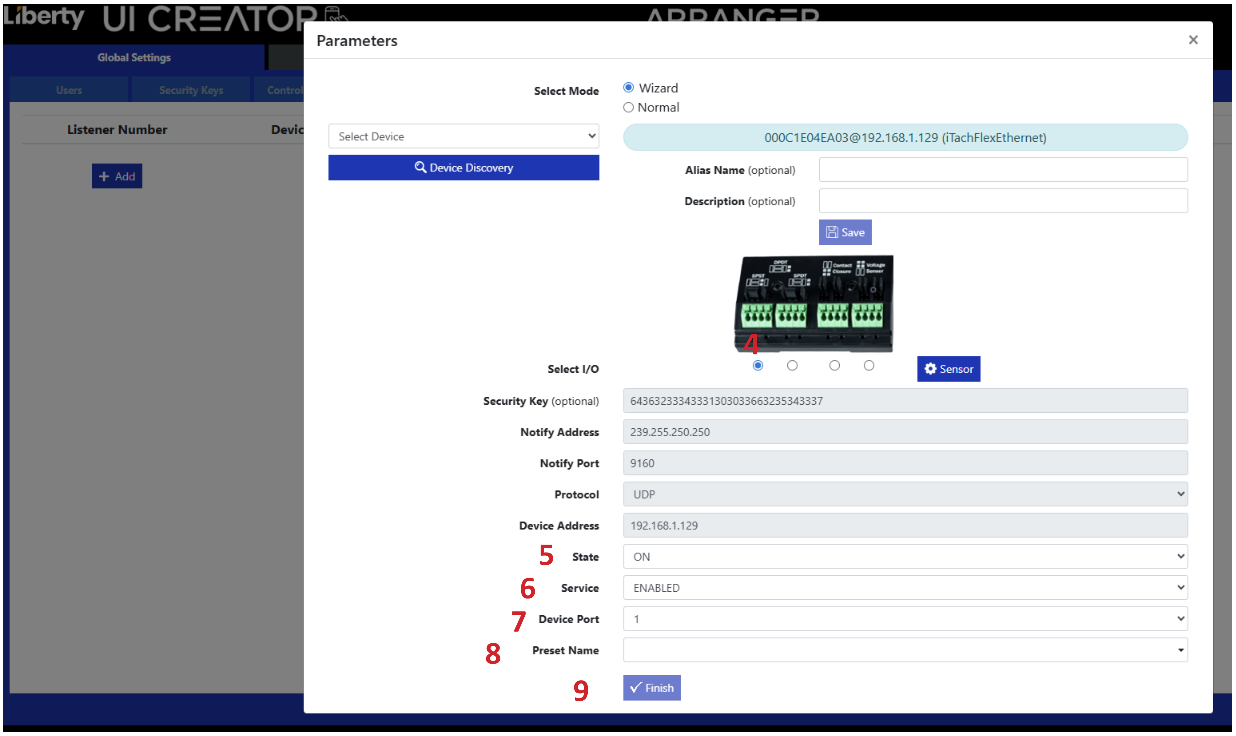
Listeners continued.....



3. Select desired device.

In this example, we will use the Global Cache IP2IR product to set an I/O port to configure a sensor input.

Listeners continued.....

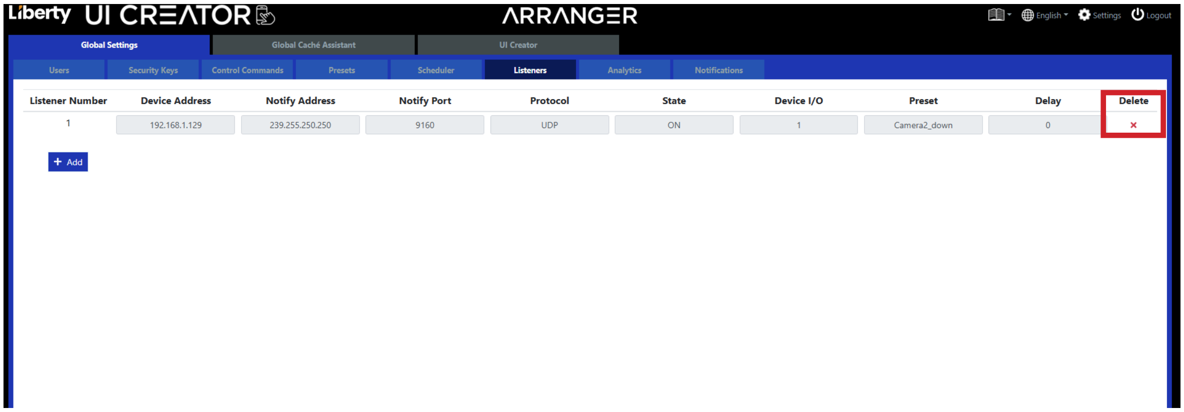


4. Configure the selected I/O port as *Sensor Notify.*

   Set the *Notify Port* to 9160.

   Set *Notify Timer* to 0.
5. Select *State*: On, Off, Any
6. Determine Listener *Service* status: Disabled or Enabled.
7. Select the *Device Port* currently selected in step 4.
8. Select the desired *Preset Name*
9. Click *Finish* when done.

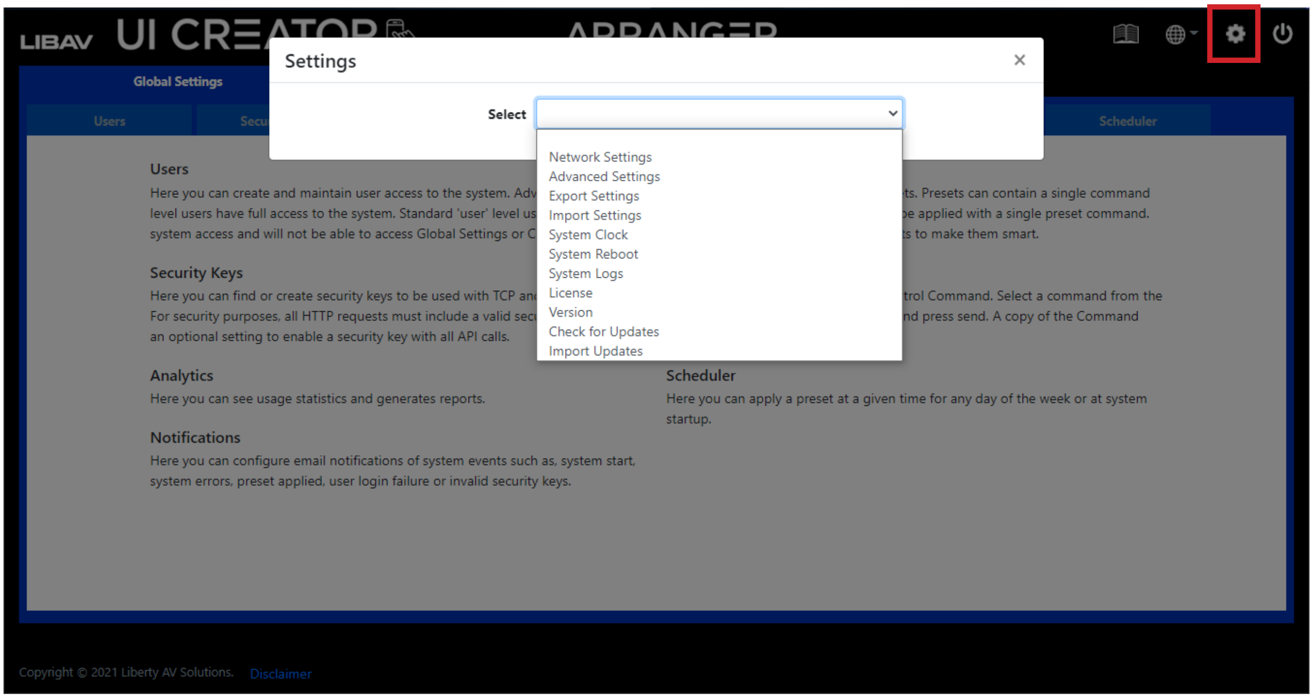
Listeners continued.....



The active *Listeners* will now be listed and can be removed either via the API command ***'set listener'*** or clicking the red *Delete* cross icon notated above.

System / Controller Settings

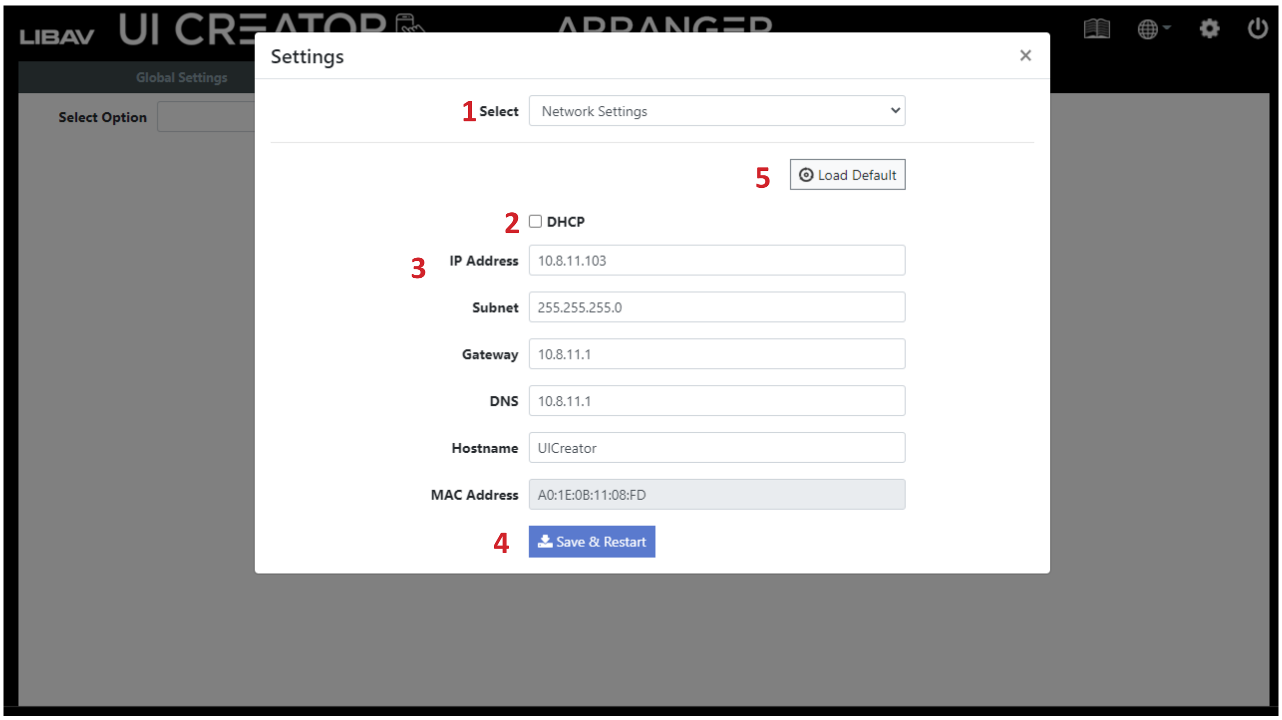
All the controller's system level settings can be accessed by admin level users via drop-down menu by clicking the gear icon on the top right of the Arranger software banner.



Network Settings
Single Network Mode

NOTE: By default the Arranger controller operates in *Single Network Mode.* Connect the RJ45 connection of the Arranger controller directly to the AV LAN for this operation. See next page for *Dual Network Mode.*

Here you can change the IP configuration of the Arranger Controller. This address must be set in the same range as the devices being controller via TCP. By default the Arranger Controller will be found at 169.254.1.1.



1. Select *Network Settings.*
2. DHCP - check this box if you want the IP address for the controller to be set automatically by a third-party router or DHCP server.
3. If using a static IP address simply enter the static address, subnet, gateway and name of hostname if required.
4. Click *Save and Restart*.
5. Click *Load Default* to reset device to the factory default IP address of 169.254.1.1.

Dual Network Mode

When using *Dual Network Mode* you can use the Arranger controller to bridge a Client or primary LAN to access or control the AV LAN without converging the two networks. This allows for a physical dedicated network for the AV components so multicast doesn't have to be managed on the Client or primary LAN.

A USB to Gigabit Ethernet NIC Network Adapter can be attached to the controller providing a second dedicated AV Endpoint network. Approved adapters include Insignia NS-PU98505, Tripp-Lite U236-000-GBW, Cable Matters 202013,. Cable Matters 202023 and USBA2RJ45.

The controllers NIC (RJ45) can be dedicated to Client LAN Communication while the secondary NIC (USB) is dedicated to Digi IP endpoint communication / subnet. Peripheral TCP devices can be controlled from either.

By default the Arranger Controller will be found on the primary NIC (RJ45) at 192.168.1.1 while maintaining 169.254.1.1 IP for USB to LAN adapter.

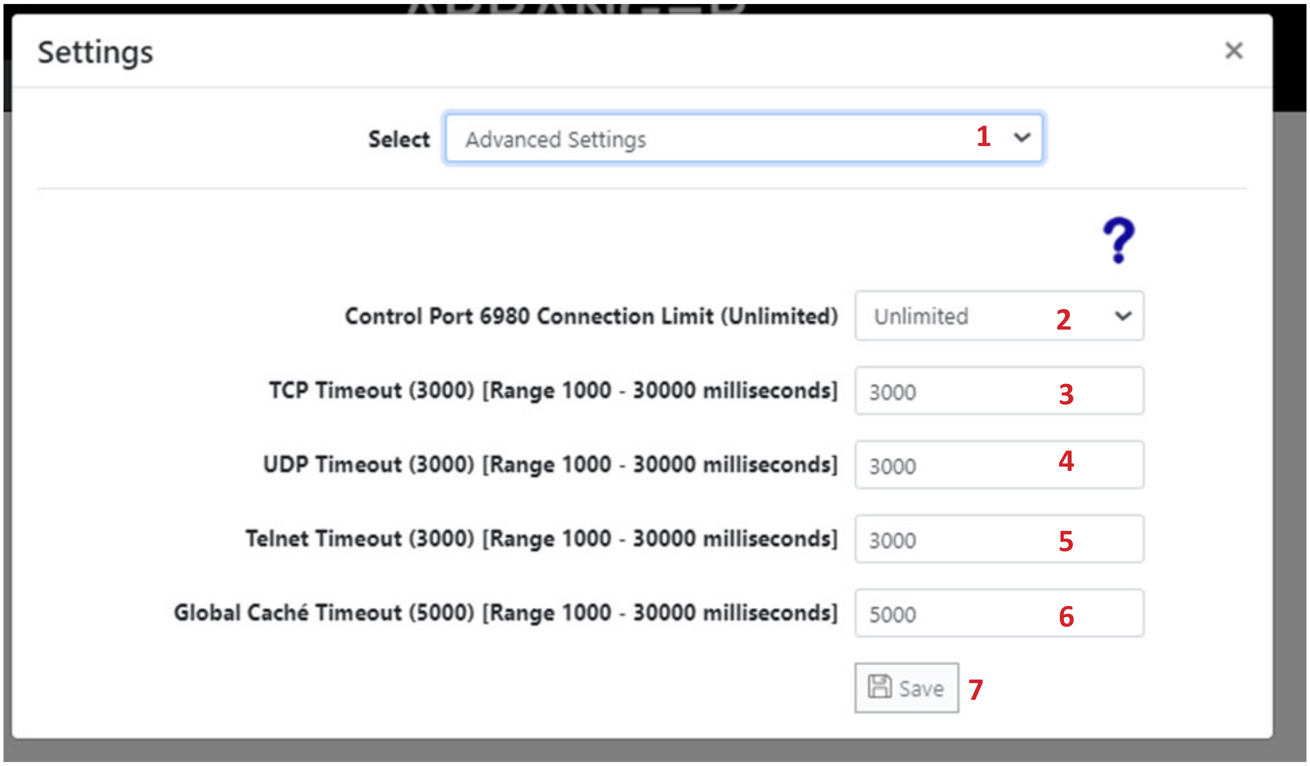
NOTE: Only a static IP address can be applied to the secondary NIC, the primary NIC also supports DHCP.

Adding in second network adapter.

1. Power on the Arranger Controller and wait for at least 1 minute before continuing.
2. Plug the USB to Gigabit Ethernet NIC Network Adapter into a USB port of the controller.
3. Perform a Factory Reset: Insert a USB key into the Arranger controller with ONLY a text file titled *factoryreset.txt* for 10 seconds, then remove the USB key.
4. Browse to the controllers default IP address at 192.168.1.1 and configure the controllers network by navigating to *1)* the settings icon on Arranger *2)* choose Network Settings from drop down menu

Advanced Settings

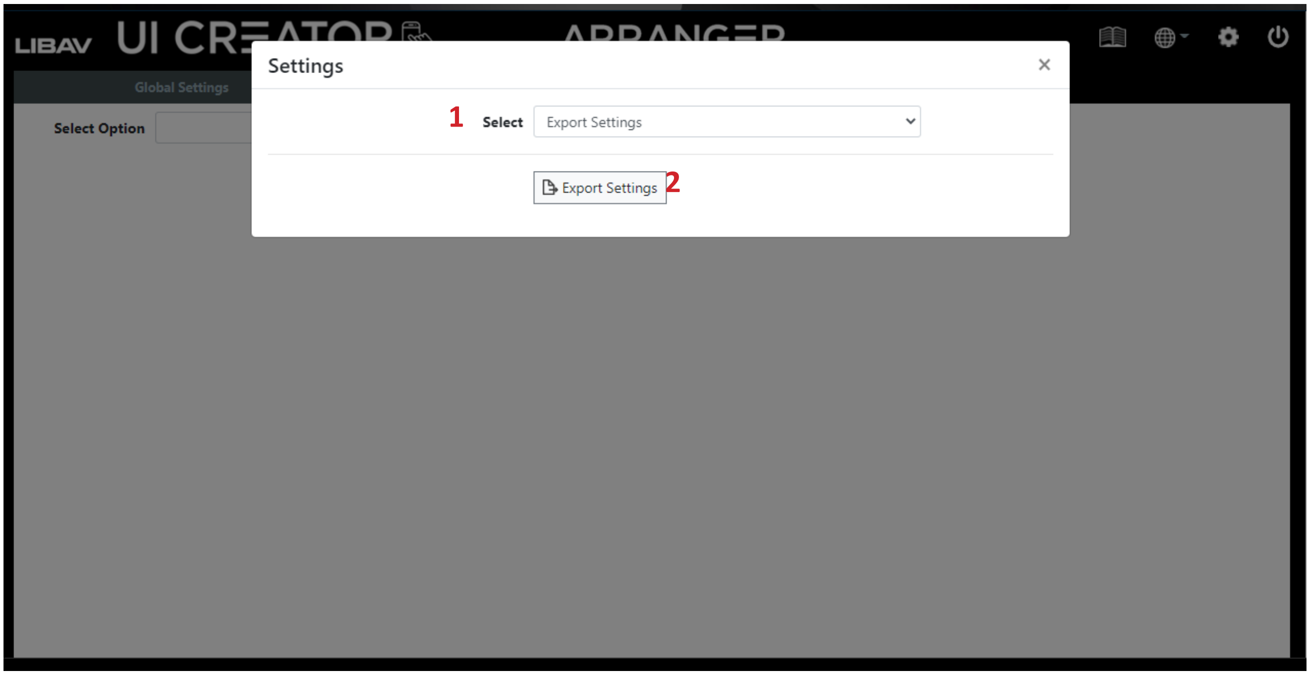
The *Advanced Settings* section contains the timing, leave subscriptions and Telnet port restriction settings of the controller.



1. Select *Advanced Settings*
2. **Telnet Port 6980 Connection limitation** - Here you can set the number of simultaneous connections to the Telnet TCP control port 6980 to unlimited or from 1 to 10 connections.
3. **TCP Timeout** - *TCP Timeout* is the maximum time in milliseconds the Arranger controller will wait for a response from a TCP-controlled device. The default is 3000 = 3 seconds with a range of 1000 – 30000.
4. **UDP Timeout** - *UDP Timeout is the maximum time in milliseconds the Director Controller will wait for a response from a UDP controlled device. The default is 3000 = 3 seconds with a range of 1000 – 30000*
5. **Telnet Timeout** -Telnet Timeout is the maximum time in milliseconds the Director Controller will wait for a response from a Telnet controlled device. The default is 3000 = 3 seconds with a range of 1000 – 30000.
6. **Global Cache Timeout** - *Global Cache Timeout* is the maximum time in milliseconds the Arranger controller will wait for a response from a Global Cache device. The default is 5000 = 5 seconds with a range of 1000 – 30000.
7. Click ***Save***.

Export Settings

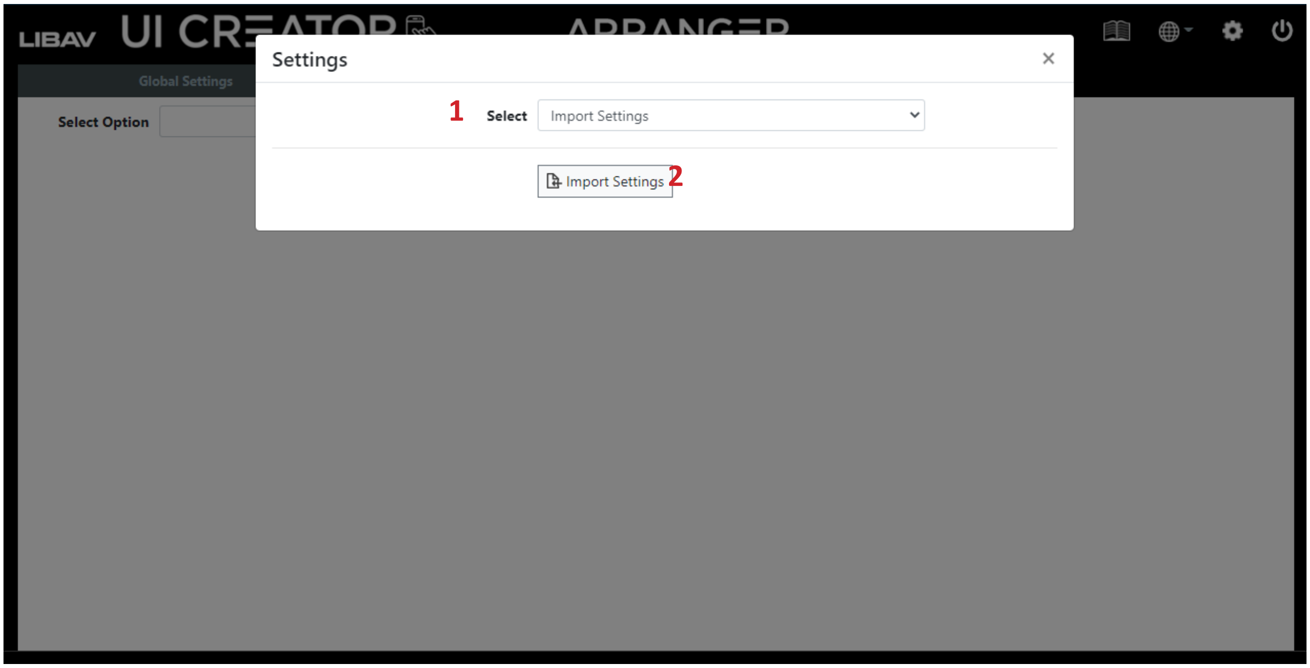
Export Settings will save a file named UIsettings.exp to your downloads folder. This file contains all the settings of the Arranger controller. Use this exported file as a configuration backup that can be imported back into the system to restore the current configuration.



1. Select *Export Settings.*
2. Click *Export Settings.*

Import Settings

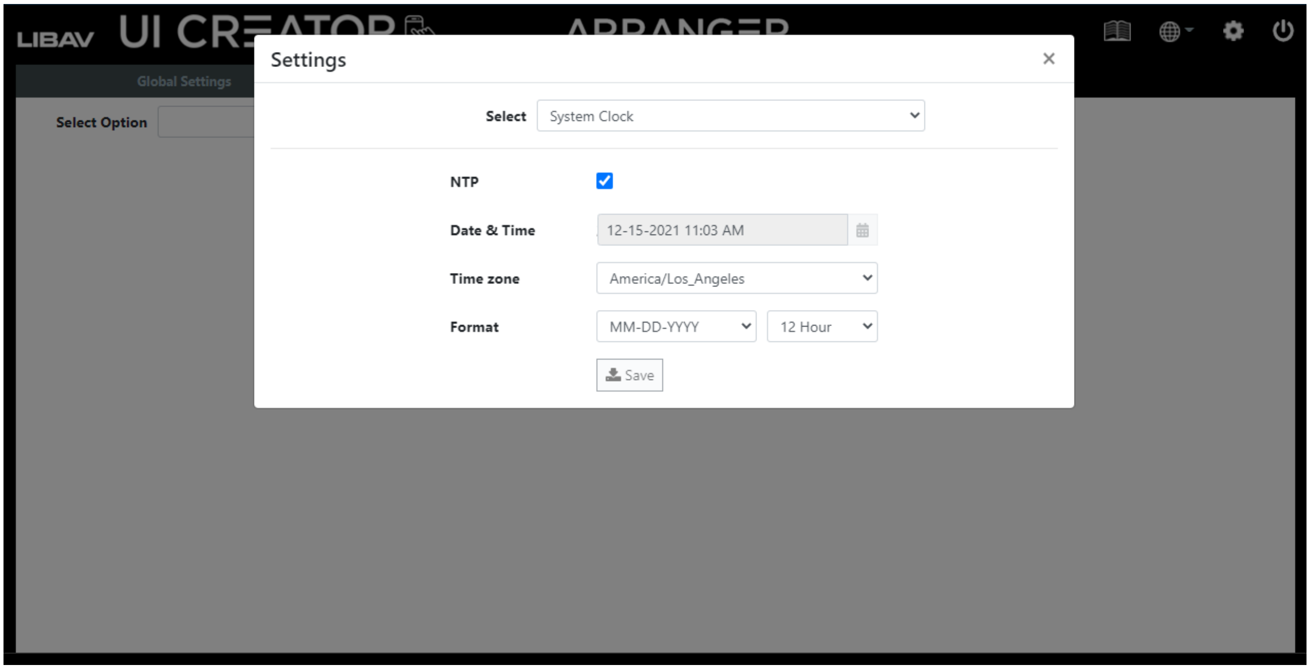
Use Import Settings to load an exported UIsettings.exp file which will restore the Arranger controller's settings.



1. Select *Import Settings.*
2. Click *Import Settings.*

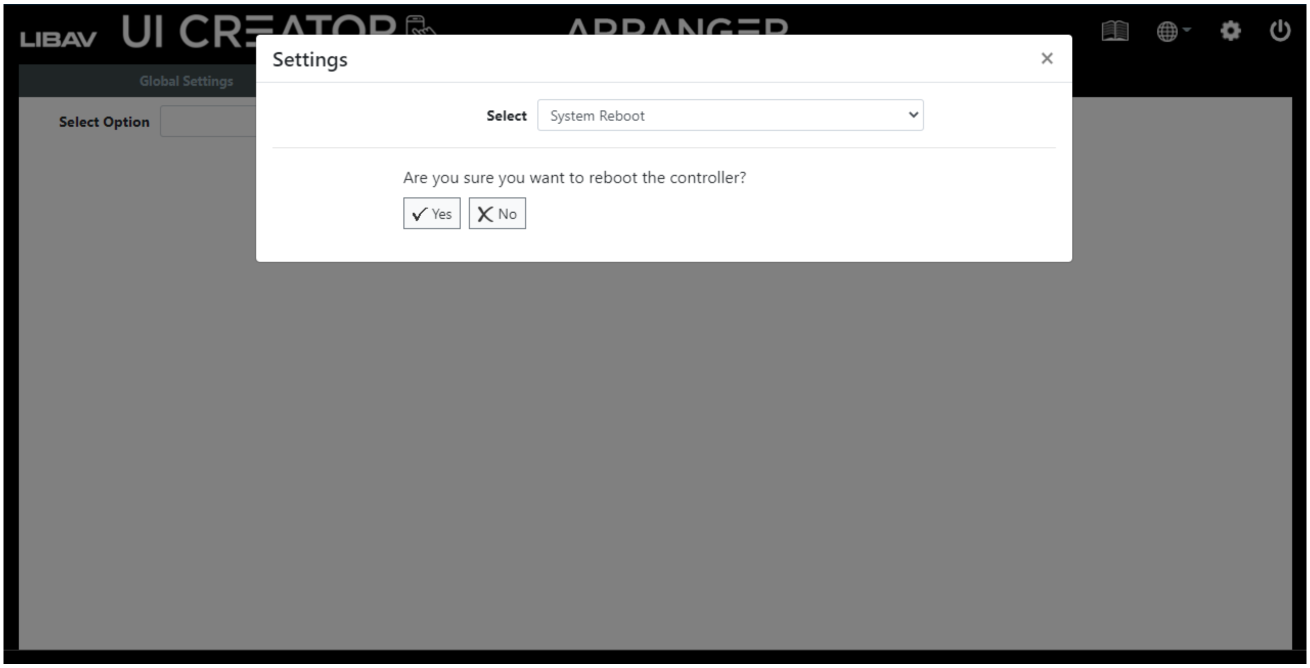
System Clock

The Arranger controller contains a RTC (Real Time Clock) to maintain the correct time and date. Set your local time and date here and click the Save button to apply the changes. The system clock is used for the scheduler and also time stamping the log entries.



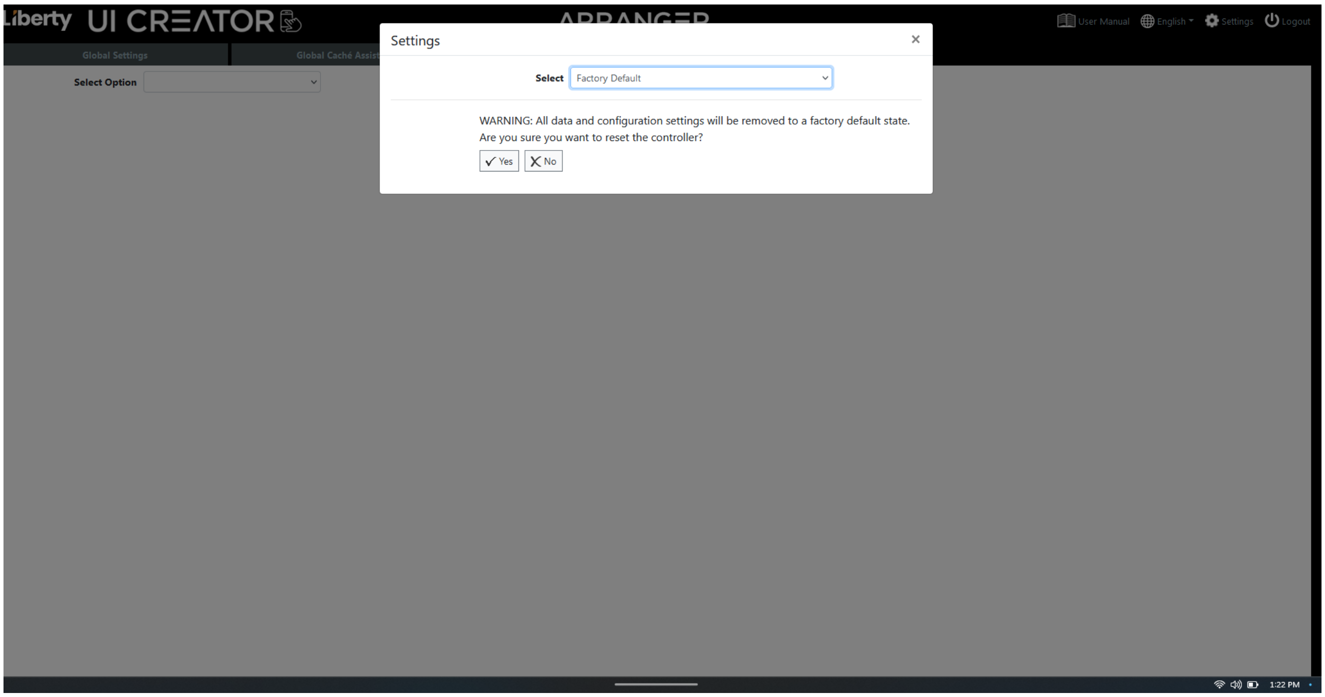
System Reboot

Here you can reboot the Arranger controller. It takes 90 seconds for the controller to reboot.



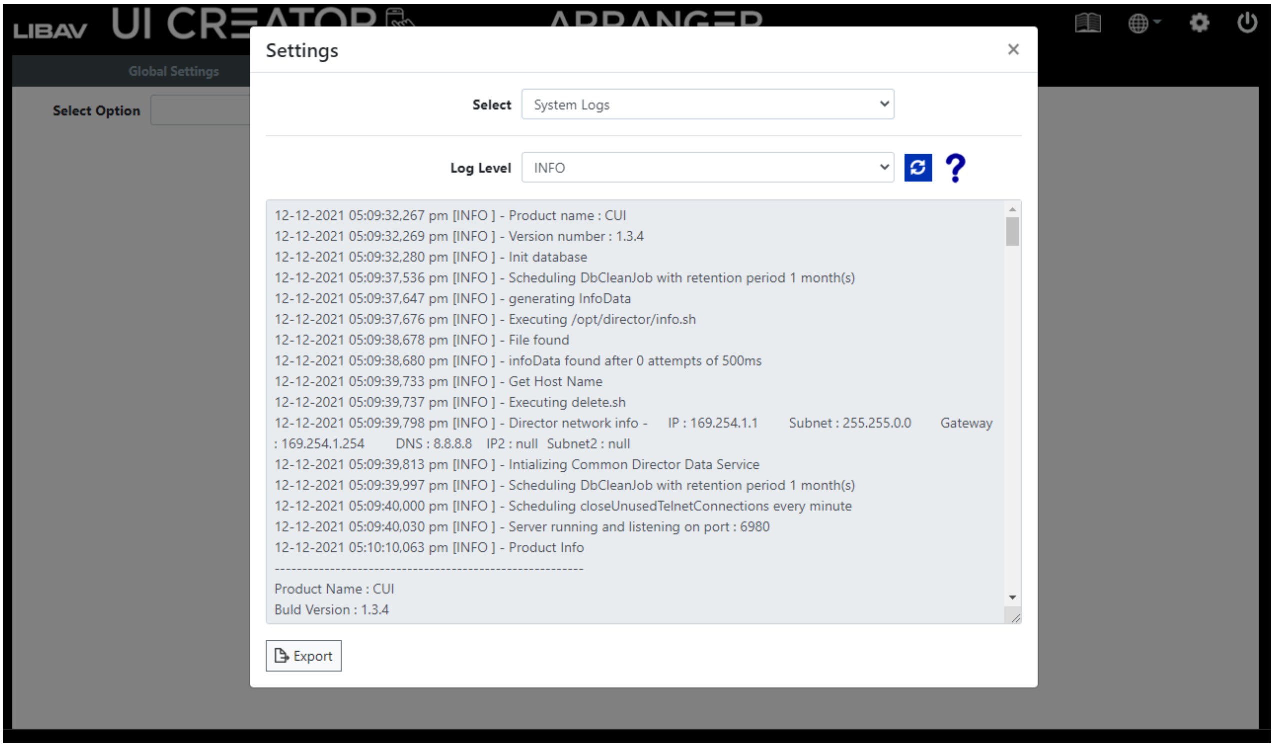
Factory Default

Here you can reset the Arranger Controller back to factory default. Warning if you press yes all data including presets, UIs and license will be lost. Please backup data before pressing yes.



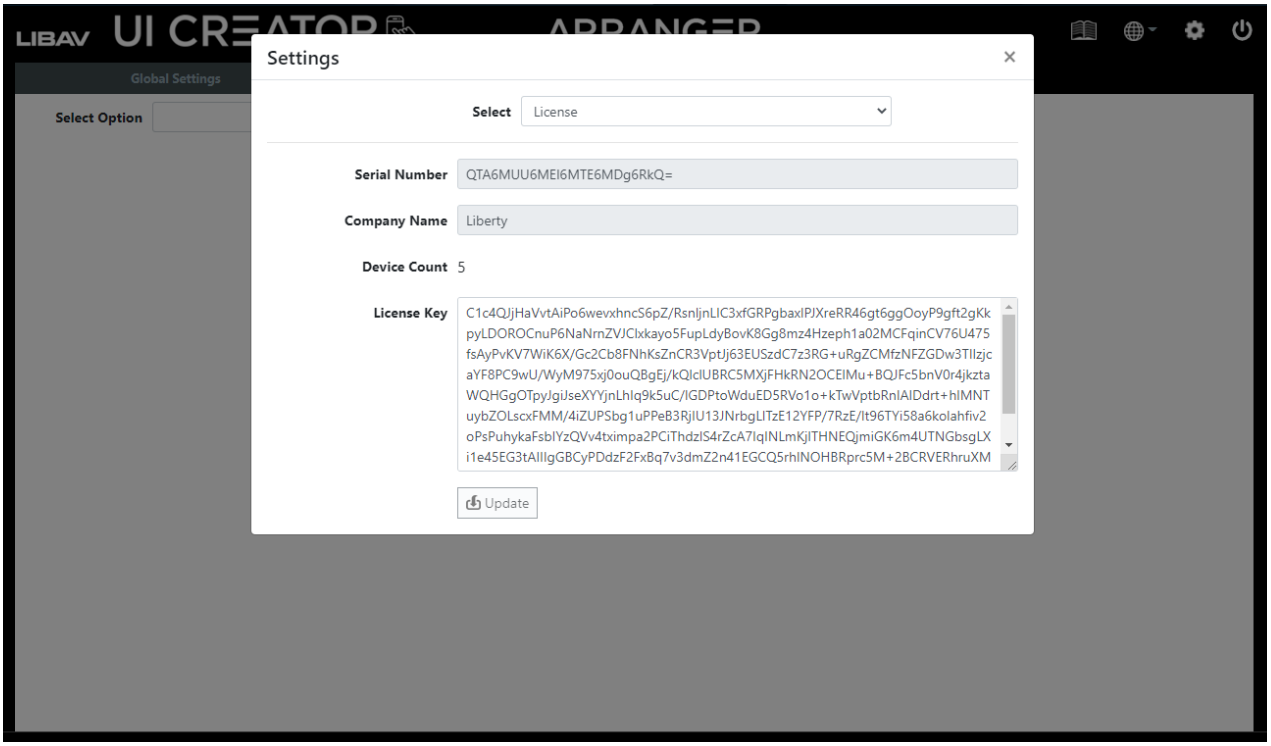
System Logs

The system keeps a log of all system activities. The level of logged information can be set from the Log Level selection. Click the *Export* button to export the log. A file named softwareLog.exp will be saved to your downloads folder. This file has zip compression.



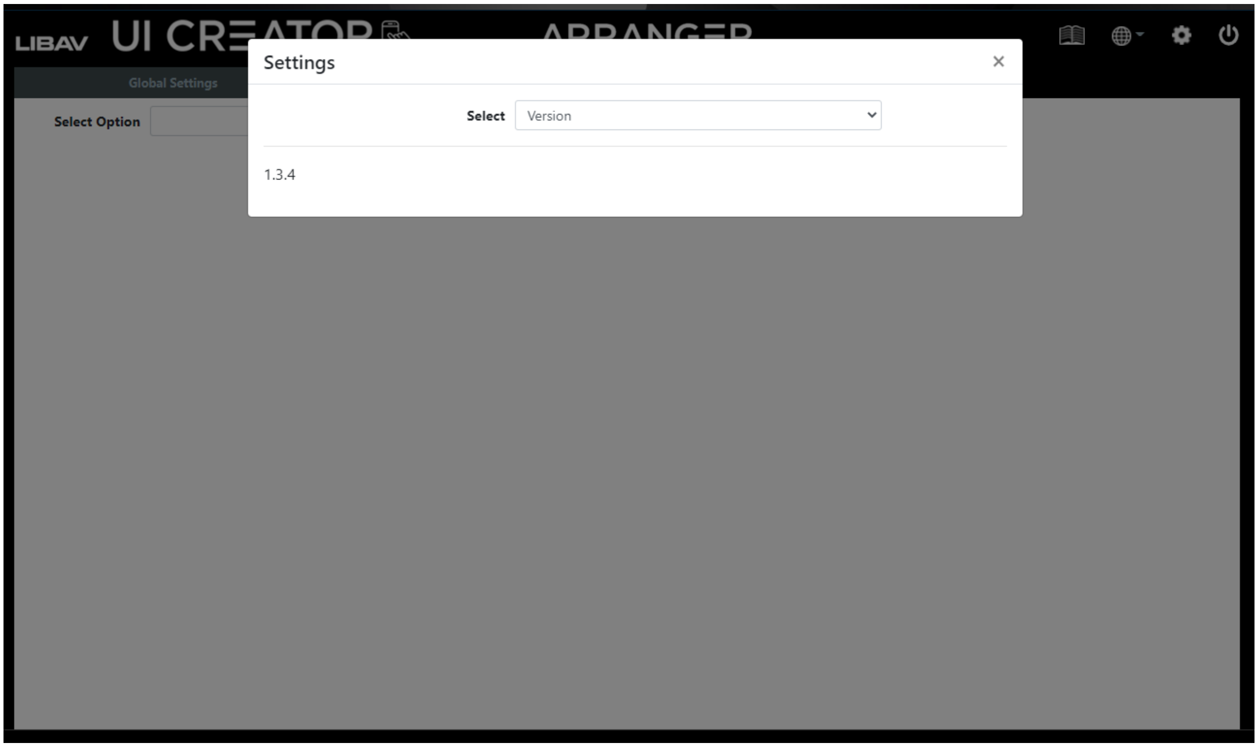
License

The Arranger controller will not operate without a valid license. When the Arranger controller is used for the first time you will be prompted to enter a license key. If a license key has already been issued it can be entered into the system from here. Contact your Liberty AV sales rep or distributor for all licensing requirements.



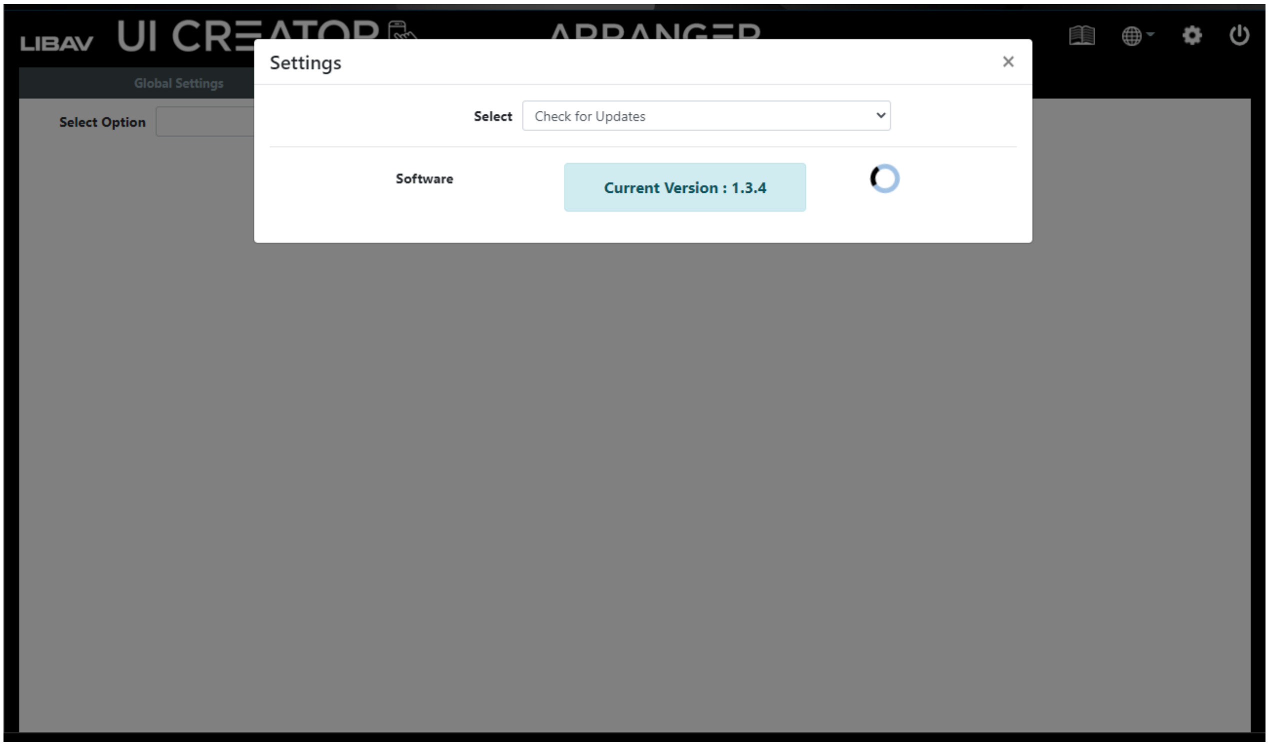
Software Version

Here you can find the current software version.



Check for Updates

Check for Updates will contact an ftp server over the Internet to obtain the latest releases.



Import Updates

When no Internet access is available or a specific update is required, the files will be provided to manually update the system.

Select the SOFTWARE option in the drop down menu.
Then click the browse button to select the required file from the file dialog pop-up.

Security Features

The Arranger software has many security features built in which will be described in detail below. Some of these features are optional and can be enabled or disabled depending on your system security requirements.

1. Required security key with all HTTP requests

The API of the system is accessible via HTTP PUT & GET requests which are protected with the addition of a security key that must be passed with each request.
The security key is accessible from the Global Settings – Security Keys tab.

2. Optional security key with all TCP commands

The API of the system is accessible via TCP port 6980 which can be optionally protected with a security key that must be passed with each command.
The security key is accessible from the Global Settings – Security Keys tab.

3. User Login Failure

This is an optional feature that is part of the system *Notifications* functions available from the Global Settings – *Notifications* tab.
An email can be sent after three (3) failed login attempts to the system.

4. Limiting simultaneous TCP connections to control port 6980

By default there is no limitation to the number of simultaneous TCP connections to control port 6980.
The number of simultaneous TCP connections can be limited between 1 and 10 from the UI Settings *Advanced* tab Connections Limit.

Using Google Assistant

The Arranger controller can be controlled with voice commands via Google Assistant. In this example we will run through the requirements to do so. An IFTTT account will also be required. Visit www.ifttt.com for more information.

To find out how to link your Google account with IFTTT visit this Google help page.

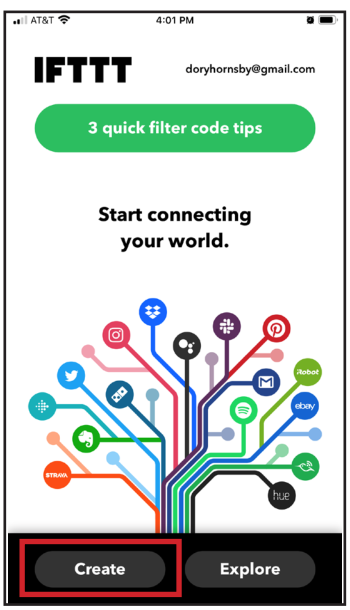
https://support.google.com/googlenest/answer/7194656

First step is to set up the Arranger controller so it is accessible from the Internet. Port 80 of the Arranger controller will need to be port-forwarded to an external IP port. For this example we are going to use an external IP address of 123.456.789.100 and a controller IP address of 169.254.1.1. From your router, port forward from internal 169.254.1.1 port 80 to external, say port 9999 (you can use any unused port number here). Every router is configured differently, so consult your router's owner's manual if required.

Now externally you should be able to access the Arranger controller's webpage by entering '123.456.789.100:9999' into a browser.

Now we can set up IFTTT to activate any command on the Arranger controller but we strongly advise that Arranger *Presets* only are specified for this activity.

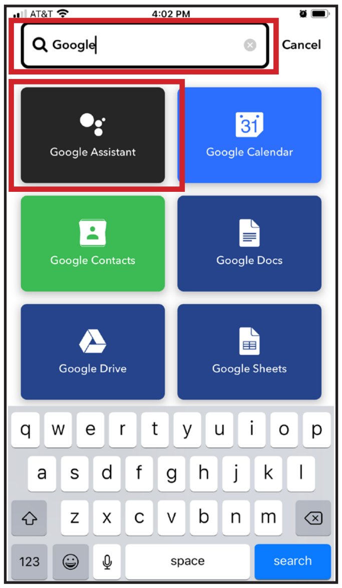
Open your IFTTT account and *Create* a new applet



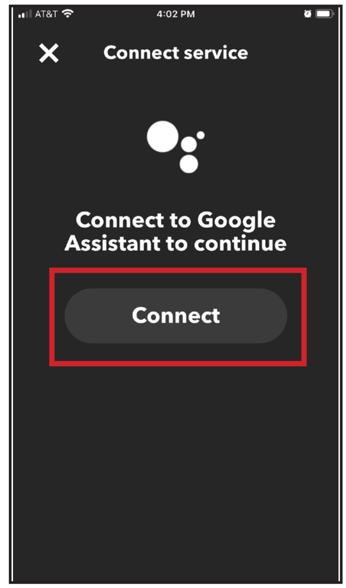
Click *if This Add*

Using Google Assistant continued....

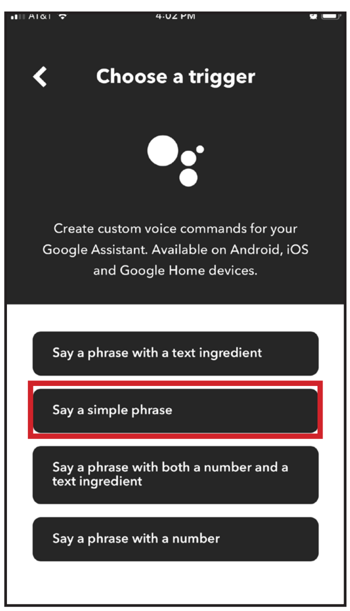
Search and select *Google Assistant* when asked to choose a service in the IFTTT applet wizard



You will be prompted for your Google account login, click *Connect* to proceed



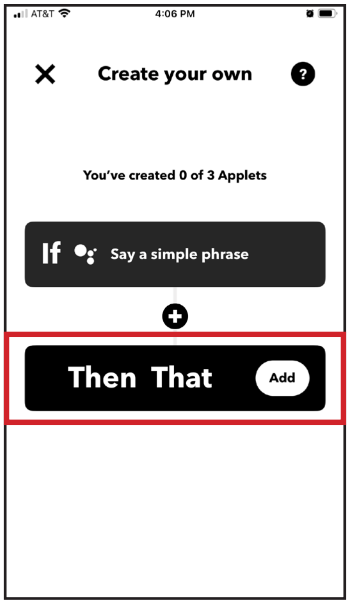
Choose the *Say a Simple Phrase* when asked for a trigger



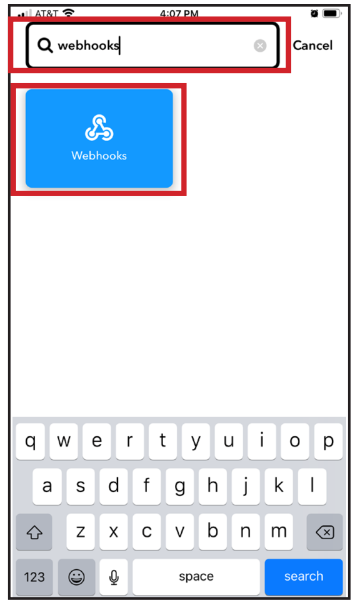
Choose what you will say, the Google response and click *Create Trigger*

Using Google Assistant continued....

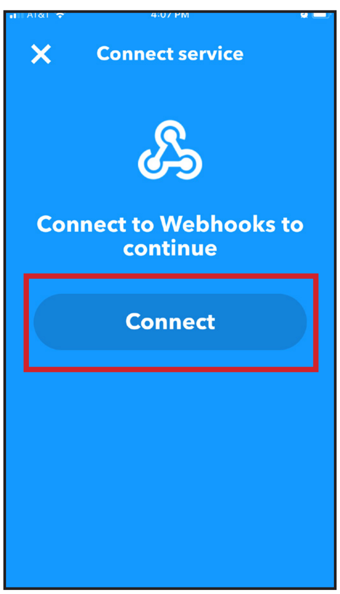
Click *Then That Add*



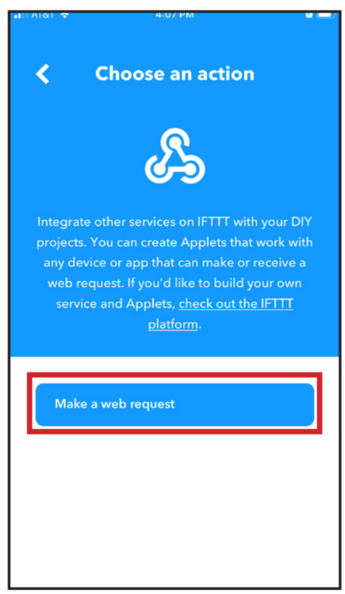
Search and choose *Webhooks* when asked for a service



You will be prompted to connect to *Webhooks* click *Connect* to proceed



Click *Make a web request*



Using Google Assistant continued....

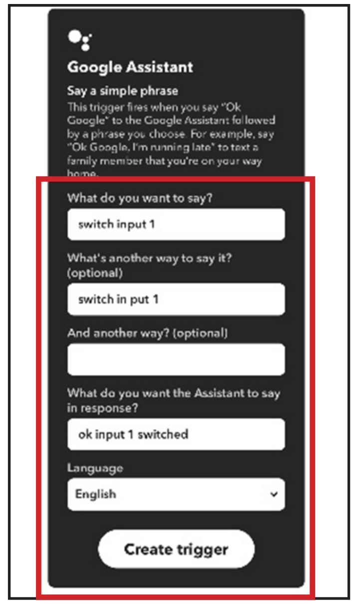
• Now enter the URL which is the external IP address and port established in the router. In our previous example; '123.456.789.100:9999'

• Select Method: 'POST'

• Select Content Type: 'application/json'

• Enter Body with desired command (see below)

• Click 'Create action'



{"cmd":"<command>","key":"<security_key>"}
=
{"cmd":"preset load <preset>","key:"<security_key>"}

where:
*<preset> = Name given to the preset to be applied
^<security_key> = HTTP Security key

Example:
{"cmd":"preset load MyPreset","key":"ts2Xe1sn1Nk1rm1>l1$p1Tk18q14e"}

*For more information on building and using Presets, see *Global Settings > Presets*

^For more information on generating and using HTTP Security keys, see *Global Settings > Security Keys > HTTP Security Key*

How to HTTP Request

- GET = http://<controllerURL>/api/command/<ARRANGER_API_COMMAND>/<KEY>
- POST = http://<controllerURL>/api/command/{'cmd': '<ARRANGER_API_COMMAND>', 'key': '<KEY>'}

Example 1 - POST - ajax

```
<script language='JavaScript' type='text/javascript'>

var controllerIP = '169.254.1.1'; *change this to the same IP address as the
controller var BaseURL = 'http://' + controllerIP + '/api/command/';
var MAXIMUM_WAITING_TIME = 5000; *timeout in milliseconds
var CheckStatusTimer; var key = '123xyz'; *replace this with the generated
security key var command = '123xyz'; *replace with any Arranger API command

     $.ajax({
     type: 'POST',
     crossDomain: true,
     contentType: 'application/json; charset=utf-8',
     dataType: 'text',
     url: BaseURL, data: "{'cmd':' + command + ','key':' + key + '}",
     timeout: MAXIMUM_WAITING_TIME,
     success: function(data, textStatus, XMLHttpRequest){
             console.log(data);
     },

     error: function (XMLHttpRequest, textStatus, errorThrown) {
            console.log('ERROR = ' + errorThrown;
     }
  });

</script>
```

Example 2 - POST - xhr

```
<script language='JavaScript' type='text/javascript'>

var controllerIP = '169.254.1.1'; *change this to the same IP address as the
controller
var BaseURL = 'http:// ' + controllerIP + '/api/command/';
var key = '123xyz'; *replace this with the generated security key
var command = '123xyz'; *replace with any Arranger API command
var xmlRequest = new XMLHttpRequest(; xmlRequest.open('POST', BaseURL,
true);
var params = "{'cmd':' +command+ ', 'key':' + key + '}";
var MAXIMUM_WAITING_TIME = 5000;
xmlRequest.onreadystatechange = function () {

  if (this.readyState == 4) {
      clearTimeout(xmlTimer);
      if(this.status == 200){
            console.log(this.responseText);

      }else{
            console.log('ERROR = ' + this.status + ' ' + this.statusText);
       }
  }else{
      if(this.status != 200){
            console.log('ERROR = ' + this.status + ' ' + this.statusText);
      }
     };
xmlRequest.send(params);
var xmlTimer = setTimeout(function() {
      xmlRequest.abort();
      console.log('ERROR = timeout');
}, MAXIMUM_WAITING_TIME);

</script>
```

Preset Logic

Within presets the following logical operators are available:

- & Used to append strings together.
- && Used as a logical AndAlso for multiple boolean conditional statements.
- || Used as a logical Or for multiple boolean conditional statements.
- == (equal to) Used to compare numbers or strings that equal.
- != (not equal to) Used to compare numbers or strings that do not equal.
- < (less than) Used to compare numbers.
- > (greater than) Used to compare numbers.
- <= (less than or equal to) Used to compare numbers.
- >= (greater than or equal to) Used to compare numbers.
- not Used to invert an if statement boolean result.

Within a UI preset the following system read only variables are available:

- <<ui_name>> contains the UI name as a string.
- <<button_name>> contains the active button name as a string.
- <<button_position>> contains the active button position as a string.
- <<slider_name>> contains the active slider name as a string.
- <<slider_value>> contains the active slider value as an integer.
- <<encoder_name>> contains the active Encoder button assigned device name as string.
- <<decoder_name>> contains the active Decoder button assigned device name as string.

substr() with instr() example.

```
Combining substr() with instr() to extract a value from a return string as follows:

(substr(<string>,<startIndex>,<optional_numOfCharacters>))
(instr(<string>,<searchString>,<optional_offset>))
set var LevelTmp (trim(send tcp 10.1.1.10 4998 "get_LED_LIGHTING,1:3\x0D" reply))
    LevelTmp = LED_LIGHTING,1:3,25,50
set var LevelTmp (substr((get var LevelTmp),18))
    LevelTmp = 25,50
set var LevelTmp (substr((get var LevelTmp),0,(instr((get var LevelTmp),",",-1))))
    LevelTmp = 25
set ui_slider myUI mySldr value (get var LevelTmp)
if ((get var LevelTmp) > 0) {
    set ui_slider myUI mySldr state enabled
    set var LightLevel (get var LevelTmp)
}
set var LevelTmp [delete]
```

```
The following get commands will return a string value that can be used with:
o     get var
o     get ui
o     get ui_button (state | position)
o     get ui_slider (state)
o     get events (state)

•     == (equal to)
•     != (not equal to)
```

Example 1:
```
if ((get var myVar) == "<string>") {
    <action>
} else {
    <action>
}
select case ((get var myVar)) {
    case "<string>"
        <action>
    case else
        <action>
}
```
Example 2:
```
if ((get var myVar) != "<string>") {
    <action>
} else {
    <action>
}
```
Example 3:
```
if (get ui myUI == "disabled") {
    <action>
}
select case ((get ui myUI)) {
    case "disabled"
        <action>
    }
}
```
Example 4:
```
if (get ui_button myUI myBtn position == "down") {
    <action>
}
if (get ui_button myUI myBtn state == "enabled") {
    <action>
}
```
Example 5:
```
if (get ui_slider myUI mySldr state == "enabled") {
    <action>
}
```
Example 6:
```
if (get events myEvent state != "enabled") {
    <action>
}
```

Command results continued.

The following get commands will return an integer value that can be used with:

- **get var**
- **get indicator (value)**
- **get slider (value)**

- **== (equal to)**
- **!= (not equal to)**
- **< (less than)**
- **> (greater than)**
- **<= (less than or equal to)**
- **>= (greater than or equal to)**

Example 1:

```
if ((get var myVar) == 100) {
    <action>
} else {
    <action>
}
select case ((get var myVar)) {
    case "100"
        <action>
    case else
        <action>
}
```

Example 2:

```
if ((get var myVar) != 100) {
    <action>
} else {
    <action>
}
```

Example 3:

```
if ((get ui_indicator myUI mySldr value) == 50) {
    <action>
} else {
    <action>
}
select case ((get ui_indicator myUI mySldr value)) {
    case "50"
        <action>
    case else
        <action>
}
```

Example 4:

```
if ((get ui_indicator myUI myInd value) < 100) {
    <action>
} else {
    <action>
}
```

Example 5:

```
if ((get ui_slider myUI mySldr value) > 100) {
    <action>
} else {
    <action>
}
```

Command results continued.

The following get command will return a Boolean value that can be used with:

o get var

- *not*

Example 1:

```
if (get var myVar) {
    <action>
} elseif (get var myVar2) {
    <action>
}

select case ((get var myVar)) {
    case "true"
        <action>
    case else
        select case ((get var myVar2)) {
            case "true"
                <action>
        }
}
```

Example 2:

```
if not (get var myVar) {
    <action>
}
select case ((get var myVar)) {
    case "false"
        <action>
    }
}
```

Command results continued.

```
The following send commands will return a string value that can be used with:
o     send tcp <feedback> = reply
o     send udp <feedback> = reply
o     send telnet <feedback> = reply
o     send gc > (port 4999 / 5000) <feedback> = reply
o     send gc > (port 4998) command get

•     == (equal to)
•     != (not equal to)
```

Example 1:

```
if (send tcp 10.1.1.10 4998 "setstate,1:1,1\x0D" reply == "state,1:1,1\x0D") {
    <action>
} else {
    <action>
}
if ((trim(send tcp 10.1.1.10 4998 "setstate,1:1,1\x0D" reply) == "state,1:1,1") {
    <action>
} else {
    <action>
}
select case ((send tcp 10.1.1.10 4998 "setstate,1:1,1\x0D" reply)) {
    case "state,1:1,1\x0D"
        <action>
    case else
        <action>
}
```

Example 2:

```
if (send gc 169.254.1.70 4999 "My String" reply == "This String") {
    <action>
}
```

Example 3:

```
if (send gc 169.254.1.70 4998 getstate,1:1 == "state,1:1,1\x0D") {
    <action>
}
if (trim(send gc 169.254.1.70 4998 getstate,1:1) == "state,1:1,1") {
    <action>
}
```

Example 4:

```
if (send udp 255.255.255.255 10000 "My String" reply == "This String") {
    <action>
}
```

Example 5:

```
if (send telnet 10.1.1.10 23 username "Username:" "MyLogin" password "Password:"
"MyPassword" ...
"My String" reply == "This String") {
    <action>
}
```

Using Command Assistant

When dealing with direct API control commands or creating presets, the *Command Assistant* wizard is available for all commands to help make the construction of command strings as simple as possible.

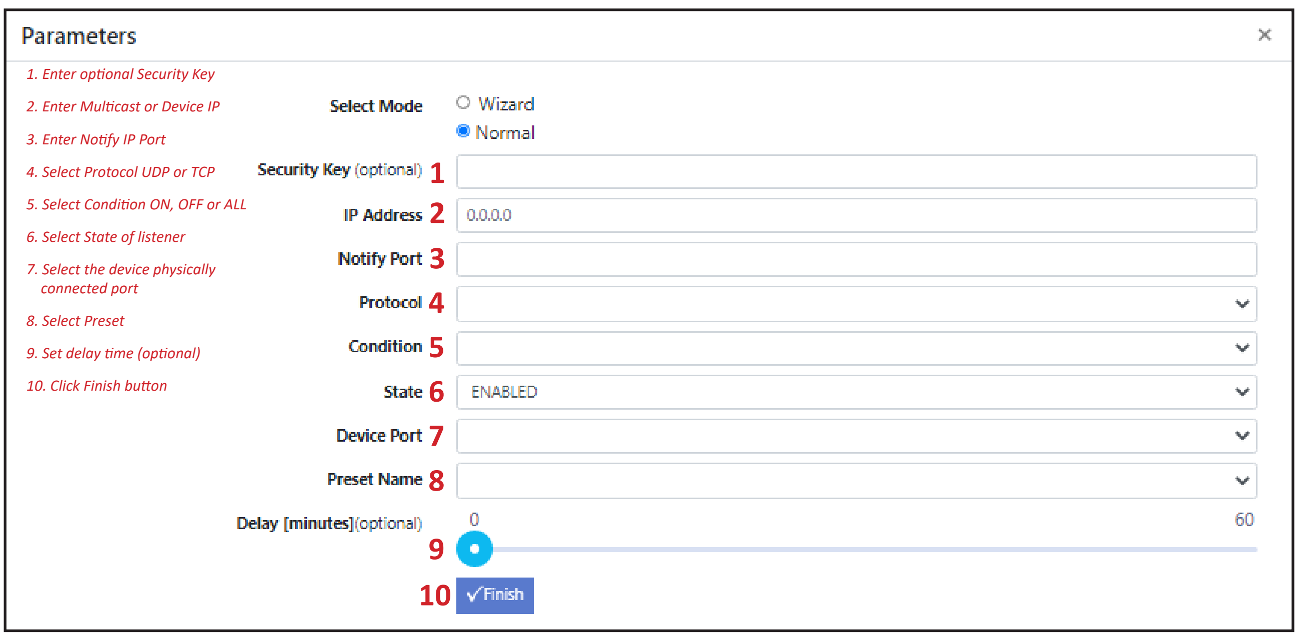
Most commands have a *Normal* and *Wizard* mode of creation. In *Normal* mode most parameters are set by entering the details into the various text boxes manually, while in *Wizard* mode parameters are mostly set with drop-down selections to ensure the syntax of the command is correct.

Below are explanations for all commands using *Normal* and *Wizard* mode.

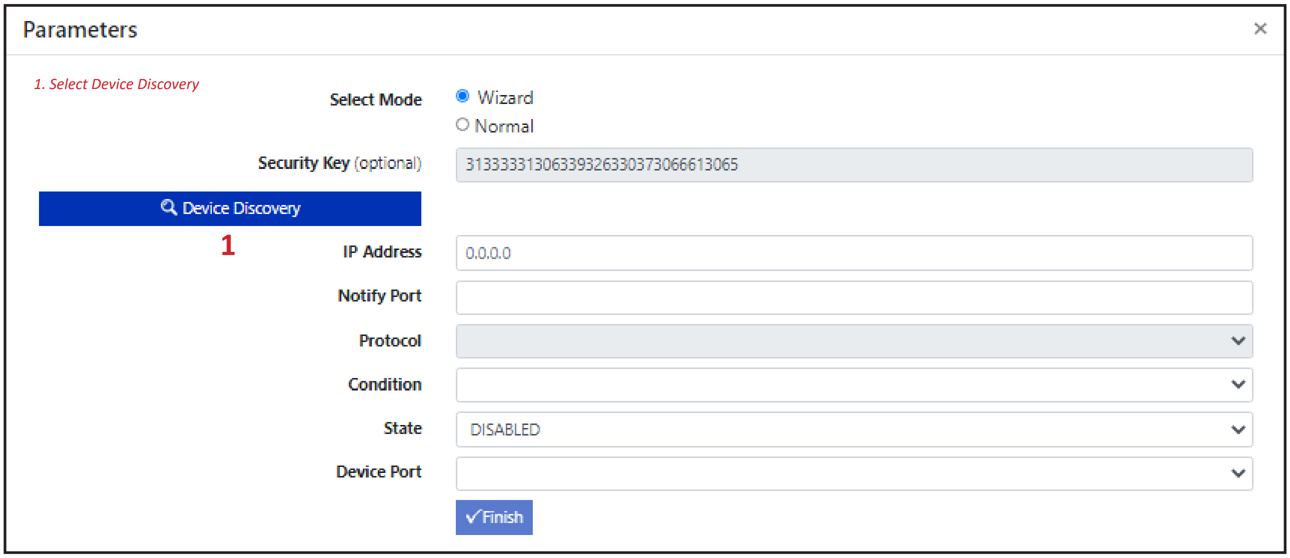
Command: set listener - Normal Mode

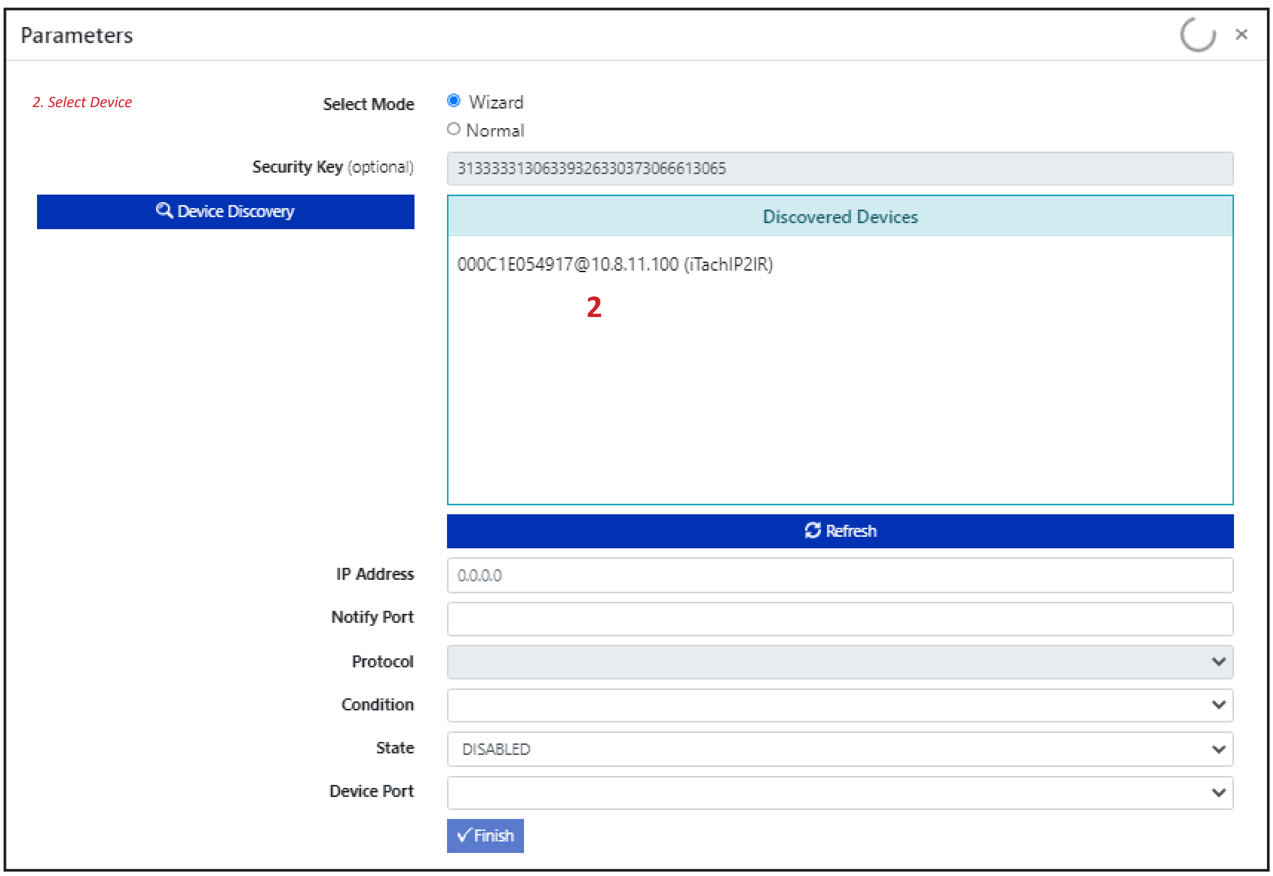
Listener is OFF (disabled)

Listener is ON (enabled)



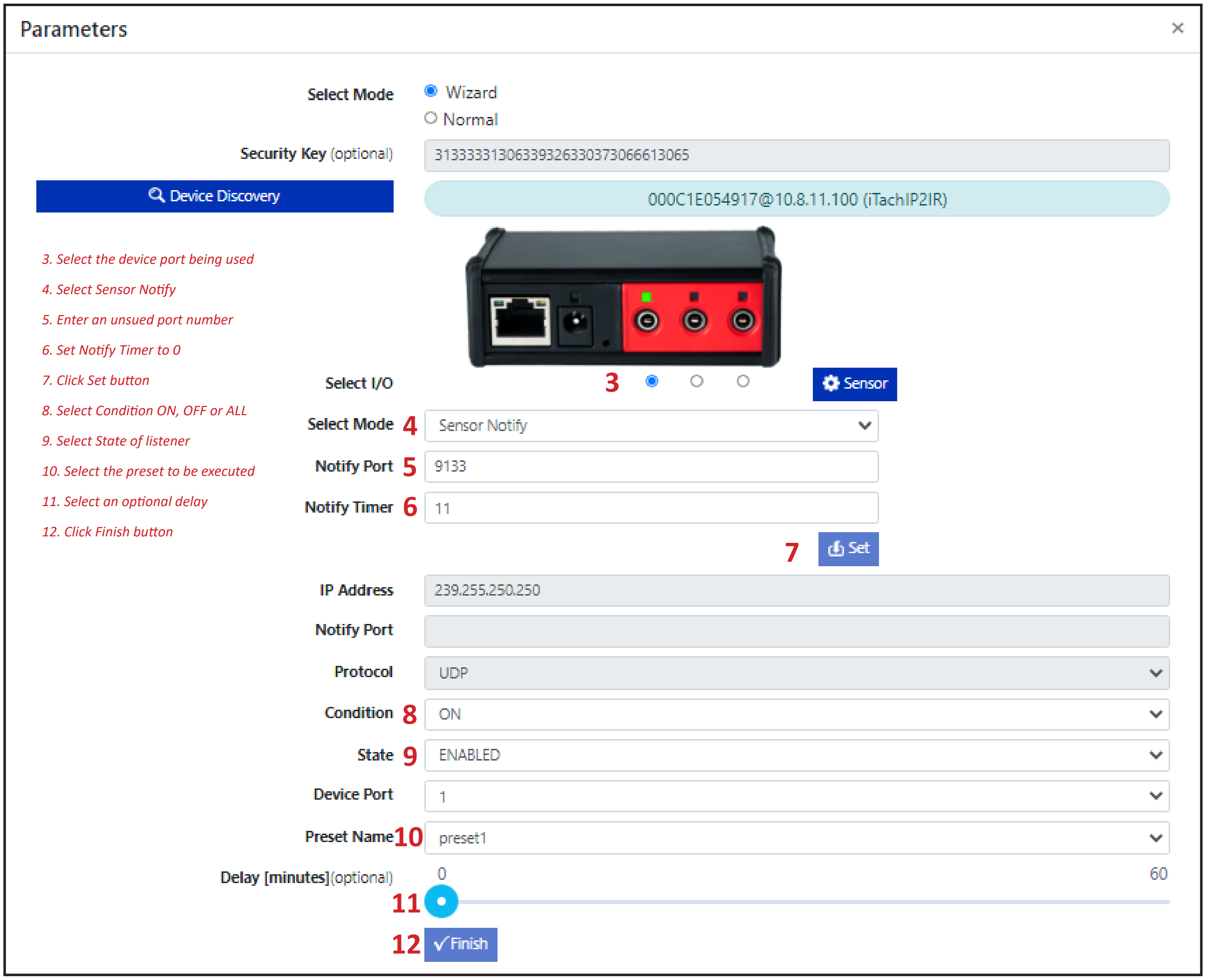
Command: set listener - Wizard Mode





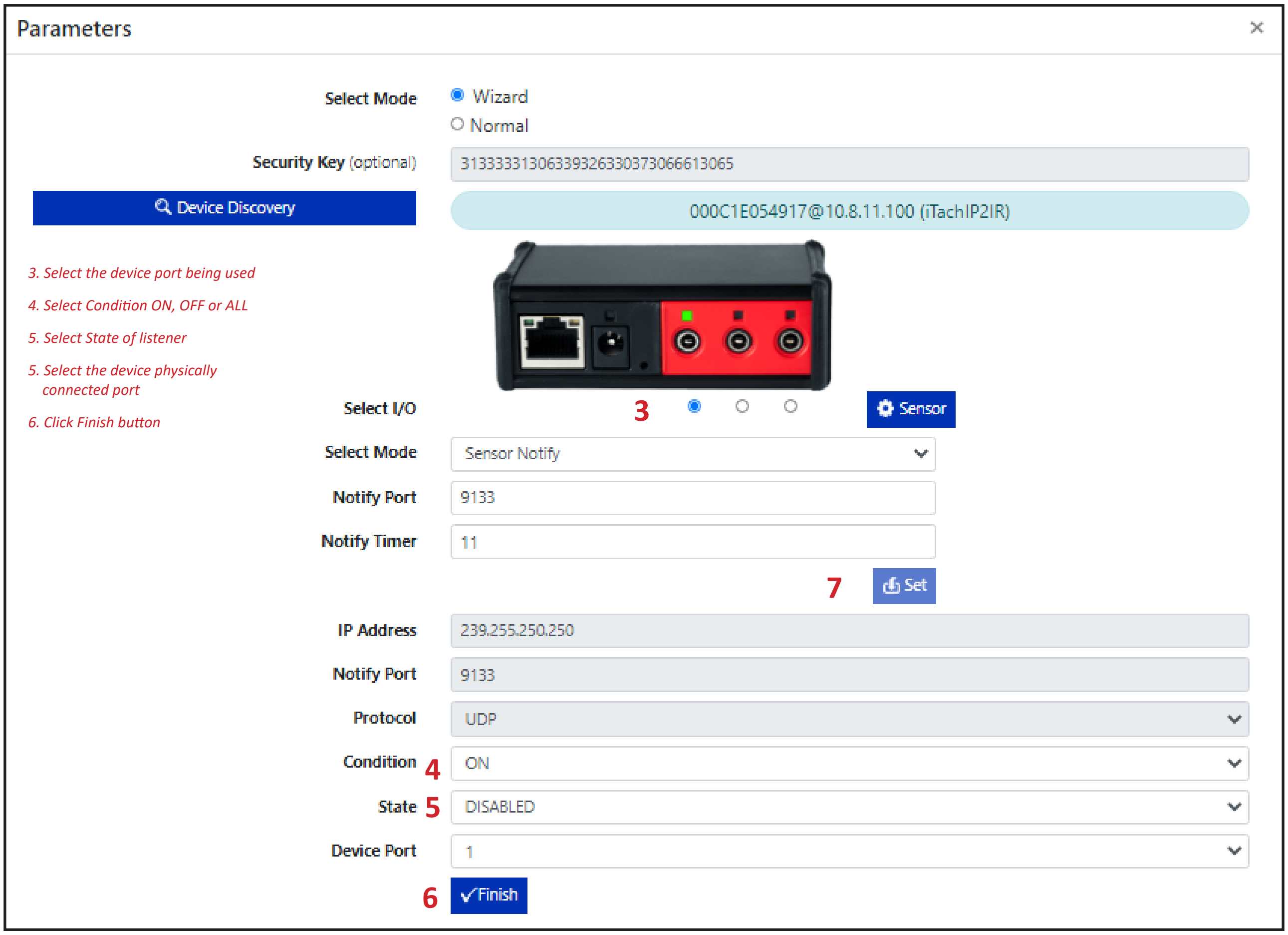
Command: set listener - Wizard Mode continued...

Listener is ON (enabled)

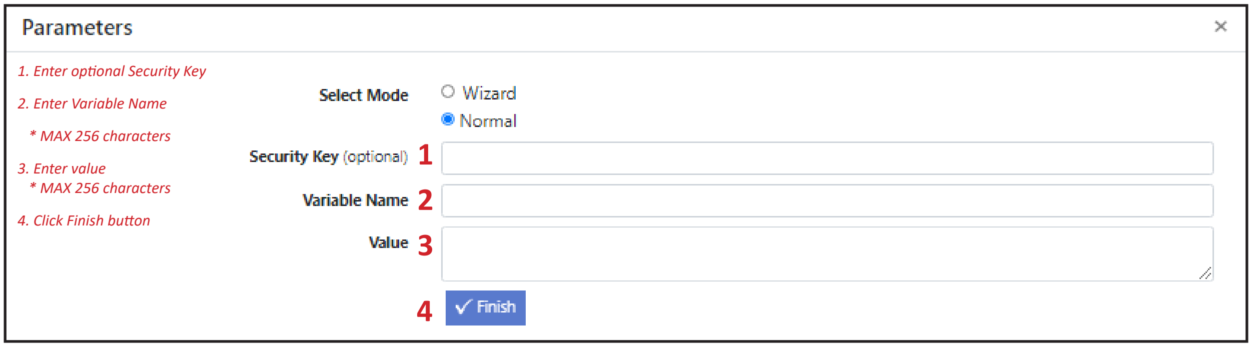


Command: set listener - Wizard Mode continued...

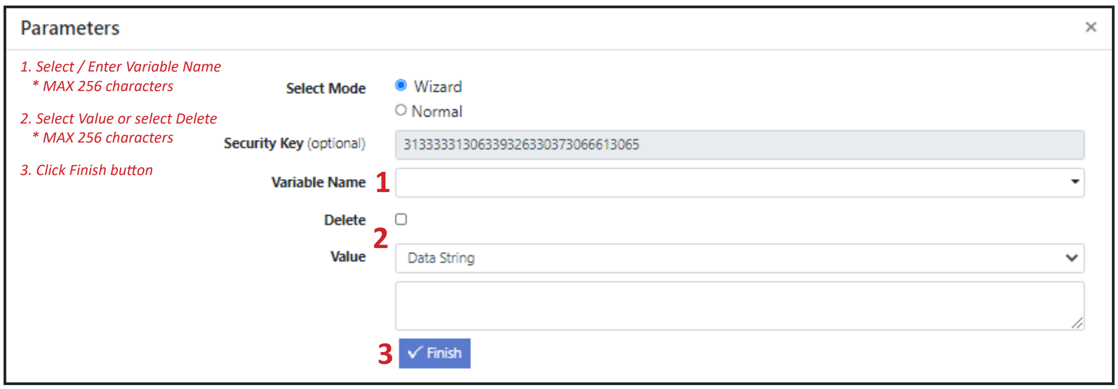
Listener is OFF (disabled)



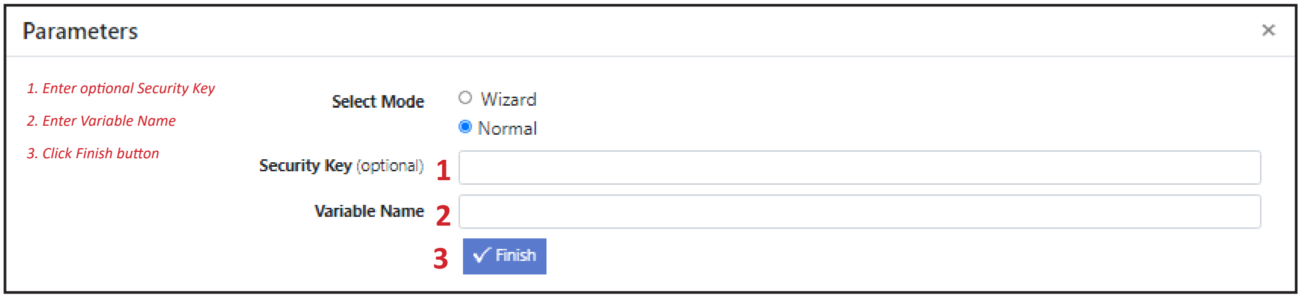
Command: set var - Normal Mode



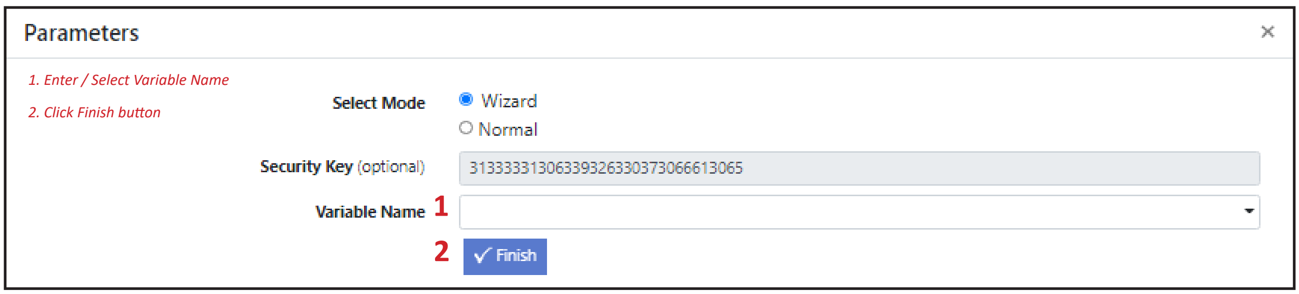
Command: set var - Wizard Mode



Command: get var - Normal Mode

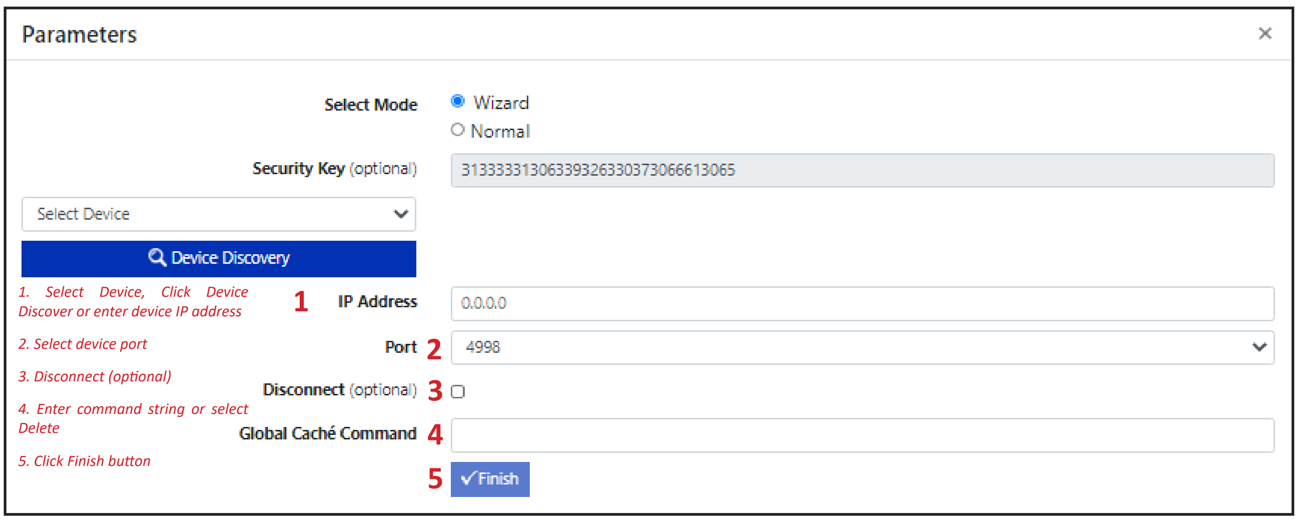


Command: get var - Wizard Mode

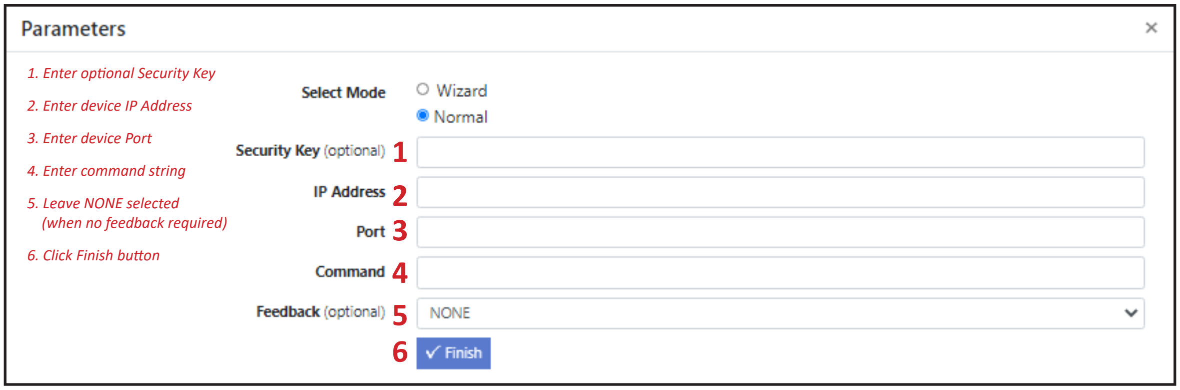


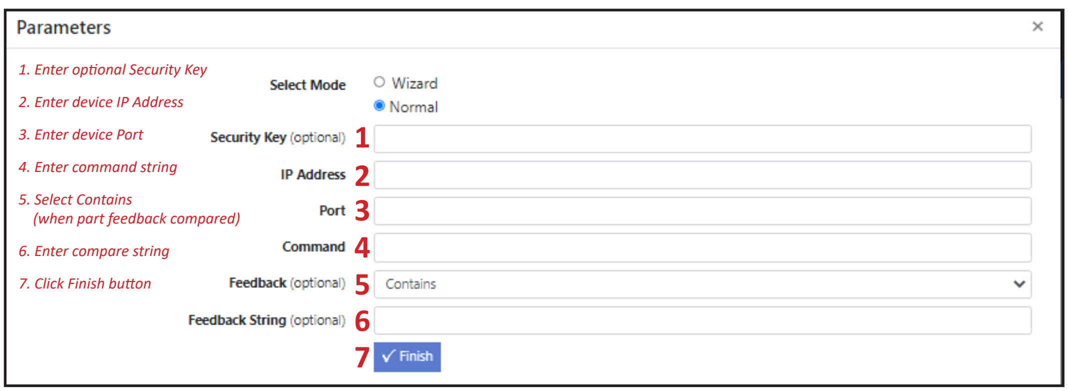
Command: send gc - Normal Mode

Command: send gc - Wizard Mode

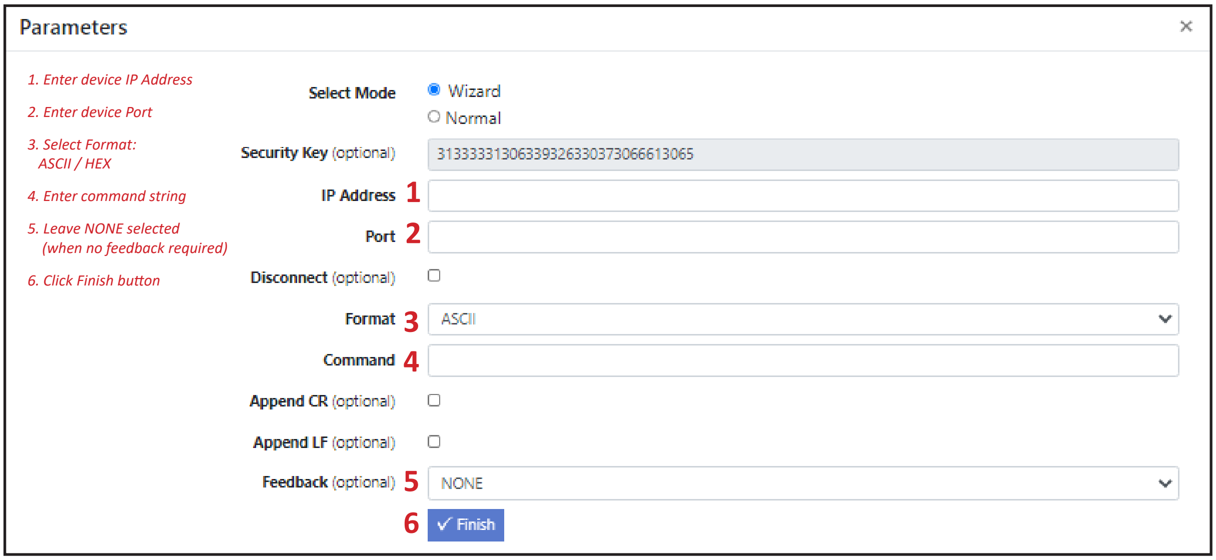


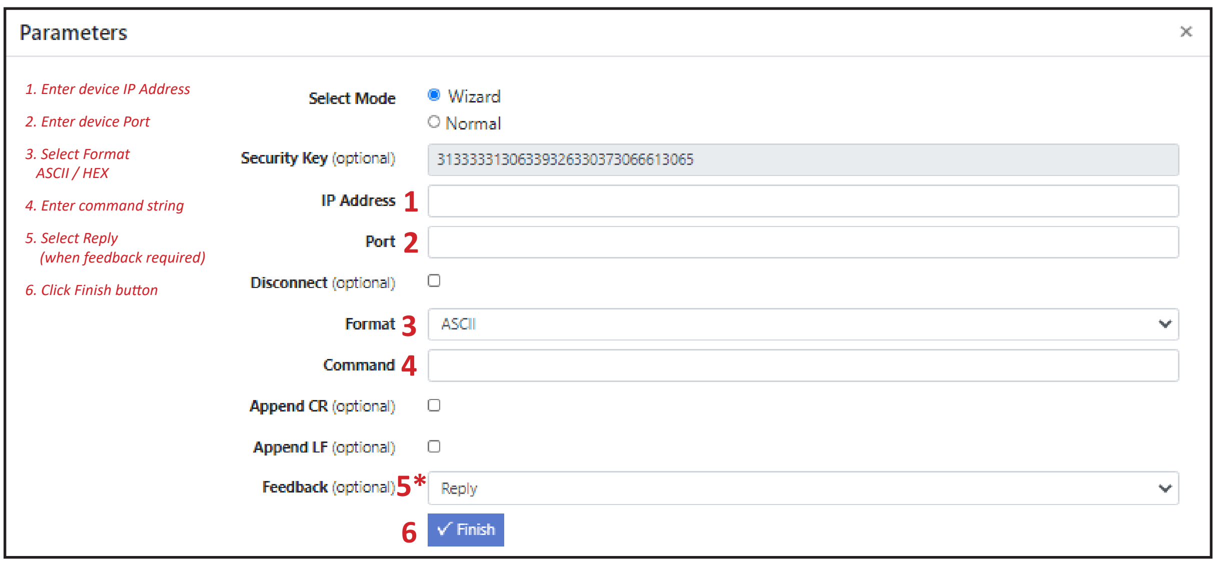
Command: send tcp - Normal Mode





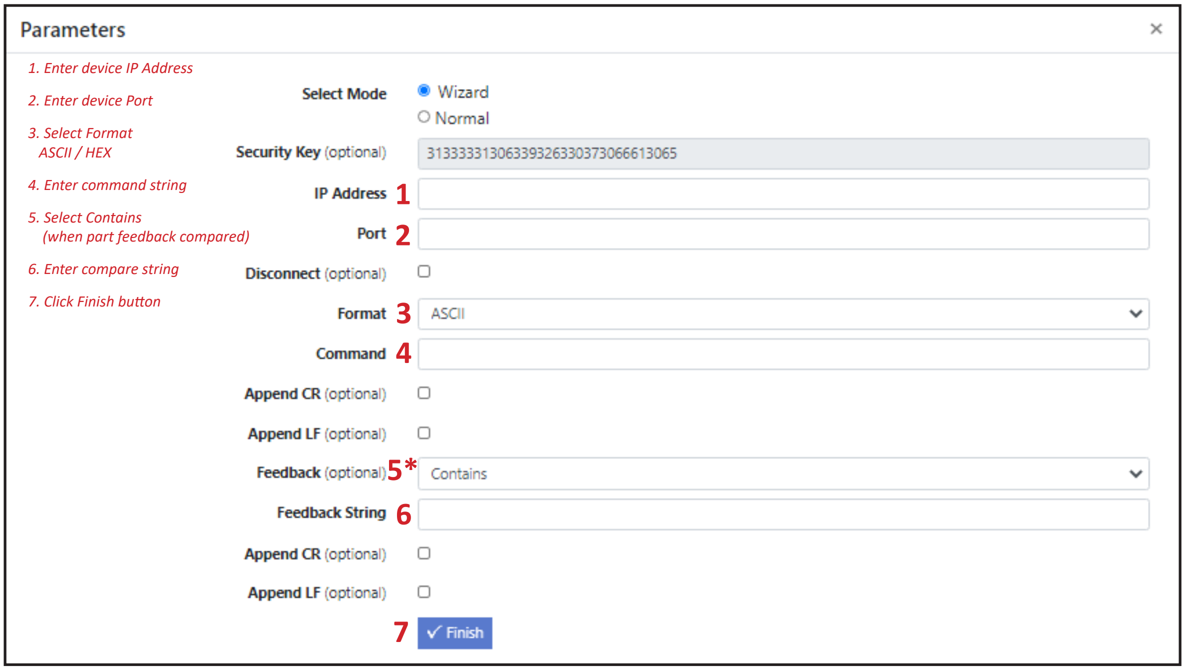
Command: send tcp - Wizard Mode



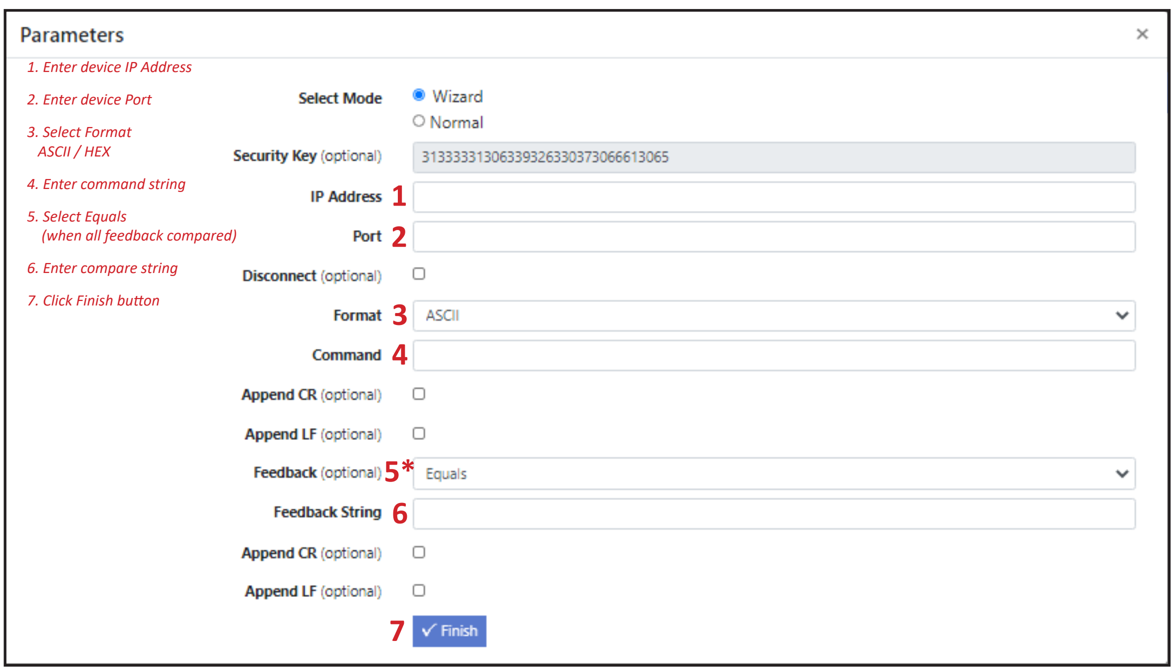


**Note:* when using the *REPLY* option the feedback generated from the TCP device will pass through

Command: send tcp - Wizard Mode continued...



**Note:* When using the *CONTAINS* option, you can tell Arranger to look for a string or part of a string in the feedback that signifies the sent command was successful and the device is in fact in the desired state

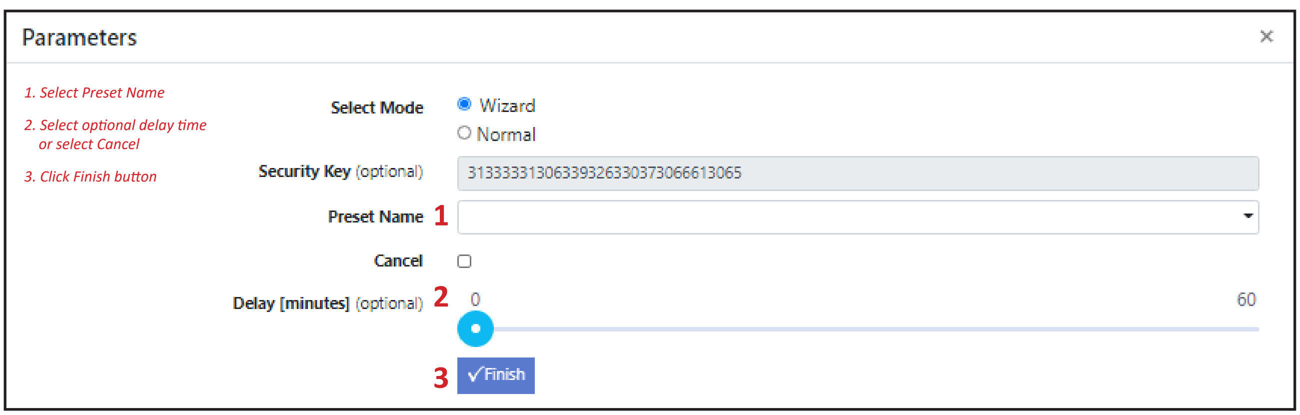


**Note:* When using the *EQUALS* option, you can tell Arranger to look for an exact string in the feedback that signifies the sent command was successful and the device is in fact in the desired state

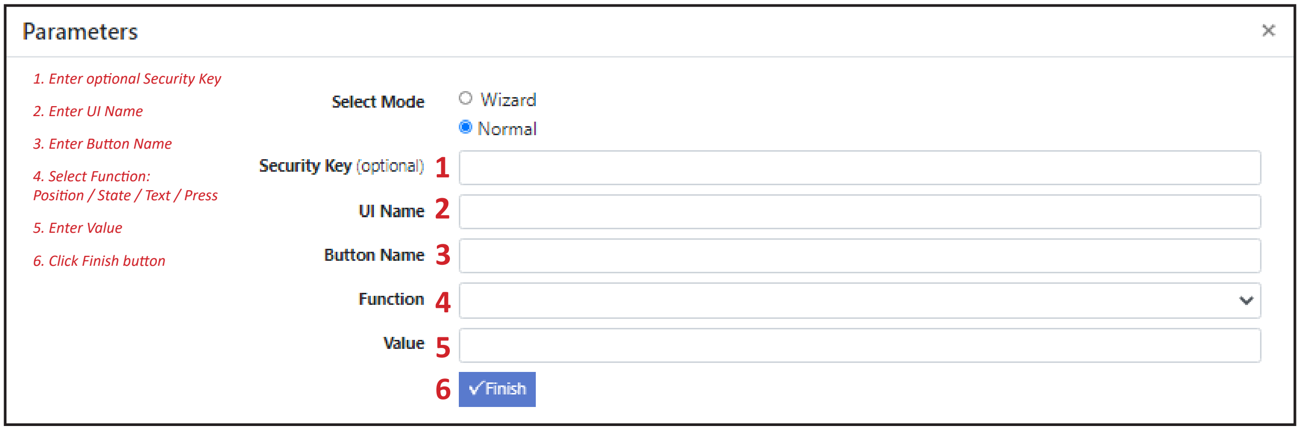
Command: preset add

Command: preset delete

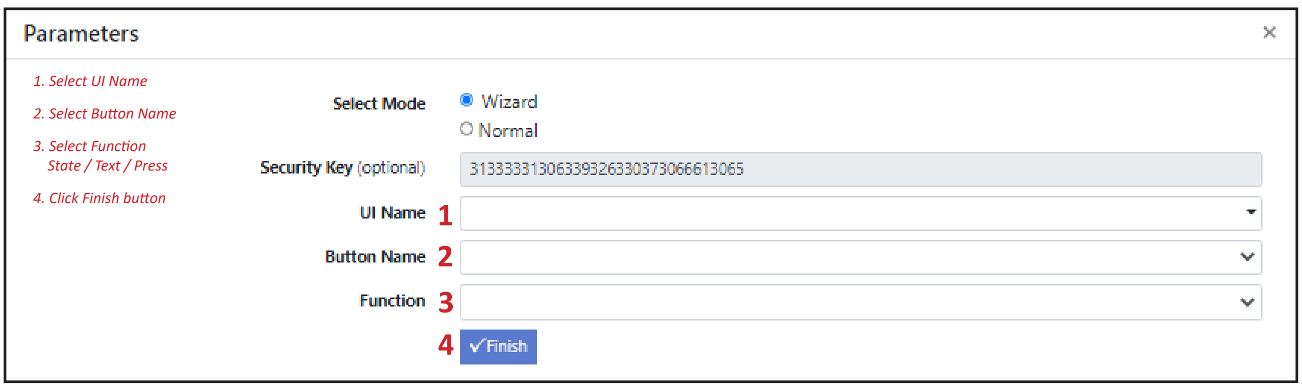
Command: preset load



Command: set ui_button - Normal Mode

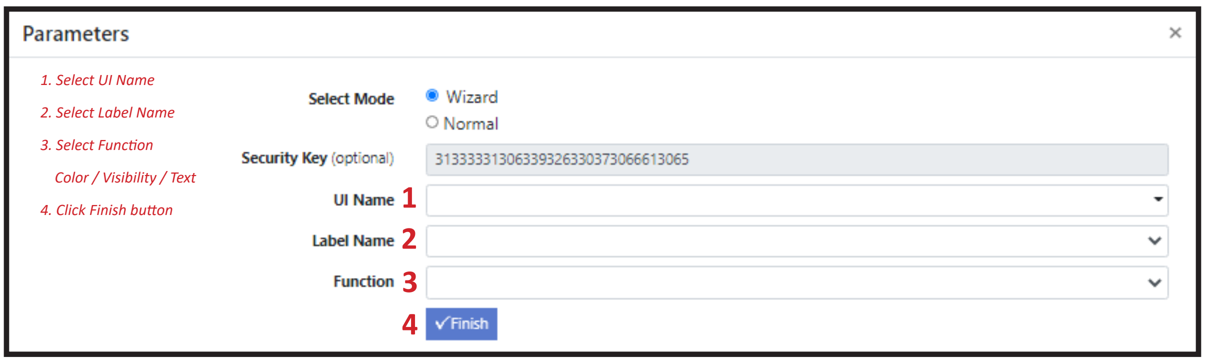


Command: set ui_button - Wizard Mode

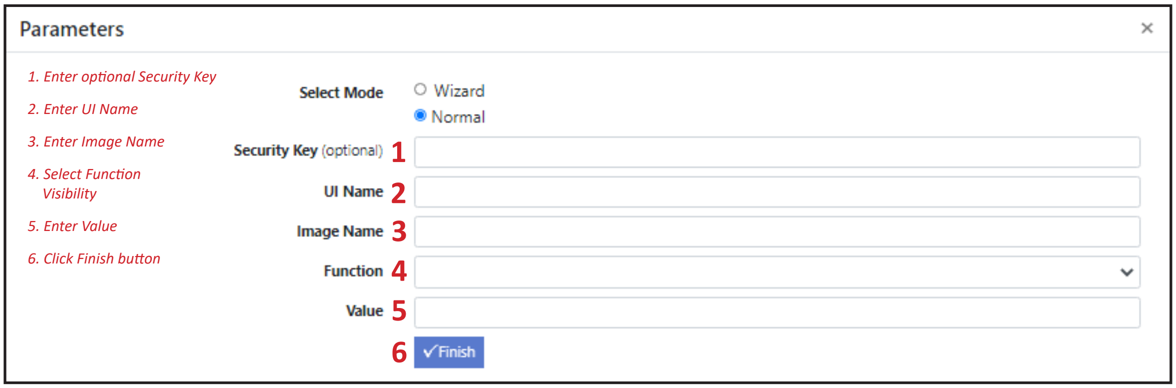


Command: set ui_label - Normal Mode

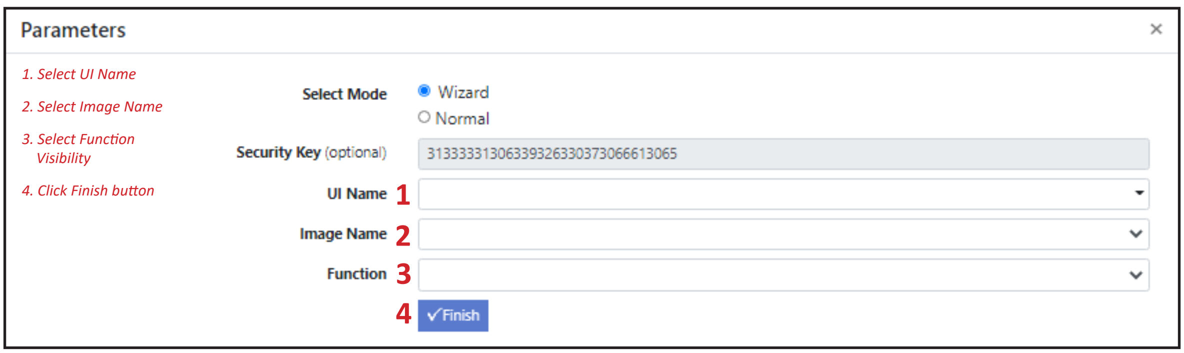
Command: set ui_label - Wizard Mode



Command: set ui_image - Normal Mode

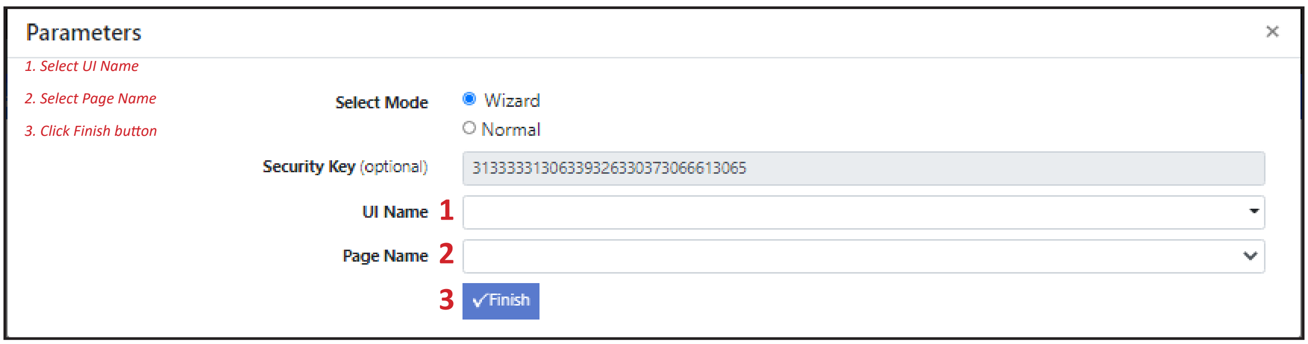


Command: set ui_image - Wizard Mode

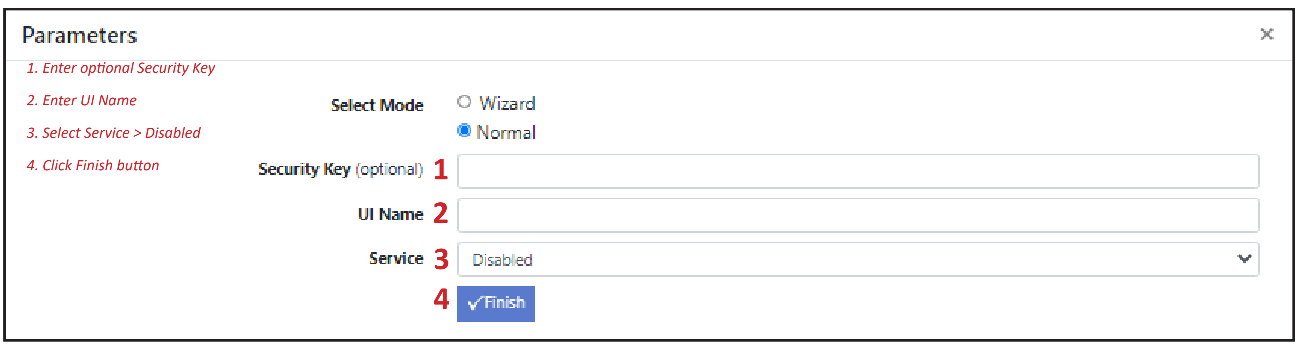


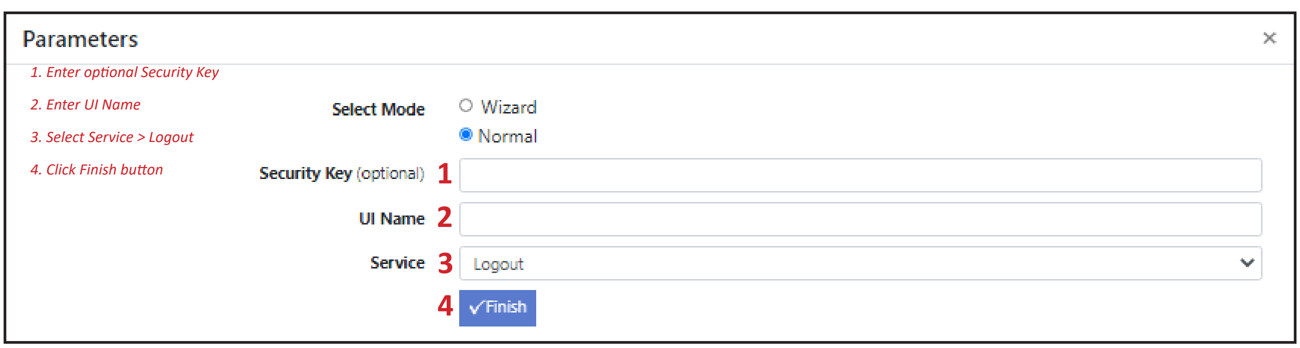
Command: set ui_page- Normal Mode

Command: set ui_page - Wizard Mode

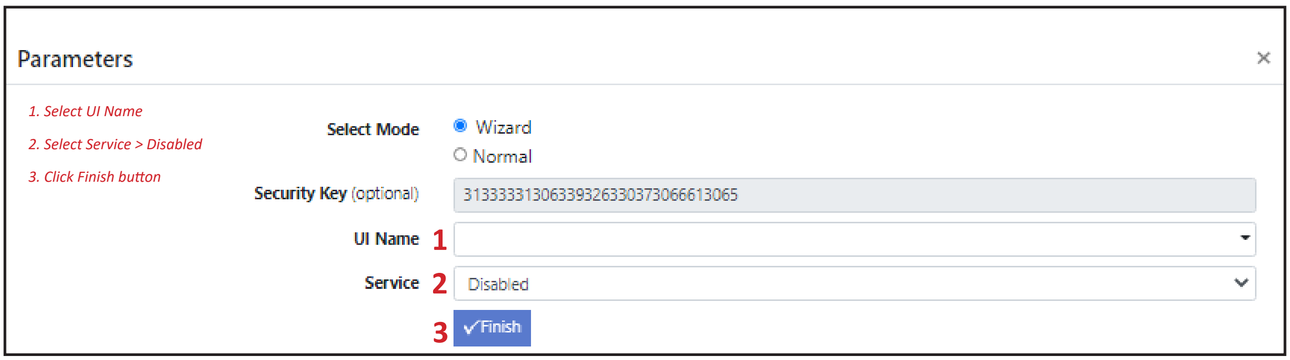


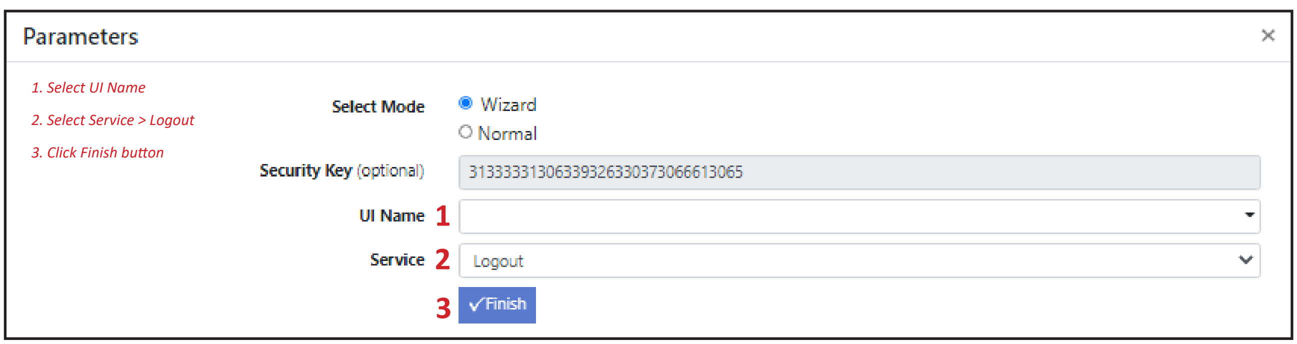
Command: set ui - Normal Mode





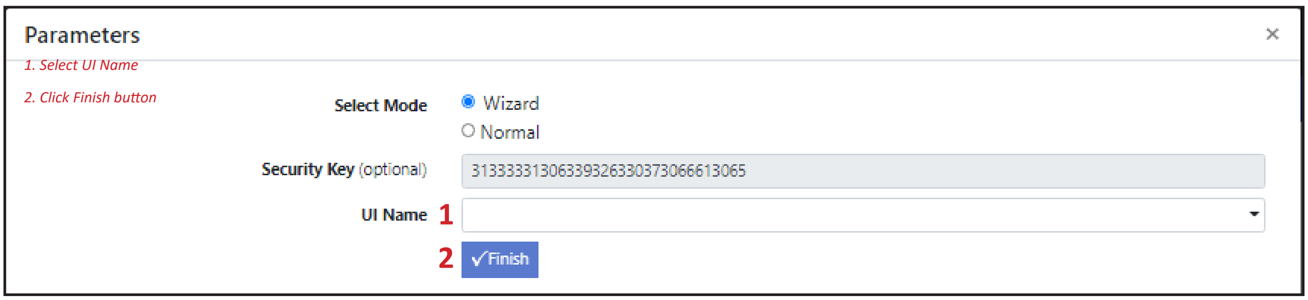
Command: set ui - Wizard Mode





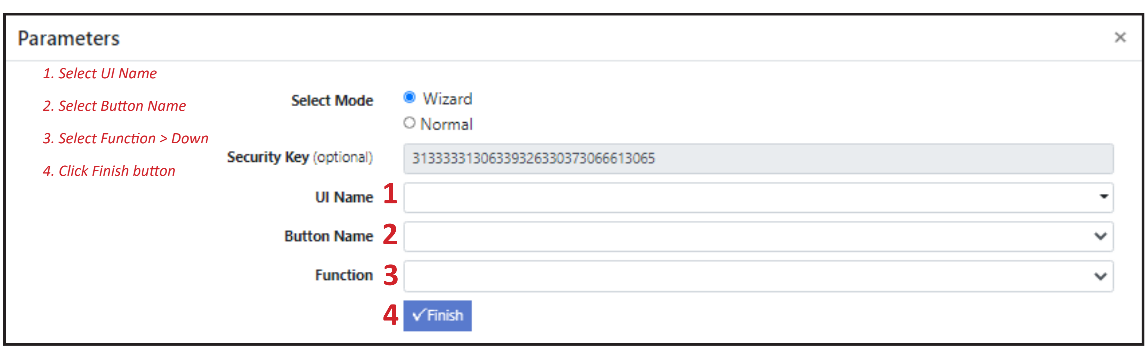
Command: get ui - Normal Mode

Command: get ui - Wizard Mode



Command: get ui_button - Normal Mode

Command: get ui_button - Wizard Mode



Command: Device Drivers

The assistant for all send commands provides a Device Driver selection allowing device commands to be stored as a driver for future use. The device command and feedback along with communication parameters are all saved.

The Device Driver selection consists in the following:

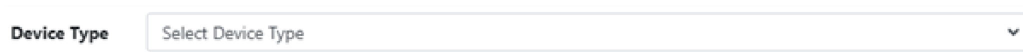
Device Type	Television
Device	LG
Function Type	Power
Function	ON

Create a Device Type entry as a manufacture's name or device type such as Matrix, TV and so on.

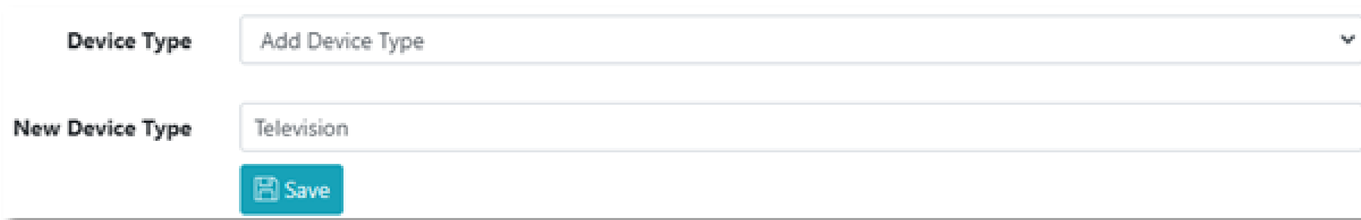
A Device is usually created as the model number of the device.

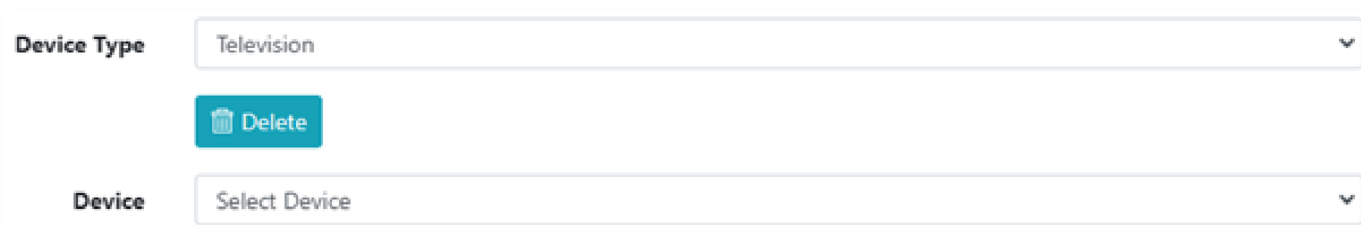
The Function Type is an operation such as Power, Source, Routing, Control and so on…

Arriving at Function types such as ON, OFF and so on…



- To add a device select Add Device Type and enter the New Device Type (Television) then Save.

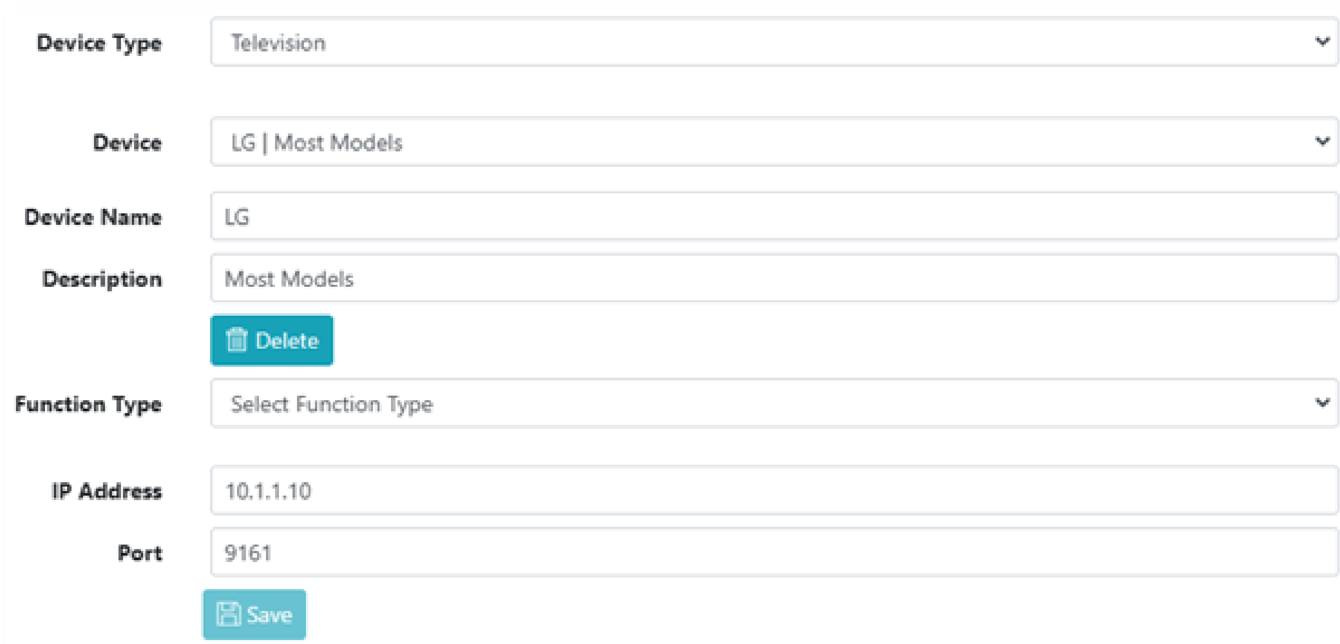




Command: Device Drivers

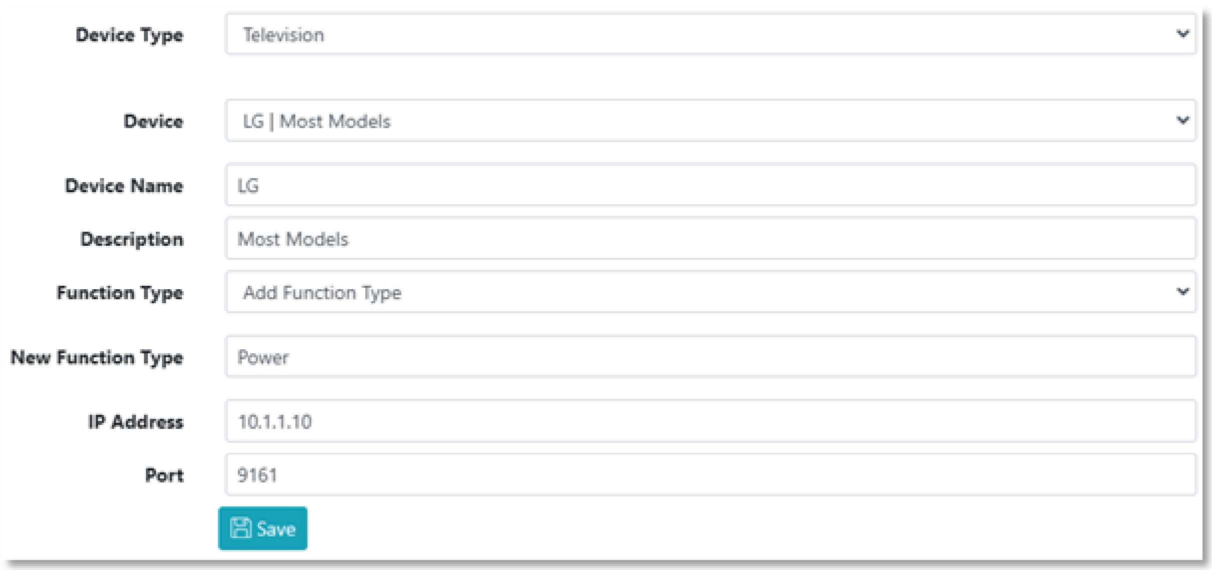
- *Select Add Device and enter a New Device (LG) along with Description (Most Models)*

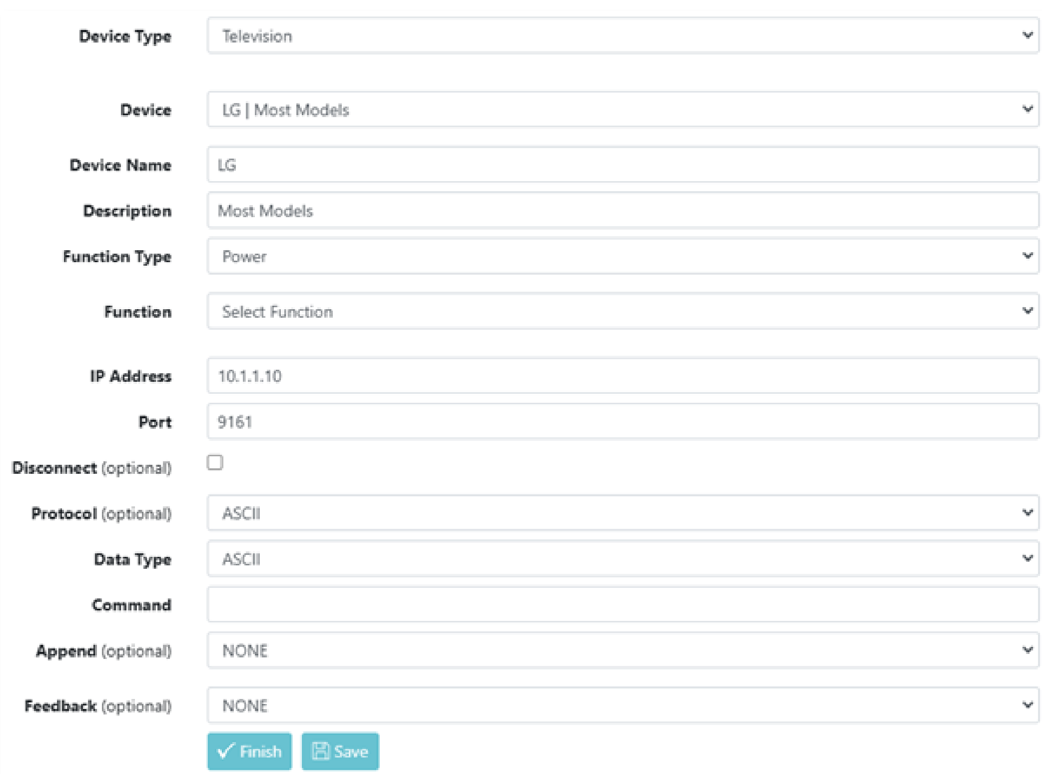
Default Terminators (CR) and IP Address (10.1.1.10) with Port (9161) then Save.



Command: Device Drivers

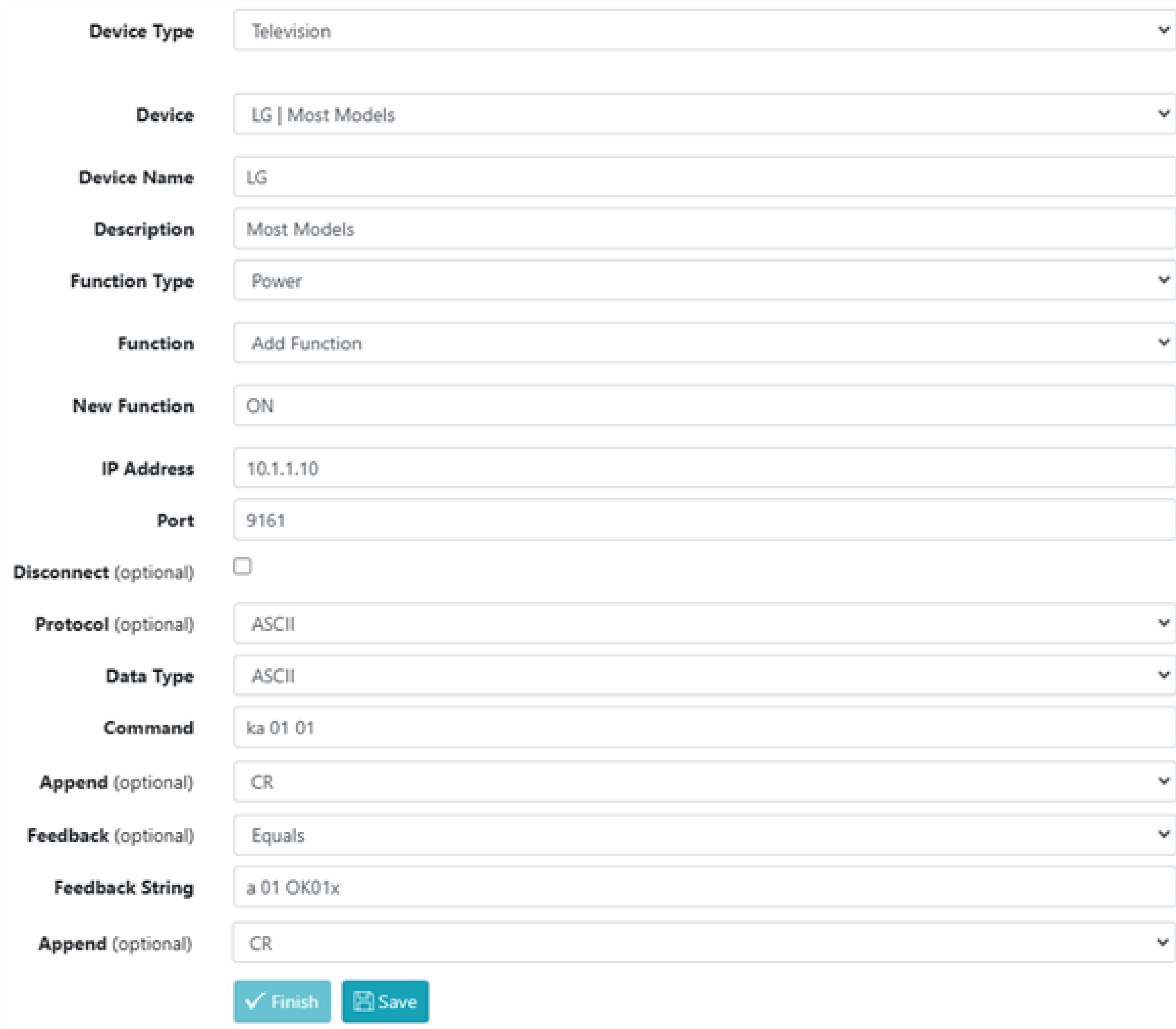
• Select Add Function Type and enter a New Function Type (Power) then Save.





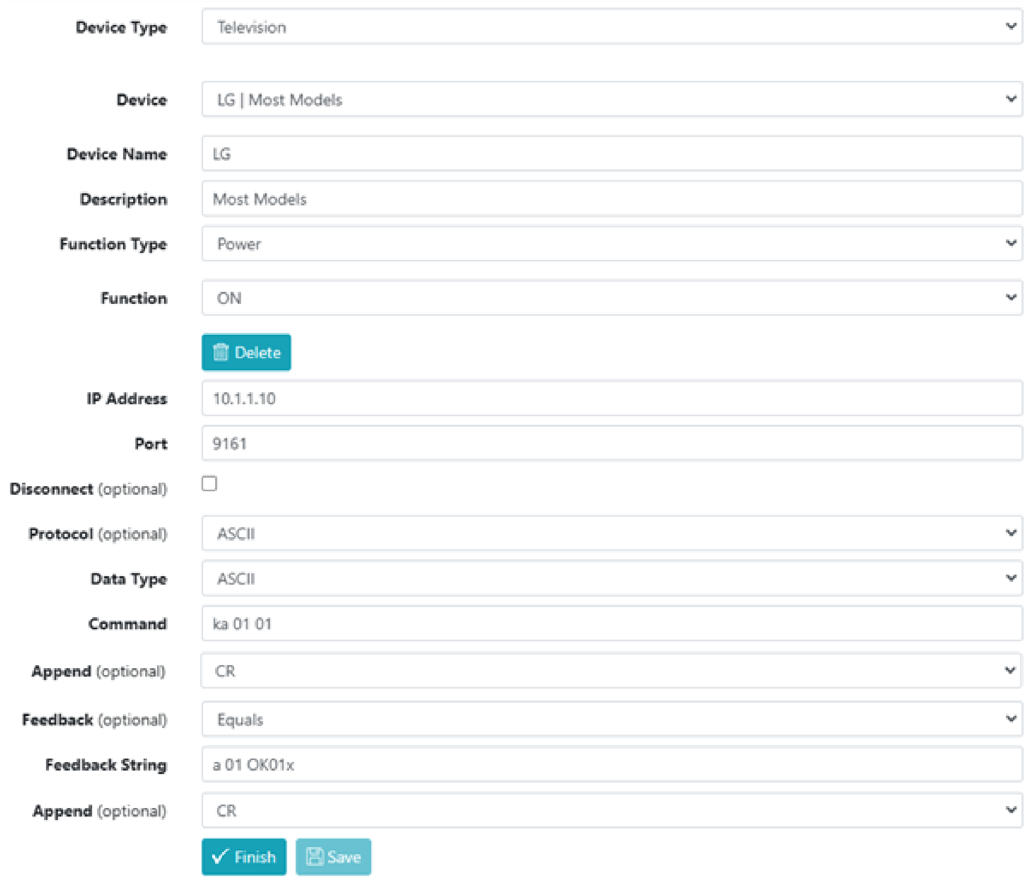
Command: Device Drivers

• Select Add Function enter a New Function (On) along with Command, Append and Feedback then Save.



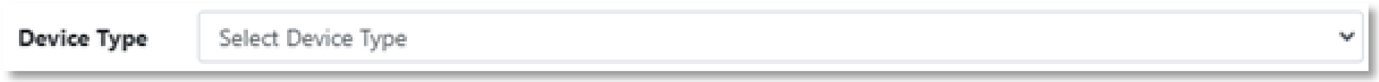
Command: Device Drivers

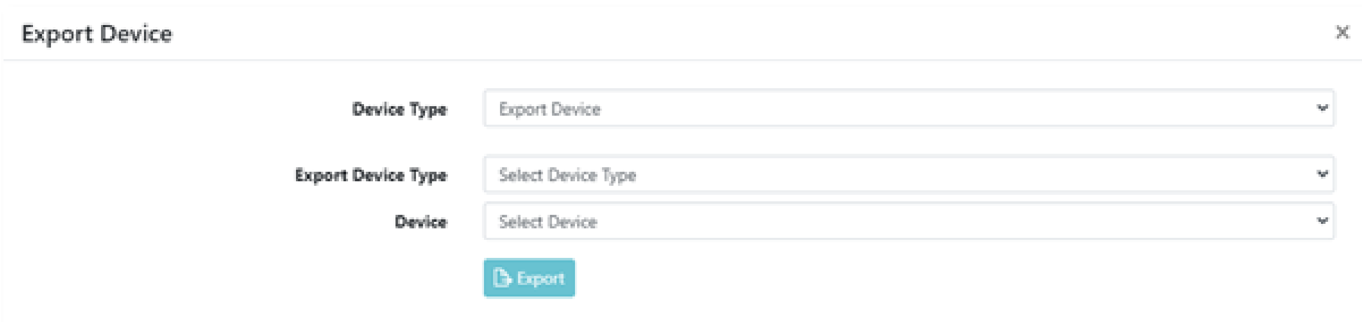
• Now the Function is saved into the Device and ready for use.



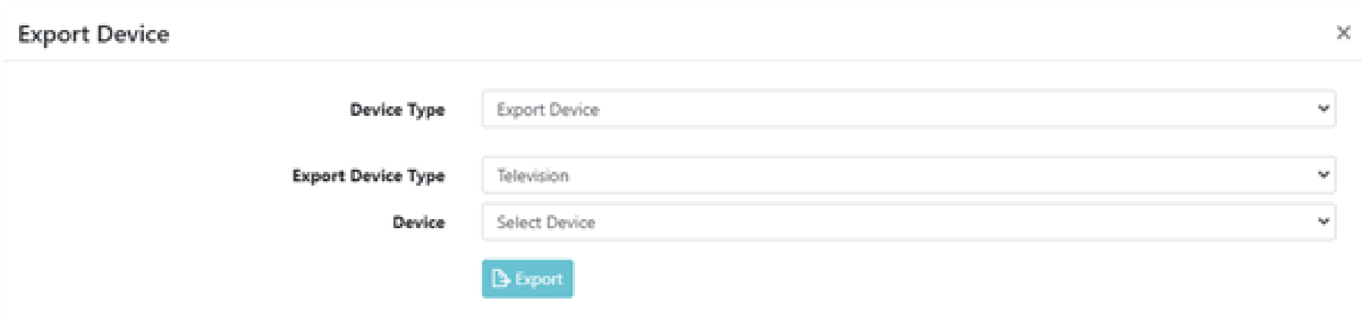
Command: Device Drivers

• To export a device from Device Type select Export Device.

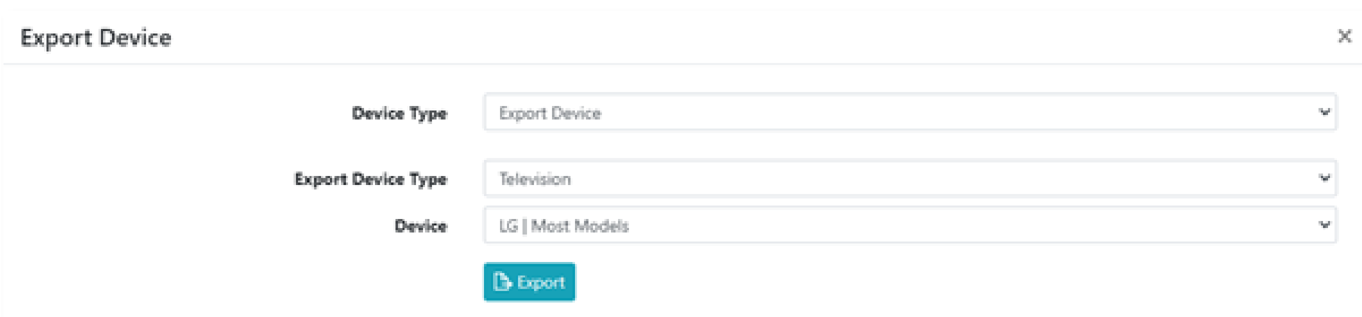




• From Export Device Type select Device Type (Television).

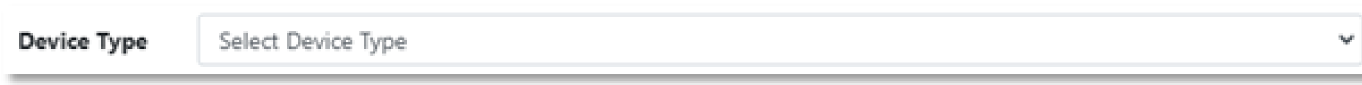


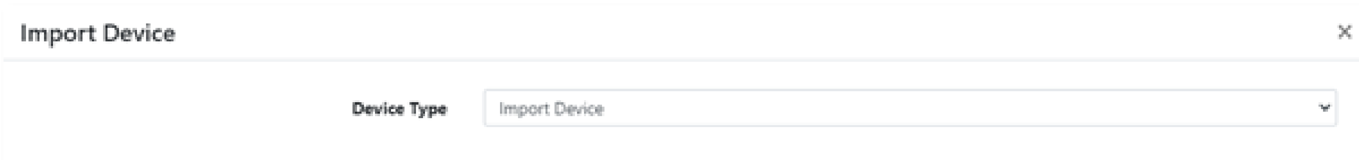
• From Device select Device (LG) and click Export.



Command: Device Drivers

• To import a device from Device Type select Import Device.





• • From the file dialog popup select the required driver *.exp file.
The driver will then be imported or if already exists a conformation popup will appear.

Liberty

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